The Binary Totient Series

Totient sections, rational comparisons and arithmetic conditions

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Abstract

Rationality of $S = \sum_{n\geq 1} \varphi(n)2^{-n}$ forces a positive shift for which every sufficiently late difference of scaled tails is an integer. We develop finite residue tests for a failure of this integrality and prove their completeness and propagation laws. The tests yield a finite denominator exclusion, but their required supply at arbitrarily large indices remains unproved. Rational comparison sequences explain why coefficient rank, finite data and specified analytic properties do not supply it. A first-harmonic route isolates the remaining weighted cancellation estimate; its group-mean bound is recorded with a prime-number-theorem input. The irrationality of S remains open.

Organisation

The argument starts from $2^N S = \Phi_N + R_N$, with $\Phi_N \in \mathbb{Z}$. A rational S gives a shift $h > 0$ for which $R_{N+h} - R_N \in \mathbb{Z}$ for every sufficiently large N. Three questions organise the record.

What finite calculation can refute that integrality? The residue test in Definition 1.2 encloses the tail difference in an explicit error interval. Section 5 gives the identities and the independent Farey-window exclusion. Appendix A supplies the complete propagation, Mersenne-residue and rank-one arguments cited in the short paper.

Which information is insufficient to supply such calculations? The short paper determines all finite-level totient relations, with a basis of $k^e + 1$ sections in every base $k \geq 2$. Its integral relation proof recovers the coefficients of every relation. Those are statements about sequences, not their weighted infinite sums. Section 10.8 explains the distinction through carry rank; Section 12.6 constructs a rational 5/4 series retaining the odd totients. Section 2 gives other comparisons, each limited to its stated preserved properties.

What additional estimate would suffice? Section 3 gives the quantified conditions. Section 12.3 separates the weighted exponential-sum estimate from the counting and group-mean bounds, corresponding to the four-estimate problem in the short paper. A second sufficient condition estimates only the average over good indices and needs no group-mean input. Appendix B records the formal declarations; an implication proved there does not establish its remaining hypothesis.

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1 Totient sections and tail differences

Put $\varphi(0) = 0$ and $\mathbb{N} = \{0, 1, 2, \ldots\}$. At base two, $\varphi(4n) = 2\varphi(2n)$ and $\varphi(4n+2) = \varphi(2n+1)$ leave five independent sequences among the seven sections through level two. More generally, the sections through level $e \geq 1$ have the basis $\varphi(n)$, $\varphi(2n)$ and $\varphi(2^j n + r)$ with $1 \leq j \leq e$ and odd $0 < r < 2^j$, of size $2^e + 1$. The all-base theorem in the short paper gives rank $k^e + 1$ and integral coordinates. Its retained positive residues satisfy $k \nmid r$; coprimality with k is unnecessary.

The coefficient relations are finite linear identities. The series question asks whether a particular *infinite* weighted sum can be rational. Put

$$S := \sum_{n \geq 1} \frac{\varphi(n)}{2^n} = \tfrac{1}{2} + \tfrac{1}{4} + \tfrac{2}{8} + \tfrac{2}{16} + \tfrac{4}{32} + \cdots = 1.3676308019850223\ldots$$

The series converges absolutely because $\varphi(n) \leq n$. Erdős Problem #249 asks whether S is irrational, which remains open. The comparisons below test proposed passages from coefficient structure to the value of this sum. Each retains specified properties while making the corresponding sum rational. It therefore defeats an argument using only those properties, not every argument that might also use the arithmetic of the totient.

Throughout, a cofinal condition means that a witness exists beyond each threshold. It need not hold at every sufficiently large index. Each comparison sequence is used only for the properties it preserves; agreement with a fixed totient prefix, for example, imposes no multiplicativity condition on the continuation.

Remark 1.1 (Earlier coefficient results). Erdős and Graham record the irrationality question for S [10, p. 61]. An integer sequence is k-regular when the $\mathbb{Z}$-module generated by its sections $n \mapsto c(k^j n + r)$, $j \geq 0$ and $0 \leq r < k^j$, is finitely generated. Coons proved that φ is not k-regular for any $k \geq 2$ [2, Theorem 3.2]. Martin's Theorem 1 implies that the functions $n \mapsto \varphi(a_i n + b_i)$ are linearly independent for pairwise nonproportional affine forms with positive integer slopes [4, Theorem 1]. The exact ranks in §10.8, together with the all-base normal form in the short paper, identify the retained sections and every relation among them. The independence itself is not claimed as new. The CRT–Dirichlet separation method is already present in Yazdani's Theorem 2 and proof, which explicitly credit Shallit [30, pp. 652–653]; our evaluation matrix uses one such separation for each column. Yazdani also proves non-automaticity of $\varphi(n)$ mod m for every $m \geq 3$ in every base [30, Cor. 4]. Allouche, Shallit and Yassawi give a survey account [31, Thm. 3, Ex. 4 and Rem. 7].

To test rationality, we remove the finite prefix, which becomes an integer after multiplication by 2^N. Write $2^N S = \Phi_N + R_N$ with $\Phi_N := \sum_{n=1}^{N} \varphi(n) 2^{N-n} \in \mathbb{N}$ and $R_N := \sum_{j \geq 0} \varphi(N+1+j)/2^{j+1} \in [0, \infty)$ (Lean source, Lean source). If $S = a/(2^c v)$ with $c \geq 0$ and $v \geq 1$ odd, Euler's theorem gives $h = \varphi(v) > 0$ and $N_0 = c$ such that $R_{N+h} - R_N \in \mathbb{Z}$ for every $N \geq N_0$ (Lean source). Indeed,

$$R_{N+h} - R_N \equiv 2^N (2^h - 1) S \pmod{\mathbb{Z}},$$

and the right side is integral for those choices.

Thus it suffices, for every positive shift h, to find arbitrarily large N for which $R_{N+h} - R_N$ is not an integer. A finite test is obtained by truncating this difference. Throughout, $a \bmod m$ denotes the least nonnegative residue, and an unqualified index in a sum is a nonnegative integer. For $h, N, L \in \mathbb{N}$, write

$$D(h,N,L) \;:=\; \sum_{j<L} \left(\varphi(N{+}h{+}1{+}j) - \varphi(N{+}1{+}j)\right) 2^{L-1-j} \in \mathbb{Z},$$

the depth-L truncation of $2^L(R_{N+h} - R_N)$, and

$$C(h,N,L) \;:\Longleftrightarrow\; (N{+}h{+}L{+}2) \;<\; D(h,N,L) \bmod 2^L \;<\; 2^L - (N{+}h{+}L{+}2),$$

a decidable condition (Lean source, Lean source).
The omitted tails satisfy the exact identity

$$2^L(R_{N+h} - R_N) - D(h,N,L) = R_{N+h+L} - R_{N+L}.$$

Both tails are nonnegative, so their difference is bounded in absolute value by the larger bound $B = N + h + L + 2$, not the sum of their bounds. The two strict inequalities in $C(h,N,L)$ keep the resulting error interval away from every multiple of 2^L; hence $R_{N+h} - R_N \notin \mathbb{Z}$. Conversely, a fixed noninteger difference has a fixed positive distance δ from $\mathbb{Z}$. Since $2^L\delta > 2B$ eventually, some depth detects it. Completeness concerns detection of an existing difference, not existence of the differences required below (Lean source). The resulting quantified condition is the following.

Definition 1.2 (The quantified residue condition)**.**

$$\mathrm{Sep} \;:\Longleftrightarrow\; \forall h \geq 1\; \forall N_0\; \exists N \geq N_0\; \exists L,\; C(h,N,L).$$

$\mathrm{Sep} \iff S \notin \mathbb{Q}$ (Lean source). The equivalence identifies the missing quantified input; it supplies no certificate by itself.

One successful (h,N,L) excludes reduced denominators dividing $2^N(2^h - 1)$. To exclude an arbitrary rational value, its odd denominator first determines a shift and its power of two determines a threshold; only then may we choose the witness N, L. This is the order in Sep. The depth must grow with N, since success requires $2(N+h+L+2) < 2^L$. A large enough depth makes the interval test possible, but does not make the actual residue lie inside it. A failed test at one depth likewise does not prove integrality. Lean source

1.1 Unconditional results

The finite denominator exclusion, the ranks of the totient sections and the tail-positivity statements below require no additional arithmetic hypothesis. To distinguish their conclusions, note that positivity excludes a zero tail difference, whereas irrationality requires a nonintegral one. The rank calculation concerns the coefficients and supplies neither conclusion about the value by itself.

Theorem 1.3 (Denominator exclusion from a fixed Farey window)**.** *If $S \in \mathbb{Q}$ then its reduced denominator exceeds*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

Equivalently, S differs from every rational number whose reduced denominator is at most Q_0. The bound is sharp for this window: $q = Q_0 + 1$ is the exact first failing denominator. It is the denominator of the mediant of two explicit unimodular Farey neighbours.

This is obtained from a classical Stern–Brocot gap lemma (Lean source) applied at window $K = 240$. Of the recorded bounds 4838, 2^{22}, approximately 2.49×10^{17}, and Q_0, this last is the largest. It is proved without assuming Sep or using the tail-residue tests. To deduce irrationality by this argument, the analogous denominator bounds would have to be unbounded as the window grows; this has not been proved.

Proposition 1.4 (Finite certificate computations)**.** *$\mathbb{C}$ has been verified at: the 28 diagonal instances of the least-common-multiple diagonal through $t = 64$; all shifts $h \in [1, 16]$ simultaneously at $(N, L) = (14, 9)$, by* `decide`*; and eight further period examples at $N = 300$. These are historical subsets of the examples. The supplied source also proves a complete diagonal band for $1 \le t \le 82$, with no omitted scales. This is bounded coverage, not a cofinal diagonal family. Note the quantifier order: the simultaneous small-shift example uses one pair N, L for every $1 \le h \le 16$. In symbols, it proves $\exists N\, \exists L\, \forall h\, (1 \le h \le 16 \Rightarrow \mathbb{C}(h, N, L))$. In contrast,* Sep *requires arbitrarily large N for each positive h; neither the shift range nor the basepoint threshold is bounded.*

Proposition 1.5 (Positivity does not exclude an integer)**.** *Put $H_a = \operatorname{lcm}(1, \dots, 2^a)$. For every $a \ge 8$ and $J \ge 0$ with $J + (a + 6) < 2 \cdot 2^a$,*

$$R_{2H_a+J} - R_{H_a+J} > 0.$$

This needs no irrationality hypothesis. If this difference is an integer and $K \ge 0$ satisfies

$$J + K + (a + 6) < 2 \cdot 2^a, \qquad 2H_a + J + K + 2 < 2^K,$$

then

$$D(H_a, H_a + J, K) \bmod 2^K = 2^K - (R_{2H_a+J+K} - R_{H_a+J+K})$$

lies strictly between $2^K - (2H_a + J + K + 2)$ and 2^K. The later tail difference is a positive integer. Its negative, not the carry itself, is the representative near zero. Thus positivity locates the residue near the upper endpoint; it does not establish the central-residue inequalities. Both restrictions on K are part of this conclusion.

For the next two propositions, a rational value $S = a/v$, with integers a and $v \ge 1$, would give the integer carry

$$u_N = vR_N = 2^N a - v\Phi_N, \qquad u_{N+1} = 2u_N - v\varphi(N + 1), \qquad 0 \le u_N \le v(N + 2).$$

Here v is any positive integer clearing the denominator, so it need not be odd. The linear bound implies $u_N/2^N \to 0$. The dyadic sections of u are the sequences $n \mapsto u_{2^j n+r}$.

Proposition 1.6 (Rationality forces unbounded carry rank)**.** *If S is rational then, for every e, the carry sections $n \mapsto u_{2^j n+r}$ with $1 \le j \le e$ and $0 \le r < 2^j$ span a rational vector space of dimension at least $2^e - 1$. The lower bound holds at every depth. It comes from the linear independence of the $2^e + 1$ retained dyadic totient sections for $e \ge 1$, proved using the Chinese remainder theorem and Dirichlet's theorem, so the full family spans an infinite-dimensional space, the case $k = 2$ of Coons's non-regularity theorem (§10.8).*

Proposition 1.7 (Periodicity modulo an integer does not bound rational rank)**.** *If S is rational, the same carry u has unbounded rational section rank and one eventual period modulo v valid for all its dyadic sections. More precisely, there are $h \ge 1$ and N_0 such that, for every $j, r \ge 0$ and $n \ge N_0$,*

$$u_{2^j(n+h)+r} \equiv u_{2^j n+r} \pmod{v}.$$

The rank and periodicity assertions hold together. This conditional theorem alone is not a counterexample to a general periodicity-to-rank implication: its antecedent is not established. The rational comparison sequence with sum $5/4$ gives a counterexample to a rank upper bound based on rationality alone; see Section 10.8.

Proposition 1.8 (Equivalent certificate conditions)**.** *Put $H_t = \operatorname{lcm}(1, \dots, t)$. Several variations of Definition 1.2 are equivalent to $S \notin \mathbb{Q}$. One may allow a positive multiple of each prescribed shift, still requiring certificates beyond every basepoint threshold (Theorem 6.48). Alternatively, one may use the original quantified condition or restrict to $h = N = H_t$ at arbitrarily large t (Theorem 6.49). Replacing the symmetric residue test by*

$$N + L + 2 \le D(h, N, L) \bmod 2^L \le 2^L - (N + h + L + 2)$$

also gives equivalent quantified conditions, including its restriction to $h = N = H_t$ at arbitrarily large t. Finally, the counted phase-separation condition in Proposition 2.7(b) is equivalent to irrationality. These equivalences change the form of the arithmetic question, not its logical strength.

Proposition 1.9 (A rational sequence preserving size bounds, parity and aperiodicity)**.** *There is $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \le 6$ and $c(n) \le n$ for all n, $c(n) \equiv \varphi(n) \pmod 2$ for every n, and c not eventually periodic; indeed for every N, G, K there are K pairs of coefficients $(6, 0)$ starting beyond N, with starting positions pairwise separated by more than G; and $\sum_n c(n)/2^n = 3/2 \in \mathbb{Q}$.*

These propositions separate several logical obligations. Positivity leaves an upper-endpoint condition; a large rational carry rank is not a contradiction; and exact equivalences still require a quantified certificate condition. The stated size bounds, parity and aperiodicity also hold for a rational comparison series. Each conclusion concerns its stated hypotheses, not all possible arithmetic information about φ.

1.2 Reduced coefficients

Reduction modulo m restores the boundedness missing from φ. The short paper isolates a prescribed nonzero coefficient between two long zero blocks: a hypothetical rational

denominator forces the right tail to vanish, after which the left block makes the nonzero centre too small. CRT creates the blocks by imposing a separate prime divisor congruent to $1 \pmod m$ at each neighbour. We extend that proof in two directions. Changing the integer base changes only the geometric tail bound. Combining dilations is subtler: one must first ensure that their contributions cannot cancel at the chosen centre.

Proposition 1.10 (Residue series in every integer base)**.** *Let $t \geq 2$ be an integer.*

(a) *For every $m \geq 3$, $\sum_{n\geq 1}(\varphi(n) \bmod m)\, t^{-n}$ is irrational* (`radix_residue_series_irrational`).

(b) *For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \to \mathbb{Q}$, the series $\sum_{n\geq 1} f(\varphi(n) \bmod 2^k)\, t^{-n}$ is rational exactly when f is constant on the even residue classes, and its value is then $(t+1)f(1)/t^2 + f(0)/(t^2(t-1))$* (`rational_zmod_radix_observable_iff`, `positiveRadixValue_eq_of_even_constant`).

Proof. For coefficients bounded by C, the right-block tail is at most $Ct^{-L}/(t-1) \leq Ct^{-L}$. Choose L with $Ct^{-L} < 1/q$; the rational-tail lattice forces that tail to be zero. The left-block tail is then the isolated coefficient times t^{-L-1}, giving the same contradiction as in the short paper. In (b), the two exceptional terms contribute $(t+1)f(1)/t^2$ and the constant tail contributes $f(0)/(t^2(t-1))$, since all totients after $n = 2$ are even. □

For example, in base 2 the function modulo 4 with $f(0) = f(2) = 5, f(1) = 7$ and $f(3) = -2$ gives the rational value $3 \cdot 7/4 + 5/4 = 13/2$: the class 3 never occurs. In contrast, the indicator of the class $2 \pmod 4$ gives an irrational value. This classification uses the fact that the prime centres $s - 1$, with s a unit, cover every even class modulo 2^k. Modulo 6 they cover only 0 and 4. The least-residue assertion needs one nonzero accessible centre, but the same argument does not classify all rational-valued functions for a general modulus.

Proposition 1.11 (Finitely many dilations)**.** *Let $t \geq 2$ be an integer.*

(a) *For every $m \geq 3$, the numbers 1 and $\sum_{n\geq 1}(\varphi(n) \bmod m)\, t^{-dn}$, $d = 1, 2, \ldots$, are linearly independent over $\mathbb{Q}$* (`linearIndependent_one_and_least_residue_values`).

(b) *Let D be a finite set of positive integers, and for $d \in D$ let $k_d \geq 1$ and $f_d : \mathbb{Z}/2^{k_d}\mathbb{Z} \to \mathbb{Q}$. For every $c \in \mathbb{Q}$, the number*

$$c + \sum_{d\in D} \sum_{n\geq 1} f_d(\varphi(n) \bmod 2^{k_d})\, t^{-dn}$$

is rational exactly when every f_d is constant on the even residue classes (`rational_mixed_moduli_with_constant_iff`).

Proof. (a) A relation gives integers c_0 and c_d, d in a finite set D of positive integers, with some $c_d \neq 0$ and $c_0 + \sum_{N\geq 1} b_N t^{-N} = 0$, where $b_N = \sum_{d\in D,\, d\mid N} c_d(\varphi(N/d) \bmod m)$ is bounded. Discard the d with $c_d = 0$ and let δ be the least remaining one. Different dilations can contribute to the same exponent N, so isolating a prime in one summand is not enough. Choose instead $N = \delta p$ with $p > \max D$ prime. A contributing d divides

δp; since $p \nmid d$, it divides δ. Minimality then forces $d = \delta$. The centre now receives exactly one active contribution, while CRT will remove all contributions at its neighbours.

Given L, choose distinct primes $q_j \equiv 1 \pmod{m}$, $0 < |j| \le L$, each larger than L and than every element of D. Impose

$$p \equiv -1 \pmod{m}, \qquad p \equiv -j\delta^{-1} \pmod{q_j} \quad (0 < |j| \le L).$$

The inverse exists because $q_j > \delta$, and the latter residue is nonzero because $q_j > |j|$. Together with the unit residue -1 modulo m, these congruences define a reduced progression. CRT and Dirichlet's theorem give a prime p as large as needed. At the centre, $b_{\delta p} = c_\delta(m-2) \neq 0$. At a neighbour $\delta p + j$, every contributing dilation d is coprime to q_j, so $q_j \mid (\delta p + j)/d$ and $m \mid \varphi((\delta p + j)/d)$. Hence $b_{\delta p + j} = 0$. The isolated coefficient makes $\sum_N b_N t^{-N}$ irrational, contradicting $c_0 \in \mathbb{Z}$.

(b) Each f_d that is constant on the even classes contributes a rational number, by Proposition 1.10(b) in base t^d. Remove these contributions, lift the remaining functions to one modulus 2^K, subtract their values at zero, and clear a common denominator. None of these operations changes rationality. For the least remaining dilation δ, choose an even residue r with $f_\delta(r) \neq f_\delta(0)$. Use the construction in (a), now with $q_j \equiv 1 \pmod{2^K}$ and $p \equiv r + 1 \pmod{2^K}$. The central residue is allowed because $r + 1$ is odd. Minimality again leaves just the nonzero contribution from δ at the centre, while every neighbouring totient is zero modulo 2^K and hence contributes zero after the subtraction. The same bounded integer-coefficient argument applies. □

Remark 1.12 (Slowly growing dyadic moduli). Let $\kappa : \mathbb{N} \to \mathbb{N}$ satisfy $2^{\kappa(n)} = o(\log n)$, and put $a_n = \varphi(n) \bmod 2^{\kappa(n)}$. Then $\sum_{n \ge 1} a_n 2^{-n}$ is rational exactly when $a_n = 0$ for all large n. In particular the sum is irrational when $\kappa(3^k) \ge 2$ for infinitely many k.

Ordinary proof; not formalised in Lean.

Proof. If a_n vanishes eventually the sum is finite. Suppose instead that $a_n \neq 0$ infinitely often and that the sum equals u/v. Then $T_N = \sum_{j \ge 1} a_{N+j} 2^{-j}$ lies in $v^{-1}\mathbb{Z}$ for every N. For $1 \le j \le L$ let q_j be a prime factor of the Fermat number $2^{2^j} + 1$; then $q_j \equiv 1 \pmod{2^{j+1}}$, and the q_j are distinct because distinct Fermat numbers are coprime. Put $Q = q_1 \cdots q_L$, so $Q < 2^{2^{L+1}}$. These factors give a size bound as well as the needed divisibility: $\log Q = O(2^L)$. The factor 2^{-L} in the remaining tail will offset this growth, leaving the arbitrarily small constant from $2^{\kappa(n)} = o(\log n)$. Given M, the Chinese remainder theorem gives $N \in [M, M + Q)$ with $q_j \mid N + j$ for $1 \le j \le L$. Then $2^{j+1} \mid \varphi(N + j)$, so $2^j \mid a_{N+j}$: either $\kappa(N+j) \le j$ and $a_{N+j} = 0$, or 2^j divides both $\varphi(N+j)$ and the modulus. Hence T_N differs from an integer by $\sum_{j>L} a_{N+j} 2^{-j}$, which is positive because $a_n \neq 0$ infinitely often, and at most $\sum_{j>L} 2^{\kappa(N+j)-j}$. Choose $\epsilon > 0$ with $3\epsilon < 1/v$, then M with $2^{\kappa(n)} \le \epsilon \log n$ for $n \ge M$, then $L \ge 2$ so large that $Q \ge M$. Then $N < M + Q \le 2Q$ and $\log(N + j) \le \log(2Q) + j \le 2^{L+1} + j$, so the tail is at most $\epsilon 2^{-L}(2^{L+1} + L + 2) \le 3\epsilon < 1/v$. It is a positive element of $v^{-1}\mathbb{Z}$ below $1/v$, a contradiction. Finally $\varphi(3^k) = 2 \cdot 3^{k-1}$ is not divisible by 4, so $a_{3^k} \neq 0$ whenever $\kappa(3^k) \ge 2$. □

Remark 1.13 (Earlier results and the reduced coefficients). Proposition 1.10(a), and with it the first assertion of the short paper's residue theorem, already follows from Erdős's Lemma 1 of [6, p. 213]: if integers $a_k \geq 0$ have $\sum_{k\leq n} a_k = O(n)$, infinitely many $a_k \neq 0$, and $\liminf_n \#\{k \leq n : a_k > 0\}/n = 0$, then $\sum_k a_k t^{-k}$ is irrational for every integer $t \geq 2$. Erdős notes that this is the case $b_k = 0$ of his Lemma 4 (stated pp. 215–216, proved pp. 216–218). For $a_n = \varphi(n) \bmod m$ the hypotheses hold: $a_n \neq 0$ forces n to have no prime factor $p \equiv 1 \pmod m$, and such n have density zero because $\sum_{p\equiv 1 \ (m)} 1/p$ diverges, while $a_p = m - 2$ for every prime $p \equiv -1 \pmod m$. Erdős's Lemma 4′ (p. 218, proof omitted there) and Theorem 3 of Kaneko, Suzuki and Tachiya [7, p. 5] give the same conclusion; the latter paper proves Lemma 4′ as its Theorem A (pp. 2 and 18). The same argument covers Proposition 1.10(b) and Proposition 1.11 when the values $f(r) - f(0)$, respectively the coefficients c_d, have one sign. With both signs present, Lemmas 4 and 4′ and Theorem 3 of [7] require an additional spacing condition on the negative terms, which is not checked here; the isolated coefficient needs none. Theorem 3 of [7], with $b = 0$ and $z_n = \log_2 \log x_n$, also gives the irrational direction of Remark 1.12 once the $n < x$ with $v_2(\varphi(n)) < \log_2 \log x$ are shown to number $o(x/\log\log x)$; this record does not write out that density estimate. Wong's base-equals-modulus theorem [5] is the case $t = m$ of Proposition 1.10(a).

2 Rational comparison sequences

Rational series can agree with the totient series on a fixed prefix, on parity at every index, or on specified linear data. The following constructions keep these requirements separate. Each gives a rational comparison satisfying the data named in its statement, and thus identifies what those data alone leave undetermined. The properties in different rows below belong to different constructions; they must not be combined into one comparison theorem.

Construction	**Data kept while the value is rational**
Prescribed prefix (Theorem 2.4)	Any fixed totient prefix and the bound $0 \leq c(n) \leq n$.
Parity (Proposition 1.9)	Parity at every index, aperiodicity and the stated size bound.
Value 5/4 (Section 12.6)	Every odd totient, changes of at most 2 at even indices, and unbounded carry rank.
Signed interpolation (Section 10.10)	A prefix, all odd totients, and the specified moment and analytic data; not the full radial germ or all arithmetic identities.

2.1 Finite data and rational continuations

We begin with a rational continuation of an arbitrary finite prefix (Theorem 2.4). Proposition 1.9 instead preserves parity at every index, a linear size bound and aperiodicity. The carry and linear descriptions in Propositions 2.11 and 2.12 impose different require-

ments again. Their counterexamples concern the displayed descriptions only; they do not exclude all finite-state or finite-dimensional arguments.

Two other limitations do not require a rational comparison sequence. A congruence construction can increase the index so much that its tail error exceeds the residue it produces (Proposition 2.9). An exact reformulation or a finite list of examples still needs the quantified arithmetic assertion (Proposition 2.7).

The finite certificates in Proposition 1.4 and the $K = 240$ Farey calculation in Theorem 1.3 have bounded ranges. Theorem 2.4 excludes an inference from one such fixed range uniformly over the whole coefficient class. It does not exclude a proof using the arithmetic of φ at arbitrarily large horizons. Likewise, the cofinal positivity theorem does not by itself separate the residue from the top edge: that is the additional information identified by Proposition 1.5.

2.2 A rational sequence with any prescribed totient prefix

A finite prefix leaves the remaining coefficients free. We construct a rational continuation satisfying the same linear size bound, then show why it cannot satisfy the quantified residue condition. First we record the two tail facts needed for a general coefficient sequence. For $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \le n$, write $T_c := \sum_{n\ge1} c(n)/2^n$, $R_N^c := \sum_{j\ge0} c(N{+}1{+}j)/2^{j+1}$, $D_c(h,N,L) := \sum_{j<L}(c(N{+}h{+}1{+}j){-}c(N{+}1{+}j))2^{L-1-j}$, and let $C_c(h,N,L)$ and Sep_c be C and Sep with D_c in place of D. For $c = \varphi$ these are the objects of Definition 1.2.

Lemma 2.1 (A residue certificate excludes an integral tail difference)**.** *Let $c : \mathbb{N} \to \mathbb{N}$ satisfy $c(n) \le n$ for all n. For $h, N, L \in \mathbb{N}$, $C_c(h,N,L) \Rightarrow R^c_{N+h} - R^c_N \notin \mathbb{Z}$.*

Proof. Splitting both tails after L terms gives

$$2^L(R^c_{N+h} - R^c_N) - D_c(h,N,L) = R^c_{N+h+L} - R^c_{N+L}.$$

Both tails on the right are nonnegative, and $R^c_M \le \sum_{j\ge1}(M+j)2^{-j} = M+2$. Their difference therefore has absolute value at most $N+h+L+2$. The corresponding totient bound is formalised at Lean source and used there to prove Lean source. If $R^c_{N+h}{-}R^c_N = k \in \mathbb{Z}$ then $2^Lk \equiv 0 \pmod{2^L}$ and D_c lies within $N{+}h{+}L{+}2$ of 2^Lk, so $D_c \bmod 2^L$ lies within $N{+}h{+}L{+}2$ of 0 or of 2^L, contradicting C_c. □

Lemma 2.2 (Generic tail-period law)**.** *Let $c : \mathbb{N} \to \mathbb{N}$ satisfy $c(n) \le n$ for all n. If $T_c = p/(2^em)$ with $p \in \mathbb{Z}$, $e \ge 0$ and m a positive odd integer, and if $h \ge 1$ satisfies $m \mid 2^h - 1$, then $R^c_{N+h} - R^c_N \in \mathbb{Z}$ for every $N \ge e$.*

Proof. Since $2^NT_c - R^c_N = \sum_{n=1}^N c(n)2^{N-n}$ is an integer,

$$R^c_{N+h} - R^c_N \equiv 2^N(2^h-1)T_c \pmod{\mathbb{Z}}.$$

The right side is integral because $2^e \mid 2^N$ and $m \mid 2^h - 1$. □

Remark 2.3. Lemmas 2.1 and 2.2 are also checked for every coefficient sequence satisfying $c(n) \le n$. The formal proofs use the true-tail recurrence and the bound $0 \le R^c_N \le$

$N+2$; the residue certificate keeps the full finite discrepancy. The tail-period proof uses $m \mid 2^h - 1$ directly, so oddness of m is only needed when choosing a period from the denominator. The following rational continuation and its finite-prefix consequence are also checked.

Theorem 2.4 (A rational sequence agreeing with any finite totient prefix)**.** *Let $B \geq 1$ and $P > B$ be integers. Define $\gamma : \mathbb{N} \to \mathbb{N}$ by*

$$\gamma(n) := \varphi(n) \ \ (n \leq B), \qquad \gamma(n) := \begin{cases} n-1, & P \mid n \\ n, & P \nmid n \end{cases} \ \ (n > B).$$

Then:

(*i*) *$\gamma(n) \leq n$ for all n, so γ lies in the same coefficient class as φ;*

(*ii*) *$\gamma(n) = \varphi(n)$ for every $n \leq B$, and consequently $D_\gamma(h,N,L) = D_\varphi(h,N,L)$ for every (h,N,L) with $N + h + L \leq B$;*

(*iii*) *$T_\gamma = 2 - \sum_{n \leq B}(n - \varphi(n))/2^n - 1/(2^P - 1) \in \mathbb{Q}$, and the odd part of its reduced denominator is exactly $2^P - 1$;*

(*iv*) *Sep_γ is false; indeed it fails already at $h = P$.*

Hence for every B there is a coefficient sequence in the same class, agreeing with φ on all of $[1, B]$, whose quantified condition is false.

Proof. (i) is immediate; (ii) holds because $D_\varphi(h,N,L)$ reads φ only at indices $\leq N+h+L$. For (iii), since $P > B$ every multiple of P exceeds B, so

$$T_\gamma = \sum_{n \geq 1} \frac{n}{2^n} - \sum_{n \leq B} \frac{n - \varphi(n)}{2^n} - \sum_{k \geq 1} 2^{-kP} = 2 - \sum_{n \leq B} \frac{n - \varphi(n)}{2^n} - \frac{1}{2^P - 1}.$$

The first two terms have denominator dividing 2^B. On putting the expression over $2^B(2^P - 1)$, its numerator is congruent to -2^B modulo $2^P - 1$. Since $\gcd(2^B, 2^P - 1) = 1$, no odd factor cancels. Thus the reduced denominator has odd part exactly $2^P - 1$ and its power-of-two part divides 2^B. For (iv), apply Lemma 2.2 with $h = P$: $R^\gamma_{N+P} - R^\gamma_N \in \mathbb{Z}$ for every $N \geq B$, so by Lemma 2.1 no $C_\gamma(P,N,L)$ holds for any $N \geq B$ and any L. Taking $N_0 = B$ refutes the inner existential of Sep_γ at $h = P$. □

Corollary 2.5 (The limit of a finite-prefix argument)**.** *No proof rule uniform over all $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \leq n$ can establish* Sep *from a single fixed prefix $\{c(n) : n \leq B\}$: Theorem 2.4 supplies a rational countermodel with that same prefix. This does not invalidate an argument that uses the fixed arithmetic sequence φ together with compatible information at arbitrarily large horizons; the theorem gives a different γ_B for each B, not one sequence agreeing with φ at every B.*

Remark 2.6 (Four distinct finite restrictions). Theorem 2.4 concerns the information in a finite prefix. It does not identify φ with one of the comparison sequences. It also does not exclude a proof that combines finite computation with an arithmetic theorem applying at all larger indices. Four different restrictions must be kept separate: a bound

on the indices inspected, a bound on the depth L, a bound on the carry states retained, and a bound on matrix rank. The depth inequality $2(N + h + L + 2) < 2^L$, for example, concerns the second restriction, not all four.

Proposition 1.9 gives another rational comparison sequence, this time preserving parity and specified aperiodicity properties. The two examples test different assumptions and must not be combined into a single stronger example.

2.3 Limits of specific reductions and estimates

We test several proposed deductions by retaining their assumptions and changing the conclusion. The constructions concern specified congruences, recurrences or estimates, so their force is limited to the corresponding deductions.

Proposition 2.7 (Three particular equivalences)**.** (*a*) *Certificate completeness. The complete residue tests considered here are equivalent to the corresponding nonintegrality assertions. Rewriting the quantified condition using one of these equivalences does not weaken it.*

(*b*) *A selectable two-point sample. The condition is as follows. For every $h \geq 1$ and X_0, choose $X, L \in \mathbb{N}$ with $X \geq \max(X_0, 1)$ and $16(2X + h + L + 2) \leq 2^L$, a nonempty set $T \subseteq [X, 2X) \cap \mathbb{N}$, a set $P \subseteq T \times T$ of ordered pairs, and a real number $\delta \geq 0$. With $E_N = \exp(2\pi i D(h, N, L)/2^L)$, require*

$$|E_i - E_j| \geq \delta \quad ((i,j) \in P), \qquad \frac{2|T|^2}{5} \leq |P|\delta^2.$$

This condition is equivalent to irrationality. For the converse, irrationality and the doubling map allow $T = \{N, N+1\}$ and $P = \{(N, N+1), (N+1, N)\}$ at arbitrarily large N, with $\delta = 9/10$. The numerical requirement is then $8/5 \leq 2(9/10)^2$. The freedom to select T matters: this is not an estimate on a sample prescribed in advance, such as all prime positions.

(*c*) *Four integral tail differences. For $H \geq 0$ and positive integers p, q, the conjunction*

$$R_{2kH} - R_{kH} \in \mathbb{Z} \qquad (k \in \{1, p, q, pq\})$$

is equivalent to $R_{2H} - R_H \in \mathbb{Z}$ alone. The affine transport identity for $H \mapsto kH$ preserves integrality, so the three additional conditions add no restriction. Neither p nor q need be prime.

These three equivalences explain why these particular reformulations retain the original arithmetic question. They do not rule out useful weaker intermediate lemmas. The full-block and fixed-margin conditions in §3 ask for additional quantitative information that the three arguments above do not supply.

Remark 2.8. Second-difference certificates need not occur at a smaller depth than first-difference certificates. At $(h, N) = (1, 8)$, the first-difference test holds at depth 8, whereas no second-difference certificate exists at a depth at most 8 (Proposition 6.55).

This one example refutes a uniform claim that second differences always save depth; it does not give a uniform ordering in the opposite direction.

Separately, unbounded growth of the denominator exclusions produced by the Farey windows would imply irrationality. The record does not establish a converse for that specific sequence of windows, so the proposed growth condition is used only as a sufficient condition.

Proposition 2.9 (Failure of a specified two-adic congruence construction)**.** *A certificate requires the residue to lie farther than $N + h + L + 2$ from either endpoint modulo 2^L. In the specific construction below, the residue equals 2^{K-1} modulo 2^K, but the defining congruence is $p \equiv 1 + 2^{K-1} \pmod{2^K}$. Hence $p \geq 1 + 2^{K-1}$. At $N = p - K$, $h = H$, and $L = K$, the error bound is $p + H + 2 > 2^{K-1}$, so the residue does not satisfy the certificate inequalities. This calculation defeats this construction at every depth. It does not prove a corresponding statement for every use of the Chinese Remainder Theorem or every prescribed totient pattern.*

Remark 2.10 (Two further prescribed patterns). Two related results concern particular prescribed patterns. Under the stated size condition, the complete terminal divisibility pattern is impossible: its last totient difference would be positive, below a modulus, and divisible by that modulus (Lean source). For the pattern with the last condition removed, the penultimate difference is forced to be 2^{m-1}, with $2^m < 2(2H + J + K + 2)$.

A different sufficient condition would prescribe the accumulated sum rather than an individual difference: $D(h, N, L) \equiv 2^{L-1} \pmod{2^L}$ and $2^{L-1} > N + h + L + 2$. These two inequalities would give a certificate. They have not been supplied at the required quantified scope. The failure of the earlier patterns does not prove that this replacement is necessary.

Proposition 2.11 (Information lost by specific carry descriptions)**.** (*i*) *Fix $m \geq 2$ and write $R = \lfloor (m+1)/2 \rfloor$. For $r = 0, \ldots, R$, let c_r vanish except at $c_r(m) = R - r$ and $c_r(m+1) = 2r$. Each sequence satisfies $0 \leq c_r(n) \leq n$, has the same binary sum $R2^{-m}$, and has the same coefficients and scaled tails before position m. Its scaled tail at position m is r. The common history therefore does not determine that tail. A finite set of labels that determines the tail for each member of this family must contain at least $R + 1 = \lfloor (m+1)/2 \rfloor + 1$ elements, one for each value of r.*

(*ii*) *Reduction modulo 2^L forgets the initial value after the same L recurrence steps: two affine binary orbits with different seeds satisfy $\mathrm{orbit}_u(L) - \mathrm{orbit}_v(L) = 2^L(u_0 - v_0)$, so the endpoint residue mod 2^L is independent of the initial carry.*

(*iii*) *Fix a precision $u \geq 1$, a finite list of nonnegative valuations ν_j and odd integers a_j, and an initial integer e_0. There are integers z_j such that*

$$c_j = 2^{\nu_j}(a_j + 2^u z_j), \qquad e_{j+1} = 2e_j + c_j, \qquad |e_{j+1}| \leq 2^{\nu_j + u - 1}.$$

At each step, choose the centred representative of $2e_j + 2^{\nu_j} a_j$ modulo $2^{\nu_j + u}$. The unrestricted integers z_j are essential: the assertion does not prescribe the full coefficients c_j.

These statements exclude decoders using only the specified common history, distinctions using only the endpoint residue, and contradictions using only the stated local symbols. They do

not prove that the actual totient expansion cannot be studied by automata. A proposed application must verify that it uses no additional arithmetic information that the comparison families fail to preserve.

Proposition 2.12 (Four limits of particular linear constructions)**.** *The following four constructions have different limitations. They must not be read as a claim that every finite family of totient sections is independent: the full family has the explicit relations described in the short paper.*

(*i*) *Dyadic sections and an integer identity. The retained family of* $2^e + 1$ *dyadic sections is linearly independent over* $\mathbb{Q}$ *for* $e \geq 1$ (*Proposition 1.6*). *Separately, positive integers* Q, v *and integers* A, b *cannot satisfy*

$$A \neq 0, \qquad QvA = b, \qquad |b| < Qv,$$

because $|A| \geq 1$ *gives* $|b| \geq Qv$. *This elementary incompatibility concerns the displayed requirements; it does not exclude every argument using an adjugate matrix.*

(*ii*) *Möbius incidence.* $U_N(i,j) = \mu((i{+}1)/(j{+}1))$ *when* $(j{+}1) \mid (i{+}1)$ *and* 0 *otherwise is lower triangular with unit diagonal, so* $\det U_N = 1$ *for every* N. *Hence multiplication by* U_N *is injective; the corresponding coefficient transformation cannot create a nonzero vector in its kernel.*

(*iii*) *Adjugate reconstruction. Let* $w_i \in \mathbb{Q}$ *and* $x_i \in \mathbb{N}$ *be finite families with* $\sum_i w_i \varphi(x_i) = 1$. *Reconstructing each value from two tails and bounding them separately gives the error bound* $\sum_i |w_i|(3x_i + 4)$. *It is at least* 3, *since*

$$1 \leq \sum_i |w_i| \varphi(x_i) \leq \sum_i |w_i| x_i.$$

Consequently this particular bound cannot be less than 1, *whatever the size of the evaluation matrix.*

(*iv*) *Finite combinations of shifts. The synthetic sequence in Observation 6.115 has the prescribed differences* $a_{(q-1)H-1} = \varphi(H)$ *for* $2 \leq q < t$, *where* $H = \operatorname{lcm}(1, \dots, t)$. *It is of the form* $a_i = 2c_i - c_{i+1}$, *and every finite integer combination of its shifts has the same form with a correspondingly shifted state. Thus these linear operations alone do not remove the compatible carry recurrence. Their uniform bounds depend on the absolute coefficient sum, as made explicit in that observation. The construction does not assert that* a_i *equals the actual totient difference at other indices.*

A further limitation concerns quotients of the finite sums $t(Y, r) = \sum_{d=1}^{Y} \mu(d)/(2^d - 1)^r$. *For* $e \geq 1$ *and* $Y \geq 4$,

$$\frac{t(Y, e+2)^2}{t(Y, 2e+2)} - (S - \tfrac{1}{2}) > \frac{1}{480}.$$

The proof bounds each infinite sum $\sum_{d \geq 1} \mu(d)/(2^d - 1)^r$, $r \geq 3$, *between* 1429/1512 *and* 1, *and its truncation error after* $Y \geq 4$ *terms by* 1/3584. *Thus the lower bound holds for every*

stated pair e, Y, not just a finite list of computed examples. These results concern the listed matrices, estimates, and comparison sequence. The fourth item has the explicit construction and source theorem given in Observation 6.115. A different finite-dimensional argument, or one using further arithmetic assumptions, is not excluded. Proposition 1.7 is conditional on rationality of S and is not, by itself, a counterexample. The separate rational $5/4$ comparison supplies the counterexample to the generic rank bound.

Proposition 2.13 (The coefficient properties of the rational example)**.** *The rational sequence in Proposition 1.9 satisfies uniform boundedness, $c(n) \le n$, agreement with $\varphi(n)$ modulo 2 at every index, and the stated separated-carry form of aperiodicity. Consequently, those properties alone cannot imply irrationality of a dyadic series. The example does not exclude arguments using further totient identities, including its multiplicative relations.*

2.4 Consequences of the comparisons

The residue margin and the truncation error must be controlled together. Enlarging a congruence may enlarge both, while a finite prefix constrains no later coefficients. Periodicity modulo an integer likewise places no general upper bound on rational rank. These comparisons leave the equivalent residue conditions intact, with their original requirements for witnesses at unbounded indices.

These are limitations of specified arguments, not an exhaustive classification of possible methods. In particular, an averaging estimate is one possible source of the missing information, not the only possible source. A new arithmetic identity or a propagation theorem could also be useful. The next section describes the precise assumptions of several approaches considered here.

3 Conditions that would imply irrationality

Below are quantitative sufficient conditions, an equivalent residue formulation, and a proposed generic rank bound. The rank bound is false, as its rational counterexample shows. The other arithmetic conditions remain open for the totient; the proofs establish their stated implications. They are examples of possible approaches, not an exhaustive list.

3.1 Cancellation on a full block

If every window residue lies close to an endpoint, its complex phase lies close to 1. A saving in the norm of a block sum must therefore produce a certificate.

Definition 3.1 (The full-block norm condition)**.** Write $e(x) := \exp(2\pi i x)$ and $E(h, N, L) := e((D(h, N, L) \bmod 2^L)/2^L)$. The block norm condition asserts that

$$\forall h \ge 1\ \forall X_0\ \exists X, L: \quad \max(X_0, 1) \le X, \quad 16(2X + h + L + 2) \le 2^L, \quad \Big\| \sum_{N \in [X, 2X)} E(h, N, L) \Big\| \le \tfrac{21}{25} X.$$

It implies $S \notin \mathbb{Q}$. Lean source Lean source

A bound on the real part also suffices: it is enough that $\sum_{N \in [X, 2X)} \operatorname{Re} E(h, N, L) \le \frac{9}{10} X$ (Lean source; for the implication from $21/25$ to $9/10$, Lean source). Unpacked, the

required object is a constant-saving cancellation estimate for the dyadically weighted totient-difference exponential sum

$$\sum_{X\le N<2X} e\left(\frac{\sum_{j<L}(\varphi(N+h+1+j)-\varphi(N+1+j))2^{L-1-j}}{2^L}\right), \qquad L \text{ subject to the displayed size inequality.}$$

No theorem in this record establishes this estimate with the required quantifiers. The displayed size inequality imposes a lower bound on L; it does not require $L = \log_2 X + O(1)$. A depth of that order would be a possible regime for the estimate, not an additional proved conclusion. A finite instance would not suffice. The norm hypothesis requires cancellation: if all phases are 1, the norm is X and the bound fails; a balanced set of opposite phases has sum zero. If instead every phase is -1, the real part is $-X$ and satisfies the real-part bound, while the norm is X and fails the norm bound. Thus a real-part estimate is strictly weaker for general phase families; these examples do not distinguish the two quantified conditions for the actual totient sequence. The same depth L must work across the whole block.

Comparison with the preceding counterexamples. The earlier counterexamples do not establish or refute this block estimate. The following distinctions explain why those particular arguments do not settle it.

- The parameters grow. The size condition forces the truncation depth to increase with X, so this is not a test of a fixed prefix.
- The sample is prescribed: all $N \in [X, 2X)$ are summed at one depth. Choosing two favourable indices does not establish this block estimate.
- The sum uses the accumulated totient differences. It does not prescribe individual values through a congruence construction.
- The estimate concerns the actual phases. Neither a bound on carry rank nor agreement only in parity gives it.
- The proved implication uses the real-part threshold $9/10$ and the constant in the size bound 16. The stronger norm bound $21/25$ implies that real-part bound. These constants do not assert any distribution theorem for the totient phases.

Formal sources for these implications: Lean source Lean source

Remark 3.2 (The finite-prefix example does not choose a method). The finite-prefix example does not select an analytic method. It shows only that a proof valid for every sequence $0 \le c(n) \le n$ cannot deduce the required assertion from one fixed prefix. An argument using additional identities of φ, a theorem propagating a finite base case, a Farey estimate, or a suitable exponential sum could provide information not contained in that prefix. The earlier finite audit cannot rank all such possibilities.

Remark 3.3 (Finite certificate observations). In the historical table of 28 diagonal certificates through $t = 64$, the reported depths exceed the necessary depth bound by a small additive constant. A separate enumeration at the indices where the least common multiple increases records a closest margin from the central interval's boundary of approximately 0.000221 of the modulus, at $t = 100$. Both are finite observations. Neither establishes nor refutes equidistribution or the estimate at arbitrarily large scales.

3.2 A four-term bound for the real part

We isolate the phase of one term $\varphi(mp)$ with $p > m$ prime, then group indices with the same cofactor m. The resulting identity separates three tasks: count the excluded indices, bound the group means, and estimate the remaining weighted, mean-subtracted sum. Only the last of these remains open after the stated group-mean input. The identity itself supplies none of the estimates.

Fix $h, s \geq 1$, $L \geq s + h$, $X \geq 1$ and $0 < \eta < 1$, and put $t = L - s + 1$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and put $m_N = (N + t)/p_N$. Assign the index N to this factorisation when

$$m_N \leq \left\lfloor \frac{\sqrt{X}}{2} \right\rfloor, \qquad p_N > 2\lfloor \sqrt{X} \rfloor.$$

Write $\mathcal{A}$ for the assigned indices and $\mathcal{G} = \{N \in \mathcal{A} : \varphi(m_N) \geq \eta m_N\}$. The two inequalities ensure $p_N > m_N \geq 1$, so $\varphi(N + t) = \varphi(m_N)(p_N - 1)$.

Since $h < t \leq L$, the selected term occurs in both truncated tails: it has coefficient $2^{-(t-h)}$ in the shifted tail and -2^{-t} in the unshifted one. Their sum is $(2^h - 1)2^{-t}$. On the assigned indices, set

$$z_N = e\left(\frac{(2^h - 1)\varphi(m_N)(p_N - 1)}{2^t} \right), \qquad w_N = \frac{E(h, N, L)}{z_N}.$$

Both factors have absolute value 1. Let $\bar{z}_m$ be the average of z_N over the assigned indices with cofactor m, taking the average to be zero when the group is empty. The four contributions are then defined by the identity

$$\sum_{N=X}^{2X-1} E(h, N, L) = \underbrace{\sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N})}_{\text{mean-subtracted contribution}} + \underbrace{\sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N}}_{\text{mean contribution}} + \underbrace{\sum_{N \in \mathcal{A} \setminus \mathcal{G}} E(h, N, L)}_{\text{excluded-cofactor contribution}} + \underbrace{\sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E(h, N, L)}_{\text{unassigned contribution}}.$$

This follows by partitioning the indices and using $E(h, N, L) = w_N z_N$. Subtracting the group mean is exact algebra; it does not make the two factors independent. The chosen prime need not be confined to the selected argument: with $X = 16$, $h = s = 1$, $L = 20$ and $N = 18$, the argument is $N + t = 38 = 2 \cdot 19$, which satisfies both assignment inequalities, but the same prime also divides $N + 1 = 19$. Thus the factorisation at $N + t$ alone gives no separation from the other totient values in the phase.

Definition 3.4 (Four simultaneous inequalities)**.** For every $h > 0$ there are $s > 0$ and $\eta \in (0, 1)$ such that for every X_0 there are X, L with $\max(X_0, 1) \leq X$, $h \leq L - s$, $16(2X + h + L + 2) \leq 2^L$, and all four of

$$\operatorname{Re}\sum_{N\in\mathcal{G}} w_N(z_N-\bar z_{m_N}) \le \tfrac{14}{25}X,$$

$$\left\|\sum_{N\in\mathcal{G}} w_N\bar z_{m_N}\right\| \le \tfrac{1}{100}X,$$

$$\left\|\sum_{N\in\mathcal{A}\setminus\mathcal{G}} E(h,N,L)\right\| \le \tfrac{1}{100}X,$$

$$\left\|\sum_{\substack{X\le N<2X\\ N\notin\mathcal{A}}} E(h,N,L)\right\| \le \tfrac{8}{25}X.$$

It implies $S \notin \mathbb{Q}$. Lean source Lean source

First fix h, then choose s, η; these must remain fixed as the threshold X_0 increases. At each threshold all four bounds must hold for one common X, L, rather than along unrelated sequences of blocks. Their constants add to $14/25 + 1/100 + 1/100 + 8/25 = 9/10$, yielding the real-part criterion. This is sufficient for a certificate and does not assert the stronger $21/25$ norm criterion. In particular, no independence of the two phase factors follows from the grouping.

The decomposition is an exact identity, proved in Lean (Lean source), as is the implication from the four bounds to the real-part bound (Lean source). The reason a new estimate is needed is visible within one group: $\sum_N (z_N - \bar z_m) = 0$, whereas $\sum_N w_N(z_N - \bar z_m)$ need not vanish. The residual weight varies with the same prime that determines z_N. By contrast, a uniform bound for $|\bar z_m|$ immediately controls the mean contribution because $|w_N| = 1$. Thus small means and weighted cancellation are different statements. A two-point group makes this explicit:

$$(z_1, z_2) = (1, -1), \quad (w_1, w_2) = (1, -1), \quad \bar z = 0, \qquad \sum_{j=1}^{2} w_j(z_j - \bar z) = 2.$$

The unweighted mean vanishes while the weighted terms agree exactly. This example explains why an implication is missing; it is not a counterexample to the estimate for the actual totient windows. For the latter, only the real-part bound $14X/25$ is required, not the earlier norm bound $X/2$. For $X \ge 4$ and $s = L - h$, the assigned indices *with cofactor* $m_N = 1$ are exactly those for which $N + h + 1$ is prime. This describes one cofactor group, not all assigned indices. The source proves this membership equality (Lean source), using $\varphi(mp) = \varphi(m)(p-1)$ and $p > m$. This special choice has $s = L - h$ varying with L; it illustrates the arithmetic of the groups but does not establish the condition with s fixed before X_0.

The four inequalities have not been established together. Propositions 12.6 and 12.7 give the unassigned and excluded-cofactor bounds at minimal admissible depth. At $s = 26$, $\eta = 1/1000$, the prime-number-theorem argument discussed in Section 3.3 also bounds the mean contribution for all large X. With that input, the missing estimate is the real-part bound for the mean-subtracted contribution, at the same parameters and

depth, on arbitrarily large blocks. Theorem 3.5 instead asks for one bound on $\mathcal{G}$, without a group-mean input.

A nonzero determinant alone does not prove the needed correlation estimate. The comparison in the linked source (Lean source) preserves such a determinant while fixing an entire row of weighted phases to 1. A successful application must distinguish that comparison, for example by additional arithmetic information, rather than only by column rescaling. The exact four-term identity supplies no prime-distribution estimate.

3.3 One bound on the good indices

Two counting estimates allow us to restrict the cancellation question to $\mathcal{G}$. For fixed h, s, let $L(X)$ be the least L satisfying $h \le L - s$ and $16(2X + h + L + 2) \le 2^L$, as in Proposition 12.6. With $s = 26$ and $\eta = 1/1000$, more than $67X/100$ indices remain in $\mathcal{G}$ for large X. A suitable saving on this set alone would therefore imply irrationality.

Theorem 3.5 (Irrationality from one bound on the good indices)**.** (*a*) *Suppose that for every $h \ge 1$ and every A there are X, L with $16(2X + h + L + 2) \le 2^L$ and a nonempty finite set T of integers in $[A, 2X)$ with*

$$\sum_{N \in T} \operatorname{Re} E(h, N, L) \le \frac{9}{10} \#T.$$

Then $S \notin \mathbb{Q}$ (`irrational_totient_series_of_support_gap`).

(*b*) *Take $s = 26$, $\eta = 1/1000$ and $L = L(X)$. Suppose that for every $h \ge 1$ there are arbitrarily large X with*

$$\operatorname{Re} \sum_{N \in \mathcal{G}} E(h, N, L(X)) \le \frac{603}{1000} X.$$

Then $S \notin \mathbb{Q}$ (`irrational_totient_series_of_goodBase_gap`).

Proof. (a) At a depth satisfying the size inequality, a nonempty set of indices below $2X$ whose average of $\operatorname{Re} E(h, N, L)$ is at most $9/10$ contains an index N with $C(h, N, L)$ (Lean source). The hypothesis therefore supplies, for every $h \ge 1$, certificates beyond every threshold, which is Definition 1.2; that condition is equivalent to $S \notin \mathbb{Q}$.

(b) For all large X, fewer than $8X/25$ indices in $[X, 2X)$ are unassigned (Proposition 12.6), and fewer than $X/100$ assigned indices have $\varphi(m_N) < m_N/1000$, by the Chebyshev form of Proposition 12.7 given after its proof. These two sets and $\mathcal{G}$ partition $[X, 2X)$, so $\#\mathcal{G} > 67X/100$ (`eventually_card_pivotGoodBases_gt`). Since $603/1000 = (9/10)(67/100)$, each large X in the hypothesis gives $\sum_{N \in \mathcal{G}} \operatorname{Re} E(h, N, L(X)) < \frac{9}{10} \#\mathcal{G}$. For $X \ge A$ the set $T = \mathcal{G}$ is nonempty and lies in $[A, 2X)$, and $L(X)$ satisfies the size inequality, so (a) applies. □

The trivial bound for the sum in (b) is $\#\mathcal{G}$, so the hypothesis asks for a fixed saving: along a sequence of blocks, the average of $\cos(2\pi D(h, N, L)/2^L)$ over $\mathcal{G}$ must be at most $603X/(1000\,\#\mathcal{G})$, a number below $9/10$. Equidistributed phases would make the sum

$o(X)$. A rational value $S = a/(2^c v)$ with v odd makes the bound fail for $h = \varphi(v)$ and all large X: every large index then has an integral tail difference, hence no certificate, and an index at a depth satisfying the size condition and having no certificate has $\operatorname{Re} E > 9/10$. The hypothesis is sufficient and is not known to be necessary: from irrationality one obtains nonintegral tail differences, and no argument turns them into a fixed saving on $\mathcal{G}$. It is open. No computation in this record bears on it, and the count $\#\mathcal{G} > 67X/100$ holds only for large X.

When the first two inequalities of Definition 3.4 hold with $s = 26$, $\eta = 1/1000$ and $L = L(X)$, they add to $\operatorname{Re} \sum_{N\in\mathcal{G}} E(h,N,L(X)) \le \frac{57}{100}X$, so the four-term condition at these parameters implies the hypothesis of (b). At these same parameters, the recorded argument takes the prime number theorem as an input and bounds every group mean $|\bar{z}_m|$ on $\mathcal{G}$ by $1/100$ for all sufficiently large X, for each fixed h (`fiberMean_le_of_primeNumberTheorem`). Since $|w_N| = 1$, this gives $|\sum_{N\in\mathcal{G}} w_N \bar{z}_{m_N}| \le \#\mathcal{G}/100 \le X/100$, regardless of the dependence of w_N on the prime. It supplies the mean bound, not the mean-subtracted cancellation. Given this mean bound, a bound $\frac{11}{20}X$ on the good-index sum, on arbitrarily large blocks for each h, yields the mean-subtracted inequality and hence irrationality (`dtw_of_fiberMean_and_centered`, `irrational_totient_series_of_primeNumberTheorem_and_goodBase_gap`). Theorem 3.5(b) supersedes that route. It assumes neither the prime number theorem nor a bound on the group means, and asks only for $603X/1000$, rather than $11X/20$. The saving is measured against $\#\mathcal{G}$, not against the full block size: since $\#\mathcal{G} > 67X/100$, the good-index average is below $9/10$. There is no need to add the discarded indices back into the sum.

For clarity, the two sufficient estimates have different uses. The centred route controls all four pieces and so bounds the sum on the whole block. The $603X/1000$ route uses only the good indices and their cardinality. Bounding the discarded part trivially would give only $(603/1000 + 33/100)X = 933X/1000$, not $9X/10$; adding it back would lose the conclusion one was trying to prove.

There is also a useful distinction between eventual and cofinal hypotheses. At $s = 26$, $\eta = 1/1000$ and the prescribed $L(X)$, fix h and take the maximum of the thresholds for the two counting bounds and the PNT-input mean bound. Any arbitrarily large sequence of blocks on which the centred bound holds eventually passes this maximum. Thus it supplies common blocks for all four estimates. Merely knowing that several different estimates each hold on some arbitrarily large blocks would not justify that intersection.

3.4 A fixed separation at prime positions

Instead of averaging over a block, one may ask for a fixed gap at prime positions. The lower bound on the prime ensures that the corresponding tail indices lie beyond the required threshold.

Definition 3.6 (A fixed gap at prime positions)**.** For every $h \ge 1$ and every N_0 there is a prime p with $\max(N_0+h+1,\, h+5) \le p$ and $\operatorname{Re} e(R_{p-1} - R_{p-h-1}) < \frac{9}{10}$, where $e(x) = \exp(2\pi i x)$. Equivalently: cofinally many primes p at which the totient tail difference across the shift h stays a *fixed* distance $> \arccos(9/10)/(2\pi) \approx 0.0718$ from every integer. It implies $S \notin \mathbb{Q}$. Lean source Lean source

This asks for more than nonintegrality. A nonintegral real number can be arbitrarily close to an integer, whereas the displayed condition requires a fixed positive distance, along prime positions and for each shift h. For example, phases at a half-integer satisfy the numerical gap and phases converging to an integer eventually fail it. These are illustrations of the hypothesis, not assertions about the totient tails. No estimate for these tail differences at prime positions with the stated quantifiers is proved here. Certificate completeness alone does not supply it.

3.5 Certificates with depth equal to the shift

The shift and truncation depth can be required to agree, provided the shift is allowed to run through positive multiples of a fixed d. The propagation argument in Appendix A.1 explains why this restriction preserves the equivalence below.

Definition 3.7 (Depth equal to a shift multiple)**.** For every $d \geq 1$ and every N there is $t \geq 1$ with $C(td, N, td)$; unpacked,

$$N + 2td + 2 \;<\; D(td, N, td) \bmod 2^{td} \;<\; 2^{td} - (N + 2td + 2).$$

It is equivalent to $S \notin \mathbb{Q}$. Lean source Lean source

For a fixed pair (d, N), a nonintegral tail difference supplies a certificate in every sufficiently late adjacent pair of multipliers. Appendix A.1 proves this propagation and hence the equivalence in Definition 3.7 (Lean source). The finite examples verify the starting hypothesis for individual pairs, not for every d, N. That universal supply of nonintegral differences is the missing arithmetic input. With these quantifiers, the condition has the full strength of the irrationality question; it is not a mild regularity assumption. The open problem remains unproved.
Formal sources: Lean source

3.6 A false generic rank bound

The coefficient rank might suggest a bound on carries under rationality. The next proposed assertion makes that suggestion precise; the rational comparison following it disproves both versions.

Definition 3.8 (A counterfactual rank criterion)**.** The proposed assertion is a constant C bounding the rank for every $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \leq n$ and $T_c \in \mathbb{Q}$, every integer $v > 0$, and every integer sequence u satisfying

$$u(n+1) = 2u(n) - vc(n+1), \qquad u(n)/2^n \longrightarrow 0.$$

Specifically, the proposed bound is

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{n \mapsto u(2^j n + r) : 1 \leq j \leq e,\ 0 \leq r < 2^j\} \leq C \qquad (e \geq 0).$$

The alternative replaces C by a function $g(e)$ growing strictly slower than $2^e - 1$, with the same quantifiers. Either assertion would contradict the lower bound of Proposition 1.6 under rationality of S. Both assertions are false, as the following comparison shows.

Both assertions in Definition 3.8 are false. The comparison sequence with sum $5/4$ in Section 10.8 satisfies $0 \le c(n) \le n$ and has an integral carry with $u(n)/2^n \to 0$ and rank at least $2^e - 1$ at every depth (Section 12.6). It therefore contradicts both proposed upper bounds. This leaves open bounds under further hypotheses specific to the totient.

Periodicity modulo v alone does not provide the additional rank bound.

4 Guide to the detailed statements

The remainder records the formulas and proofs behind the preceding arguments. To use a statement, distinguish its conclusion from its input. A finite certificate excludes a family of denominators; a cofinal supply excludes every rational value. Positivity of a tail difference is not nonintegrality, and a conditional implication still needs its arithmetic premise.

The following entry points answer different questions; they are not successive steps of one proof of irrationality.

Question	**Where to read**
Which section identities hold?	Section 10.8: independent sections and integral relations.
What does a finite computation exclude?	Section 5: tail residues; Section 5.3: the denominator cutoff.
Can one nonintegral difference yield more certificates?	Appendix A.1: propagation with growing shift and depth.
Which data still allow a rational value?	Section 2: comparison sequences, with their hypotheses kept separate.
Which estimate would suffice?	Section 12.3: the four-term decomposition; Theorem 3.5: the weaker good-index condition.
What was formally checked?	Appendix B: the declaration and source-version guide.

The detailed statements retain their source references so that a proof can be recovered without treating a source label as another hypothesis.

5 Series identities and finite exclusions

The prefix-tail identity supports two different finite exclusions. Residue certificates test a tail difference modulo $\mathbb{Z}$; the Farey calculation in Section 5.3 encloses S itself in an interval. Möbius inversion then gives other expressions for the same value, useful only to the extent that they permit new estimates. An identity alone does not provide a denominator exclusion.

5.1 The series

We begin with the value and the prefix-tail decomposition used by all the subsequent tests.

Definition 5.1 (The Erdős–249 constant)**.** Let φ denote Euler's totient function. Define

$$S := \sum_{n\geq 1} \frac{\varphi(n)}{2^n} = \sum_{n:\mathbb{N}} \frac{\varphi(n)}{2^n},$$

the two forms coinciding because $\varphi(0) = 0$; the second, $\mathbb{N}$-indexed form is the one carried in the Lean source. The series converges absolutely since $\varphi(n) \leq n$. The irrationality of this value is the question under consideration.

Remark 5.2 (Status). No proof or disproof of $S \notin \mathbb{Q}$ exists anywhere in the record, formal or informal. What exists is: one large unconditional finite denominator exclusion (§5.3), several exact reformulations of S that relocate the same open question onto different coordinates without touching its truth value (§5.4–§5.6), and a certificate apparatus whose cofinal quantified condition is the precise open target (Definition 1.2; only its finite, checked instances are catalogued here, §5.7).

5.2 The tail-shift identity

Multiplying by a power of two makes the finite prefix integral and leaves a scaled tail. The tail values themselves may grow; it is their classes modulo $\mathbb{Z}$, rather than their absolute sizes, that encode a possible rational period.

Proposition 5.3 (Digit-shift identity)**.** *For every $N : \mathbb{N}$,*

$$2^N \cdot S = \Phi_N + R_N,$$

where $\Phi_N := \sum_{n\leq N} \varphi(n) \cdot 2^{N-n} \in \mathbb{N}$ (the integer prefix) and $R_N := \sum_{j\geq 0} \varphi(N+1+j)/2^{j+1}$ (the scaled tail, R in Lean).

Consequence for rationality. Subtracting the prefix identities at $N + h$ and N gives

$$R_{N+h} - R_N \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}.$$

If $S = a/(2^c v)$ with $v \geq 1$ odd, Euler's theorem supplies $h = \varphi(v) > 0$ and $N_0 = c$, for which the difference is integral for all $N \geq N_0$ (Lean source). Conversely, an integral difference for even one $h > 0, N \geq 0$ forces S to be rational, because $2^N(2^h - 1)$ is a nonzero integer. Thus the unresolved question is not this converse, but whether every such tail difference is nonintegral. Definition 1.2 expresses that question through finite residues. □

5.3 The finite denominator bound

An interval containing S excludes rational numbers whose denominators are too small to occur inside it. The Farey argument below makes this precise using a finite totient prefix and a bound on its tail. It is independent of the period and residue criteria.

Lemma 5.4 (Farey gap, fully general)**.** *For integers a, b, c, d, r, s with $b > 0$, $d > 0$, $bc - ad = 1$ (i.e. a/b and c/d are unimodular Farey neighbours), and $as < rb$, $rd < cs$ (i.e. r/s lies strictly between them): $b + d \leq s$.*

Indeed, the two strict inequalities make $cs - dr$ and $br - as$ positive integers, and

$$s = b(cs - dr) + d(br - as) \geq b + d,$$

using $bc - ad = 1$. This also proves $s > 0$, which the formulation did not need to assume separately. The lemma is the classical Stern–Brocot denominator bound. The mediant $(a + c)/(b + d)$ lies between the two neighbours and is reduced: any common divisor of $a + c$ and $b + d$ divides $b(a + c) - a(b + d) = bc - ad = 1$. It attains denominator $b + d$, so the general interval bound is sharp. A particular tail interval still requires its own endpoint and first-failure checks.

Proposition 5.5 (The Farey certificate at $K = 240$)**.** *For the window $(N, K) = (1, 240)$, put*

$$V = \left(\sum_{r=1}^{240} \varphi(1 + r) 2^{240-r} \right) \bmod 2^{240}.$$

For every $q : \mathbb{N}$ with

$$0 < q \;\leq\; Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \;\; (\approx 7.96 \times 10^{34}),$$

$$(q \cdot V) \bmod 2^{240} \;+\; 243\,q \;<\; 2^{240}.$$

This bound is sharp: $q = Q_0 + 1 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,054$ is the exact first failing denominator, obtained as the denominator of the mediant of two explicit unimodular Farey neighbours.

Here "failing" means that this particular enclosure can no longer exclude the denominator. It does not mean that the mediant equals S, or that another argument could not exclude that denominator too. Sharpness concerns the finite interval test, not a smallest possible denominator for the unknown value.

Theorem 5.6 (Denominator exclusion)**.** *For every $p \in \mathbb{Q}$ whose reduced denominator is at most Q_0,*

$$S \;\neq\; p.$$

Equivalently: if S is rational, its reduced denominator exceeds $Q_0 \approx 7.96 \times 10^{34}$.

Remark 5.7 (What this does and does not say). Theorem 5.6 is a complete, unconditional finite fact: no rational of denominator at most Q_0 equals S. A single such bound does not exclude arbitrarily large denominators. Proving unbounded exclusions by the same mechanism would, however, establish irrationality. The question this leaves is whether the first failing denominator $(b + d)(K)$ at window K is unbounded as $K \to \infty$. An affirmative answer would prove irrationality by Proposition 6.75, independently of the certificate condition discussed after Proposition 5.3. The question is open; it is neither claimed nor proved anywhere in the record.

Remark 5.8 (Earlier finite denominator bounds). The recorded Farey computations gave bounds 4838, 2^{22}, approximately 2.49×10^{17} at $K = 120$, and Q_0 at $K = 240$. Each applies the finite mechanism of Lemma 5.4 and Proposition 5.5. These values do not prove that the bounds are unbounded as K increases.

5.4 Equivalent expressions and their denominator bounds

The finite bound in Theorem 5.6 also applies to the two expressions for $S - 1/2$ below, with half the denominator bound. Amiram Eldar posted the coprimality interpretation of Proposition 5.9 in revision 28 of OEIS A256936 on 15 March 2026, and the Möbius-square formula of Proposition 5.14 in revision 31 [8]. Steve Fan posted the latter formula on the problem's forum thread on 16 May 2026 [9]. We derive both identities by Möbius inversion and geometric summation.

Proposition 5.9 (Fair-coin coprimality form)**.** *Let X, Y be independent random variables with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ for $n \geq 1$ (independent fair-coin waiting times). Then*

$$S \;=\; \tfrac{1}{2} \;+\; \Pr(\gcd(X, Y) = 1) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b\geq 1 \\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Equivalently, on the visible lattice: summing $2^{-(a+b)}$ over the half-open coprime pairs ($a \geq 1$, $b \geq 0$, $\gcd(a, b) = 1$) recovers $\sum_n \varphi(n)/2^n$ exactly, with no boundary correction, because the visible-point count on the half-open antidiagonal at height n equals $\varphi(n)$ for every n, including $n = 0, 1$.

Proposition 5.10 (The gcd-layer normalisation)**.** *For independent fair-coin waiting times as above, $\sum_{g\geq 1} \Pr(\gcd(X, Y) = g) = 1$ exactly; and for every $d > 0$, $\Pr(d \mid X \wedge d \mid Y) = 1/(2^d - 1)^2$.*

Theorem 5.11 (Denominator exclusion for the coprimality-probability form)**.** *Let*

$$Q_1 := \left\lfloor \frac{Q_0}{2} \right\rfloor = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526.$$

For every $a \in \mathbb{Z}$, $d \in \mathbb{N}$ with $0 < d \leq Q_1$: the visible coprime-pair probability $\Pr(\gcd(X, Y) = 1)$ is not equal to a/d.

Theorem 5.12 (Denominator exclusion for the Möbius-square form)**.** *With Q_1 as in Theorem 5.11: for every $a \in \mathbb{Z}$, $d \in \mathbb{N}$ with $0 < d \leq Q_1$, the signed series $T := \sum_{m\geq 1} \mu(m)/(2^m - 1)^2 = S - \frac{1}{2}$ (see §5.5) is not equal to a/d.*

Remark 5.13 (Why the bound halves, and why this is not new information). $Q_1 = \lfloor Q_0/2 \rfloor$ arithmetically: Q_0 is odd ($Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$), so $Q_0/2 = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526.5$ and Q_1 is its floor. The halving is the cost of transporting the known bound through the affine shift $S = \frac{1}{2} + T$ (resp. $S = \frac{1}{2} + \Pr(\gcd(X, Y) = 1)$) on a denominator-exclusion statement: excluding all denominators $\leq Q_1$ for T follows from excluding all denominators $\leq Q_0$ for S (a denominator-d value of T with $d \leq Q_1$ yields a denominator dividing $2d \leq Q_0$ for S). Theorems 5.11 and 5.12 are therefore the *same* finite Farey record as Theorem 5.6, transported through Proposition 5.9 and Proposition 5.14 respectively; not independent evidence. The converse finite implication is not asserted: subtracting $1/2$ can double a denominator, so the Q_1 exclusion for T alone need not recover the full Q_0 exclusion for S.

5.5 The Möbius-weighted identity

Subtracting $1/2$ converts the totient series to a Möbius-weighted squared-Lambert series without changing rationality.

Proposition 5.14 (Möbius-square reduction)**.**

$$S = \sum_{n\geq 1} \frac{\varphi(n)}{2^n} = \frac{1}{2} + \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2},$$

where μ is the Möbius function, so $\mu(d) \in \{-1, 0, 1\}$ for every d. Consequently Erdős #249 $\iff$ $T := \sum_{d\geq 1} \mu(d)/(2^d-1)^2 \notin \mathbb{Q}$, since adding a rational number does not change irrationality.

Remark 5.15 (Why the bounded coefficients matter; the Erdős-1948 regime). The identity of Proposition 5.14 rewrites S (equivalently T) as a Lambert-type series with squared denominators and bounded Möbius weights. To see the constant $1/2$ directly, expand

$$\sum_{d\geq 1} \frac{\mu(d)}{2^d-1} = \sum_{n\geq 1} \frac{\sum_{d\mid n} \mu(d)}{2^n} = \frac{1}{2}.$$

Combining this identity with $\varphi = \mu * \mathrm{Id}$ and $2^d = (2^d-1)+1$ proves the formula for T; absolute convergence justifies the rearrangements. Thus the corresponding first-power sum $L(\mu) := \sum_d \mu(d)/(2^d-1)$ is rational, whereas $L(1) = \sum_{d\geq 1} 1/(2^d-1) = E$ is the Erdős–Borwein constant. Erdős proved its irrationality in 1948 [11], and it is also proved in Lean (Lean source).

In contrast, the coefficients of the original power series satisfy $0 \leq \varphi(n) \leq n$ and are unbounded. Bounded Möbius weights do not make the new denominators powers of a fixed base, so they do not turn this representation into a bounded-digit expansion. The function φ has average order $6n/\pi^2$, equivalently $\sum_{k\leq x} \varphi(k) \sim 3x^2/\pi^2$, but there is no pointwise estimate $\varphi(n) = \Theta(n)$. The near-integer irrationality criterion (formalised generically as Lean source and its base-power specialisation Lean source) applies to either representation if its nonzero approximation errors can be established. Boundedness of the coefficients does not itself establish those errors. Erdős's 1948 argument is itself a proof for the specific Lambert series: it chooses arguments by a system of congruences so that the divisor counts along a block are divisible by increasing powers of the base, and a tail bound then forces a long run of zero digits [11, pp. 63–66]. Boundedness of a signed weight supplies none of that divisibility input.

The rational identity $L(\mu) = 1/2$ is already a warning: bounded signed coefficients alone do not imply irrationality. The later denominator calculations retain the Möbius terms whose indices do not divide the chosen LCM; those terms cannot be discarded when transferring an estimate.

5.6 A general squared-Lambert identity

The same geometric summation applies to any weight with the stated linear bound.

Proposition 5.16 (The squared-Lambert identity)**.** *For $w : \mathbb{N} \to \mathbb{R}$ with $|w(d)| \le d$ for all $d > 0$, and $0 \le r < 1$:*

$$\sum_{d\ge1} w(d)\left(\frac{r^d}{1-r^d}\right)^2 = \sum_{n\ge1}\left(\sum_{e\mid n} w(e)\left(\tfrac{n}{e}-1\right)\right) r^n.$$

The identity follows by expanding $r^d/(1-r^d)$ as a geometric series and interchanging absolutely convergent sums. It is the squared analogue of the classical divisor-sum Lambert identity $\sum_n a_n q^n/(1-q^n) = \sum_m (\sum_{d\mid m} a_d) q^m$ [16, (1), p. 2].

At $r = 1/2$, the choices $w = 1$ and $w = \varphi$ give the following two identities. They help distinguish the role of the weight from that of the squared denominator.

Proposition 5.17 (The divisor-sum identity)**.**

$$\sum_{d\ge1} \frac{1}{(2^d-1)^2} = \sum_{n\ge1} \frac{\sigma(n)-\tau(n)}{2^n} = \zeta_q(2) - \zeta_q(1) \text{ at } q = \tfrac{1}{2},$$

where σ is the sum-of-divisors function and τ the number-of-divisors function. The identity is machine-checked. With $\zeta_q(s) = \sum_{n\ge1} n^{s-1} q^n/(1-q^n)$, Postelmans and Van Assche prove that $1, \zeta_q(1), \zeta_q(2)$ are linearly independent over $\mathbb{Q}$ for $q = 1/p$ with an integer $p \ge 2$ [12, Theorem 1.3, p. 3]; at $q = \frac{1}{2}$ this gives the irrationality of the value $\zeta_q(2) - \zeta_q(1)$. Two irrational numbers can have a rational difference, so the separate irrationality of the two values would not suffice.

The Lean proof of the irrationality assertion assumes the linear independence theorem of Postelmans and Van Assche quoted above; the displayed identity is proved without it.

Proposition 5.18 (The gcd-moment identity)**.**

$$\sum_{d\ge1} \frac{\varphi(d)}{(2^d-1)^2} = \sum_{n\ge1} (P(n)-n)\cdot 2^{-n} = \mathbb{E}[\gcd(X,Y)],$$

where $P(n) := \sum_{e\mid n} \varphi(e)\cdot(n/e) = (\varphi * \mathrm{Id})(n)$ is Pillai's gcd-sum function $P(n) = \sum_{k\le n} \gcd(k,n)$ [15, §1, (1)–(2)] and X, Y are the independent fair-coin waiting times of Proposition 5.9. The identity itself is machine-checked; irrationality of $\mathbb{E}[\gcd(X,Y)]$ is not proved here. This is a different series from S.

Remark 5.19 (Related series). Writing $L(f) := \sum_d f(d)/(2^d-1)$ and $L_2(f) := \sum_d f(d)/(2^d-1)^2$, the preceding identities give $L(\mu) = \frac{1}{2}$ (rational), $L(1) = E$ (Erdős–Borwein, irrational), $L(\varphi) = 2$ (rational); $L_2(\mu) = S - \frac{1}{2}$ (irrationality open; this is #249), $L_2(1) = \zeta_q(2) - \zeta_q(1)$ (irrational by the cited theorem, Proposition 5.17), $L_2(\varphi) = \mathbb{E}[\gcd(X,Y)]$ (irrationality open, Proposition 5.18). These examples show why the denominator shape alone does not determine irrationality: changing the weight can produce either rational or irrational values.

5.7 Diagonal certificates through scale 82

Each row is a finite instance of Definition 1.2, with a link to the computation producing its witness. The aggregate result covers every integer scale $t \le 82$. It contains no assertion at $t = 83$ or at arbitrarily large scales.

Definition of the finite residue test For $h, N, L : \mathbb{N}$, recall the truncated difference $D(h,N,L) := \sum_{j<L}(\varphi(N+h+1+j) - \varphi(N+1+j)) \cdot 2^{L-1-j} \in \mathbb{Z}$ (Lean source), and

$$C(h,N,L) \;:\equiv\; (N+h+L+2) < D(h,N,L) \bmod 2^L < 2^L - (N+h+L+2),$$

(Lean source; a decidable condition). By completeness (Lean source),

$$(\exists L,\ C(h,N,L)) \iff R_{N+h} - R_N \notin \mathbb{Z}$$

for every h, N. Requiring such certificates beyond every basepoint threshold for each fixed positive h is precisely Sep and is equivalent to $S \notin \mathbb{Q}$. The table records finite instances, indexed either by (h,N,L) or by t; it does not establish that quantified condition.

Computation	Exact statement of what was checked	Lean source
Fixed-window example, depth 16	$C(h,12,16)$ holds for every $h \in \{1,\dots,8\}$ (by direct evaluation, 256 evaluations with totient arguments below 37); consequently $S \neq r$ for every $r \in \mathbb{Q}$ whose reduced denominator divides $2^{12}(2^h-1)$ for some $1 \le h \le 8$.	Lean source Lean source
Fixed-window example, depth 9, wider net	$C(h,14,9)$ holds for every $h \in \{1,\dots,16\}$: $S \neq r$ for every $r \in \mathbb{Q}$ whose reduced denominator divides $2^{14}(2^h-1)$ for some $1 \le h \le 16$.	Lean source
Diagonal certificates, 12 base scales	$C(H_t,H_t,L_t)$, $H_t := \mathrm{lcm}(1,\dots,t)$, holds (by explicit evaluation of the totient values, via factored prime-power blocks with Lucas-primality certificates) for $t \in \{1,2,3,4,5,7,8,9,11,13,16,17\}$ with certificate depths $L_t \in \{6,5,7,7,9,14,15,14,21,22,23,26\}$ respectively (in the same order).	Lean source Lean source Lean source
Diagonal certificates, further scales through $t = 64$	The same predicate $C(H_t,H_t,L_t)$ is additionally checked, one Lean file per scale, at $t \in \{19,23,25,27,29,31,32,37,41,43,47,49,53,59,61,64\}$ (16 further explicit values). Together with the base 12 scales above this is **28 explicit values of t in total**, the complete list being $\{1,2,3,4,5,7,8,9,11,13,16,17,19,23,25,27,29,31,32,37,41,43,47,49,53,59,61,64\}$. This does *not* establish $C(H_t,H_t,\cdot)$ for infinitely many t; these 28 examples are a historical strict subset of the contiguous band in the next row.	One Lean file per scale; endpoint Lean source

Computation	Exact statement of what was checked	Lean source
Contiguous lcm-diagonal band through $t \le 82$	For every natural $t \le 82$ there exists a depth L with $C(H_t, H_t, L)$. This closes every scale in the finite interval, including plateau transfers, with no holes. The next lcm jump is at the prime 83; no certificate at $t = 83$ is claimed, and any such certificate must have depth at least 125.	Lean source Lean source
Farey denominator bound	Every $q \in \mathbb{N}$ with $0 < q \le Q_0 = 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$ satisfies the window-$(N, K) = (1, 240)$ gap certificate; $q = Q_0 + 1$ is the exact first failure.	Lean source
Coprimality-probability / Möbius-square Farey bound	Every $d \in \mathbb{N}$ with $0 < d \le Q_1 = 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (exactly $\lfloor Q_0/2 \rfloor$) is excluded as a denominator of $\Pr(\gcd(X, Y) = 1)$ and, separately, of $T = \sum_d \mu(d)/(2^d - 1)^2$.	Lean source Lean source

Remark 5.20 (Scope of the table). No row above is cofinal in its indexing parameter (h, t, or the denominator bound), and no row is claimed to extrapolate. The historical diagonal depths are irregular and nonmonotone $(6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, \dots)$; no closed-form growth rate for L_t is proved or conjectured in the record. This table records the historical 28-scale list through $t = 64$; the later aggregate theorem for every $t \le 82$ is stated above and is not itemised row by row here. Neither finite record supplies the quantified condition of Definition 1.2.

6 Detailed statements

The following statements give the exact hypotheses of the preceding arguments. We begin with unconditional identities, rank bounds and finite computations, then pass to the residue and approximation conditions. A Lean declaration proving a conditional implication retains that condition as a hypothesis; it does not supply the missing estimate for φ.

6.1 Unconditional structure and finite examples

The rank formulas hold at every stated level. The residue tests and Farey interval below instead give conclusions on specified finite sets of shifts, scales or denominators. This distinction matters when applying the later conditions, which require witnesses at unbounded indices.

Theorem 6.1 (Rationality gives an eventual tail period). *If $S = a/(2^c v)$ with $a \in \mathbb{Z}$, $c \in \mathbb{N}$ and v a positive odd integer, then*

$$R_{N+h} - R_N \in \mathbb{Z} \qquad (N \ge c), \qquad h = \varphi(v) > 0.$$

Euler's theorem gives $v \mid 2^h - 1$, and the prefix-tail identity then makes $2^N(2^h - 1)S$ integral. Thus rationality supplies a positive shift for which all sufficiently late tail differences are integral.

Theorem 6.2 (Rationality forces unbounded carry rank)**.** *If $S \in \mathbb{Q}$, there are an integer $v > 0$ and an integer sequence u such that*

$$u(N+1) = 2u(N) - v\varphi(N+1), \qquad u(N)/2^N \longrightarrow 0,$$

and, for every $e \geq 0$, the sequences $n \mapsto u(2^j n + r)$ with $1 \leq j \leq e$ and $0 \leq r < 2^j$ span a space of $\mathbb{Q}$-dimension at least $2^e - 1$.

The recurrence follows from the scaled tails after clearing a hypothetical denominator of S. The only growth condition imposed here is the displayed limit $u(N)/2^N \to 0$; subexponential growth is not assumed. The rank lower bound is a necessary consequence of rationality and yields no contradiction by itself. An upper bound using additional properties of the actual totient coefficients would be needed; the later rational comparison rules out such a bound for arbitrary rational coefficient series.

Theorem 6.3 (Independence of the retained dyadic sections)**.** *For every integer $e \geq 0$, the family*

$$\varphi(n), \quad \varphi(2n), \quad \varphi(2^j n + r) \quad (1 \leq j \leq e,\ 0 < r < 2^j,\ r \text{ odd})$$

is linearly independent over $\mathbb{Q}$ and has $2^e + 1$ members. At $e = 0$ this family still contains $\varphi(2n)$; it is not the full level-zero truncation, which consists only of $\varphi(n)$. For $e \geq 1$ it is the basis of the full truncated dyadic family.

The proof separates one section at a time: the Chinese remainder theorem forces suitable prime divisors in the other affine arguments, while Dirichlet's theorem makes the selected argument prime. This proves independence, not just the displayed cardinality. In particular, the span of all dyadic sections is infinite-dimensional; that conclusion alone says nothing about rationality of their weighted sum.

Proposition 6.4 (Denominator divisibility forces tail integrality)**.** *If $S = a/q$ in lowest terms with $q > 0$ and $q \mid 2^N(2^h - 1)$, then $R_{N+h} - R_N \in \mathbb{Z}$. Indeed, the prefix identity expresses that difference as $2^N(2^h - 1)S$ minus an integer. This is the contradiction used by a nonintegrality certificate to exclude the rational value a/q.*

The denominator results use the following polynomial, defined for $r \geq 1$:

$$P_r(X) = \sum_{d \mid r} \mu(d) \frac{r}{d} \sum_{j=0}^{r/d-1} X^{dj}.$$

For example, $P_p(X) = p(1 + X + \cdots + X^{p-1}) - 1$ when p is prime. Thus $P_p(2) = p(2^p - 1) - 1$ is coprime to $2^p - 1$. For general $r \geq 1$ and $H \geq 1$, write

$$b_r = \frac{P_r(2)}{2^r - 1}, \qquad \beta_H = \frac{b_{\operatorname{rad}(H)}}{\operatorname{rad}(H)}, \qquad r_t = \operatorname{rad}(H_t).$$

Here $\operatorname{rad}(H)$ is the product of the distinct prime factors of H, and $\operatorname{den}(x)$ denotes the positive reduced denominator of $x \in \mathbb{Q}$. These rational numbers are finite rational contributions to the Möbius decomposition; no convergence to S is asserted for them. Also put $\mathcal{P}_t = \{p \text{ prime} : t/2 < p \leq t\}$. The hypotheses below specify which factors of $2^{r_t} - 1$ remain in the reduced denominator after cancellation.

Proposition 6.5 (A cyclotomic factor remains after cancellation)**.** *If r is squarefree, then*

$$\gcd(|P_r(2)|, |\Phi_r(2)|) = 1, \qquad |\Phi_r(2)| \mid \operatorname{den}(b_r).$$

Here Φ_r is the r-th cyclotomic polynomial. Proposition 6.40 gives $P_r \equiv \mu(r) \pmod{\Phi_r}$, and $\mu(r) = \pm 1$ because r is squarefree. This proves that the indicated cyclotomic factor cannot cancel from b_r.

Proposition 6.6 (Primes in the upper half remain after cancellation)**.** *For $t \geq 5$ the whole product*

$$\prod_{p \in \mathcal{P}_t} (2^p - 1) \mid \operatorname{den}(H_t \beta_{H_t})$$

remains after cancellation. More generally, the source gives the same product-divisibility conclusion for a multiplier coprime to that product; it also supplies a sufficient condition on the multiplier's prime factors. For the multiplier H_t/r_t, those prime factors are at most t, and the LCM result above follows.

The restriction $t \geq 5$ belongs to the sufficient prime-factor condition, not just to the LCM example. At $t = 3$, one has $r_t = 6$, $b_6 = 298/63$ and $\mathcal{P}_3 = \{2, 3\}$. The multiplier 9 has only prime factors at most 3, but $\operatorname{den}(9b_6) = 7$ is not divisible by $(2^2 - 1)(2^3 - 1) = 21$. Thus the prime-factor condition alone does not imply survival of the product at every scale. This is a denominator statement about a finite rational contribution, not a nonintegrality assertion about a real tail.

Proposition 6.7 (A lower bound for the reduced denominator)**.** *For every integer $t \geq 5$,*

$$2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{P}_t} (2^p - 1) \leq \operatorname{den}(H_t \beta_{H_t}).$$

Bertrand's postulate supplies a prime in $\mathcal{P}_t$; its factor is at least $2^{\lfloor t/2 \rfloor}$. Proposition 6.6 then gives the second inequality. The complementary term in the real tail decomposition can still cancel this rational contribution, so denominator growth alone does not prove irrationality of S.

Proposition 6.8 (The exact reduced denominator)**.** *At every integer scale $t \geq 0$,*

$$\operatorname{den}(H_t \beta_{H_t}) = \frac{2^{r_t} - 1}{\gcd\left(2^{r_t} - 1, \dfrac{H_t}{r_t} \prod_{\substack{p \mid r_t \\ p \text{ odd prime}}} (p^2 - 1)\right)}.$$

The product is 1 when there is no odd prime divisor. This gives the exact cancellation, including the multiplier H_t/r_t, rather than merely a lower bound for the denominator. For example, $H_5 = 60$ and $r_5 = 30$, and the denominator is $(2^{30} - 1)/3$.

Proposition 6.9 (Nonvanishing of a signed dyadic sum)**.** *Let I be finite, let $u_i \in \mathbb{Z}$ and $e_i \in \mathbb{N}$, and suppose that $m \in I$ is the unique index with maximal exponent e_m. If u_m is odd, then*

$$2^{e_m} \sum_{i \in I} \frac{u_i}{2^{e_i}} = \sum_{i \in I} u_i 2^{e_m - e_i} \equiv 1 \pmod 2.$$

Indeed, every summand other than u_m is even. The sum is therefore nonzero. In the determinant application, the rectangular Cauchy–Binet formula produces such finite signed sums from truncated moments. The result tests a specified configuration; it does not show that a suitable maximal-exponent term exists at arbitrarily large scales.

Proposition 6.10 (Small certificates)**.** *For each integer $1 \le h \le 8$, the finite test $\mathbb{C}(h, 12, 16)$ holds. Each discrepancy uses two 16-term windows. Across all eight shifts, only the 24 distinct totient values at $13 \le n \le 36$ are needed, since the windows overlap. The source verifies the eight integer residue inequalities by exact computation.*

Consequently, if a/b is a reduced fraction with $b > 0$ and $b \mid 2^{12}(2^h - 1)$ for at least one $1 \le h \le 8$, then $S \ne a/b$. This follows from Propositions 6.18 and 6.4.

Proposition 6.11 (A common certificate for sixteen shifts)**.** *For each integer $1 \le h \le 16$, one has $\mathbb{C}(h, 14, 9)$. The common basepoint is 14 and the depth is 9; these parameters have different roles. The two windows for each shift use only the 25 distinct totient values at $15 \le n \le 39$ across the whole family. Thus $S \ne a/b$ for every reduced fraction with $b > 0$ such that $b \mid 2^{14}(2^h - 1)$ for at least one $1 \le h \le 16$.*

The sixteen residue inequalities are a finite computation, checked by exact evaluation.

Proposition 6.12 (Historical diagonal examples and the complete band through 82)**.** *Let $H_t = \operatorname{lcm}(1, \dots, t)$ and let*

$$P(t) \quad :\Longleftrightarrow \quad \exists L,\ \mathbb{C}(H_t, H_t, L).$$

The checked certificate table proves $P(t)$ at 28 explicit indices, ending at $t = 64$. For the initial indices

$$t = 1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots,$$

the corresponding depths begin $6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, \dots$. Each entry is a finite kernel computation on explicit totient values. The factorisations use checked prime-power blocks and Lucas primality certificates. Thus the table proves a finite list of instances of $P(t)$; it does not prove that $P(t)$ holds for infinitely many t.

The historical 28 examples are now a strict subset of the aggregate theorem $\forall t \le 82,\ P(t)$. That theorem closes the finite interval without holes but supplies neither $P(83)$ nor a cofinal family.

6.2 Definitions and elementary identities

The prefix-tail identity and the truncation formula give both directions of the finite residue test. We then specialise the test to LCM shifts and express it in terms of rational approximation. Möbius inversion provides the alternative coefficient weights used in the subsequent estimates.

6.2.1 Initial implications

Definition 6.13 (The scaled tail)**.** For $N \in \mathbb{N}$,

$$R_N = \sum_{j \geq 0} \frac{\varphi(N+1+j)}{2^{j+1}}.$$

The series converges because $\varphi(n) \leq n$. It is the tail of S multiplied by 2^N, not the fractional part of $2^N S$; indeed, R_N need not lie in $[0,1)$. Lean source

Definition 6.14 (The truncated tail difference)**.** For $h, N, L \in \mathbb{N}$, define the integer

$$D(h,N,L) = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j}.$$

The sum is empty when $L = 0$. Dividing it by 2^L gives the first L terms of $R_{N+h} - R_N$. Computing the finitely many totients therefore determines $D(h,N,L)$ exactly; the residue inequality below is the decidable condition. Lean source

Definition 6.15 (The central-residue inequality)**.** For $h, N, L \in \mathbb{N}$, the finite condition $C(h,N,L)$ is

$$N + h + L + 2 < D(h,N,L) \bmod 2^L < 2^L - (N + h + L + 2).$$

It puts the residue outside both endpoint intervals of width $N + h + L + 2$. After division by 2^L, the excluded width tends to zero as L grows with h, N fixed. No analytic hypothesis is part of this finite test; the separate condition Sep quantifies over such tests. Lean source

Proposition 6.16 (The prefix-tail identity)**.** *For every $N \in \mathbb{N}$,*

$$2^N S = \Phi_N + R_N, \qquad \Phi_N = \sum_{n \leq N} \varphi(n) 2^{N-n} \in \mathbb{Z}.$$

Thus R_N and $2^N S$ have the same fractional part. Subtracting the identities at $N + h$ and N gives the tail-difference congruence used in the irrationality criterion.

Lemma 6.17 (The necessary depth inequality)**.** *For $h, N, L \in \mathbb{N}$, the condition $C(h,N,L)$ implies*

$$2(N + h + L + 2) < 2^L.$$

Indeed, its lower residue bound must be smaller than its upper bound. Thus $L > 1 + \log_2(N + h + L + 2)$ whenever a certificate holds. In particular, a fixed depth cannot accommodate unbounded $N + h$. This is a necessary depth bound, not an upper bound for finding a certificate.

Proposition 6.18 (A certificate implies nonintegrality)**.** *For all $h, N, L \in \mathbb{N}$,*

$$C(h,N,L) \implies R_{N+h} - R_N \notin \mathbb{Z}.$$

The scaled truncation error satisfies $|2^L(R_{N+h} - R_N) - D(h,N,L)| \leq N + h + L + 2$. If the tail difference were integral, $D(h,N,L)$ would therefore lie within that distance of a multiple of 2^L, contrary to the two strict residue inequalities.

Theorem 6.19 (Nonintegrality gives a certificate at some depth)**.** *For every $h, N \in \mathbb{N}$,*

$$(\exists L \in \mathbb{N},\ C(h,N,L)) \quad \Longleftrightarrow \quad R_{N+h} - R_N \notin \mathbb{Z}.$$

For the converse to Proposition 6.18, fix a positive distance from the nonintegral tail difference to the nearest integer. The normalised error $(N+h+L+2)/2^L$ tends to zero, so a sufficiently large truncation preserves that distance. This is a pointwise equivalence; irrationality requires the stated quantifiers over the shift and basepoint.

Definition 6.20 (The least common multiple of the first integers)**.** Put $H(0) = 1$ and $H(t+1) = \operatorname{lcm}(H(t), t+1)$. Then $H(t) = \operatorname{lcm}(1, \dots, t)$, $t \le H(t)$, and every positive integer $h \le t$ divides $H(t)$. This is why a single LCM parameter can absorb any fixed hypothetical period once t is large enough. Lean source Lean source

Proposition 6.21 (The Farey gap lemma)**.** *Let $a, b, c, d, r, s \in \mathbb{Z}$, with $b, d > 0$ and $bc - ad = 1$. If*

$$as < rb \quad \text{and} \quad rd < cs,$$

then $b + d \le s$. The two cross-products $rb - as$ and $cs - rd$ are positive integers, and

$$s = d(rb - as) + b(cs - rd) \ge d + b.$$

In particular, a rational r/s strictly between the Farey neighbours a/b and c/d, with positive denominator, has denominator at least $b + d$.

Proposition 6.22 (An irrationality criterion from rational approximations)**.** *Let $x \in \mathbb{R}$ and let $u_k = p_k/q_k \in \mathbb{Q}$ be in lowest terms, with $q_k > 0$. If $u_k \neq x$ for every sufficiently large k and $q_k|x - u_k| \to 0$, then x is irrational. Indeed, if $x = a/b$ with $a \in \mathbb{Z}$ and $b \ge 1$, a nonzero integer numerator gives $q_k|x - u_k| = |aq_k - bp_k|/b \ge 1/b$, a contradiction. This elementary criterion requires both nonvanishing and the scaled error estimate. Convergence $u_k \to x$ alone is insufficient, and the approximants need not be continued-fraction convergents.*

Proposition 6.23 (An irrationality criterion from near integers)**.** *Let $\xi \in \mathbb{R}$. Suppose that for every integer $q \ge 1$ there are integers m, z with*

$$0 < |m\xi - z| < 1/q.$$

Then ξ is irrational: if $\xi = a/b$ with $b \ge 1$, every nonzero such difference is at least $1/b$. Restricting the multiplier to powers $m = b_0^n$ of a fixed integer base $b_0 \ge 2$ gives a sufficient special case, not a hypothesis satisfied by every irrational number. The strict lower bound excludes exact integer hits; the upper bound must be available for arbitrarily large q.

For a binary example, form a number by concatenating the blocks 10 *at square indices $k \ge 1$ and* 01 *at the other indices. Its binary expansion is not eventually periodic, since the block sequence has increasingly long gaps between the square indices, so the number is irrational. There are no three consecutive equal digits. Every fractional part after a binary shift therefore lies in $[1/8, 7/8]$, and the multiples $2^n\xi$ do not approach the integers. This verifies that the restriction to base powers is genuinely stronger. This example is an ordinary mathematical argument, not an additional claim about the linked formalisation.*

Proposition 6.24 (A nonsingular evaluation matrix gives independence)**.** *Let I be a finite index set and let $f_j : \mathbb{N} \to \mathbb{Q}$ for $j \in I$. If there are evaluation points $n_i \in \mathbb{N}$ such that*

$$\det(f_j(n_i))_{i,j\in I} \neq 0,$$

then the family $(f_j)_{j\in I}$ is linearly independent over $\mathbb{Q}$. A relation among the functions, evaluated at the n_i, gives a vector in the kernel of this nonsingular matrix, so all coefficients vanish. Finiteness is required to form the displayed determinant. In the totient application the arithmetic work is to construct these evaluation points; the implication itself is ordinary linear algebra.

Proposition 6.25 (The gap between distinct rational numbers)**.** *If $a/b < c/d$ are reduced rational numbers with $b, d > 0$, then*

$$\frac{c}{d} - \frac{a}{b} = \frac{bc - ad}{bd} \geq \frac{1}{bd}.$$

The numerator $bc - ad$ is a positive integer. Thus an upper bound on a positive rational error gives a lower bound on the product of the reduced denominators. A large denominator in an unreduced displayed expression need not be the reduced denominator appearing in this conclusion.

Proposition 6.26 (The Möbius identity for S)**.** *The absolutely convergent series satisfy*

$$S = \sum_{n\geq 1} \frac{\varphi(n)}{2^n} = \sum_{d\geq 1} \frac{\mu(d)2^d}{(2^d-1)^2} = \frac{1}{2} + \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2}.$$

For the first rearrangement use $\varphi(n) = \sum_{d|n} \mu(d)n/d$. For the second use $2^d = (2^d-1)+1$ and the classical identity $\sum_{d\geq 1} \mu(d)/(2^d-1) = 1/2$. These are identities of values, not transfers of irrationality from a Lambert series with a different weight.

Corollary 6.27 (An equivalent irrationality question)**.** *Subtracting the rational number $1/2$ gives*

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2} \notin \mathbb{Q}.$$

This is the same irrationality question in a different series representation. The notation $L_2(\mu)$ for the right-hand series is introduced in Definition 6.29.

Proposition 6.28 (The squared-Lambert identity)**.** *Let $w : \mathbb{N} \to \mathbb{R}$ satisfy $|w(d)| \leq d$ for $d \geq 1$, and let $0 \leq r < 1$. Then*

$$\sum_{d\geq 1} w(d) \left(\frac{r^d}{1-r^d}\right)^2 = \sum_{n\geq 1} \left(\sum_{d|n} w(d) \left(\frac{n}{d} - 1\right)\right) r^n.$$

Expand $(x/(1-x))^2 = \sum_{j\geq 1}(j-1)x^j$ and group terms by $n = dj$. Absolute convergence follows from the bound on w and geometric decay. The cases $w = \mu$, $w = 1$ and $w = \varphi$ give the next three series; the same algebra applies to each, but their arithmetic values differ.

Definition 6.29 (The squared-Lambert series with weight μ)**.** Put

$$L_2(\mu) = \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2} = S - \frac{1}{2}.$$

The equality follows from Proposition 6.26 and Proposition 6.28 with $w = \mu$. Subtracting $1/2$ preserves rationality, so the irrationality of this series is exactly the question about S, not a separate easier claim. The formula is credited in §5.4. LEAN SOURCE

Proposition 6.30 (Weight one and divisor sums)**.** $\sum'_{d:\mathbb{N}^+} 1/(2^d-1)^2 = \sum'_{n:\mathbb{N}^+} (\sigma(n) - \tau(n)) \cdot (1/2)^n = \zeta_q(2) - \zeta_q(1)$ *at* $q = 1/2$*. The displayed identity is formalised. Irrationality of its value follows from the cited linear independence result of Postelmans and Van Assche, as explained in Proposition 5.17; that literature result is not formalised here. Replacing the weight* 1 *by* μ *changes the value to that of Definition 6.29. Observation 6.32 compares the two choices of weight.*

The Lean proof of the irrationality assertion assumes the linear independence theorem of Postelmans and Van Assche.

Proposition 6.31 (Totient weight and gcd moments)**.**

$$\sum_{d:\mathbb{N}^+}{}' \varphi(d)/(2^d-1)^2 = \sum_{n:\mathbb{N}^+}{}' (P(n)-n)\cdot(1/2)^n,$$

where $P = \varphi * \mathrm{Id}$ *(Pillai's gcd-sum function). It also equals* $\mathbb{E}[\gcd(X,Y)]$ *when* X, Y *are independent and* $\mathbb{P}(X = n) = \mathbb{P}(Y = n) = 2^{-n}$ *for* $n \geq 1$*; see Proposition 5.18. This is a different weighted series. Its rationality is not settled by the identities proved here.*

Observation 6.32 (Comparison of weighted series). For $L(f) = \sum_{d\geq 1} f(d)/(2^d-1)$, the three weights give

$$L(\mu) = \frac{1}{2}, \qquad L(1) = E, \qquad L(\varphi) = 2,$$

where E is the irrational Erdős–Borwein constant. For the squared denominators, they give

$$L_2(\mu) = S - \frac{1}{2}, \qquad L_2(1) = \zeta_{1/2}(2) - \zeta_{1/2}(1), \qquad L_2(\varphi) = \mathbb{E}[\gcd(X,Y)].$$

The first equality is Definition 6.29; the second and third are Propositions 6.30 and 6.31. The middle value is irrational by the cited linear independence theorem. The other two rationality questions are not settled in this record. These examples show that the choice of weight matters: an identity with the same denominators does not transfer an irrationality theorem from one weight to another. They do not classify all weighted Lambert series.

Proposition 6.33 (The weight of pairs divisible by a fixed integer)**.** *Let* X, Y *be independent random variables with* $\mathbb{P}(X = n) = \mathbb{P}(Y = n) = 2^{-n}$ *for* $n \geq 1$*. For every* $d \geq 1$*,*

$$\mathbb{P}(d \mid X,\ d \mid Y) = \frac{1}{(2^d-1)^2}.$$

This follows by multiplying the two geometric sums $\sum_{k\geq 1} 2^{-dk} = 1/(2^d-1)$. It explains the squared denominator in the preceding Möbius identity.

Proposition 6.34 (The sum over coprime directions)**.** *The sum over positive coprime pairs satisfies*

$$\sum_{\substack{a,b\geq 1\\ \gcd(a,b)=1}} \frac{1}{2^{a+b}-1} = 1.$$

To see the normalisation, write every pair of positive integers uniquely as (ka, kb) with $\gcd(a,b) = 1$, and sum $2^{-k(a+b)}$ over $k \geq 1$. The total is $(\sum_{n\geq 1} 2^{-n})^2 = 1$. This base-two normalisation is used in Proposition 6.35.

Proposition 6.35 (A Stern–Brocot recursion with stopping)**.** *For positive integers a, b, put*

$$M(a,b) = \frac{1}{(2^a-1)(2^b-1)}.$$

Then

$$M(a,b) = \frac{1}{2^{a+b}-1} + M(a+b,b) + M(a,a+b).$$

Multiplication by the common denominator proves the identity. At the root $(1,1)$, the three terms on the right are each $1/3$, and $M(1,1) = 1$.

The first term is a mass retained at the current node, not passed to either child. Thus the probabilistic interpretation is a branching process with stopping: after division by $M(a,b)$, the stopping and two transition probabilities are, respectively,

$$\frac{(2^a-1)(2^b-1)}{2^{a+b}-1}, \qquad \frac{2^a-1}{2^{a+b}-1}, \qquad \frac{2^b-1}{2^{a+b}-1}.$$

They sum to one. The stopping probability is at least $1/3$, which gives the next proposition's geometric error bound. Also $M(a,b) = \mathbb{P}(a \mid X)\mathbb{P}(b \mid Y)$ for the independent geometric variables used above. The identity does not describe a conservative mass split between the two children alone.

Proposition 6.36 (Convergence with an explicit error)**.** *In the splitting identity of Proposition 6.35, the sum of the two child terms is at most $2/3$ of the parent term. If $M_d(a,b)$ is the sum of the contributions removed during the first d levels, then*

$$|M(a,b) - M_d(a,b)| \leq (2/3)^d M(a,b).$$

In particular, $M_d(a,b) \to M(a,b)$. This convergence statement is separate from identifying the limit with a sum over any independently defined infinite set of descendants; no additional identification is asserted here.

Proposition 6.37 (A numerator polynomial and its coefficients)**.** *For squarefree $r \geq 1$, the polynomial defined above has coefficients*

$$[X^k]P_r(X) = \begin{cases} \dfrac{r}{\gcd(r,k)}\varphi(\gcd(r,k)), & 0 \leq k < r, \\ 0, & k \geq r. \end{cases}$$

For $k < r$, extracting the coefficient gives $\sum_{d\mid\gcd(r,k)} \mu(d)r/d$; the usual divisor formula for φ evaluates this sum. All coefficients in the range $0 \leq k < r$ are positive. Thus a signed sum of geometric polynomials has an explicit positive-coefficient expression. The squarefree hypothesis is retained as in the cited formal statements.

Proposition 6.38 (Evaluation at two)**.** *For squarefree $r \geq 1$, evaluation of P_r at two gives the integer*

$$P_r(2) = \sum_{d\mid r} \mu(d)\frac{r}{d}\frac{2^r-1}{2^d-1}.$$

Each quotient is an integer because $d \mid r$. The formal definition sums over subsets of the prime divisors of r; for squarefree r these are exactly its divisors. This identifies the polynomial calculation with the numerator used in b_r.

Proposition 6.39 (Decomposition by the radical)**.** *For $H \geq 1$ and $r \geq 1$,*

$$H\beta_H = \frac{H}{\operatorname{rad}(H)}b_{\operatorname{rad}(H)}, \qquad \operatorname{den}(b_r) = \frac{2^r-1}{\gcd(|P_r(2)|, 2^r-1)}.$$

The first equality follows from the definition of β_H; the second is ordinary reduction of an integer fraction. Multiplication by $H/\operatorname{rad}(H)$ can cause further cancellation. No coprimality assumption is included in either identity.

The constant in Proposition 6.40 is the value of P_r at a primitive m-th root of unity. Since the X^k coefficient of P_r depends only on $\gcd(k,r)$, the value is a discrete Fourier coefficient of a function of the greatest common divisor, which Schramm expresses through Ramanujan sums [14, Theorem and (2), p. 2]; Proposition 6.40 specialises this to the present numerator polynomial and proves the integral cyclotomic consequence.

Proposition 6.40 (A cyclotomic congruence)**.** *Let r be squarefree and $m \mid r$. In $\mathbb{Z}[X]$,*

$$P_r(X) \equiv \mu(m)J_2(r/m) \pmod{\Phi_m(X)},$$

where $J_2(n) = n^2\prod_{p\mid n}(1-p^{-2})$ is the second Jordan totient. Evaluation at two gives

$$\Phi_m(2) \mid P_r(2) - \mu(m)J_2(r/m).$$

Proof. To prove the congruence, let ζ be a primitive m-th root of unity. Since $m \mid r$, the geometric sum in the definition of P_r is r/d when $m \mid d$ and zero otherwise. Hence

$$P_r(\zeta) = \sum_{\substack{d\mid r \\ m\mid d}} \mu(d)(r/d)^2.$$

Write $r = mu$. Squarefreeness gives $\gcd(m, u) = 1$, so substituting $d = me$ reduces this to $\mu(m)\sum_{e\mid u}\mu(e)(u/e)^2 = \mu(m)J_2(u)$. The minimal polynomial Φ_m therefore divides the difference in $\mathbb{Q}[X]$; because it is monic, polynomial division gives the quotient in $\mathbb{Z}[X]$. Taking $m = r$ gives the coprimality in Proposition 6.5. □

Squarefreeness cannot simply be dropped from this constant formula. For $r = 4$ and $m = 2$, $P_4(X) = 2 + 4X + 2X^2 + 4X^3$, so $P_4(-1) = -4$, whereas $\mu(2)J_2(2) = -3$. Here $P_4(2) = 50$ and $\Phi_4(2) = 5$; the latter factor cancels from $b_4 = 50/15 = 10/3$. Positive polynomial coefficients alone therefore do not prevent denominator cancellation.

Proposition 6.41 (Adjoining a prime: new divisors)**.** *Let $r \geq 1$, let $p \nmid r$ be prime, and let $m \mid r$. Then*

$$P_{rp}(X) \equiv -P_r(X^p) \pmod{\Phi_{mp}(X)}.$$

Proof. Splitting the divisors of rp into d and pd, with $d \mid r$, gives the exact polynomial identity

$$P_{rp}(X) = p\left(\sum_{j=0}^{p-1} X^{jr}\right)P_r(X) - P_r(X^p).$$

At a primitive mp-th root ζ, the number ζ^r is a nontrivial p-th root, because $m \mid r$ and $p \nmid r$. The geometric factor vanishes, proving the stated congruence. No squarefree hypothesis on r is needed for this identity. If r is squarefree, ζ^p is a primitive m-th root, and Proposition 6.40 gives the constant remainder $-\mu(m)J_2(r/m)$. □

Proposition 6.42 (Adjoining a prime: existing divisors)**.** *Let r be squarefree, let $p \nmid r$ be prime, and let $m \mid r$. For the existing divisors m, the corresponding identity is*

$$P_{rp}(X) \equiv (p^2 - 1)P_r(X) \pmod{\Phi_m(X)}.$$

Proof. Apply Proposition 6.40 to both r and rp. Since $p \nmid r/m$, multiplicativity gives $J_2(rp/m) = (p^2 - 1)J_2(r/m)$, which proves the identity. The squarefree hypothesis is retained as in the Lean statement. This compares the remainders at the existing divisors m; the preceding proposition treats the new divisors mp. □

Proposition 6.43 (Lambert-series identities)**.** *For an arithmetic function f for which the sums converge absolutely, write $L(f) = \sum_{n\geq 1} f(n)/(2^n - 1)$. Expanding each geometric series gives*

$$L(f) = \sum_{m\geq 1} \frac{(f * 1)(m)}{2^m}.$$

In particular, if $\alpha = \varphi * \mu$, then $\alpha * 1 = \varphi$ and $L(\alpha) = S$. The table distinguishes this value from four other Lambert-series values. The identity changes the coefficients and the form of the denominator; it does not transfer an irrationality theorem for one weight to another.

Table 2: Five Lambert-series values and the support for their stated arithmetic status.

Weight f	Value $L(f)$	Status
μ	$L(\mu) = 1/2$	rational, trivial
φ	$L(\varphi) = 2$	rational
Id	$L(\mathrm{Id}) = \sum_m \sigma(m)/2^m$	transcendental, since $1 - 24L(\mathrm{Id})$ is Ramanujan's $P(1/2)$ (Nesterenko [13, Cor. 2, p. 1320]; the Lean proof takes his theorem as a hypothesis)
1	$L(1) = E$, Erdős–Borwein constant	irrational (Erdős [11]), also proved in Lean (Lean source)
$\alpha = \varphi * \mu$	$L(\alpha) = S$	**OPEN**; this is #249

Remark 6.44 (How the identities differ). The coefficients matter: a result for the constant weight 1 does not settle the series with weight $\alpha = \varphi * \mu$. A theorem expressed in Lambert-series form applies here only after its assumptions have been checked for that weight.

The equality between the two indexing conventions for S is proved in Lean (Lean source). The table keeps apart a cited theorem, an ordinary identity and a formal declaration.

Proposition 6.45 (An affine combination annihilates the specified divisor terms)**.** *Let $H \geq 1$ and $d \mid H$, with $d > 0$. The dth term in the Möbius expansion of R_{mH} is $\mu(d)K_d(mH)$, where*

$$K_d(mH) = \frac{mH}{d(2^d - 1)} + \frac{2^d}{(2^d - 1)^2}.$$

This is affine in m. Consequently, for any finite family of real coefficients c_i and nonnegative integers m_i with $\sum_i c_i = \sum_i c_i m_i = 0$, one has $\sum_i c_i K_d(m_i H) = 0$.

For the four multipliers $(1, 3, 5, 15)$ and coefficients $(4, -3, -2, 1)$,

$$K_d(15H) - 3K_d(3H) - 2K_d(5H) + 4K_d(H) = 0.$$

The zeroth and first moments are zero, while the second is $15^2 - 3 \cdot 3^2 - 2 \cdot 5^2 + 4 = 152$. Equivalently, $XY - 3X - 2Y + 4$ takes the values $0, 0, 152$ at $(1, 1), (3, 5), (9, 25)$. This explains the coefficients: they cancel the affine contribution of each divisor of H, without cancelling a general quadratic function of the multiplier.

For a finite truncation of depth $L \geq 0$, write

$$U = \sum_{j=1}^{L} (\varphi(15H + j) - 3\varphi(3H + j) - 2\varphi(5H + j) + 4\varphi(H + j))2^{L-j}.$$

The difference between $2^L(R_{15H} - 3R_{3H} - 2R_{5H} + 4R_H)$ *and* U *has absolute value at most* $19H + 5L + 5$. *Indeed, the error equals*

$$(R_{15H+L} + 4R_{H+L}) - (3R_{3H+L} + 2R_{5H+L}),$$

and both parenthesised terms lie between 0 *and* $19H + 5L + 5$ *by* $0 \le R_n \le n + 1$ *for* $n \ge 1$. *Thus*

$$19H + 5L + 5 < U \bmod 2^L < 2^L - (19H + 5L + 5)$$

is sufficient for this four-tail combination to be nonintegral. The bound is derived from these tail enclosures; no optimality for the actual totient tails is asserted. Obtaining such certificates at the required unbounded family of LCM heights remains unproved, as in Theorems 6.60 and 6.52.

Proposition 6.46 (The error under composite dilation)**.** *For* $A \subseteq \mathbb{N}$ *and* $n \ge 1$, *let* $d_A(n) = \#\{d \ge 1 : d \mid n,\ d \in A\}$. *If* $a \in A$ *and* $a, x \ge 1$, *then*

$$d_A(ax) = d_A(x) + \mathbf{1}_{a \nmid x} + \#\{d \mid ax : d \in A,\ d \nmid x,\ d \ne a\}.$$

Indeed, partition the divisors of ax *into those already dividing* x, *the possible new divisor* a, *and all other new divisors. If every element of* A *is prime, the last set is empty, since a prime dividing* ax *but not* x *must equal the prime* a. *For composite* a *it can be nonempty: at* $A = \mathbb{N}$, $a = 6$, $x = 1$, *the other new divisors are* 2 *and* 3.

This is an unweighted counting identity. The associated Lambert series satisfies

$$\sum_{\substack{a \in A \\ a \ge 1}} \frac{1}{2^a - 1} = \sum_{n \ge 1} \frac{d_A(n)}{2^n}.$$

For the totient series the required divisor-convolution weight is instead $\alpha = \varphi * \mu$, *since* $\alpha * 1 = \varphi$, *as in Proposition 6.43. Weighting all divisors by* φ *would give* $\sum_{d \mid n} \varphi(d) = n$, *not* $\varphi(n)$; *already at* $n = 2$ *these values are* 2 *and* 1. *A weighted version must retain the weights of the new divisors, so it cannot be obtained by substituting* $A = \mathbb{N}$ *in this unweighted formula. Also* $\alpha(p) = p - 2$ *for primes* p, *so* α *is unbounded and is not a periodic weight.*

Iterated prime dilations of divisor counts. Multiplying an argument by a new prime adds exactly one new family of divisors. For the divisor count d_A above and $n \ge 1$, put $\Delta_p f(n) = f(pn) - f(n)$. If $p_1, \dots, p_r$ are distinct primes, $P = p_1 \cdots p_r$ and $\gcd(n, P) = 1$, then

$$\Delta_{p_1} \cdots \Delta_{p_r} d_A(n) = \#\{d \mid n : Pd \in A\}.$$

For one prime, the divisors of pn not dividing n are exactly pd with $d \mid n$, because $p \nmid n$. Iteration gives the displayed identity: distinctness ensures that each new prime is coprime to the arguments produced at the earlier steps. Equivalently, inclusion–exclusion cancels every divisor not containing all the new prime factors.

Let $\Omega(a)$ count the prime factors of $a \ge 1$ with multiplicity, and let $K \ge 0$ be an integer. If $\Omega(a) \le K$ for every positive $a \in A$ and $r > K$, the set on the right is empty, since $\Omega(Pd) = r + \Omega(d) > K$. This proves the stated vanishing with all its hypotheses.

LEAN SOURCE The source also proves vanishing of the right-hand divisor count for a list with repeated primes. That is a statement about the condition $Pd \in A$, not an identity with repeated dilation differences.

For example, with $A = \{2\}$ and $n = 1$,

$$\Delta_2^2 d_A(1) = d_A(4) - 2d_A(2) + d_A(1) = -1, \qquad \#\{d \mid 1 : 4d \in A\} = 0.$$

Thus the bound $\Omega(a) \leq 1$ and two prime factors do not by themselves make the repeated difference vanish. The strict inequality $r > K$ is also material: $\Delta_2 d_{\{2\}}(1) = 1$ when $r = K = 1$. Finally, at $n = 2$ the one-prime identity without the coprimality hypothesis would give $\Delta_2 d_{\{2\}}(2) = 0$ on the left and $\#\{d \mid 2 : 2d \in \{2\}\} = 1$ on the right. This last example concerns the identity, not the necessity of coprimality for every possible vanishing theorem. These divisor counts belong to the unweighted support problem; they are not the totient coefficients.

6.2.2 Equivalent quantified certificate conditions

Completeness identifies the base residue condition and the LCM-diagonal condition with irrationality. The following statements spell out the quantifiers and the implications between them. The existence of the required certificates beyond every threshold remains unproved.

Theorem 6.47 (The quantified certificate condition is equivalent to irrationality)**.** $(\forall h : \mathbb{N},\ 0 < h \to \forall N_0 : \mathbb{N},\ \exists N \geq N_0,\ \exists L,\ C(h,N,L)) \leftrightarrow S \notin \mathbb{Q}$. *The quantified-condition side is exactly $C(h,N,L)$ quantified as $\forall h \geq 1\ \forall N_0 \geq 0\ \exists N \geq N_0\ \exists L$. The universally quantified assertion remains unproved. The equivalence identifies the finite witnesses that suffice; it does not establish their existence beyond every threshold.*

Theorem 6.48 (It suffices to use multiples of a period)**.** $(\forall h_0 > 0,\ \forall N_0,\ \exists m > 0,\ \exists N \geq N_0,\ \exists L,\ C(m \cdot h_0, N, L)) \to S \notin \mathbb{Q}$. *The shift may be any positive multiple of a prescribed period. This enlarges the choice of finite witness: Theorem 6.47 gives the condition with $m = 1$. Conversely, rationality would make all such tail differences integral after a fixed starting index. The displayed implication therefore makes this quantified condition equivalent to irrationality; its truth remains unproved.*

Theorem 6.49 (The diagonal condition is equivalent to irrationality)**.** $(\forall t_0 : \mathbb{N},\ \exists t \geq t_0,\ \exists L,\ C(H(t),H(t),L)) \leftrightarrow S \notin \mathbb{Q}$. *For a fixed hypothetical rational value, sufficiently large t makes both its dyadic denominator and its odd-part period admissible at $N = h = H(t)$. Conversely, irrationality and pointwise completeness supply a witness for every prescribed t. This is an equivalent condition with one scale parameter; the verified cases $t \leq 82$ do not establish it at arbitrarily large scales.*

Lemma 6.50 (Nondivisors in a short LCM window)**.** *Let $t \geq 1$ and $1 \leq j < 2t$ be integers. If $j \nmid H(t)$, then $j = p^a > t$ for a prime p and an integer $a \geq 1$. Indeed, a prime-power divisor of j must exceed t, and $j < 2t$ leaves no room for a cofactor larger than 1. This classifies the exceptional offsets in this short window; it does not bound their contribution to a weighted sum of totient differences.*

Proposition 6.51 (Totient factorisation on an LCM progression)**.** *Let $t \geq 1$, $j \geq 1$ and $q \geq 0$ be integers. Suppose $j \mid H(t)$ and every prime divisor of j also divides $H(t)/j$. Then*

$$\varphi(qH(t) + j) = \varphi(j)\,\varphi(q(H(t)/j) + 1).$$

The two factors in $qH(t) + j = j(q(H(t)/j) + 1)$ are coprime: the second is 1 modulo each prime dividing j. Totient multiplicativity gives the identity. The extra prime-divisor hypothesis is not automatic from $j \mid H(t)$; for example, it fails at $t = j = 2$, where $\varphi(H(2) + 2) = 2$ but the proposed product would be 1.

Theorem 6.52 (Rationality forces tail integrality on an LCM grid)**.** *If $S \in \mathbb{Q}$, there is $t_1 \in \mathbb{N}$ such that for all $t \geq t_1$, $q \geq 1$ and $m \geq 0$,*

$$R_{(q+m)H(t)} - R_{qH(t)} \in \mathbb{Z}.$$

Thus all tails at positive multiples of $H(t)$ have the same fractional part once $t \geq t_1$. A certificate at one fixed scale does not refute rationality: the threshold t_1 depends on the hypothetical rational value. At a shift $mH(t) > 0$ and basepoint $qH(t)$, such a certificate excludes rational values whose reduced denominator divides $2^{qH(t)}(2^{mH(t)} - 1)$. Certificates at arbitrarily large scales, as required in Theorem 6.53, exclude every hypothetical rational value.

Theorem 6.53 (A sufficient nonintegrality condition on the grid)**.** *Suppose that for every $t_0 \in \mathbb{N}$ there are integers $t \geq t_0$, $q \geq 1$, $m \geq 0$ and $L \geq 0$ such that*

$$C(mH_t, qH_t, L).$$

Then $S \notin \mathbb{Q}$. Rationality would make the corresponding tail difference integral at every sufficiently large scale, contradicting Proposition 6.18. A successful certificate necessarily has $m > 0$, since the discrepancy vanishes at shift zero.

The diagonal choice is $q = m = 1$; consecutive grid points have $m = 1$, and for $p \geq 2$ the pair (H_t, pH_t) has $(q, m) = (1, p - 1)$. The diagonal equivalence in Theorem 6.49 also gives the converse from irrationality to the displayed quantified condition; that condition remains unproved.

Corollary 6.54 (Equivalent conditions stated without certificates)**.** *The series S is irrational if and only if*

$$R_{2H_t} - R_{H_t} \notin \mathbb{Z} \qquad \text{for arbitrarily large integers } t.$$

Equivalently, for arbitrarily large t there are integers $q \geq 1$ and $m \geq 1$ with $R_{(q+m)H_t} - R_{qH_t} \notin \mathbb{Z}$. Apply the pointwise equivalence of Theorem 6.19 to Theorems 6.49 and 6.53, respectively. These are equivalent statements of the irrationality question; the required unbounded sets of scales are not established.

Proposition 6.55 (Soundness of a second-difference certificate)**.** *Let $h, N, L \in \mathbb{N}$. If*

$$\begin{aligned} 2(N + 2h + L + 2) &< (D(h, N + h, L) - D(h, N, L)) \bmod 2^L \\ &< 2^L - 2(N + 2h + L + 2), \end{aligned}$$

then $R_{N+2h}-2R_{N+h}+R_N \notin \mathbb{Z}$. The numerator is the second difference of the finite windows. Subtracting the two tail identities leaves an error of absolute value at most $2(N+2h+L+2)$, which explains the radius in the hypothesis. At $(h,N)=(1,8)$, the first-difference test holds at depth 8, whereas the second-difference test fails at every depth $L \le 8$ and holds at $L=9$. A historical probe over $t \le 20$ reported that the first-difference test was at least as shallow in 30 of 40 sampled cases. This finite comparison is not a uniform ordering of the depths.

For $H, L \in \mathbb{N}$ and a finite set Q of multipliers, put

$$A_q = \sum_{j=1}^{L} \varphi(qH+j)2^{L-j}, \qquad B_q = qH+L+2 \quad (q \in Q).$$

The finite-grid test compares these integer numerators modulo 2^L.

Theorem 6.56 (A finite-grid condition gives a nonintegral pair)**.** *Let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. Suppose that $B_q < 2^L$ for every $q \in Q$ and that*

$$\forall q_i \in Q\ \exists q_j \in Q, \qquad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L.$$

Then $R_{q_jH} - R_{q_iH} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$.

Proof. The tail identity and its elementary bound give

$$2^L R_{qH} = A_q + R_{qH+L}, \qquad 0 \le R_{qH+L} \le B_q.$$

Suppose all pairwise tail differences are integers. Choose $q_* \in Q$ minimising R_{qH+L}. For every $q \in Q$, the difference $R_{qH+L} - R_{q_*H+L}$ is then an integer in $[0, B_q]$, congruent to $A_{q_*} - A_q$ modulo 2^L. Since $B_q < 2^L$, it is the least nonnegative residue. This contradicts the displayed condition at q_*. □

For example, take $H = 840$, $L = 12$ and $Q = \{1,2,3,4\}$. The following exact values verify one failed arc containment in each row:

q_i	$A_{q_i} \bmod 4096$	B_{q_i}	q_j	$(A_{q_i} - A_{q_j}) \bmod 4096$
1	2920	854	3	3484
2	2024	1694	1	3200
3	3532	2534	4	3832
4	3796	3374	1	876

The last column exceeds B_{q_j}, not necessarily B_{q_i}. On the circle $\mathbb{R}/4096\mathbb{Z}$, every two arcs $[A_q, A_q + B_q]$ intersect, but the four arcs have no common point. Thus a test using all four points can exclude a common fractional part without any pair of these arcs being disjoint. The theorem does not require pairwise compatibility, or require $|Q| \ge 3$.

The size hypothesis is $2^L > \max_{q \in Q} B_q$. In contrast, the symmetric test C for a pair $q_i < q_j$ needs $2^L > 2B_{q_j}$ before its two-sided interval can contain a residue. These are inequalities for the modulus 2^L. They do not say that the new test uses half as many binary digits, or give a uniform comparison of the first depths at which the tests succeed.

Theorem 6.57 (A sufficient quantified finite-grid condition)**.** *Suppose that for every t_0 there are $t \geq t_0$, a depth L, and a finite nonempty $Q \subseteq \mathbb{N}_{>0}$ such that, with $H = H_t$, all the hypotheses of Theorem 6.56 hold. Then $S \notin \mathbb{Q}$. Indeed, rationality would make all tail differences on every sufficiently large LCM grid integral, whereas the finite-grid theorem supplies a nonintegral pair on such a grid. The example above verifies one grid; it does not establish the quantified hypothesis.*

Proposition 6.58 (A finite carry test implies nonintegrality)**.** *For each integer z with $|z| \leq N + h + 1$, define*

$$c_z(0) = z, \qquad c_z(i+1) = 2c_z(i) - (\varphi(N+h+i+1) - \varphi(N+i+1)).$$

Suppose that for every such z there is an $i \leq K$ with

$$|c_z(i)| \geq N + i + h + 2.$$

Then $R_{N+h} - R_N \notin \mathbb{Z}$. Indeed, an integral value of the tail difference would be one of the $2(N+h+1)+1$ initial candidates. Its recurrence would satisfy $c_z(i) = R_{N+h+i} - R_{N+i}$ and hence $|c_z(i)| < N + i + h + 2$ at every step, contrary to the finite test. The excluded strip is open: equality at its boundary already suffices. This proposition proves soundness for a specified h, N, K; an irrationality argument still needs the quantified family of tests.

Theorem 6.59 (The block norm condition implies irrationality)**.** *Suppose that for every integer $h \geq 1$ and every threshold $X_0 \in \mathbb{N}$ there are $X, L \in \mathbb{N}$ with*

$$X \geq \max(X_0, 1), \qquad 16(2X + h + L + 2) \leq 2^L,$$

and

$$\left\| \sum_{X \leq N < 2X} E(h, N, L) \right\| \leq \frac{21}{25} X.$$

Then $S \notin \mathbb{Q}$. Indeed, the norm bound implies the real-part bound $\operatorname{Re} \sum E(h, N, L) \leq (9/10)X$. Under the displayed size condition, this latter bound already gives a finite residue certificate, by $\cos(\pi/8) > 9/10$ and averaging. The argument uses this specific numerical gap; an unspecified saving below X is not the stated hypothesis.

The hypothesis, the block norm condition of Definition 3.1, is unproved.

Theorem 6.60 (A four-tail residue criterion)**.** *For nonnegative integers H, p, L, define the four-tail combination, its finite numerator and its error bound by*

$$\begin{aligned} J(H,p) &= R_{2pH} - R_{pH} - pR_{2H} + pR_H, \\ W(H,p,L) &= D(pH, pH, L) - pD(H,H,L), \\ B(H,p,L) &= 3pH + (p+1)(L+2). \end{aligned}$$

Then

$$|2^L J(H,p) - W(H,p,L)| \leq B(H,p,L).$$

Consequently the finite condition

$$B(H,p,L) < W(H,p,L) \bmod 2^L < 2^L - B(H,p,L)$$

implies $J(H,p) \notin \mathbb{Z}$. *If, for every* $t_0 \in \mathbb{N}$, *there are* $t \geq t_0$, $p \geq 1$ *and* $L \geq 0$ *satisfying this condition with* $H = H(t)$, *then* $S \notin \mathbb{Q}$. *Rationality would instead make both diagonal differences in* $J(H(t),p)$ *integral for every sufficiently large* t.

The error bound groups the remainder as

$$(R_{2pH+L} + pR_{H+L}) - (R_{pH+L} + pR_{2H+L}).$$

Each parenthesis is nonnegative and at most $B(H,p,L)$, *by* $0 \leq R_n \leq n+2$. *Their difference therefore has absolute value at most* $B(H,p,L)$. *This proves the bound without asserting optimality for the actual totient tails. Primality of* p *is not required by the criterion; a prime* p *is useful when comparing consecutive LCM heights. The checked instance is* $(H,p,L) = (12,5,15)$, *for which* $W(H,p,L) \bmod 2^L = 18834$ *and* $B(H,p,L) = 282$.

The checked instance is a finite computation. The hypothesis for every t_0 is unproved.

6.3 Consequences

The next conditions concern a single LCM-diagonal tail difference. For $t, j, a \in \mathbb{N}$, write

$$\delta_t(j) = \varphi(2H(t)+j) - \varphi(H(t)+j), \qquad \Omega_a = R_{2H(2^a)} - R_{H(2^a)}.$$

For $h, N \in \mathbb{N}$ and $z \in \mathbb{Z}$, define the integer recurrence

$$c_{h,N,z}(0) = z, \qquad c_{h,N,z}(k+1) = 2c_{h,N,z}(k) - (\varphi(N+h+k+1) - \varphi(N+k+1)).$$

If $z = R_{N+h} - R_N$, then $c_{h,N,z}(k) = R_{N+h+k} - R_{N+k}$. This separates an arbitrary integer initial value from the actual tail.

Two finite quantities will also recur. With $H = H_{2^a}$, put

$$\rho_{a,q} = \frac{D(H,H,2q+1) + \delta_{2^a}(2q+2)}{2^{2q+1}}, \qquad \varepsilon_{a,q} = \frac{2H+2q+3}{2^{2q+1}}.$$

The prescribed odd depth used below is $2q_a + 1$, where

$$q_a = \left\lfloor \frac{\lfloor \log_2 H \rfloor + 10}{2} \right\rfloor.$$

Thus $2q_a + 1$ is the least odd integer at least $\lfloor \log_2 H \rfloor + 10$. The error bound is $|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q}$. For $a \geq 2$, let $u_{a,q}$ be the centred integer representative of

$$\frac{D(H,H,2q+1) + \delta_{2^a}(2q+2)}{2} \pmod{4^q}, \qquad -\frac{4^q}{2} < u_{a,q} \leq \frac{4^q}{2}.$$

The numerator is even, since the totient arguments exceed 2; at the midpoint this convention chooses the positive representative. The normalised block $\rho_{a,q}$ approximates Ω_a, whereas $u_{a,q}$ records a residue. They must not be confused with the real tail difference or with the integer recurrence.

Proposition 6.61 (An equivalent diagonal nonintegrality condition)**.** *The following statements are equivalent:*

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad \forall a_0 \in \mathbb{N}\ \exists a \geq a_0,\ \Omega_a \notin \mathbb{Z}.$$

This is an exact reformulation of irrationality, not a weaker theorem about a finite set of scales. The later sufficient conditions seek a finite inequality that establishes the right-hand side.

Proposition 6.62 (A sufficient short-window condition)**.** *Suppose that for every $a_0 \in \mathbb{N}$ there are $a \geq a_0$ and $L < 2 \cdot 2^a$ with $C(H_{2^a}, H_{2^a}, L)$. Then $S \notin \mathbb{Q}$. Writing out the certificate, the required inequality is*

$$2H_{2^a} + L + 2 < D(H_{2^a}, H_{2^a}, L) \bmod 2^L < 2^L - (2H_{2^a} + L + 2).$$

It places the residue away from both endpoints of the modulus interval. The examples at $a = 4, 6$ establish the bounded statement in Proposition 6.67, not the condition for every threshold. Pointwise completeness allows a sufficiently large depth but does not supply a depth below $2 \cdot 2^a$.

The hypothesis is unproved.

Proposition 6.63 (A sufficient approximation condition)**.** *Use the prescribed index q_a and the error ε_{a,q_a} defined above. Suppose*

$$\forall a_0 \in \mathbb{N}\ \exists a \geq \max(2, a_0)\ \forall z \in \mathbb{Z}, \qquad |\Omega_a - z| \geq \frac{1}{32} + \varepsilon_{a,q_a}.$$

Then $S \notin \mathbb{Q}$. The estimate $|\Omega_a - \rho_{a,q_a}| < \varepsilon_{a,q_a}$ implies $|\rho_{a,q_a} - z| > 1/32$ for every integer z, and hence the finite residue separation from which the proof derives irrationality. This is a fixed separation condition along an unbounded family of LCM scales. It is not an irrationality-measure theorem for arbitrary rational approximants, and nonintegrality alone does not give the stated margin. The depth is prescribed, not an additional free witness.

The separation hypothesis is unproved.

Proposition 6.64 (A sufficient upper-endpoint separation)**.** *Let $a, J, K, m \in \mathbb{N}$, $a \geq 8$, and $H = H(2^a)$. Assume*

$$J + K + (a + 6) < 2 \cdot 2^a.$$

Define the finite condition

$$\begin{aligned} \mathcal{G}_a(J, K, m) \quad :\Longleftrightarrow \quad & m \leq K, \qquad 2H + J + K + 2 < 2^m, \\ & D(H, H + J, K) \bmod 2^m \leq 2^m - (2H + J + K + 2). \end{aligned}$$

Then $\mathcal{G}_a(J, K, m)$ implies $R_{2H+J} - R_{H+J} \notin \mathbb{Z}$. In particular, irrationality follows if for every a_0 there are $a \geq \max(a_0, 8)$ and K, m satisfying $K + (a + 6) < 2 \cdot 2^a$ and $\mathcal{G}_a(0, K, m)$.

The sign theorem forces an integral tail to give a residue in the upper endpoint interval. This one-sided test excludes that interval; it does not require a lower residue bound. Both the sign-range hypothesis and the strict modulus bound are essential premises of this implication. The required unbounded family is not established.

Proposition 6.65 (Relations among the sufficient conditions)**.** *The sufficient conditions do not form a single linear chain. The proved implications are*

$$\begin{aligned}
&\textit{odd-prefix interval at the prescribed depth} \iff \textit{centred-residue bound at the prescribed depth},\\
&\textit{centred-residue bound at the prescribed depth} \implies \textit{centred-residue bound at a variable depth},\\
&\textit{centred-residue bound at a variable depth} \implies \textit{adjacent-suffix band} \implies \textit{upper-endpoint test}.
\end{aligned}$$

There is also a separate sufficient argument: terminal dominance and flexible centred magnitude each imply the two-sided exclusion in Proposition 6.66, which implies diagonal nonintegrality. These statements refer to the quantified versions, with the bounds on a, q and the depth retained. The detailed conditions and sources appear in Proposition 9.14. No reverse implication between terminal dominance and flexible magnitude is asserted, and none of the required unbounded families is proved here.

Proposition 6.66 (An exact endpoint formula)**.** *Let $a, q \in \mathbb{N}$, $a \geq 8$, and $H = H(2^a)$. Assume*

$$2q + 2 + (a + 6) < 2 \cdot 2^a, \qquad 2(H + q + 2) \leq 4^q.$$

If $z \in \mathbb{Z}$ and $z = \Omega_a$, then

$$2u_{a,q} = \delta_{2^a}(2q + 2) - c_{H,H,z}(2q + 1).$$

Consequently either of the inequalities

$$\begin{aligned}
2u_{a,q} &\leq \delta_{2^a}(2q + 2) - (2H + 2q + 3),\\
\delta_{2^a}(2q + 2) &\leq 2u_{a,q}
\end{aligned}$$

is sufficient for $\Omega_a \notin \mathbb{Z}$. Under integrality the recurrence equals a positive tail difference strictly smaller than $2H + 2q + 3$, and the identity rules out both displayed inequalities. The identity is exact; the resulting nonintegrality criterion is only sufficient. Its converse is not asserted.

Proposition 6.67 (A sufficient extension of the examples through exponent 6)**.** *The supplied finite result is*

$$\forall a_0 \leq 6\ \exists a, L \in \mathbb{N}, \qquad a \geq a_0, \quad L < 2 \cdot 2^a, \quad C(H_{2^a}, H_{2^a}, L).$$

The witnesses at $(a, L) = (4, 23)$ and $(6, 93)$ give this bounded family. A witness with $a \geq 7$ would extend the verified range, but any finite extension would still leave the assertion for arbitrarily large thresholds in Proposition 6.62 unproved.

Proposition 6.68 (A sufficient extremal-order condition)**.** *Let $H \geq 1$ and $j \geq 0$. If*

$$\varphi(2H + j) < \min\{\varphi(H + j), \varphi(3H + j)\},$$

then

$$\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) > 0.$$

If instead the middle value is strictly larger than both outer values, the displayed second difference is negative. In each case the claim follows by adding the two strict differences. No lower bound on their sizes is needed.

For $H = H_{2^a}$, this reduces nonvanishing of one second difference to an ordering of three totient values. For a prescribed j, the remaining question is whether either strict ordering occurs for arbitrarily large a. The elementary implication does not establish those occurrences.

Proposition 6.69 (A directed certificate condition)**.** *For fixed $h, N \in \mathbb{N}$, a depth $L \in \mathbb{N}$ satisfying*

$$N + L + 2 \;\le\; D(h, N, L) \bmod 2^L \;\le\; 2^L - (N + h + L + 2)$$

exists if and only if $R_{N+h} - R_N \notin \mathbb{Z}$. The endpoint inequalities are non-strict.

Proof. The exact truncation identity leaves an error $R_{N+h+L} - R_{N+L}$ strictly between $-(N + L + 2)$ and $N + h + L + 2$. An integral tail difference would therefore put the residue in $[0, N + L + 2)$ or $(2^L - (N + h + L + 2), 2^L)$, contrary to the displayed test. Conversely, a nonintegral difference has a symmetric certificate by Theorem 6.19, and every symmetric certificate satisfies this wider interval condition. □

At $H_3 = 6$ and $L = 6$, the discrepancy is $D(6, 6, 6) = 270$, with residue 14 modulo 64. The directed interval is $[14, 44]$, so its lower endpoint succeeds. The symmetric interval is $(20, 44)$ and fails; its first successful depth is 7. Thus the wider test can improve the first successful depth, but no uniform depth saving is asserted. Replacing C by the displayed inequality in Theorem 6.49 gives the equivalent directed diagonal condition, whose occurrence at arbitrarily large scales remains unproved.

Proposition 6.70 (A sufficient condition at the prescribed mod-four pulses)**.** *Suppose that for every positive integer h and every $B \in \mathbb{N}$ there are a prime $p > B$ and $K \in \mathbb{N}$ such that*

$$\varphi(p + 4h) - \varphi(p) \equiv 2 \pmod 4$$

and every integer z with $|z| \le p + 4h + 1$ and $z \equiv 2 \pmod 4$ has an index $0 \le i \le K$ satisfying

$$|c_{4h,p,z}(i)| \ge p + i + 4h + 2.$$

Then $S \notin \mathbb{Q}$. Under a hypothetical eventual period h, the proved congruence argument at a sufficiently large such prime forces $R_{p+4h} - R_p$ to be an integer in the tested residue class. The finite test excludes every candidate in that class, a contradiction.

The mod-four congruence is supplied by Proposition 6.91; exclusion at the same prime is not. Restricting to $2 \pmod 4$ retains one residue class out of four, not necessarily exactly one quarter of the candidates in a finite interval. The quantifier over every bound B cannot be replaced by one fixed prime for each h.

Proposition 6.71 (The full quantified condition)**.**

$$(\forall h \ge 1,\ \forall N_0,\ \exists N \ge N_0,\ \exists L,\ C\,h\,N\,L) \iff S \notin \mathbb{Q}.$$

This is exactly the quantifier structure $\forall h \ge 1\, \forall N_0\, \exists N \ge N_0\, \exists L\, C(h, N, L)$. The quantified condition is not supplied here. The equivalence specifies exactly what a residue-based proof would need beyond the finite examples.

Proposition 6.72 (One diagonal parameter suffices)**.** *The following condition is equivalent to irrationality:*

$$(\forall t_0 \in \mathbb{N}\ \exists t \geq t_0\ \exists L \geq 0,\ C(H(t), H(t), L)) \quad \Longleftrightarrow \quad S \notin \mathbb{Q}.$$

For a hypothetical rational value, let h_0 and N_0 be its eventual tail period and starting index. Taking $t \geq \max(h_0, N_0)$ ensures $h_0 \mid H(t)$ and $H(t) \geq N_0$. The corresponding diagonal difference is then integral, contradicting a certificate at that scale. Conversely, irrationality and pointwise completeness provide a witness at every prescribed t. The reduction uses one scale parameter; it neither establishes the certificate condition nor makes one fixed scale sufficient for all rational values.

Proposition 6.73 (A sufficient condition on the LCM grid)**.**

$$(\forall t_0,\ \exists t \geq t_0,\ \exists q\, m\, L,\ 0 < q \wedge C(m \cdot H\, t)(q \cdot H\, t)\ L) \implies S \notin \mathbb{Q}.$$

One residue certificate anywhere on the two-multiplier LCM cone, at arbitrarily large t, suffices; the diagonal choice $q = m = 1$ recovers Proposition 6.72.

The hypothesis is unproved.

Proposition 6.74 (A finite-grid certificate condition)**.** *Let $H, L \in \mathbb{N}$ and let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. Define $A_q = \sum_{j=1}^{L} \varphi(qH + j)2^{L-j}$ and $B_q = qH + L + 2$. If $B_q < 2^L$ for every $q \in Q$ and*

$$\forall q_i \in Q\ \exists q_j \in Q, \qquad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L,$$

then $R_{q_j H} - R_{q_i H} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$. Such a test at $H = H_t$ for arbitrarily large t implies $S \notin \mathbb{Q}$. Neither $|Q| \geq 3$ nor compatibility of every pair is a hypothesis. The proof, by choosing a minimum tail in the finite set, is given in Theorem 6.56.

The existence of such tests at arbitrarily large t is unproved.

Proposition 6.75 (Unbounded Farey bounds imply irrationality)**.** *The Farey-gap denominator bound at window K is currently $\sim 7.96 \times 10^{34}$ at $K = 240$ (Prop. 6.99). If*

$$\sup_K (b + d)(K) = \infty$$

(that is, the proved exclusion bounds are arbitrarily large), then #249 follows via the denominator-exclusion implication, without assuming the certificate supply in Proposition 6.71. No unboundedness claim is proved or attempted in the record.

Proposition 6.76 (The lower bound and a false proposed upper bound)**.** *By Prop. 6.98, the canonical dyadic totient-kernel family is unconditionally $(2^e + 1)$-dimensional at every level $e \geq 1$. Rationality of S forces an associated carry orbit with $\mathbb{Q}$-rank $\geq 2^e - 1$ at every level (Prop. 6.90). An upper bound independent of e for these same carry ranks would contradict the lower bound. More generally, a bound $g(e)$ with $g(e) < 2^e - 1$ at some level $e \geq 1$ would suffice, provided it applies to the actual carry under the hypothetical rationality of S. A bound that merely grows with e need not contradict anything. The generic assertion for arbitrary rational coefficient*

series is false: the $5/4$ *comparison sequence in Section 10.8 has an integer carry satisfying the growth condition and the same rank lower bound. The incompatible integer identity and the finite-shift counterexample (Observation 6.115) remain separate counterexample results. None gives a rank upper bound from rationality alone. The Lean proof covers the rank lower bound and counterexample results, not the counterfactual upper bound.*

Proposition 6.77 (Soundness of the finite carry test)**.** *For fixed* $h, N, K \in \mathbb{N}$*, use the integer recurrences* $c_{h,N,z}$ *defined above. Test each of the* $2(N+h+1)+1$ *candidates* $z \in \mathbb{Z}$ *with* $|z| \le N+h+1$*. If every candidate has some* $i \le K$ *for which*

$$|c_{h,N,z}(i)| \ge N+i+h+2,$$

then $R_{N+h} - R_N \notin \mathbb{Z}$*. An integral tail difference would give one of the initial candidates and would remain in the open strip* $|c_{h,N,z}(i)| < N+i+h+2$ *at every step. Equality at a boundary already excludes a candidate. This is the soundness implication for a specified finite test; no bound for a successful search depth is asserted.*

6.4 Further exact identities

We now bound the errors in expansions of tail differences, without assuming residue separation. Such bounds determine whether a proposed rational centre stays far enough from the integers after the omitted terms are restored.

Proposition 6.78 (The exact totient difference on an LCM progression)**.** *Put* $H = H(t)$*. For every* $j \ge 0$*, the arithmetic expression in the piecewise form given below equals* $\delta_t(j) = \varphi(2H+j) - \varphi(H+j)$*. For a divisor* $j \mid H$*, this assertion contains a useful product formula, not merely a change of notation. Set*

$$a = H/j, \qquad g_1 = \gcd(j, a+1), \qquad g_2 = \gcd(j, 2a+1).$$

Then

$$\delta_t(j) = \varphi(j)\left(\frac{g_2\varphi(2a+1)}{\varphi(g_2)} - \frac{g_1\varphi(a+1)}{\varphi(g_1)}\right).$$

Here $j \ge 1$ *because* $H > 0$*, so both denominators are nonzero. The identity follows from* $\varphi(jx) = \varphi(j)\varphi(x)\gcd(j,x)/\varphi(\gcd(j,x))$*, which retains all primes shared by* j *and* x*. If every prime dividing* j *also divides* a*, then* $g_1 = g_2 = 1$*, and the formula reduces to* $\varphi(j)(\varphi(2a+1) - \varphi(a+1))$*. For example, at* $H = j = 2$ *the two gcds are* $g_1 = 2$ *and* $g_2 = 1$*; the exact difference is* $\varphi(6) - \varphi(4) = 0$*, whereas discarding the gcd factors would give* 1*. For* $j \nmid H$*, that expression is the literal totient difference. Thus no coprimality assumption is imposed on the full expression.*

Proposition 6.79 (A lower bound for the totient of a rough integer)**.** *Let* $a \ge 8$*, set* $t = 2^a$*, and let* $n > 0$ *be an integer all of whose prime factors exceed* t*. If* $n < 2^{2t}$*, then* n *has fewer than* $t/4$ *distinct prime factors and*

$$\varphi(n) > \frac{3}{4}n.$$

If n *has* r *distinct prime factors, then* $n \ge t^r = 2^{ar}$*, so* $ar < 2t$ *and* $r < t/4$ *because* $a \ge 8$*. For the second, use* $\varphi(n)/n = \prod_{p\mid n}(1-1/p) \ge 1 - \sum_{p\mid n} 1/p > 3/4$*. These are bounds for rough integers in the stated size range; they are not pointwise bounds for arbitrary totient values.*

Proposition 6.80 (Positivity of the short-window differences)**.** *For $a \geq 8$ and $1 \leq j < 2{\cdot}2^a$,*

$$\delta_{2^a}(j) = \varphi(2H_{2^a} + j) - \varphi(H_{2^a} + j) > 0.$$

The proof treats divisor offsets and the exceptional prime-power offsets separately, using multiplicativity and the rough-integer estimate of Proposition 6.79. No rationality hypothesis is used. Positivity of these finitely many coefficients contributes to the later sign theorem; a bound for the remaining infinite tail is still needed to obtain that theorem.

Proposition 6.81 (The accumulated sum is the discrepancy)**.** *For all $t, L \in \mathbb{N}$, the weighted diagonal sum is exactly*

$$\sum_{j=1}^{L} \delta_t(j) 2^{L-j} = D(H(t), H(t), L).$$

Thus its two-sided residue inequality is precisely $C(H(t), H(t), L)$. The equality follows term by term from the definition of δ_t. A separate restriction on L, when present, remains part of the hypothesis; this identity does not supply such a restricted-depth witness.

Proposition 6.82 (A global-to-local identity)**.**

$$\Omega_a \;=\; 2^H(2^H - 1)\Big(\textstyle\sum_n' \varphi(n)/2^n\Big) - (\Phi_{2H} - \Phi_H), \quad H = H(2^a).$$

Thus a separation estimate for Ω_a is an estimate for this integer translate of a multiple of S. The identity does not require other rational-approximation arguments to use this particular translate.

Proposition 6.83 (A rational approximation with an error bound)**.** *For all $a, q \in \mathbb{N}$, with $H = H_{2^a}$ and the finite $\rho_{a,q}$ defined above,*

$$|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q} = \frac{2H + 2q + 3}{2^{2q+1}}.$$

The triangle inequality transfers a separation estimate between the real tail and its rational approximation at the cost of this error. The bound tends to zero with q for fixed a; no assertion of optimality is made.

Proposition 6.84 (Unconditional positivity)**.** *For $a \geq 8$, $J + (a + 6) < 2 \cdot 2^a$:*

$$0 < R_{2H+J} - R_{H+J}, \qquad H = H(2^a),$$

with no irrationality hypothesis; the true, infinite, real translated tail difference is strictly positive throughout almost the entire short window, proved unconditionally from Prop. 6.80 plus a directed one-sided tail bound. Any argument about the sign of the actual orbit (not merely its residue mod 2^L) must agree with this; specialised at $J = 0$ this gives $0 < \Omega_a$.

Proposition 6.85 (The residue forced by integrality)**.** *Let $a, J, K \in \mathbb{N}$, $a \geq 8$, $H = H(2^a)$, and assume*

$$J + K + (a + 6) < 2 \cdot 2^a, \qquad 2H + J + K + 2 < 2^K.$$

If $R_{2H+J} - R_{H+J} \in \mathbb{Z}$, then

$$D(H, H+J, K) \bmod 2^K = 2^K - (R_{2H+J+K} - R_{H+J+K}),$$
$$0 < R_{2H+J+K} - R_{H+J+K} < 2H + J + K + 2.$$

The later tail difference is a positive integer. Its negative is the representative near zero, while the least nonnegative residue lies near 2^K. Positivity alone therefore does not prove nonintegrality; a separate estimate must exclude this upper interval.

Proposition 6.86 (The penultimate term of a partial divisibility pattern)**.** *Let $a, J, K, m \in \mathbb{N}$, $a \geq 8$, $H = H_{2^a}$ and $B = 2H + J + K + 2$. Suppose that*

$$0 < m \leq K, \qquad J + K + a + 6 < 2 \cdot 2^a,$$
$$B < 2^m, \qquad \delta_{2^a}(J + K) \leq 2^m - B,$$
$$2^{r+1} \mid \delta_{2^a}(J + K - m + r + 1) \quad (0 \leq r,\ r + 1 < m).$$

Then

$$\delta_{2^a}(J + K - 1) = 2^{m-1}, \qquad 2^m < 2B.$$

Indeed, $B \geq 4$ and $B < 2^m$ force $m \geq 3$. The penultimate divisibility, positivity, and the bound $\delta_{2^a}(J + K - 1) < B < 2^m$ leave 2^{m-1} as its only possible value. Thus the modulus lies in the strict interval $B < 2^m < 2B$; there is at most one possible power of two. The value fixed at 2^{m-1} is the penultimate difference. The terminal difference $\delta_{2^a}(J + K)$ instead satisfies the separate upper bound in the hypotheses. These restrictions are necessary for the specified pattern; the proposition does not construct such patterns at arbitrarily large a.

Proposition 6.87 (An equivalent test using two residue bits)**.** *For every $h, N \in \mathbb{N}$, existence of a certificate at some depth is equivalent to the following explicit condition:*

$$\exists s, b \in \mathbb{N}, \quad C(h, N + s, b + 1) \quad \text{or} \quad \begin{cases} N + s + h + b + 4 < 2^b, \\ 2^b \leq D(h, N + s, b + 2) \bmod 2^{b+2} < 3 \cdot 2^b. \end{cases}$$

The last interval says that the two leading bits are 01 or 10. The proof uses the first successful certificate of depth L and may take $b = \lfloor \log_2(N + h + L + 2) \rfloor + 1$. Since b depends on that unknown L, the equivalence is not an a priori bound on the search depth. The proof uses the finite tail recurrence and its size bound; it supplies no totient-specific assertion that a witness exists.

Proposition 6.88 (The factor in a second difference)**.** *Let $a \geq 4$ and $j \geq 1$ be integers with $j^2 \leq 2^a$, and put $H_a = H(2^a) = \operatorname{lcm}(1, \ldots, 2^a)$. Then*

$$2\varphi(j) \mid \varphi(3H_a + j) - 2\varphi(2H_a + j) + \varphi(H_a + j).$$

To see the factor, set $A = H_a/j$. The square bound ensures that $j \mid H_a$ and that every prime divisor p of j also divides A: indeed, $pj \leq j^2 \leq 2^a$ implies $pj \mid H_a$. Consequently $\gcd(j, qA+1) = 1$ for $q = 1, 2, 3$, and the displayed difference equals

$$\varphi(j)(\varphi(3A + 1) - 2\varphi(2A + 1) + \varphi(A + 1)).$$

Also $2j \le 2^a$, so $2j \mid H_a$ and $A \ge 2$ is even. The three arguments $qA+1$ are therefore odd and greater than two, and all three totients are even. This proves the extra factor of two. The divisibility is a lower bound, not an exact valuation at each LCM height. The different family in Observation 6.108 does not satisfy these LCM and square-window hypotheses, so it does not establish sharpness within this restricted family.

Proposition 6.89 (Carry displacement and tail integrality)**.** *For a positive integer v and an integral totient carry u with $u(N)/2^N \to 0$,*

$$v \mid u(N+k) - u(N) \quad \Longleftrightarrow \quad R_{N+k} - R_N \in \mathbb{Z}.$$

The recurrence and the limit identify $u(N) = vR_N$, so the difference is $v(R_{N+k} - R_N)$. This tests divisibility of a carry difference and gives no dimension bound for the carry sections. For another coefficient sequence, the same argument applies once its carry is identified with its scaled tail under the corresponding recurrence and limit.

Proposition 6.90 (A consequence of rationality and its limitation)**.** *If $S \in \mathbb{Q}$, there are a positive integer v and an integer carry sequence u, with $u(N)/2^N \to 0$, whose retained sections through every level e have rational rank at least $2^e - 1$. At the same time, those sections are uniformly eventually periodic modulo v. Periodicity after reduction modulo v is a statement in a finite quotient; it is not a rank upper bound over $\mathbb{Q}$. Proposition 6.76 gives the lower bound, while the rational comparison sequence explains why a general rank upper bound cannot be deduced from these recurrence assumptions alone.*

Proposition 6.91 (A totient difference congruent to two modulo four)**.** *For every positive integer h and every $B \in \mathbb{N}$, there is a prime $p > B$ such that*

$$\varphi(p+4h) - \varphi(p) \equiv 2 \pmod 4.$$

To construct p, set $H = 4h$, choose a prime $r > H$ with $r \equiv 1 \pmod 4$, and use Dirichlet's theorem in the progression $p \equiv 3r - H \pmod{4r}$. This progression is reduced: its residue is odd and is $-H \not\equiv 0 \pmod r$. Then $p \equiv 3 \pmod 4$, so $\varphi(p) = p - 1 \equiv 2 \pmod 4$, whereas $r \mid p+H$ gives $4 \mid \varphi(p+H)$. Subtracting gives the stated congruence. This supplies the arithmetic congruence at arbitrarily large primes; the additional finite exclusion in Proposition 6.70 remains a separate hypothesis.

Proposition 6.92 (A prescribed residue after a long divisibility block)**.** *Let $K \ge 2$ and $H > K$ be integers. For every $B \in \mathbb{N}$ there is a prime $p > \max(B, H+K)$ such that*

$$\begin{aligned} &p \equiv 1 + 2^{K-1} \pmod{2^K}, \quad 2^K \mid \varphi(p+H), \\ &2^K \mid \varphi(p-j), \qquad\qquad 2^K \mid \varphi(p-j+H) \qquad (1 \le j < K). \end{aligned}$$

The construction uses $2K-1$ distinct auxiliary primes congruent to $1 \pmod{2^K}$: one for $p+H$, and two for each of the $K-1$ preceding positions. Choose them larger than $\max(H+K, 2^K)$. For the first prime impose $p \equiv -H$; for the pair at position j impose $p \equiv j$ and $p \equiv j - H$, respectively. These nonzero residues, together with the displayed odd residue modulo 2^K, combine by the Chinese remainder theorem into a reduced progression. Dirichlet's theorem then supplies

arbitrarily large prime values of p. Each auxiliary prime contributes the required factor 2^K to the corresponding totient.

Thus the first $K-1$ totient differences in the window starting at $p-K$ vanish modulo 2^K, while the last is 2^{K-1} modulo 2^K. Consequently

$$D(H, p-K, K) \equiv 2^{K-1} \pmod{2^K}.$$

If $R_{N+H} - R_N \in \mathbb{Z}$ for every $N \geq N_0$, choose such a prime with $p \geq N_0 + K$. Iterating the carry recurrence from $p-K$ then gives

$$R_{p+H} - R_p \in \mathbb{Z}, \qquad R_{p+H} - R_p \equiv 2^{K-1} \pmod{2^K}.$$

The initial integer is multiplied by 2^K and disappears only after reduction modulo 2^K. This does not bound its higher quotient. The condition $H > K$ is part of the construction; it does not provide arbitrarily large K for one fixed shift H.

Proposition 6.93 (A Möbius-inversion formula for the tail)**.** *For $N \in \mathbb{N}$ and $d \geq 1$, let $r_d(N) = d - (N \bmod d)$ and $q_d(N) = \lfloor N/d \rfloor + 1$. Thus $r_d(N)$ is the distance to the next strictly larger multiple of d, so $1 \leq r_d(N) \leq d$. Möbius inversion gives*

$$R_N = \sum_{d \geq 1} \mu(d) 2^{d - r_d(N)} \left(\frac{q_d(N)}{2^d - 1} + \frac{1}{(2^d-1)^2} \right).$$

Indeed, substitute $\varphi(n) = \sum_{d|n} \mu(d) n/d$ and group by d. Its multiples after N have quotients $q_d(N) + \ell$ and forward distances $r_d(N) + d\ell$, for $\ell \geq 0$; summing the two geometric series gives the formula. Absolute convergence justifies the regrouping, since the sum of the absolute divisor contributions at n is at most $\sum_{d|n} n/d \leq n^2$, and $\sum_{n>N} n^2 2^{N-n} < \infty$. This identity is available for later transformations; it does not assert that every Möbius or Lambert-series argument must use it.

Proposition 6.94 (Explicit finite examples)**.** *The finite certificates $C(H_{16}, H_{16}, 23)$ and $C(H_{64}, H_{64}, 93)$ imply $\Omega_4, \Omega_6 \notin \mathbb{Z}$. Both satisfy the short-window restriction, since $23 < 32$ and $93 < 128$. They are the certificates at $t = 16$ and $t = 64$ in the diagonal table of Section 5.7. These are two explicit instances of the desired condition, not the start of a proved induction or a claim that no other finite instances can be checked.*

Proposition 6.95 (The necessary depth)**.** *Every certificate $C(h, N, L)$ satisfies*

$$2(N + h + L + 2) < 2^L.$$

The two endpoint margins must fit inside an interval of length 2^L. In particular, successful depths grow at least logarithmically with $N + h$. This is a necessary lower bound, not an upper bound for the first successful depth. The later two-bit reformulation applies at its stated modulus and does not reverse this implication.

Proposition 6.96 (Rationality gives an eventual tail period)**.** *Suppose $S = a/(2^c v)$ in lowest terms, where $a \in \mathbb{Z}$, $c \geq 0$ and $v \geq 1$ is odd. Take $h = \varphi(v)$, including $h = 1$ when $v = 1$. Euler's theorem gives $v \mid 2^h - 1$, so*

$$2^N(2^h-1)S \in \mathbb{Z} \qquad (N \geq c).$$

The identity relating S to its scaled tails therefore gives $R_{N+h} - R_N \in \mathbb{Z}$ for every $N \geq c$. These are explicit witnesses for the eventual period asserted in Proposition 6.71. To use the finite residue test against a hypothetical rational value, one must obtain a nonintegral difference for its period at an index beyond its preperiod. Soundness proves that the quantified certificate condition rules out rationality. Conversely, the prefix-tail identity makes every positive-shift tail difference nonintegral when S is irrational, and pointwise completeness (Theorem 6.19) then supplies a depth. Proposition 6.81 only identifies the weighted diagonal sum with D; it is not the completeness argument and supplies no depth bound.

Proposition 6.97 (Tail integrality on an LCM grid)**.** *If $S \in \mathbb{Q}$, there is $t_1 \in \mathbb{N}$ such that for all $t \geq t_1$, $q \geq 1$ and $m \geq 0$,*

$$R_{(q+m)H(t)} - R_{qH(t)} \in \mathbb{Z}.$$

The threshold may depend on the hypothetical rational value. A certificate at one fixed scale therefore gives a finite denominator exclusion, not irrationality. Proposition 6.73 requires certificates at arbitrarily large scales, so that one lies beyond the threshold supplied by any such rational value.

Proposition 6.98 (Independence of the retained dyadic family)**.** *For every $e \geq 0$, the auxiliary canonical dyadic family has $2^e + 1$ indexed sections, and these sections are linearly independent over $\mathbb{Q}$. The CRT–Dirichlet argument is explained in Section 10.8. For $e \geq 1$ the family is a basis for all sections through level e. At $e = 0$ the auxiliary family still contains both $\varphi(n)$ and $\varphi(2n)$, so it is not the actual level-zero truncation, whose dimension is one. No rationality hypothesis on S is used.*

Proposition 6.99 (The exact range of the stated Farey-gap inequality)**.** *Let V be the totient residue at $(N, K) = (1, 240)$ defined in Section 5.3. For every integer q satisfying*

$$1 \leq q \leq 79639646646701375323355774875831053,$$

one has

$$(qV) \bmod 2^{240} + 243q < 2^{240}.$$

The first failure occurs at

$$q = 79639646646701375323355774875831054,$$

obtained from the stated Farey mediant. Thus the range is exact for this particular inequality. It is not an optimality claim among all denominator-exclusion tests. The preceding range gives the exclusion in Proposition 6.100, approximately 7.96×10^{34}.

The range and its first failure are an exact finite computation at the window $K = 240$.

Proposition 6.100 (The resulting denominator exclusion)**.** *For every reduced fraction a/q, with $a \in \mathbb{Z}$ and $q \geq 1$,*

$$q \leq 7963964664670137532335574875831053 \quad \Longrightarrow \quad S \neq a/q.$$

This is an unconditional finite denominator exclusion. It does not assume certificates at arbitrarily large scales, as required by Proposition 6.71. Conversely, excluding this finite range does not rule out rational values with larger reduced denominators.

Proposition 6.101 (Two general irrationality criteria)**.** *Let $x \in \mathbb{R}$. One sufficient condition is a sequence of reduced fractions $u_j = a_j/q_j$, $q_j \geq 1$, with $u_j \neq x$ for all sufficiently large j and*

$$q_j|x - u_j| \longrightarrow 0.$$

Another is that, for every integer $Q \geq 1$, there exist $m, z \in \mathbb{Z}$ such that

$$0 < |mx - z| < 1/Q.$$

Indeed, if $x = a/b$ is reduced with $b > 0$, every nonzero error in the first condition is at least $1/(bq_j)$, and every nonzero error in the second is at least $1/b$. The strict lower bounds exclude exact rational hits. Requiring $m = b_0^n$ for a fixed integer base $b_0 \geq 2$ is a stronger sufficient condition, not a reformulation valid for every irrational number. The binary example in Proposition 6.23 explains this distinction.

Proposition 6.102 (Lambert identities involving S)**.** *For $L(f) = \sum_{n\geq 1} f(n)/(2^n - 1)$, the identities are $L(\mu) = 1/2$, $L(\varphi) = 2$, $L(1) = E$ and $L(\varphi * \mu) = S$. The last identity rewrites the same unknown value; it does not deduce its irrationality from that of E. The value $L(\mathrm{Id}) = \sum_{m\geq 1} \sigma(m)/2^m$ is transcendental by Nesterenko [13, Cor. 2, p. 1320]. These examples show that Lambert-series form alone does not determine arithmetic status. The individual comparisons are in Proposition 6.43 and its table.*

The Lean proof of the transcendence of $L(\mathrm{Id})$ assumes Nesterenko's theorem on the algebraic independence of the three Eisenstein series values at an algebraic point of the punctured unit disc. The four Lambert identities and the identification of $L(\mathrm{Id})$ with $\sum_{m\geq 1} \sigma(m)/2^m$ are proved without it.

Proposition 6.103 (A general rational gap bound)**.** *If $a/b < c/d$ are reduced fractions with $b, d > 0$, then*

$$\frac{1}{bd} \leq \frac{c}{d} - \frac{a}{b},$$

since $bc - ad$ is a positive integer. In particular, if $S = a/b$ and $p_N/q_N < S$ is a reduced rational approximation with positive error at most ε_N, then

$$b \geq \frac{1}{q_N \varepsilon_N}.$$

These lower bounds tend to infinity precisely when $q_N\varepsilon_N \to 0$; a subsequence with this property is enough to contradict a fixed b. A tail bound by itself does not imply this. For example, the ordinary dyadic prefix has denominator dividing 2^N and tail at most $(N+2)2^{-N}$. Using only these two bounds yields $b \geq 1/(N+2)$, which is vacuous. The rational-spacing lemma can convert sharper error and denominator information into an exclusion; it does not automatically improve the Farey bound in Proposition 6.99.

Proposition 6.104 (Diagonal certificates through 82)**.** *The current aggregate establishes*

$$\forall t \in \mathbb{N}, \quad t \le 82 \implies P\,t,$$

where $P\,t$ is the diagonal certificate predicate in Proposition 6.72. The earlier aggregate covered 28 explicit cases through 64, including $1,2,3,4,5,7,8,9,11,13,16,17$; the later theorem fills the gaps and extends the range through 82. Each instance reduces to exact finite arithmetic with the displayed totient values, using the prime-power factorisations and Lucas primality certificates in the source. This proves neither $P\,83$ nor infinitely many instances, and therefore does not establish the quantified conditions in Propositions 6.71, 6.72 and 6.73.

Proposition 6.105 (A coprime-pair expression for S)**.** *For $n \ge 0$, the number of integer pairs $a \ge 1$, $b \ge 0$ with $a + b = n$ and $\gcd(a,b) = 1$ is $\varphi(n)$, with $\varphi(0) = 0$. Indeed, $b = n - a$ and $\gcd(a, n-a) = \gcd(a,n)$. The boundary case $n = 1$ contributes $(a,b) = (1,0)$; it disappears when both coordinates are required to be positive. Consequently,*

$$\sum_{\substack{a,b\ge 1\\ \gcd(a,b)=1}} 2^{-(a+b)} = S - \frac{1}{2}.$$

For independent random variables X, Y with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$, $n \ge 1$, the left side is $\Pr(\gcd(X,Y) = 1)$. Thus $S - 1/2$ has a coprimality-probability interpretation. This identity does not supply an estimate for the binary digits or tail residues used elsewhere in the paper.

6.5 Counterexamples and limitations

A residue or coefficient condition may hold for a rational comparison series as well as for S. Each example below records the information it preserves and the deduction it disproves. The irrationality of S remains open, and an argument using further totient identities is unaffected by a comparison that does not preserve them.

Observation 6.106 (The sign of the actual endpoint representative). Assume $a \ge 8$, $H = H_{2^a}$ and $J + K + a + 6 < 2 \cdot 2^a$, as in Proposition 6.84. If $d = R_{2H+J} - R_{H+J}$ is an integer, define

$$c_0 = d, \qquad c_{j+1} = 2c_j - (\varphi(2H + J + j + 1) - \varphi(H + J + j + 1)).$$

The tail recurrence gives $c_K = R_{2H+J+K} - R_{H+J+K} > 0$. Its negative is an integer representative satisfying

$$|-c_K| \le 2H + J + K + 2, \qquad -c_K \equiv D(H, H + J, K) \pmod{2^K}.$$

Thus the representative corresponding to the actual tail is negative. The stated size bound allows it to remain in the endpoint interval, so its sign alone does not contradict integrality. Other integers in that interval with the same residue need not represent the actual tail. Proposition 6.85 gives the further residue separation that would exclude this case. LEAN SOURCE

Observation 6.107 (The complete divisibility pattern is impossible). Let $a \geq 8$, $H = H_{2^a}$ and $J, K, m \in \mathbb{N}$. Assume

$$m \leq K, \qquad J + K + a + 6 < 2 \cdot 2^a, \qquad 2H + J + K + 2 < 2^m.$$

The simultaneous divisibilities

$$2^{r+1} \mid \delta_{2^a}(J + K - m + r + 1) \qquad (0 \leq r < m)$$

cannot hold. The terminal difference would be positive by Proposition 6.80, smaller than 2^m, and divisible by 2^m, a contradiction. The obstruction applies to this complete pattern under the stated size bounds. Omitting its last divisibility condition leads to the necessary restrictions in Proposition 6.86; the residue inequality in Proposition 6.64 is another sufficient condition. Neither statement here supplies such witnesses at arbitrarily large scales. Lean source

Observation 6.108 (An exact valuation in a dyadically scaled family). For every integer $n \geq 0$, take $H = 2^{n+2}$ and $j = 3 \cdot 2^{n+1}$. Then

$$\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) = -2^{n+1}.$$

Indeed, the three arguments are $9 \cdot 2^{n+1}$, $7 \cdot 2^{n+1}$ and $5 \cdot 2^{n+1}$. Their totients are respectively $6 \cdot 2^n$, $6 \cdot 2^n$ and $4 \cdot 2^n$, which give the identity. Thus the difference has 2-adic valuation exactly $n+1$. This example limits a valuation bound for arbitrary even heights and scaled odd offsets. Here $j > H$, whereas the offset in Proposition 6.88 is at most $2^{a/2}$ and the height is $H(2^a)$. The example therefore does not exclude additional divisibility on that LCM family. Lean source

Observation 6.109 (A rational sequence with matching parity). Start with the coefficient sequence $0, 1, 1, 2, 4, 4, 4, \ldots$, indexed from 0. For each $m = 2^{k+3}$, $k \geq 0$, replace the pair at positions $m, m+1$ from $(4, 4)$ to $(6, 0)$, leaving all other coefficients unchanged. These pairs are disjoint. Each replacement preserves the binary sum because

$$2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0.$$

The resulting sequence has the same sum $3/2$ and the same parity at every index as φ. It is not eventually periodic, since the gaps between its occurrences of 6 tend to infinity. This construction uses the particular initial parity pattern and the binary weights; it does not establish an analogous result for arbitrary prescribed parities or other summand weights. Lean source

Observation 6.110 (What the parity example disproves). The sequence c in Observation 6.109 satisfies

$$0 \leq c(n) \leq \min(n, 6), \qquad c(n) \equiv \varphi(n) \pmod 2, \qquad \sum_{n \geq 0} c(n) 2^{-n} = \frac{3}{2},$$

and is not eventually periodic. More precisely, for every $N, G, K \in \mathbb{N}$, it has K of the displayed $(6, 0)$ pairs starting beyond N, with successive starting positions separated by more than G. Choose sufficiently late powers of two in the construction to obtain

these pairs. Hence the growth bound, the exact parity pattern, nonperiodicity and these arbitrarily separated occurrences are compatible with a rational sum. They cannot by themselves prove irrationality. The construction does not preserve the complete totient sequence or its multiplicativity, and does not specify which additional arithmetic hypothesis would suffice. LEAN SOURCE

Observation 6.111 (A Mersenne factor does not force tail-difference divisibility). For the completely multiplicative sequence $c(n) = n$, the scaled tail is

$$T_N = \sum_{j\geq 1}(N+j)2^{-j} = N+2, \qquad T_{N+K} - T_N = K.$$

If $q > K > 0$ and $q \mid 2^K - 1$, then every such tail difference is integral but is not divisible by q. For example, take $K = 3$ and $q = 7$; the multiplicative order of 2 modulo 7 is 3. Thus even this prime divisor of the homogeneous recurrence multiplier does not force divisibility of the inhomogeneous tail difference. An application to the totient sequence needs a further relation between the prime and the coefficient residues or the tail's size. This example supplies no general obstruction for Problem #257 merely because that problem also involves Mersenne denominators. Compare the specific totient construction in Proposition 6.91. LEAN SOURCE

Observation 6.112 (A lower bound for the stated reconstruction cost). Let I be finite, $w_i \in \mathbb{Q}$ and $x_i \in \mathbb{N}$, and assume $\sum_{i\in I} w_i\varphi(x_i) = 1$. Then

$$\sum_{i\in I} |w_i|(2(x_i+1) + (x_i+2)) \geq 3.$$

Indeed, $1 \leq \sum_i |w_i|\varphi(x_i) \leq \sum_i |w_i|x_i$, and the displayed cost is $3\sum_i |w_i|x_i + 4\sum_i |w_i|$. For $x \geq 1$, the reconstruction $\varphi(x) = 2R_{x-1} - R_x$ and the separate estimates $R_M \leq M+2$ give exactly this cost. Terms at $x = 0$ contribute nothing to the normalisation, since $\varphi(0) = 0$. This particular termwise triangle bound cannot be smaller than one. The conclusion is about that bound, not the true combined tail error; cancellation or a different estimate is not excluded. The same elementary calculation applies to any nonnegative sequence $c(n) \leq n$ under the corresponding normalisation $\sum_i w_i c(x_i) = 1$. LEAN SOURCE

Observation 6.113 (A second-difference test need not use less depth). For $h, N, L \in \mathbb{N}$, put

$$D_2 = D(h, N+h, L) - D(h, N, L), \qquad B_2 = 2(N + 2h + L + 2).$$

The second-difference certificate is the finite inequality

$$B_2 < D_2 \bmod 2^L < 2^L - B_2.$$

It implies $R_{N+2h} - 2R_{N+h} + R_N \notin \mathbb{Z}$. The two first-difference truncation errors have bounds $N + 2h + L + 2$ and $N + h + L + 2$ after multiplication by 2^L; their sum is at most the stated B_2. This explains the radius used by the linked test, without claiming it is optimal. At $(h, N) = (1, 8)$, the first-difference test holds at depth 8, whereas the second-difference test fails at every depth at most 8 and first holds at 9. The earlier 40-cell probe

for $t \le 20$ reported the first test at least as shallow in 30 cells; that aggregate is a reported computation, not part of the single-cell theorem linked below. These comparisons do not give a uniform ordering of the depths. LEAN SOURCE

Observation 6.114 (The prime-power restriction in the unit-gap refinement). The ordinary interval test excludes a rational denominator when its possible numerator interval contains no integer. Reduced fractions permit a refinement: every numerator is coprime to its denominator, so an interval containing only nonunits can also be excluded. For a candidate $S = a/(2^c u)$ in lowest terms, with $u > 0$ odd and $N \ge c$, the integer $uR_N = 2^{N-c}a - u\Phi_N$ is coprime to u. For such an N and any $K \ge 0$, define

$$\begin{aligned} V &= \left(\sum_{r=1}^{K} \varphi(N+r) 2^{K-r} \right) \bmod 2^K, \\ uV &= J2^K + W, \qquad 0 \le W < 2^K, \\ L &= \left\lfloor \frac{W + u(N+K+2)}{2^K} \right\rfloor. \end{aligned}$$

After subtracting an integer multiple of u, the finite prefix of uR_N is $J + W/2^K$ and its remaining tail lies in $(0, u(N+K+2)/2^K]$. Its possible integer values are therefore $J+1, \dots, J+L$. The refined condition is $\gcd(J+t, u) > 1$ for all $1 \le t \le L$, contradicting the required coprimality of uR_N. For a prime p and $e \ge 1$, nonunits modulo p^e are exactly the multiples of p. Since two consecutive integers cannot both be multiples of p, the condition for $u = p^e$ is equivalent to

$$L = 0 \quad \text{or} \quad L = 1 \text{ and } p \mid J + 1.$$

Thus a *successful* refined certificate at a prime-power denominator has at most one candidate integer. This is not a bound for an arbitrary window: $(u, N, K) = (3, 3, 3)$ gives $V = 2, J = 0, W = 6$ and $L = 3$, and the test fails because $1, 2, 3$ are not all nonunits modulo 3. The prime-power restriction also matters. With the actual totient window $(u, N, K) = (15, 13, 8)$, one gets

$$V = 148, \quad J = 8, \quad W = 172, \quad L = 2.$$

Both candidates $9, 10$ are nonunits modulo 15. This interval passes the refined test but not the empty-interval test. The prime-power bound therefore neither applies to all composite denominators nor bounds the possible improvement in the final denominator cutoff. The prime-power criterion is proved in Lean (LEAN SOURCE).

Observation 6.115 (A synthetic sequence preserving the stated LCM differences). Here is the recurrence behind the comparison. Fix $t \ge 3$, put $H = \operatorname{lcm}(1, \dots, t)$ and $A = \varphi(H)$, and define

$$c_i = \begin{cases} -A, & i = (q-1)H \text{ for some } 2 \le q < t, \\ 0, & \text{otherwise}, \end{cases} \qquad a_i = 2c_i - c_{i+1}.$$

The index i here corresponds to the argument $H + i + 1$ of a totient difference. For $2 \le q < t$ this gives

$$a_{(q-1)H-1} = \varphi((q+1)H) - \varphi(qH) = A.$$

The equality follows because all prime divisors of q and $q+1$ divide H: hence $\varphi(qH) = qA$ and $\varphi((q+1)H) = (q+1)A$. Every a_i is divisible by A, and $|c_i| \le A$, $|a_i| \le 2A$. For finitely supported integer weights $(\lambda_j)_{j\ge 0}$, put $d_i = \sum_j \lambda_j c_{i+j}$ and $b_i = \sum_j \lambda_j a_{i+j}$. Linearity and telescoping give

$$b_i = 2d_i - d_{i+1}, \qquad \sum_{r=0}^{L-1} 2^{L-1-r} b_{i+r} = 2^L d_i - d_{i+L}.$$

The triangle inequality gives the precise uniform bounds

$$|d_i| \le A \sum_j |\lambda_j|, \qquad |b_i| \le 2A \sum_j |\lambda_j|.$$

These are uniform in i, not in the weights. The factor $\sum_j |\lambda_j|$ cannot be omitted: taking only $\lambda_0 = 2$ doubles the nonzero state and coefficient maxima. Divisibility by A and by $\varphi(d)$ for $d \mid H$ is preserved, since $\varphi(d) \mid \varphi(H)$.

Finite shift polynomials therefore preserve the recurrence, but not the original numerical bounds independently of their coefficients. This comparison does not identify a_i with actual totient differences away from the prescribed indices, or bound the terms discarded by a proposed projection. The construction and the weight-dependent bounds are in Lean source. The decomposition of the actual totient into divisor contributions is proved in Lean (Lean source).

Observation 6.116 (Compatible carries for fixed-precision local symbols). For example, prescribing an odd coefficient modulo 8 permits any coefficient of the form $a + 8z$ with a odd and $z \in \mathbb{Z}$. Whatever the incoming integer carry e, one can choose z so that $2e+a+8z$ lies between -4 and 3. More generally, fix $u \ge 1$ and a finite list of integers $v_j \ge 0$ and odd integers a_j. Starting from any integer e_0, there are integers z_j and coefficients

$$c_j = 2^{v_j}(a_j + 2^u z_j), \qquad e_{j+1} = 2e_j + c_j, \qquad |e_{j+1}| \le 2^{v_j+u-1}.$$

At each step, reduce $2e_j + 2^{v_j} a_j$ to its centred representative modulo 2^{v_j+u}, choosing the interval

$$[-2^{v_j+u-1}, 2^{v_j+u-1}).$$

Solving for z_j gives the stated next carry. Iterate this choice. The bounds apply to the resulting states, not to the arbitrary initial state e_0. The unrestricted higher quotients z_j are essential: the construction prescribes a valuation and a finite unit residue, not the full coefficients or their signs. It therefore does not realise a prescribed totient-difference sequence. A contradiction using only these local symbols cannot exclude all compatible integer carries; further restrictions on the coefficients need a separate argument. Lean source

Observation 6.117 (Correction-free factorisation does not ensure nonvanishing). Consider the two finite differences

$$\varphi(65) - \varphi(65) = 0, \qquad \varphi(78) - \varphi(70) = 24 - 24 = 0.$$

They arise from the square-congruence choice $52 \equiv 52 \pmod{13^2}$ and $52 \equiv 2 \pmod{5^2}$. The four arguments factor as $13 \cdot 5$, $5 \cdot 13$, $13 \cdot 6$ and $5 \cdot 14$; in each product the prime and cofactor are coprime. Thus all four totients have the correction-free form $\varphi(pm) = (p-1)\varphi(m)$, yet both differences vanish. Nonvanishing is also possible: the choices $27 \equiv 0 \pmod{3^2}$ and $27 \equiv 2 \pmod{5^2}$ give

$$\varphi(30) - \varphi(30) = 0, \qquad \varphi(33) - \varphi(35) = 20 - 24 = -4,$$

again with coprime factors at every term. The square-CRT construction removes the non-coprime multiplicativity correction on the prescribed finite range. That property alone does not imply nonvanishing. These are explicit witnesses of the two possibilities, not a claim that the first is a smallest counterexample. LEAN SOURCE

Observation 6.118 (No fixed integer clears every primitive denominator). Let $A = \varphi * \mu$ be the Dirichlet convolution used in Proposition 6.102. At prime powers,

$$A(p) = p - 2, \qquad A(p^r) = p^{r-2}(p-1)^2 \quad (r \geq 2).$$

These follow by subtracting consecutive prime-power values of φ. For every odd prime p, the reduced denominator of $A(p)/p$ is p. For every prime p, the reduced denominator of $A(p^2)/p^2$ is p^2, including the denominator 4 at $p = 2$. Consequently any positive integer D for which $DA(n)/n \in \mathbb{Z}$ for all $1 \leq n \leq N$ must be divisible by every odd prime $p \leq N$, every prime square $p^2 \leq N$, and by 4 once $N \geq 4$. No fixed D can do this for all n, since there are infinitely many odd primes. In periodic-point counting, the quantities $A(n)/n$ would count orbits of length n; their integrality and nonnegativity characterise realisable periodic-point sequences [28, Lemma 3.1, p. 5]. Here only the denominator obstruction is used. It excludes a fixed integer clearing all these particular coordinates, not every possible Euler-product representation or an argument with a multiplier growing with N. It is not an irrationality proof for S. LEAN SOURCE

Observation 6.119 (A finite exclusion is not a membership test). A successful residue certificate proves the exclusion stated in its soundness theorem. Failure of that sufficient condition at finitely many depths does not prove integrality of the tail difference. In particular, a finite unsuccessful search cannot be substituted for the eventual-period assertion needed to prove rationality. The same logical distinction occurs for the linked greedy test in a separate Mersenne achievement-set problem: its finite inequality implies nonmembership. The linked object is a definition, with a soundness theorem in the same file, not a theorem asserting a universal impossibility of finite rationality proofs for Problem #249. Membership or rationality established by some additional exact identity is not excluded by this observation. Likewise, the successful finite ranges in Propositions 6.94 and 6.104 remain finite ranges until an unbounded supply is proved. LEAN SOURCE

6.6 Related structural results

We apply the Möbius-weighted identities first to log-concavity and then to integer carries, rational approximations and factorisations. The objects differ, but each proof starts from the identities already given for that object.

6.6.1 Log-concavity of the Möbius-weighted series

Put $\Theta_r = \sum_{n\geq 0} \mu(n+1)/(2^{n+1}-1)^r$, so that $\Theta_2 = S - 1/2$. We prove strict log-concavity for every integer $r \geq 1$. This orders the values of the family without determining their rationality.

Definition 6.120. For each integer $r \geq 1$, put

$$\Theta_r = \sum_{n\geq 0} \frac{\mu(n+1)}{(2^{n+1}-1)^r} = 1 - 3^{-r} + \sum_{n\geq 0} \frac{\mu(n+3)}{(2^{n+3}-1)^r}.$$

The first two summands isolate the dominant contribution. The remaining absolute sum is at most $1/(2^r(2^r-1))$, by comparison with $\sum_{n\geq 2} 2^{-rn}$. In particular, $\Theta_r > 0$ for $r \geq 1$: $1 - 3^{-r} \geq 2/3$ and this tail bound is at most $1/2$. Lean source Lean source Lean source

Theorem 6.121 (The values at exponents one and two)**.** *The first two values are* $\Theta_1 = 1/2$ *and* $\Theta_2 = S - 1/2$.

The second equality is the Möbius-square identity credited in §5.4; whether Θ_2 is rational is #249 itself.

Theorem 6.122 (The first two summands give a positive Hankel gap)**.** $(1 - 3^{-(r+1)})^2 - (1 - 3^{-r})(1 - 3^{-(r+2)}) = 4/3^{r+2}$, *so the rational sequence* $1 - 3^{-r}$ *is strictly log-concave for every integer* $r \geq 1$.

Theorem 6.123 (Strict log-concavity for all integer $r \geq 1$)**.** *For every integer* $r \geq 1$,

$$\Theta_r \Theta_{r+2} < \Theta_{r+1}^2.$$

Equivalently, the determinant of $\left(\begin{smallmatrix} \Theta_r & \Theta_{r+1} \\ \Theta_{r+1} & \Theta_{r+2} \end{smallmatrix}\right)$ *is negative. The two-summand comparison gives an elementary proof, detailed below; the same strict inequality is also proved in Lean.*

A quantitative comparison. For this proof put $a_r = 1 - 3^{-r}$ and $\varepsilon_r = 1/(2^r(2^r-1))$. The preceding bound gives $|\Theta_r - a_r| \leq \varepsilon_r$. Since $0 < a_r < 1$, expansion of the two products gives

$$\Theta_{r+1}^2 - \Theta_r \Theta_{r+2} \geq \frac{4}{3^{r+2}} - B_r, \qquad B_r = 2\varepsilon_{r+1} + \varepsilon_{r+1}^2 + \varepsilon_r + \varepsilon_{r+2} + \varepsilon_r \varepsilon_{r+2}.$$

Here $\varepsilon_{r+1} < \varepsilon_r/4$, so $B_{r+1} < B_r/4$. The rational comparison

$$B_5 = \frac{50106593}{32001951744} < \frac{4}{3^7}$$

therefore proves the inequality for all $r \geq 5$: the error decreases faster than the positive main term. For $r = 1, 2, 3, 4$, truncate each Θ_j at $n = 8$. Its omitted absolute sum is at most

$$\frac{1}{(2^9-1)^j(1-2^{-j})}.$$

Using $\Theta_1 = 1/2$ and these rational intervals, the four gaps $\Theta_{r+1}^2 - \Theta_r \Theta_{r+2}$ are respectively greater than $1/4$, $1/16$, $1/64$ and $1/200$. These finite rational comparisons complete the ordinary proof. □

Observation 6.124. The inequality excludes a representation $\Theta_r = \int_0^\infty x^r \, d\nu(x)$ by a positive measure with these finite moments: Cauchy–Schwarz would give $\Theta_{r+1}^2 \le \Theta_r \Theta_{r+2}$. It does not by itself imply irrationality of Θ_2. Indeed, the rational sequence $1 - 3^{-r}$ satisfies the same strict log-concavity inequalities. Thus the determinant sign supplies an obstruction to a positive-moment representation, not a rational-approximation estimate.

The accompanying formal results include a rectangular determinant expansion derived from the Leibniz formula (Lean source), and the following elementary parity fact: in a finite sum of dyadic fractions, if the largest denominator exponent occurs once and its numerator is odd, clearing that denominator gives an odd numerator (Lean source).

6.6.2 Integer recurrences and rational binary series

For coefficients $0 \le c(n) \le n$, rationality is equivalent to the existence of an integer carry with $u(N)/2^N \to 0$. The recurrence alone is not restrictive: every integer initial value produces an integer sequence. The limit condition selects the initial value that represents the infinite sum.

Theorem 6.125 (Integer recurrence criterion for a binary series)**.** *Let $c : \mathbb{N} \to \mathbb{N}$ satisfy $c(n) \le n$. Then $\sum_{n\ge1} c(n)2^{-n}$ is rational if and only if there are an integer $v > 0$ and an integer sequence u such that*

$$u(N+1) = 2u(N) - v\,c(N+1), \qquad \frac{u(N)}{2^N} \longrightarrow 0.$$

For each fixed v, such a sequence is unique and equals

$$u(N) = vR_N^c, \qquad R_N^c = \sum_{j\ge1} \frac{c(N+j)}{2^j}.$$

Proof. Iterating the recurrence gives the finite identity

$$\frac{u(N)}{2^N} = u(0) - v \sum_{n=1}^{N} \frac{c(n)}{2^n}.$$

Taking $N \to \infty$ gives $S_c = u(0)/v$, where $S_c = \sum_{n\ge1} c(n)2^{-n}$. Subtracting the finite prefix then gives $u(N) = vR_N^c$.

Conversely, suppose S_c is rational. Choose an integer $v > 0$ with $vS_c \in \mathbb{Z}$, set $u(0) = vS_c$, and iterate the recurrence. The terms are integers by construction; the displayed identity gives $u(N) = vR_N^c$. Finally $0 \le R_N^c \le N + 2$ proves the required limit. □

Without the boundary condition, any integer $u(0)$ gives an integer solution of the recurrence, whether the series is rational or irrational. Two solutions with the same v differ by 2^N times their initial difference. Thus the limit, not the recurrence alone, determines the initial value. Changing v rescales the canonical solution; uniqueness is asserted only for a fixed multiplier.

Proposition 6.126 (Uniqueness under the growth condition)**.** *A real sequence d with $d(N+1) = 2d(N)$ and $d(N) = o(2^N)$ is identically zero.*

Observation 6.127 (The specified finite-state decoder is impossible). Fix $m \geq 2$, put $R = \lfloor (m+1)/2 \rfloor$, and for $0 \leq r \leq R$ define

$$c_{m,r}(n) = \begin{cases} R - r, & n = m, \\ 2r, & n = m+1, \\ 0, & \text{otherwise.} \end{cases}$$

The two nonzero terms have total contribution $(R-r)2^{-m} + 2r2^{-(m+1)} = R2^{-m}$, independent of r. All coefficients and scaled tails before m are therefore the same, whereas the scaled tail at m is r. No map from a state common to this whole family can recover that tail. More generally, exact recovery of all its possible values requires at least $R+1$ distinct labels. Lean source Lean source The sequences in this family have identical histories before the isolated change but different subsequent tails. Therefore that finite history alone cannot determine the exact tail in this class of sequences. This is not a conclusion about the totient sequence.

There is also an infinite family with a common rational value. Let $a : \mathbb{N} \to \{0,1\}$ satisfy $a(0) = 0$, and put $c(2k) = k - a(k)$ and $c(2k+1) = 2a(k)$. The adjacent terms contribute $k4^{-k}$, so

$$\sum_{n \geq 1} \frac{c(n)}{2^n} = \sum_{k \geq 1} \frac{k}{4^k} = \frac{4}{9},$$

regardless of the binary sequence a. The cited result also gives an integer recurrence satisfying the growth condition (Lean source). The assumptions $a(0) = 0$ and $a(k) \in \{0,1\}$ are part of that formal statement, not an arbitrary bounded-input hypothesis. For comparison, the identity coefficients $c(n) = n$ have the explicit multiplier-one recurrence $u(N) = N + 2$ (Lean source).

6.6.3 Rational approximation with a squared-Mersenne remainder

The first-order Lambert term in the diagonal tail difference can be summed exactly. The approximation below leaves only a squared-Mersenne tail to estimate. Small error is only half of a denominator-clearing argument. If a target $\alpha = a/b$ is rational, then every unequal rational p/q satisfies $q|\alpha - p/q| = |aq - bp|/b \geq 1/b$. Thus nonzero errors tending to zero after multiplication by the reduced denominators would prove irrationality. The identity and geometric bound below give explicit errors, but not that comparison with the denominators.

Proposition 6.128 (An exact rational approximation formula). *For integers $H, D \geq 0$, with the integer prefixes Φ_N defined above,*

$$R_{2H} - R_H - \left[\Phi_H - \Phi_{2H} + 2^H(2^H - 1) \left(\frac{1}{2} + \sum_{d=1}^{D} \frac{\mu(d)}{(2^d - 1)^2} \right) \right]$$
$$= 2^H(2^H - 1) \sum_{d > D} \frac{\mu(d)}{(2^d - 1)^2}.$$

This follows by substituting the squared-denominator identity for S into $R_{2H} - R_H = 2^H(2^H - 1)S + \Phi_H - \Phi_{2H}$. There is no required ordering between D and H.

Proposition 6.129 (A geometric tail bound)**.** *For every integer $D \geq 0$,*

$$\left| \sum_{d>D} \frac{\mu(d)}{(2^d-1)^2} \right| \leq \frac{4}{3(2^{D+1}-1)^2}.$$

Indeed, $|\mu(d)| \leq 1$ and $2^{D+1+j} - 1 \geq 2^j(2^{D+1}-1)$ for $j \geq 0$. The sum of the resulting geometric majorant is $\sum_{j\geq 0} 4^{-j} = 4/3$.

Observation 6.130 (One-sided and symmetric error bounds). Suppose $n > D$, $\mu(d) = 0$ for $D < d < n$, and $|\mu(n)| = 1$. Removing that first nonzero summand gives the sharper enclosure

$$\left| \mu(n) \sum_{d>D} \frac{\mu(d)}{(2^d-1)^2} - \frac{1}{(2^n-1)^2} \right| \leq \frac{4}{3(2^{n+1}-1)^2}.$$

The radius is smaller than the positive centre, so the tail has sign $\mu(n)$. The interval's width, divided by the width of the preceding symmetric enclosure, is

$$\left(\frac{2^{D+1}-1}{2^{n+1}-1} \right)^2 < \frac{1}{4}.$$

The cited theorem gives this signed enclosure (Lean source). Multiplication by $2^H(2^H-1)$ and translation by the rational approximation above enclose $R_{2H} - R_H$. If the resulting interval contains no integer, the tail difference is nonintegral. For the symmetric bound, the linked statement requires the distance from its rational centre to every integer to exceed its error radius (Lean source). The enclosure does not establish that strict separation. In particular, shrinking an interval around a rational approximation does not, without a denominator comparison, show that the interval misses every integer.

6.6.4 Divisor terms and their complement

At $H = 6$ and $s = 2$, the difference $\varphi(2H+s) - \varphi(H+s) = \varphi(14) - \varphi(8)$ is 2, whereas the terms with divisor index $d \mid H$ sum to 3. The other indices contribute -1. The following decomposition keeps both contributions. It holds for every positive integer H, not only a least common multiple; an estimate is needed before its second sum can be discarded.

Proposition 6.131 (The sum over divisor indices)**.** *For integers $H > 0$ and $s \geq 0$, the part of the Möbius expansion indexed by divisors of H is*

$$\sum_{\substack{d \mid H \\ d \mid s}} \mu(d) \frac{H}{d} = H \frac{\varphi(\gcd(H,s))}{\gcd(H,s)}.$$

This is the standard identity $\sum_{d\mid g} \mu(d)/d = \varphi(g)/g$ at $g = \gcd(H,s) > 0$. The formal statement writes the same value after extracting $H/\operatorname{rad}(H)$, where $\operatorname{rad}(H) = \prod_{p\mid H} p$.

Proposition 6.132 (The divisor sum and its complement)**.** *For integers $H > 0$ and $s \geq 0$,*

$$\varphi(2H+s) - \varphi(H+s) = H\frac{\varphi(\gcd(H,s))}{\gcd(H,s)} + \sum_{\substack{1 \leq d \leq 2H+s \\ d \nmid H}} \mu(d)\left(\frac{2H+s}{d}\mathbf{1}_{d \mid 2H+s} - \frac{H+s}{d}\mathbf{1}_{d \mid H+s}\right).$$

Here $\mathbf{1}$ is the indicator of the stated divisibility condition. The identity follows by expanding both totients as $\varphi(n) = \sum_{d \mid n} \mu(d)n/d$ and separating the indices $d \mid H$. Such an index divides either endpoint precisely when it divides s, and its difference is then $\mu(d)H/d$.

Theorem 6.133 (Doubling the full totient difference)**.** *For nonnegative integers H, r with H even,*

$$\varphi(4H+2r) - \varphi(2H+2r) = \begin{cases} 2(\varphi(2H+r) - \varphi(H+r)), & r \text{ even}, \\ \varphi(2H+r) - \varphi(H+r), & r \text{ odd}. \end{cases}$$

This applies to the full difference, not just its divisor contribution. It follows from $\varphi(2n) = 2\varphi(n)$ for even n and $\varphi(2n) = \varphi(n)$ for odd n.

Observation 6.134 (Squarefree support of the complementary terms)*.* A nonzero summand in the complementary sum must have a squarefree index d dividing exactly one of $2H+s$ and $H+s$. Indeed, otherwise $\mu(d) = 0$ or both indicators vanish. Both endpoints cannot be divisible by d: their difference is H, and $d \nmid H$. Lean source Lean source The restriction identifies which indices can contribute. It does not control the sign or magnitude of their sum.

Proposition 6.135 (A doubling identity for two portions of the sum)**.** *Let $H > 0$, $s \geq 0$ and $d \geq 1$, with $d \nmid H$ and $d \mid H + s$. The d-summands in the complementary sums at offsets s and $2s$ are, respectively,*

$$-\mu(d)\frac{H+s}{d} \qquad \text{and} \qquad 2\mu(d)\frac{H+s}{d}.$$

For the second expression, $d \mid 2H+2s$ and $d \nmid H+2s$; otherwise d would divide their difference H. Thus the second contribution is -2 times the first, including the zero case $\mu(d) = 0$. The two terms have different binary weights in a window sum, so this identity alone is not a cancellation of their weighted contributions.

6.6.5 A numerator polynomial with explicit positive coefficients

Theorem 6.136 (Positive coefficients of the numerator polynomial)**.** *For a squarefree integer $r \geq 1$, the numerator polynomial has the explicit expression*

$$\sum_{d \mid r} \mu(d)\frac{r}{d}\sum_{j=0}^{r/d-1} X^{dj} = \sum_{k=0}^{r-1} \frac{r}{\gcd(r,k)}\varphi(\gcd(r,k))X^k.$$

Every coefficient for $0 \leq k < r$ is positive, and the higher coefficients are zero. Extracting X^k on the left gives $\sum_{d \mid \gcd(r,k)} \mu(d)r/d$, which proves the formula. The squarefree assumption identifies this divisor sum with the subset-of-primes definition in the formal source.

Corollary 6.137. *For the same squarefree r, evaluation at $X = 2$ gives the integer*

$$\sum_{d \mid r} \mu(d) \frac{r}{d} \frac{2^r - 1}{2^d - 1}.$$

Each quotient is an integer because $d \mid r$.

6.6.6 Exact dyadic comparisons and tail bounds

A finite prefix locates a dyadic rational relative to the tail. The following two formulas give the exact comparison and an upper bound for the omitted terms, so that the residue margin can be compared with its error.

Definition 6.138. For $p \in \mathbb{Z}$, $n \geq 0$ and an integer $L > 0$, the difference between $p/(2L)$ and the dyadic point $2^{-(n+1)}$ is

$$\frac{p}{2L} - 2^{-(n+1)} = \frac{2^n p - L}{2^{n+1} L}.$$

Its sign is therefore the sign of the integer numerator $2^n p - L$. For fixed p, L, the next numerator satisfies $2^{n+1} p - L = 2(2^n p - L) + L$. LEAN SOURCE LEAN SOURCE

Proposition 6.139 (The geometric remainder bound)**.** *The remainder after the first two geometric terms of $1/(2^n - 1)$ is*

$$\frac{1}{2^n - 1} - 2^{-n} - 4^{-n} = \frac{8^{-n}}{1 - 2^{-n}} \leq \frac{4}{3} 8^{-n} \qquad (n \geq 2).$$

Indeed, $1 - 2^{-n} \geq 3/4$. Summing over $n > m$, for an integer $m \geq 1$, gives

$$\sum_{n > m} \left(\frac{1}{2^n - 1} - 2^{-n} - 4^{-n} \right) \leq \frac{4}{3} \sum_{n > m} 8^{-n} = \frac{4}{21} 8^{-m}.$$

6.6.7 Squared distances between complex phases

For unit complex numbers, the identity $|z - 1|^2 = 2 - 2 \operatorname{Re} z$ converts squared distances into a real-part sum. We also sum the corresponding pairwise identity. Applying these formulas to totient phases still requires a separation estimate for those phases.

Proposition 6.140 (Squared distance from the phase one)**.**

$$\sum_{N \in T} \|E(h, N, L) - 1\|^2 = 2|T| - 2 \sum_{N \in T} \operatorname{Re} E(h, N, L).$$

Lemma 6.141 (Squared-distance bound for separated pairs)**.** *For a finite family $z : T \to \mathbb{C}$, a real number $\delta \geq 0$, and any set of pairs $P \subseteq T \times T$ each separated by $\geq \delta$, $|P| \cdot \delta^2 \leq \sum_{i,j \in T} \|z_i - z_j\|^2$.*

6.6.8 Approximating the nondivisor terms

The terms with $d \mid H$ form a finite rational contribution to $R_{2H} - R_H$. To use it, we must also approximate the terms with $d \nmid H$. The approximation and its error are explicit; for a cutoff $D \geq 2H$, only separation of the resulting rational centre from the integers remains to be checked.

For $d \geq 1$ and $N \geq 0$, let $a_d(N) = d - (N \bmod d)$ be the least positive shift from N to a multiple of d. Summing the multiples of d in the totient tail gives the rational term

$$\kappa_d(N) = \mu(d) 2^{d-a_d(N)} \left(\frac{N + a_d(N)}{d(2^d - 1)} + \frac{1}{(2^d - 1)^2} \right), \qquad R_N = \sum_{d \geq 1} \kappa_d(N).$$

Indeed, substitute $\varphi(n) = \sum_{d \mid n} \mu(d) n/d$ into R_N. For a fixed d, the admissible shifts are $a_d(N) + kd$, $k \geq 0$; their contribution is $\mu(d) \sum_{k \geq 0} ((N + a_d(N))/d + k) 2^{-a_d(N)-kd}$, which is the displayed expression. The interchange of sums is justified by

$$\sum_{j \geq 1} 2^{-j} \sum_{d \mid N+j} |\mu(d)| \frac{N + j}{d} \leq \sum_{j \geq 1} (N + j)^2 2^{-j} < \infty.$$

Thus subtraction and finite splitting of these series are valid. The terms are signed: for example, $\kappa_2(0) = -4/9$. This identity is proved in Lean (Lean source).

Proposition 6.142 (The finite divisor sum)**.** *For an integer $H > 0$, the contribution of the divisor indices to $R_{2H} - R_H$ is*

$$A_H = H \sum_{d \mid H} \frac{\mu(d)}{d(2^d - 1)}.$$

For $d \mid H$, both $a_d(H)$ and $a_d(2H)$ equal d. Hence $\kappa_d(2H) - \kappa_d(H) = H\mu(d)/(d(2^d - 1))$. Summing over the divisors gives A_H, not the whole tail difference: the nondivisor terms must still be included.

For $H > 0$ and an integer cutoff $D \geq 0$, set

$$P_{H,D} = \sum_{\substack{1 \leq d \leq D \\ d \nmid H}} (\kappa_d(2H) - \kappa_d(H)),$$

$$\varepsilon_{H,D} = 2^H (2^H - 1) \left(\frac{2}{2^D} + \frac{4}{3 \cdot 4^D} \right).$$

Here A_H includes *all* divisor indices, even when $D < H$. The following implication therefore keeps the error estimate as an explicit hypothesis; it will be proved below when $D \geq 2H$.

Proposition 6.143 (Separation larger than the error implies exclusion)**.** *Suppose*

$$|(R_{2H} - R_H) - (A_H + P_{H,D})| \leq \varepsilon_{H,D}, \qquad |A_H + P_{H,D} - z| > \varepsilon_{H,D} \quad \text{for every } z \in \mathbb{Z}.$$

Then $R_{2H} - R_H \notin \mathbb{Z}$.

Proof. If $R_{2H} - R_H = z \in \mathbb{Z}$, the two hypotheses give opposite inequalities for $|A_H + P_{H,D} - z|$. □

Why the error hypothesis holds for $D \geq 2H$. Every divisor of H then lies below the cutoff. For $d > D$, both H and $2H$ are smaller than d, so the omitted increment simplifies to

$$\kappa_d(2H) - \kappa_d(H) = \mu(d)2^H(2^H - 1)\left(\frac{1}{2^d - 1} + \frac{1}{(2^d - 1)^2}\right).$$

Since $|\mu(d)| \leq 1$ and $2^d - 1 \geq 2^{d-1}$,

$$\begin{aligned}|(R_{2H} - R_H) - (A_H + P_{H,D})| &\leq 2^H(2^H - 1)\sum_{d>D}\left(\frac{2}{2^d} + \frac{4}{4^d}\right)\\ &= \varepsilon_{H,D}.\end{aligned}$$

Thus for $H > 0$ and $D \geq 2H$ the error bound needs no additional convergence hypothesis. The same estimate is proved in Lean (Lean source); together with the displayed separation inequality, it gives the cited nonintegrality consequence (Lean source).

For example, at $H = 1$ and $D = 5$ the rational centre and radius are

$$A_1 + P_{1,5} = \frac{304282}{423801}, \qquad \varepsilon_{1,5} = \frac{49}{384}, \qquad \frac{1}{2} < A_1 + P_{1,5} - \varepsilon_{1,5} < A_1 + P_{1,5} + \varepsilon_{1,5} < 1.$$

Thus this finite calculation excludes an integral value of $R_2 - R_1$. The cutoff restriction alone does not ensure success: at $H = 1, D = 4$, the centre $346/441$ is within the radius $25/96$ of 1. For fixed H the radius tends to zero as $D \to \infty$, but an irrationality argument still needs the required family of separated centres. Neither a single example nor convergence of the error supplies it.

When $D \geq 2H$, the first-order part of the omitted sum can also be included exactly: $\sum_{d>D} \mu(d)/(2^d - 1) = 1/2 - \sum_{d\leq D} \mu(d)/(2^d - 1)$. Adding $2^H(2^H - 1)$ times this rational correction to $A_H + P_{H,D}$ gives the squared-Mersenne approximation above (Lean source). This explains the relation between the two approximations and the improved squared-denominator remainder; it does not provide their integer separation.

6.6.9 Coprime-pair sums

Coprime-pair counting gives a further expression for S. The classical Lambert identity alongside it uses different weights and sums to an elementary rational function. That evaluation does not determine the rationality of the totient series.

Theorem 6.144 (Coprime-pair counting and the totient)**.** *For every $n \in \mathbb{N}$, $\#\{(a,b) \in \mathbb{N}^2 : a + b = n,\ a > 0,\ \gcd(a,b) = 1\} = \varphi(n)$. Here $b = 0$ is allowed: the boundary pair $(1,0)$ accounts for the value $\varphi(1) = 1$.*

Proposition 6.145 (Two lattice sums)**.** *For $0 \leq r < 1$, the two choices of boundary give*

$$\sum_{\substack{a\geq 1,\ b\geq 0\\ \gcd(a,b)=1}} r^{a+b} = \sum_{n\geq 1} \varphi(n)r^n,$$

$$\sum_{\substack{a,b\geq 1\\ \gcd(a,b)=1}} r^{a+b} = \sum_{n\geq 1} \varphi(n)r^n - r.$$

The removed pair is $(1,0)$*. Partitioning all strictly positive pairs by their greatest common divisor gives*

$$\sum_{g\geq 1}\sum_{\substack{a,b\geq 1\\ \gcd(a,b)=1}} r^{g(a+b)} = \left(\frac{r}{1-r}\right)^2.$$

This total is 1 *exactly when* $r = 1/2$*.*

Theorem 6.146 (The classical coprime-pair Lambert identity)**.** *For every* $0 \leq r < 1$*,* $\sum_{(a,b)\text{ coprime},a,b\geq 1} \frac{r^{a+b}}{1-r^{a+b}} = \left(\frac{r}{1-r}\right)^2$*, an elementary rational function of* r*, hence rational at every rational* r *including* $r = 1/2$*. This is the classical visible-point identity, and its Lean proof is a formalisation of it rather than a new result: writing each pair* (A,B) *of positive integers uniquely as* $g\cdot(a,b)$ *with* $\gcd(a,b) = 1$ *converts the quadrant sum* $\sum_{A,B\geq 1} r^{A+B} = (r/(1-r))^2$ *into the displayed sum over visible points. With the plain weight* r^{a+b}*, the same strictly positive index set instead sums to* $\sum_{n\geq 1}\varphi(n)r^n - r$*. Thus at* $r = 1/2$ *the Lambert-weighted sum is* 1*, whereas the plain-weight sum is* $S - 1/2$*, not* S*. Adding the boundary pair* $(1,0)$ *recovers* S *in the plain-weight sum.*

Observation 6.147. The Lambert-weighted sum is rational at every rational $r \in [0,1)$. Consequently, coprimality of the indices, the decomposition by greatest common divisor and geometric decay alone do not imply its irrationality. This comparison does not decide the arithmetic nature of the differently weighted sum $S - 1/2$, or of S. A proof using the plain-weight sum must exploit information not contained in those three properties.

6.6.10 The single-parameter diagonal condition

Taking the period to be a least common multiple reduces the certificate condition to a single scale t. The witness must still occur beyond every threshold in t. We prove the reduction and retain that arithmetic requirement explicitly.

Observation 6.148 (The diagonal implication). The irrationality of S follows if, for every integer $t_0 \geq 0$, there are $t \geq t_0$ and $L \geq 0$ such that $C(H_t, H_t, L)$, where $H_t = \operatorname{lcm}(1,\dots,t)$ and $H_0 = 1$. This implication is proved in Lean (Lean source); its hypothesis is unproved. Theorem 6.49 also records the converse. The quantified condition is therefore equivalent to irrationality, not a weaker hypothesis.

Theorem 6.149 (Nondivisors in a short LCM window)**.** *Let* $t \geq 1$*. If* $1 \leq j < 2t$ *and* $j \nmid H_t$*, then* $j = p^a > t$ *for a prime* p *and an integer* $a \geq 1$*. Every integer* $1 \leq j \leq t$ *divides* H_t*. To see the first claim, some prime-power divisor* p^a *of* j *exceeds* t*; otherwise every prime-power divisor would divide* H_t*. Since* $j < 2t < 2p^a$*, its remaining cofactor is* 1*.*

If $j \mid H_t$ *and every prime divisor of* j *also divides* H_t/j*, then, for every integer* $q \geq 0$*,*

$$\varphi(qH_t + j) = \varphi(j)\varphi(q(H_t/j) + 1).$$

Indeed, the second factor's argument is coprime to j*, so totient multiplicativity applies. The hypothesis is essential to this factorisation: at* $t = 2$*,* $j = 2$*,* $q = 1$ *the left side is* $\varphi(4) = 2$ *but the displayed product would be* 1*.*

Observation 6.150 (Finite diagonal examples). The cited file proves $C(H_t, H_t, L)$ at the following depths:

t	1	2	3	4	5	6	7	8
H_t	1	2	6	12	60	60	420	840
L	6	5	7	7	9	9	14	15

The largest totient argument is $2H_6 + 9 = 129$ for $t \le 6$, and $2H_8 + 15 = 1695$ for the full table. Lean source Lean source This is a finite verification; certificates at arbitrarily large scales remain unproved.

6.6.11 Prime divisors of Mersenne factors and finite exclusions

For $X - 2$, multiplicative orders supply infinitely many prime divisors among the Mersenne factors. They therefore satisfy the abstract prime-support hypothesis below. This does not imply the required tail-residue inequality. The accompanying finite exclusions concern only their stated periods and basepoints.

Theorem 6.151 (Unbounded prime support in Mersenne factors)**.** *Every prime divisor p of $2^q - 1$, for prime q, satisfies $q \mid p - 1$ (the order of 2 mod p is exactly q, by Fermat/Lagrange in $(\mathbb{Z}/p)^\times$); consequently the prime divisors appearing in the layers $\{2^n - 1\}$ are unbounded, unconditionally, with no cyclotomic resultant hypothesis left open.*

Theorem 6.152 (A prime satisfying the stated cyclotomic conditions)**.** *For every period $h > 0$ and threshold N_0, there exist a prime q and a prime factor p of $|\Phi_{hq}(2)|$ (the binary cyclotomic layer) with p coprime to hq, $hq \mid p - 1$, and $p - 1 \ge N_0$. The characteristic-prime exceptional case in the cyclotomic order decomposition is eliminated directly, by choosing $q > 2^h$ (rules out $p = q$) and $q > h$ (rules out $p \mid h$).*

Observation 6.153. The first theorem uses the multiplicative order of 2 modulo a prime. For the second, choose a prime $q > \max(h, N_0, 2^h)$ and then a prime divisor of $|\Phi_{hq}(2)| > 1$. The cyclotomic order formula, together with the stated inequalities on q, excludes prime divisors of hq and gives $\mathrm{ord}_p(2) = hq$. Existence of arbitrarily large primes suffices here; no prime-in-progression theorem is needed for this construction.

These results supply prime divisors with the required orders. They do not supply the tail-residue inequality at the resulting periods. The all-prime and eventual forms of the order hypothesis must also be distinguished, as explained in §6.6.12.

Observation 6.154 (Finite exclusions at period 30). The Lean kernel decides two independent certificates at the composite period $h = 30$, at the basepoints $N = 300$ and $N = 330$. The certificate at $N = 300$ gives the denominator exclusion $S \ne r$ for every $r \in \mathbb{Q}$ whose reduced denominator divides $2^{300}(2^{30} - 1)$. Lean source Lean source This is a finite exclusion; the corresponding condition at arbitrarily large scales remains unproved.

6.6.12 A sufficient hypothesis for unbounded prime support

Here the exponent k in $p^k \equiv 1 \pmod{mq}$ bounds the order of p modulo mq, or equivalently a finite-field extension degree. It does not bound the order of 2 modulo p in

the binary examples. The condition forces a prime divisor of $C(mq)$ to grow with q, provided the separate assumption $C(mq) > 1$ ensures the existence of such a divisor.

Theorem 6.155 (A sufficient order hypothesis for unbounded prime support)**.** *Let $C : \mathbb{N} \to \mathbb{N}$, and fix integers $m \geq 1$ and $d \geq 0$. Suppose that for every pair of primes q, p with $p \mid C(mq)$ there is an integer k such that*

$$1 \leq k \leq d, \qquad mq \mid p^k - 1.$$

The divisibility forces $\gcd(p, mq) = 1$. For such p, existence of k in the displayed range is equivalent to $\operatorname{ord}_{mq}(p) \leq d$. It also gives $mq < p^d$. Hence every fixed finite set of primes is disjoint from the prime divisors of $C(mq)$ for all sufficiently large prime q. If, in addition, for all sufficiently large prime q one has $C(mq) > 1$ and $\gcd(C(mq), mq) = 1$, then for every B, N_0 there are primes $q \geq N_0$ and $p > B$ with $p \mid C(mq)$.

Indeed, the divisibility gives $mq \leq p^k - 1 < p^k \leq p^d$. For a finite set of primes, take q larger than all their d-th powers; for the last assertion, take a prime divisor of the nontrivial value $C(mq)$ after this threshold. The stated coprimality is included in the formal source's layer hypothesis, although this last extraction argument uses only $C(mq) > 1$. For $C(n) = 2^n - 1$, the first theorem above supplies the hypothesis with $m = d = 1$: here $\operatorname{ord}_q(p) = 1$, whereas $\operatorname{ord}_p(2) = q$. For $C(n) = |\Phi_n(2)|$ and a general fixed $m > 0$, the source uses the eventual version: the divisibility is required only for prime q above a fixed threshold. The same proof then applies after that threshold. It is not valid to replace this by an all-prime assertion: $\Phi_6(2) = 3$, but $6 \nmid 3 - 1$. For the binary cyclotomic family, choosing $q > \max(m, 2^m)$ removes these exceptional indices. Other families require their own proof of the divisibility and nontriviality assumptions.

6.6.13 The lower bound for rank-one quotients

The finite-sum quotients below stay a positive distance above $\Theta_2 = S - 1/2$, uniformly in their exponent and truncation length. Consequently neither parameter can make the associated rational linear form tend to zero. Appendix A.4 gives the sharp minimum and its ordinary proof, strengthening the bound recorded here.

Theorem 6.156 (A uniform positive gap for rank-one quotients)**.** *For integers $r \geq 2$ and $Y \geq 1$, write*

$$\Theta_r = \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^r}, \qquad t(Y, r) = \sum_{d=1}^{Y} \frac{\mu(d)}{(2^d - 1)^r}.$$

For every $e \geq 1$ and $Y \geq 4$,

$$\frac{t(Y, e+2)^2}{t(Y, 2e+2)} - \Theta_2 > \frac{1}{480}.$$

Here $\Theta_2 = S - 1/2$. For $r \geq 3$, the proof uses $1429/1512 \leq \Theta_r < 1$ and $|t(Y, r) - \Theta_r| \leq 1/3584$ when $Y \geq 4$. In particular, these inequalities make the denominator positive.

Proposition 6.157 (The bound is preserved by positive normalised averaging)**.** *For a nonempty finite family of admissible quotients with positive weights summing to* 1*, the weighted average still exceeds* Θ_2 *by more than* $1/480$*. If an admissible quotient is* p/q *in lowest terms, with* $q > 0$*, then* $|q\Theta_2 - p| > q/480$*.*

Observation 6.158. Writing an admissible quotient in lowest terms as p/q, with $q > 0$, gives $|q\Theta_2 - p| > q/480 \geq 1/480$. Such forms cannot tend to zero. The conclusion concerns exactly the displayed quotients and their positive normalised averages. It makes no assertion about other rational approximants or about signed averages.

6.6.14 Multiples of a period

We allow a positive multiple of each prescribed period. Telescoping proves sufficiency, and completeness of the residue test proves necessity, giving another condition equivalent to irrationality. The universally quantified condition remains unproved. For the finite examples, we specify which excluded denominators already occur in the diagonal computations.

Theorem 6.159 (Concatenation and specified period multiples)**.** *For nonnegative integers* a, b, N*, define the integer block sum*

$$Q_{a,N} = \sum_{j=1}^{a} \varphi(N+j)2^{a-j}.$$

Splitting a block after its first a *terms gives*

$$Q_{a+b,N} = 2^b Q_{a,N} + Q_{b,N+a}.$$

In particular, $Q_{2h,N} = 2^h Q_{h,N} + Q_{h,N+h}$*. The corresponding cyclotomic identity is* $\Phi_4(2^h) = 2^{2h} + 1 = \Phi_2(2^{2h})$*: the order-four factor at height* h *is the order-two factor at height* $2h$*.*

The order-three factor is different: $\Phi_3(2^h) = 2^{2h} + 2^h + 1$ *comes from tripling the period. It is not generally part of the doubling sequence. For example,* $\Phi_3(2) = 7$ *divides none of* $2^{2^j} - 1$*, because the order of* 2 *modulo* 7 *is* 3*, which does not divide* 2^j*. Thus the block identities relate specified period multiples; they do not put all the order-two, order-three and order-four factors into one doubling chain.*

Observation 6.160 (An equivalent quantified condition). Both implications of this equivalence are proved in Lean (Lean source); the quantified condition itself is not established. The aggregate diagonal table gives certificates at every H_t for $t \leq 82$ (Lean source); this equivalence contributes no new information about $t = 83$ or any condition at arbitrarily large scales.

Observation 6.161 (Eight finite period certificates). The cited declarations give certificates at basepoint $N = 300$ for

$$h \in \{67, 81, 97, 101, 121, 125, 127, 128\}.$$

Each excludes every reduced fraction a/b, with $b > 0$ and $b \mid 2^{300}(2^h - 1)$. These exclusions overlap with those already obtained from the diagonal. In particular, 67 and 81 divide H_{82}, and $H_{82} > 300$, so

$$2^{300}(2^h - 1) \mid 2^{H_{82}}(2^{H_{82}} - 1) \qquad (h = 67, 81).$$

Their entire denominator classes are already covered by Proposition 6.104. The eight cases therefore cannot all be described as new denominator classes relative to that proposition. The exact comparison of the odd factors is

$$\gcd(2^h - 1, 2^{H_{82}} - 1) = 2^{\gcd(h, H_{82})} - 1,$$

by the Euclidean algorithm on the exponents. None of the remaining six periods divides H_{82}, so its full odd factor is not covered by that diagonal denominator. This compares with the $t = 82$ certificate, not with every other exclusion in the record; the case $h = 127$ includes the Mersenne prime $2^{127} - 1$.

The case $h = 67$ also succeeds at depth $L = 67$. It illustrates the restriction $L = h$ in the full-depth condition, while retaining a small basepoint. That universally quantified condition is equivalent to irrationality; one finite instance is not. Lean source Lean source Lean source Lean source These are eight finite exclusions; the condition at arbitrarily large scales remains unproved.

6.6.15 The strict prime-position criterion

A sufficient prime-index condition is that, for every integer $h \geq 1$ and every $N_0 \geq 0$, there is a prime $p \geq \max(N_0 + h + 1, h + 5)$ such that

$$\operatorname{Re} e(2^{p-h-1}(2^h - 1)S) < \frac{9}{10}.$$

Here $e(x) = \exp(2\pi i x)$. The positive difference between $9/10$ and this real part may depend on the chosen prime; the truncation depth is then chosen so that its error fits within that difference. This is a weaker phase requirement than the earlier bound $4/5$, not a larger saving from 1. The implication to irrationality is proved, but the displayed prime-index condition remains unproved. Lean source

6.6.16 Finite Euler-product identities

These are identities for the Dirichlet-convolution square $\mu * \mu$, not the pointwise square $\mu(n)^2$. For each prime p, its coefficients at $1, p, p^2$ are $1, -2, 1$, and the higher prime-power coefficients vanish. The local factors at exponents 1 and 2 are

$$1 - \frac{2}{p} + \frac{1}{p^2} = \left(1 - \frac{1}{p}\right)^2, \qquad 1 - \frac{2}{p^2} + \frac{1}{p^4} = \left(1 - \frac{1}{p^2}\right)^2.$$

For a prime p and $e \geq 0$, the corresponding convolution with the divisor-sum function σ gives

$$\sigma(p^{e+2}) - 2\sigma(p^{e+1}) + \sigma(p^e) = p^{e+1}(p - 1) = \varphi(p^{e+2}).$$

This follows by subtracting the finite geometric sums. These finite identities do not assert a transcendence result. Lean source Lean source

7 Parameter ranges and quantifiers

The distinction between a finite calculation and an unbounded family is in the quantifiers. A statement such as $\forall a \geq 8$ is uniform in a, whereas a list of computed scales is finite and a condition $\forall N_0\, \exists N \geq N_0$ asks for witnesses arbitrarily far out. The table uses these distinctions even when a uniform statement is conditional on an unproved hypothesis. A bound on a secondary variable, such as $J + (a+6) < 2 \cdot 2^a$ in the positivity row, does not bound a.

Some identities have no distinguished scale parameter. They are listed with the general identities. An equivalence can hold for all parameters without establishing either of the equivalent arithmetic assertions.

The LCM formulation of Problem 249 requires arbitrarily large scales: Lean source

$$\mathrm{Irrational}\Big(\sum_{n\geq 1} \varphi(n)/2^n\Big) \iff \forall a_0\, \exists a \geq a_0,\ \Omega_a \notin \mathbb{Z},$$

a cofinal statement in the exponent a indexing $H = H(2^a)$. A finite calculation alone does not establish it; a further theorem could propagate finite information to arbitrarily large scales.

The table groups statements by their parameter ranges, not by proof status. In particular, some assertions with cofinal quantifiers are already proved, such as the prime constructions and rational comparison sequences. Others are implications whose quantified hypotheses remain open. The horizontal rule separates the cofinal group for ease of reference; it does not mark a boundary between proved and unproved mathematics. The descriptions identify this distinction in that group. The prefixes summarise the quantifiers; complete hypotheses are given in the cited statements. A definition records an object or condition; it does not by itself prove the associated soundness or existence assertion.

In the periodic-weight rows, write $w : \mathbb{N} \to \mathbb{Z}$ and $c_w(n) = \sum_{d|n} w(d)$ for $n \geq 1$. Positivity and nonvanishing refer to these divisor coefficients, not to the weight w itself. Here "nonzero arbitrarily far out" means that for every N there is $n > N$ with $c_w(n) \neq 0$; it does not mean nonvanishing on every arithmetic progression. The nonprincipal character χ_4 modulo four is 0 on even integers and $(-1)^{(n-1)/2}$ on odd integers.

Table 3: Results and unproved conditions, grouped by their quantifiers. The dividing rule separates the cofinal group, not proof status. Complete hypotheses are given in the cited statements.

Result	Source	Quantifier prefix	Representation
Uniform statements, including unconditional identities and implications			
scaled totient tail (definition)	Lean source	$\forall N$	binary-weighted series
integer prefix and real tail	Lean source	$\forall N$	binary-weighted series
$D(h,N,L)$ (definition)	Lean source	$\forall h\,\forall N\,\forall L$	finite binary sums
$C(h,N,L)$ (definition)	Lean source	$\forall h\,\forall N\,\forall L$	finite binary sums
necessary depth inequality	Lean source	$\forall h\,\forall N\,\forall L$, given $C\,h\,N\,L$	finite binary sums
residue test soundness	Lean source	$\forall h\,\forall N\,\forall L$, given C	finite binary sums
completeness of the residue test	Lean source	$\forall h\,\forall N$	finite binary sums
integrality from denominator divisibility	Lean source	$\forall N\,\forall h\,\forall r{:}\mathbb{Q}$, given $S = r$, $r.\mathrm{den} \mid 2^N(2^h-1)$	binary-weighted series
tail-period law	Lean source	$\neg\mathrm{Irrational}\,S \to \exists h{>}0\,\exists N_0\,\forall N{\geq}N_0$	binary-weighted series
least common multiple $H(t)$	Lean source	$\forall t$	LCM periods
a nondivisor below $2t$ is a prime power	Lean source	$\forall t\,\forall j\,(0{<}j{<}2t)$, given $j \nmid H\,t$	LCM offsets
totient factorisation at divisor offsets	Lean source	$j \mid H(t)$, with the prime-divisor hypotheses of Proposition 6.51	totient factorisation
integrality on the LCM grid under rationality	Lean source	$\neg\mathrm{Irrational}\,S \to \exists t_1\,\forall t{\geq}t_1\,\forall q{>}0\,\forall m$	LCM grids
second-difference residue test	Lean source	$\forall h\,\forall N\,\forall L$, given second-difference residue condition	finite binary sums
a nonintegral pair on a finite grid	Lean source	finite $Q \neq \emptyset$; all size and residue hypotheses of Theorem 6.56	finite LCM grids
finite exclusion of integer carries	Lean source	$\forall h\,\forall N\,\forall K$, given finite carry exclusion	integer recurrences
Farey neighbour denominator law	Lean source	$a,b,c,d,r,s \in \mathbb{Z}$; $b,d>0$, $bc-ad=1$, $as<br$, $dr<cs$	rational intervals
rational-approximation criterion	Lean source	$u_j \in \mathbb{Q}$, $x \in \mathbb{R}$; eventually $u_j \neq x$, and $\mathrm{den}(u_j)\lvert u_j-x\rvert \to 0$	general criterion
near-integer irrationality criterion	Lean source	$\forall \xi{:}\mathbb{R}$, $\forall q{>}0\,\exists m\,z\,0{<}\lvert m\xi-z\rvert{<}1/q$	general criterion

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Result	Source	Quantifier prefix	Representation
independence from a nonsingular evaluation matrix	Lean source	a finite family with a square evaluation matrix of nonzero determinant	general criterion
dyadic totient-kernel rank = 2^e+1	Lean source	$\forall e \geq 1$	dyadic sections
rationality forces unbounded carry-kernel rank	Lean source	$\neg$Irrational $S \to \exists v{>}0\, \exists u\, \forall e$	carry sections
no fixed D clears all $A(n)/n$ for $A = \varphi * \mu$	Lean source	$\forall D{>}0$	Möbius sums
Lambert-series value identities	Lean source	value identities, $L(\mu){=}\frac{1}{2}$, $L(\varphi){=}2$, $L(1)$=Erdős–Borwein	Möbius sums
prime-power unit-gap criterion	Lean source	$\forall p$ prime, $e > 0$, $N, K \geq 0$	rational intervals
positive rational-difference lower bound	Lean source	$\forall$ whole pfx:$\mathbb{Q}$, pfx<whole	general criterion
$L_2(\mu) = S - \frac{1}{2}$	Lean source	unconditional	Möbius sums
weighted squared-denominator identity	Lean source	$\forall w{:}\mathbb{N} \to \mathbb{R}$ $(\lvert w(d)\rvert{\leq}d)$ $\forall r \in [0,1)$	Möbius sums
$L_2(\mu) = S - \frac{1}{2}$	Proposition 5.14	unconditional	Möbius sums
$L_2(1) = \zeta_q(2) - \zeta_q(1)$ identity	Lean source	unconditional	Möbius sums
$L_2(\varphi)$ = gcd moment	Lean source	unconditional	gcd moments
comparison of the weights	Propositions 5.14–5.18	unconditional	Möbius sums
gcd-divisibility factorizes	Lean source	$\forall d{:}\mathbb{N}$, $d{>}0$	geometric weights
sum over positive coprime pairs equals 1	Lean source	unconditional (sum over all coprime pairs)	geometric weights
Stern–Brocot recursion with stopping	Lean source	positive integers a, b	Stern–Brocot tree
positive numerator coefficients	Lean source	$\forall r$ (Squarefree r) $\forall k{<}r$	divisor and cyclotomic sums
evaluation at 2 gives the numerator	Lean source	$\forall r$ (Squarefree r)	divisor and cyclotomic sums
reduction to the radical of H	Lean source	$\forall H{>}0$ $\forall r{>}0$	Möbius sums
numerator congruence modulo Φ_m	Lean source	$\forall r$ (Squarefree r) $\forall m$ $(m\lvert r)$	divisor and cyclotomic sums
gcd of the numerator and $\Phi_r(2)$	Lean source	$\forall r$ (Squarefree r)	divisor and cyclotomic sums
Mersenne factors retained in the denominator	Lean source	$t \geq 5$, squarefree r; finite P of primes dividing r in $(t/2, t]$; prime factors of r and the multiplier are at most t	divisor and cyclotomic sums

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Result	Source	Quantifier prefix	Representation
denominator lower bound / exact value	Lean source	$\forall t$ (exact-value); $t \geq 5$ (lower bound)	Möbius sums
numerator congruence after adjoining a prime	Lean source	$r \geq 1$, p prime, $p \nmid r$, $m \mid r$	divisor and cyclotomic sums
squared-Mersenne series: partial sum plus tail	Lean source	$\forall D{:}\mathbb{N}$	Möbius sums
cancellation of the affine divisor terms	Lean source	$\forall d{>}0$ $(d \mid H)$; general: finite affine annihilator	LCM diagonal
new-divisor counting identity	Lean source	$\forall a{\in}A$ $(a{>}0)$ $\forall x{>}0$	divisor counts
bound on zero blocks in divisor counts	Lean source	$\forall \varepsilon{>}0$ $\exists B$, given rationality of the base-2 support series	divisor counts
equivalent shift condition for the target $1/2$ (Problem 257)	Lean source	iff, no extra parameter	Mersenne subsums
totient multiplication with common prime factors	Lean source	$\forall x{>}0$ (j fixed)	totient identities
totient relative-Euler-product split	Lean source	$\forall j\ x{>}0$	totient identities
formula for $\varphi(2H(t)+j) - \varphi(H(t)+j)$	Lean source	$\forall t\ \forall j$	Möbius sums
the finite diagonal sum equals $D(H(t), H(t), L)$	Lean source	$\forall t\ \forall L$	Möbius sums
#249 $\iff$ cofinal actual-orbit nonintegrality	Lean source	iff	Möbius sums
the LCM tail difference is an integer translate of a multiple of S	Lean source	$\forall a$	Möbius sums
explicit finite-block approximation	Lean source	$\forall a\ \forall q$	Möbius sums
an equivalent residue-interval condition	Lean source	$\forall h\ N$, iff	integer recurrences
sign of the three-term totient difference under the stated middle-value bound	Lean source	$\forall H\ j$, given MiddleRankTotientExtremal	totient differences
exact value on a specified dyadic family	Lean source	$\forall n$ (closed-form)	totient differences
the rational parity comparison (definition)	Lean source	$\forall n$ (definition)	binary-weighted series
R_N as a sum over positive divisor indices	Lean source	$\forall N$	divisor and cyclotomic sums
R_N as a shifted Möbius sum	Lean source	$\forall N$	divisor and cyclotomic sums

continued on next page

Result	Source	Quantifier prefix	Representation
absolute convergence and regrouping of Lambert sums	Lean source	$\forall w\, v{:}\mathbb{N} \to \mathbb{R}$ $(\lvert w(d)\rvert \le d,\ \lvert v(m)\rvert \le m)$	divisor and cyclotomic sums
carry divisibility and integral tail differences	Lean source	$\forall N\, k$, given the carry recurrence and limit	integer recurrences
rationality ⇒ periodicity & unbounded rank	Lean source	$\neg$Irrational $S \to \exists v\, u\, \forall e$	integer recurrences
the stated reconstruction cost is at least 3	Lean source	$\forall$ finite $w{:}\iota \to \mathbb{Q}\ x{:}\iota \to \mathbb{N}$, given $\sum w_i \varphi(x_i){=}1$	weighted tail bounds
directed residue interval (definition)	Lean source	$h, N, L \ge 0$; soundness and the existential-depth equivalence are proved in Proposition 6.69	integer recurrences
rationality and an integer carry with vanishing boundary term	Lean source	$\forall c{:}\mathbb{N} \to \mathbb{N}$ $(c(n) \le n)$, iff	binary-weighted series
doubling-orbit rigidity	Lean source	$\forall d{:}\mathbb{N} \to \mathbb{R}$, doubling and $d(n)/2^n \to 0$ imply $d \equiv 0$	integer recurrences
number of labels needed for exact tail recovery	Lean source	$\forall m\ \forall$State, collapsing-state → no decoder	binary-weighted series
endpoint residues do not determine the initial carry	Lean source	$\forall a{:}\mathbb{N} \to \mathbb{Z}\ \forall u_0\, v_0\ \forall L$	binary-weighted series
integer-coefficient block-certificate dichotomy	Lean source	$\forall b{\ge}2\ \forall c\ \forall q$, given full-block certificate at q	binary-weighted series
divisibility under iterated forward differences	Lean source	$d \mid f(n)$ for every n; every finite list of shifts	integer sequences
totient series as a coprime-pair sum	Lean source	$\forall n$; $\forall r \in [0, 1)$	geometric weights
factorisation through a rational-valued linear functional	Lean source	rational vector spaces V, W; surjective linear ev : $V \to \mathbb{Q}$, linear $L : V \to W$, $\ker(\mathrm{ev}) \subseteq \ker L$	divisibility and linear algebra
divisibility of a reduced denominator	Lean source	$\forall D{>}0\ \forall m \mid D\ \forall a$	divisibility and linear algebra
the complementary denominator divides the integer multiplier	Lean source	$\forall x{:}\mathbb{Q}\ \forall H$ $(H \mid x.\mathrm{den})$ $\forall c$ $((cx).\mathrm{den} \mid H)$	divisibility and linear algebra
largest-exponent nonvanishing criterion	Lean source	$\exists m \in s$ (strictly maximal, odd coeff), $\forall i \ne m\ e(i) < e(m)$	divisibility and linear algebra
column rescaling and determinant nonvanishing	Lean source	$\forall d\ \forall e{:}\mathrm{Fin}\, d \to \mathbb{N}\ \forall z{:}\mathrm{Fin}\, d \to \mathbb{C}$ (z nonzero)	complex matrices
determinant 1 for the divisor-incidence matrix	Lean source	$\forall N{:}\mathbb{N}$	Möbius sums
error bound from convergence	Lean source	$2H \le D$; convergence is supplied for $H > 0$ in Section 6.6.8	Möbius sums

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Result	Source	Quantifier prefix	Representation
carries realising fixed-precision unit residues	Lean source	$\forall u{>}0$ (fixed precision), $\forall$ finite carry word	divisibility and linear algebra
vanishing of iterated prime dilation differences	Lean source	distinct primes $p_1, \dots, p_r$, $r > K$, $\gcd(n, \prod p_i) = 1$; $\Omega(a) \le K$ on A	divisor counts
$\varphi(n) > \frac{3}{4}n$ for the stated rough integers	Lean source	$\forall a{\ge}8\ \forall n{>}0$ (n is t-rough, $t{=}2^a$, $n{<}2^{2\cdot 2^a}$)	Möbius sums
positivity of short-window totient differences	Lean source	$\forall a{\ge}8\ \forall j$ ($0{<}j{<}2\cdot 2^a$)	Möbius sums
the negative representative of the true endpoint	Lean source	$\forall a{\ge}8\ \forall J\ K$ (size bound)	Möbius sums
integrality forces upper-endpoint residue	Lean source	$\forall a{\ge}8\ \forall J\ K$ (size condition, $2H{+}J{+}K{+}2{<}2^K$)	Möbius sums
terminal dyadic staircase impossible under the size condition	Lean source	$\forall a{\ge}8\ \forall J\ K\ m$ (size condition, modulus-wide)	Möbius sums
penultimate difference in a partial divisibility pattern	Lean source	$\forall a{\ge}8\ \forall J\ K\ m$ (hypotheses of Proposition 6.86)	Möbius sums
upper-endpoint residue-gap ⇒ nonintegral	Lean source	$\forall a{\ge}8\ \forall J\ K\ m$ (size condition, one-sided)	Möbius sums
relation between the centred residue and the actual carry	Lean source	$\forall a{\ge}8\ \forall q$ (size condition, fit bound $2(H{+}q{+}2){\le}4^q$)	Möbius sums
divisibility of the three-term totient difference	Lean source	$\forall a{\ge}4\ \forall j{>}0$ ($j^2{\le}2^a$)	totient differences
equality of two finite rational approximations	Lean source	$H > 0$, $D \ge 2H$; no separation hypothesis for this equality	Möbius sums
a Mersenne divisor need not divide a tail difference	Lean source	$\forall K, q$ ($K{>}0$, $K{<}q$, $q{\mid}2^K{-}1$)	Möbius sums
finite numerator separation implies nonintegrality	Lean source	$\forall H, D$ fixed, given uniform numerator-gap bound	Möbius sums
periodic-weight Lambert dichotomy	Lean source	$\forall b{\ge}2\ \forall m{>}0\ \forall w$ ($w(n{+}m){=}w(n)$)	Möbius sums
periodic weight with nonnegative divisor coefficients	Lean source	$b \ge 2$, $m > 0$, $w(n+m) = w(n)$; $c_w(n) \ge 0$ and $c_w(n) \ne 0$ arbitrarily far out	Möbius sums
zeros on one progression for the character modulo four	Lean source	$c_{\chi_4}(n) = 0$ for every $n \equiv 3 \pmod 4$; the weight is fixed	Möbius sums
positivity of the translated LCM tail difference	Lean source	$\forall a{\ge}8\ \forall J$ ($J{+}(a{+}6){<}2\cdot 2^a$), no rationality hypothesis	Möbius sums
Finite ranges and parameter-bounded checks			

continued on next page

Result	Source	Quantifier prefix	Representation
Farey window K=240 denominator bound	Lean source	$\forall q{:}\mathbb{N},\ 0<q\le$ 79,639,646,646,701, 375,323,355,774,875,831,053	rational intervals
denominator lower bound $S\neq p/q,\ q\le Q_0$	Lean source	$\forall p{:}\mathbb{Q},\ p.\mathrm{den}\le Q_0$	rational intervals
finite LCM certificates covering thresholds $a_0 \le 6$	Lean source	$\forall a_0\le 6\ \exists a\ L$	Möbius sums
contiguous diagonal certificate band through $t\le 82$	Lean source	$\forall t\le 82\ \exists L$; no t=83 certificate claimed	LCM diagonal
Listed examples			
residue certificates for $1 \le h \le 8$, $N = 12$, $L = 16$	Lean source	$\forall h \in [1,8]$, $C\,h\,12\,16$ (direct evaluation)	finite binary sums
denominator exclusions for $1 \le h \le 16$, binary exponent 14	Lean source	$\forall h \in [1,16]$, denominator exponent 14 (direct evaluation)	finite binary sums
the second-difference test uses more depth at $(h, N) = (1, 8)$	Lean source	first-difference certificate at $L = 8$; no second-difference certificate for $1 \le L \le 8$, and one at $L = 9$	finite binary sums
28 listed diagonal certificates with $t \le 64$	Lean source	$t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots\}$ (28 explicit values)	LCM diagonal
LCM certificates at $a = 4$ and $a = 6$	Lean source	a=4, a=6 (concrete, formally checked)	Möbius sums
square-CRT coprime-factor examples: zero / nonzero differences	Lean source	witness n=52 (vanishes); witness n=27 (nonzero)	totient factorisation
Statements with cofinal quantifiers: proved constructions and conditional implications			
Cofinal statements; status specified in the description			
certificate criterion (proved implication; unproved premise)	Lean source	$\forall h{\ge}1\ \forall N_0{\ge}0\ \exists N{\ge}N_0\ \exists L,\ C\,h\,N\,L$	finite binary sums
period-multiple criterion (equivalent; unproved condition)	Lean source	$\forall h_0{>}0\ \forall N_0\ \exists m{>}0\ \exists N{\ge}N_0\ \exists L$	multiples of periods
LCM-diagonal criterion (unproved condition)	Lean source	$\forall t_0\ \exists t{\ge}t_0\ \exists L$	LCM diagonal
LCM-grid certificate criterion (unproved premise)	Lean source	$\forall t_0\ \exists t{\ge}t_0\ \exists q{>}0\ \exists m\ \exists L$	LCM grids

continued on next page

Result	Source	Quantifier prefix	Representation
LCM nonintegrality criterion (unproved condition)	LEAN SOURCE	$\forall t_0\ \exists t \geq t_0,\ R(2H_t) - R(H_t) \notin \mathbb{Z}$	real tail differences
finite-grid residue criterion (unproved premise)	LEAN SOURCE	for every t_0, data $t \geq t_0, L, Q$ satisfying Theorem 6.57	finite LCM grids
first-harmonic norm criterion (unproved premise)	LEAN SOURCE	$\forall h{>}0\ \forall X_0\ \exists X \geq \max(X_0, 1)\ \exists L$, size condition $16(2X{+}h{+}L{+}2) \leq 2^L$	binary-weighted series
four-tail residue criterion (unproved premise)	LEAN SOURCE	$\forall t_0\ \exists t \geq t_0\ \exists p, L{>}0$	binary-weighted series
short LCM-window criterion (unproved premise)	LEAN SOURCE	$\forall a_0\ \exists a\ L,\ a_0 \leq a \wedge L < 2 \cdot 2^a$	Möbius sums
LCM-tail separation criterion (unproved premise)	LEAN SOURCE	cofinal $1/32$+error separation at prescribed odd depths	Möbius sums
five endpoint criteria (unproved premises)	LEAN SOURCE	one of the five quantified conditions in Proposition 9.14	Möbius sums
proved rational counterexample to bound, parity and aperiodicity	LEAN SOURCE	$\exists c{:}\mathbb{N} \to \mathbb{N}$, $c \leq 6$, $c \leq n$, $c \equiv \varphi \pmod 2$, arbitrarily-separated blocks	binary-weighted series
proved mod-4 prime construction (CRT and Dirichlet)	LEAN SOURCE	$\forall h{>}0\ \forall B\ \exists \text{prime}\ p{>}B$	Möbius sums
sufficient mod-4 condition (unproved premise)	LEAN SOURCE	the displayed mod-four residue condition at arbitrarily large scales	Möbius sums
proved two-adic prime construction	LEAN SOURCE	$\forall K \geq 2\ \forall H{>}K\ \forall B\ \exists$cofinally many primes $p{>}B$	divisibility and linear algebra
proved implication from eventual integrality	LEAN SOURCE	$\forall K \geq 2$, given the stated two-adic prime construction and eventual integrality	divisibility and linear algebra

7.1 Finite extension and cofinal estimates

Extending a computation supplies more finite witnesses. To reach a cofinal statement, an argument must instead produce a witness beyond an arbitrary threshold. The examples below specify which of these extensions is needed; the largest computed scale is not asserted to be a limit of the method.

1. **Finite certificate tables.** Results such as the diagonal certificates at 28 explicit values of t through t=64, now a strict subset of the aggregate band $t \le 82$ (Lean source), the certificates for $h \in [1,8]$ at fixed depth 16 (Lean source), and the exclusions for $h \in [1,16]$ at exponent 14 (Lean source) are each a finite *list* of independently verified rows, not a single argument instantiated at a free parameter. The existing lists do not establish further rows; an extension needs either additional computations or an argument for a family of parameters.

2. **A bounded window inside a uniform theorem.** The rough-integer bound of Proposition 6.79 and the positivity theorem of Proposition 6.84 hold for every $a \ge 8$, subject to their stated bounds on the other parameters. In the sign theorem, for example, $J + a + 6 < 2 \cdot 2^a$. Their proofs are therefore not merely enumerations of a fixed set of exponents. The missing conclusion is a suitable residue inequality at arbitrarily large scales, not extension of the range of a in the positivity theorem. No complexity lower bound follows from these examples.

3. **Parameter dependence in an approximation estimate.** For reduced approximants p_K/q_K to S, the conditions $0 < |S - p_K/q_K| \le \varepsilon_K$ and $q_K\varepsilon_K \to 0$ imply irrationality. A growing denominator alone does not control this product. In particular, the exact reduced-denominator formula of Proposition 6.8 concerns a specified finite Möbius sum; it does not estimate the complementary tail. Likewise, a bound proved with a constant depending on a fixed auxiliary parameter is not automatically uniform when that parameter grows. Both the dependence of the constant and the error estimate must be retained in any passage to an unbounded family.

The missing arithmetic inputs include the stated constant-saving exponential-sum estimate and an upper-endpoint residue inequality. For the positivity theorem, the unresolved step is exclusion of that residue interval, not positivity at more exponents.

8 Relations between the representations

Binary tails, Möbius sums, rational approximations and exponential sums express related information about S. We give the identities that connect them and then check which assumptions survive each change. This is necessary before transferring a counterexample to another representation, as the finite-state decoding and polynomial denominator examples illustrate.

Binary-weighted series and tails.

The coefficients in $S = \sum_{n\ge1} \varphi(n)2^{-n}$ are unbounded; this expression is not the base-two digit expansion of S. The tail R_N, finite difference numerator $D(h,N,L)$ and test $C(h,N,L)$ retain this coefficient information. The identity

$$S = \frac{1}{2} + \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2}$$

(Lean source) and its partial-sum version (Lean source) express the same value by Möbius sums. An equality of values does not by itself transfer a claim about the original coefficient subsequences.

Lambert sums and finite numerator polynomials.
The Lambert identities for μ, 1 and φ give the series relations in Section 5. For a finite divisor sum, the polynomial evaluation (Lean source) and the definition (Lean source) put its numerator into integer polynomial arithmetic. This evaluates a finite contribution, not the complementary infinite tail.

Coprimality probability.
Let X, Y be independent with $\Pr(X = n) = \Pr(Y = n) = 2^{-n}$ for $n \geq 1$. Then $S - \frac{1}{2} = \Pr(\gcd(X, Y) = 1)$. The divisibility probabilities are $\Pr(d \mid X,\ d \mid Y) = (2^d - 1)^{-2}$ (Lean source). Möbius inversion converts these to the coprimality probability. The decomposition by the greatest common divisor also gives the weighted coprime-pair identity (Lean source). Counting coprime pairs by their sum gives the totient expression directly (Lean source).

Cyclotomic factors of finite denominators.
The coefficient formula for P_r (Lean source) and the cyclotomic divisibility statement (Lean source) identify factors in the reduced denominators of the specified finite sums. The product-divisibility theorem (Lean source) is used in the exact denominator formula (Lean source). Growing finite denominators alone do not exclude cancellation with the remaining tail.

Numerator divisibility and denominator bounds.
Write $x = a/b$ in lowest terms, with $b > 0$. For $c \in \mathbb{Z}$ and a positive divisor H of b, if the reduced denominator of cx divides H, then $b/H \mid c$ (Lean source). In particular, $|c| \geq b/H$ when $c \neq 0$. A related bound is

$$0 < \frac{a}{b} < \frac{2}{2^n - 1}, \quad 2^r \mid a, \quad n \geq 1 \quad \Longrightarrow \quad 2^r(2^n - 1) < 2b$$

for integers $r \geq 0$ (Lean source). Here positivity gives $a \geq 2^r$; the size inequality then gives the conclusion. These are arithmetic bounds on a reduced fraction, not irrationality criteria without an additional approximation estimate.

Farey intervals.
The denominator bound (Lean source) applied to the explicit window $(N, K) = (1, 240)$ gives (Lean source) and hence $S \neq p/q$ for integers p and $1 \leq q \leq Q_0$. The finite input is a totient numerator modulo 2^{240}. Its exact first failing denominator is $Q_0 + 1$, the denominator of the relevant Farey mediant (Lean source). This last assertion concerns this fixed window.

Differences at least-common-multiple shifts.
For $H_t = \operatorname{lcm}(1, \ldots, t)$ (Lean source), a jump in H_t can occur only at a prime

power (Lean source). The identities (Lean source) and (Lean source) identify the coefficient difference as $\delta_t(j) = \varphi(2H_t + j) - \varphi(H_t + j)$ and give $D(H_t, H_t, L) = \sum_{j=1}^{L} \delta_t(j) 2^{L-j}$. Thus these arithmetic formulas concern particular instances of the same finite residue test; additional window restrictions remain in force.

Exponential sums of the window residues.

For integers $h, X \geq 1$, the relevant block statistic is

$$\sum_{N=X}^{2X-1} \cos\left(2\pi \frac{D(h, N, L)}{2^L}\right).$$

A bound by $9X/10$, together with $L \geq h$ and $16(2X + h + L + 2) \leq 2^L$, yields a certificate in the block (Lean source). The linked irrationality implication uses the stronger complex norm bound $21X/25$, for every h on arbitrarily late blocks satisfying the same size inequality (Lean source). The finite implication uses $\cos(\pi/8) > 9/10$; the required cancellation bound is still unproved. The norm bound on the complex exponential sum is sufficient for the displayed real-part bound, but is stronger.

8.1 Binary tails and Möbius sums

Erdős's argument for $\sum_n 1/(2^n - 1)$ chooses congruences so that divisor counts along a block are divisible by increasing powers of the base. A tail bound then forces a long run of zero digits [11, pp. 63–66]. It uses no finite automaton.

A different proposal would recover each later block from a fixed finite summary of the preceding history. The counterexample of Proposition 2.11(i) (Lean source) shows why that proposal fails for the family constructed there. For $m \geq 2$, its members have the same earlier history but different successors. A state that identifies all those histories cannot recover every successor. More generally, an exact label-and-decoder pair for that family needs at least $\lfloor (m+1)/2 \rfloor + 1$ distinct labels. This is a lower bound on the number of labels, not a linear lower bound on the number of bits, and it is not a state-complexity theorem for the actual totient sequence.

The implication from the quantified certificate condition (Lean source) is a separate theorem. The counterexample above neither proves its hypothesis nor makes it the only possible route to irrationality. Likewise, Proposition 10.4 shows that inverse-phase column weights can preserve a determinant, and Theorem 6.3 gives the exact growing ranks of totient sections. These are different statements with different hypotheses; none is a general prohibition on other representations of S.

The Möbius representation instead allows the squared-Lambert identity of Proposition 5.16 to be applied to μ, since $|\mu(d)| \leq d$ for $d \geq 1$. The polynomial identities of Propositions 6.37–6.41 operate at each finite radical without an autonomous decoder. The exact reduced denominator and its lower bound are then given by Propositions 6.8 and 6.7. Those formulas concern a finite rational contribution. They do not exclude cancellation by the complementary tail and hence do not settle irrationality. The representations are equal expressions for S; what differs is the information supplied by the particular arguments.

9 Index of unproved conditions

Every condition in this index is accompanied by the implication that would turn it into irrationality of $S = \sum_{n\geq 0} \varphi(n)2^{-n}$. Those implications are proved. The arithmetic hypotheses still have to be established with the displayed quantifiers.

9.1 Scope of the index

The index compares particular quantified hypotheses and the results that use them. An unsuccessful implication between two entries therefore says nothing by itself about every argument using the same method.

Each entry states its quantifiers in full. A cofinal assertion, with a quantifier of the form $\forall a_0\ \exists a \geq a_0$, may itself be proved or may remain an unproved hypothesis; the text of each entry says which. A cofinal assertion about a different quantity, or with a different order of quantifiers, need not supply the irrationality condition.

The quantified residue condition for #249, stated in Definition 1.2 and proved sufficient (Lean source), is the following statement:

$$\forall h \geq 1,\ \forall N_0,\ \exists N \geq N_0,\ \exists L,\ C\,h\,N\,L.$$

The following catalogue compares fourteen proposed approaches to this condition. It starts with the exponential-sum estimate, then gives thirteen further approaches with their individual hypotheses and source references.

9.2 The exponential-sum criterion

We express the truncated tail difference as a phase and ask for cancellation when that phase is summed over a block. The following implications explain how a saving gives a residue certificate and, with the required quantifiers, Definition 1.2.

Definition 9.1 (The truncated difference)**.** For $h, N, L \in \mathbb{N}$,

$$D(h,N,L) \;=\; \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j)) \cdot 2^{L-1-j} \;\in \mathbb{Z}.$$

This is the depth-L truncation of $2^L \cdot (R_{N+h} - R_N)$, where $R_N = \sum_{j\geq 1} \varphi(N+j)/2^j$ is the local totient tail (Lean source). Lean source

Definition 9.2 (The central-residue condition)**.**

$$C\,h\,N\,L \;:\Leftrightarrow\; (N{+}h{+}L{+}2) < D(h,N,L) \bmod 2^L < 2^L - (N{+}h{+}L{+}2).$$

That is: the residue of $D(h,N,L)$ modulo 2^L avoids the radius-$(N{+}h{+}L{+}2)$ neighbourhood of 0. It forces the size condition $2(N{+}h{+}L{+}2) < 2^L$ (Lean source). Lean source

Definition 9.3 (The complex phase and its real part)**.** Put

$$\operatorname{Re} E(h,N,L) = \cos\!\Big(2\pi \tfrac{D(h,N,L) \bmod 2^L}{2^L}\Big), \qquad E(h,N,L) = \exp\!\Big(i\,2\pi \tfrac{D(h,N,L) \bmod 2^L}{2^L}\Big).$$

The real part of E is the displayed cosine (Lean source), and $|E(h,N,L)| = 1$ (Lean source). Thus E is the first additive character of the endpoint discrepancy modulo 2^L, a standard exponential-sum object. Lean source Lean source

Theorem 9.4 (A real-part bound gives a certificate)**.** *For all h, X, L with $0 < X$ and the size condition $16(2X+h+L+2) \le 2^L$, if*

$$\sum_{N=X}^{2X-1} \operatorname{Re} E(h,N,L) \;\le\; \tfrac{9}{10}\, X,$$

then $\exists N \in [X, 2X)$ with $C\ h\ N\ L$.

Theorem 9.5 (The same implication for a nonempty subset)**.** *For any nonempty finite $T \subseteq \mathbb{N}$ with $T \subset [0, 2X)$ and the same size condition, if*

$$\sum_{N\in T} \operatorname{Re} E(h,N,L) \;\le\; \tfrac{9}{10}\, |T|,$$

then $\exists N \in T$ with $C\ h\ N\ L$. This generalises Theorem 9.4: T can be any explicitly chosen nonempty finite subset of the dyadic block, not the whole block, and the proof (an averaging pigeonhole) never uses that T has positive density or comes from a partition.

Theorem 9.6 (A norm bound gives the real-part criterion)**.** *The complex norm bound implies the real-part bound ($|z| \ge \operatorname{Re}(z)$, and $21/25 < 9/10$ absorbs the slack), so it composes through Theorem 9.4 to the same certificate. Define*

$$\textit{block norm condition} :\Leftrightarrow\ \forall h{>}0\ \forall X_0\ \exists X, L,\ \max(X_0, 1) \le X\ \wedge\ 16(2X+h+L+2) \le 2^L\ \wedge$$

$$\Big\| \sum_{N=X}^{2X-1} E(h,N,L) \Big\| \;\le\; \tfrac{21}{25}\, X.$$

This sufficient condition is unproved. Then

$$\textit{block norm condition} \implies \text{Irrational}\Big(\sum_{n\ge 0} \tfrac{\varphi(n)}{2^n}\Big),$$

proved in full, with no further gap, by chaining Theorem 9.6→Theorem 9.4→ Theorem 6.47.

Observation 9.7 (What remains unproved). The implications above do not establish the cofinal block estimate. Three further Lean statements retain a separation or gap hypothesis: one uses the two-element set $\{N, M\}$ (Lean source), and the other two use a supplied finite set T and the singleton $\{N\}$ (Lean source, Lean source). They prove what follows from the relevant estimate; they do not provide that estimate at arbitrarily large scales. The finite numerical observations in Table 6 are a different kind of evidence and do not discharge this quantified hypothesis.

Remark 9.8 (What the reduction does and does not prove). The required savings are explicit: a real-part bound of $9X/10$, or the stronger norm bound of $21X/25$, on blocks at arbitrarily large scales for every positive shift h. An unspecified nonzero saving is not the stated hypothesis. A bound $o(X)$ would suffice, but it is more than the implication needs.

The inequality $16(2X + h + L + 2) \le 2^L$ can be met by increasing L for fixed X, h. Doing so changes the exponential sum, and hence does not preserve an estimate proved at another depth. The arithmetic estimate must hold at a depth satisfying this inequality.

The subset theorem permits any *nonempty* finite sample inside $[0, 2X)$; no density assumption is required. A prime-defined sample is one possible choice. The four-term decomposition of §3.2 isolates one prime factor and the remaining totient-dependent phase, but does not prove the required cancellation between them.

9.3 Proved results and the additional assumptions they need

Besides the exponential-sum condition, the sufficient hypotheses involve tail signs, endpoint residues, rational approximation, short windows, Farey bounds, divisibility, denominator growth and carry rank. For each, we state the missing estimate used by the proposed application. These are specific possibilities rather than a classification of methods for Problem #249.

9.3.1 Positivity and the remaining residue inequality

Definition 9.9. The *actual LCM tail orbit* at exponent a is $\Omega_a = R_{2H} - R_H$, where $H = H(2^a)$ (Lean source).

Theorem 9.10. *For every $a \ge 8$ and every J with $J + (a+6) < 2 \cdot 2^a$,*

$$0 \;<\; R(2H+J) - R(H+J), \qquad H = H(2^a).$$

Hence $0 < \Omega_a$ for every $a \ge 8$. The proof is genuinely uniform in a: it rests on two facts proved for all $a \ge 8$ and $0 < j < 2^{a+1}$, namely $\delta_{2^a}(j) > 0$ and $H < 8 \cdot 2^a\, \delta_{2^a}(j)$, by a two-case structural split (divisor letter vs. foreign prime power), with absolute constants $4, 8, 32$ and no lookup table.

Remark 9.11 (Which endpoint interval must be excluded). Under the hypotheses of Proposition 6.85, integrality would force the residue to be $2^K - e$, where $e = R_{2H+J+K} - R_{H+J+K}$ is a positive integer smaller than $2H+J+K+2$. Thus the residue lies near the upper endpoint, not near zero in its least nonnegative representation. The representative $-e$ is negative; the carry e is positive.

The one-sided test below excludes precisely that upper interval. This uses the bound $J + K + (a + 6) < 2 \cdot 2^a$, as well as the modulus bound. It does not follow from positivity that the residue already satisfies half of a nonintegrality certificate without those assumptions.

9.3.2 An exact endpoint identity

Under the bounds below, positivity places an integral tail difference at the upper residue endpoint. A one-sided inequality would exclude it. That additional inequality is a sufficient condition, and is not claimed to be strictly weaker than irrationality.

Theorem 9.12 (The proved implications)**.** *For every $a \geq 8$ and every J, K, m inside the sign corridor $(J+K+(a+6) < 2 \cdot 2^a)$, a one-sided residue gap at precision $m \leq K$ (size condition $2H+J+K+2 < 2^m$ and $D(H, H+J, K) \bmod 2^m \leq 2^m - (2H+J+K+2)$) already forces $R(2H+J) - R(H+J) \notin \mathbb{Z}$. No lower margin at all is demanded. The proof chain is complete: the theorem holds for every $a \geq 8$, and cofinal upper-endpoint condition ⇒ Irrational S is proved.*

Definition 9.13 (The unproved quantified condition)**.** The *upper-endpoint condition* is

$$\forall a_0\ \exists a, K, m,\ a_0 \leq a\ \wedge\ 8 \leq a\ \wedge\ K+(a+6) < 2 \cdot 2^a\ \wedge\ \mathcal{G}_a(0, K, m).$$

No unconditional statement anywhere in the record locates the residue of the diagonal word $D(H, H, K)$ (where $H = H(2^a)$) below the top strip at even one large a. The condition reduces further to the last m differences $\delta_{2^a}(j)$ in the sum $D(H, H, K)$ (Lean source). Lean source

Proposition 9.14 (Five sufficient conditions)**.** *Each of the five conditions below suffices for irrationality; none of them is proved. The first four imply the upper-endpoint condition; the fifth gives nonintegrality directly by the endpoint identity. These are not asserted to form a linear hierarchy. Here $H = H(2^a)$, and every condition quantifies over arbitrarily large exponents a.*

1. *Adjacent-suffix band:*

$$\forall a_0\ \exists a, m,\ a_0 \leq a \wedge 8 \leq a \wedge m+1+(a+6) < 2 \cdot 2^a \wedge 2H+m+3 < 2^m \wedge$$

$$2H+m+2 \leq \textit{suffix residue}(2^a)\, 0\, m \leq 2^m - (2H+m+2)$$

   (*a two-sided band on the adjacent-suffix residue directly, one candidate depth m*).

2. *Odd-prefix interval at the prescribed depth: at the prescribed odd depth $2q+1$, a two-sided band of half-width $H+q+2$ on the half-word residue modulo 4^q. The depth is fixed by $q = q_a$; it cannot be chosen independently of a.*

3. *Centred-residue bound at the prescribed depth, proved equivalent to the previous one:*

$$\forall a_0\ \exists a, q,\ \max(14, a_0) \leq a \wedge q = q_a \wedge H+q+2 \leq |u_{a,q}|.$$

4. *Centred-residue bound at a variable depth: the same magnitude bound with the depth restriction relaxed from the prescribed value $q = q_a$ to any odd $2q+1$ satisfying the half-cell fit $2(H+q+2) \leq 4^q$ and the positivity range $2q+2+(a+6) < 2 \cdot 2^a$.*

5. *Terminal dominance: under the same bounds as item 4, $\delta_{2^a}(2q+2) \leq 2u_{a,q}$. This is a different one-sided comparison, treated immediately below.*

For the first four conditions, the relevant implications into the upper-endpoint condition are as follows: adjacent-suffix band → upper-endpoint condition; odd-prefix interval at the prescribed depth → adjacent-suffix band; centred-residue bound at the prescribed depth ⇔ odd-prefix interval at the prescribed depth; centred-residue bound at a variable depth → adjacent-suffix band; centred-residue bound at the prescribed depth → centred-residue bound at a variable depth.

9.3.3 A bound on the last totient difference

We compare the last totient difference with a centred residue. To use the comparison for irrationality, it would have to hold for arbitrarily large a, with q in the prescribed range.

Definition 9.15. The *terminal dominance condition*, which is unproved, is the following. Writing $H = H(2^a)$,

$$\forall a_0\ \exists a, q,\ a_0 \le a \wedge 8 \le a \wedge 2q{+}2{+}(a{+}6) < 2\cdot 2^a \wedge 2(H{+}q{+}2) \le 4^q \wedge$$

$$\delta_{2^a}(2q{+}2)\ \le\ 2\cdot u_{a,q}.$$

Lean source

Theorem 9.16 (The proved implication)**.** *The dominance hypothesis at a single odd rank already excludes integrality of Ω_a, and the terminal dominance condition implies $S \notin \mathbb{Q}$.*

Observation 9.17 (How the sign and size bounds enter the contradiction). Use the hypotheses and notation of Proposition 6.66. Under the assumption $z = \Omega_a \in \mathbb{Z}$, the exact identity is

$$2u_{a,q} = \delta_{2^a}(2q+2) - c_{H,H,z}(2q+1), \qquad 0 < c_{H,H,z}(2q+1) < 2H + 2q + 3.$$

Terminal dominance $\delta_{2^a}(2q+2) \le 2u_{a,q}$ would contradict the positive lower bound on the carry. The other sufficient inequality, $2u_{a,q} \le \delta_{2^a}(2q+2) - (2H+2q+3)$, would contradict its upper bound. Both are valid ways to contradict integrality.

These bounds do not eliminate either approach: they hold under the integrality assumption that the approach is intended to refute. They neither prove nor disprove an unconditional estimate for $u_{a,q}$. Finite examples lying between the two boundaries do not establish an asymptotic exclusion of either inequality. The identities and conditional bounds are proved in Lean (Lean source, Lean source, Lean source).

Proposition 9.18 (A sufficient extension)**.** *Either of the following suffices: (i) the lower-escape branch cofinally, as displayed just above; or (ii) the two-sided magnitude form (item 4 of Proposition 9.14), which asks only $H{+}q{+}2 \le |u_{a,q}|$ and which the Lean source proves implies corridor escape via a clean sign split (positive branch escapes above the terminal letter, negative branch escapes below the directed bound), without prescribing the sign of the centred representative. Neither branch is supplied by the conditional sign identity itself.*

9.3.4 Separation of a rational approximation

Theorem 9.19. *Unconditionally, for every a and q,*

$$|\,\Omega_a - \rho_{a,q}\,| \;<\; \frac{4H + 2(2q+1) + 4}{2^{2q+2}},$$

where $\rho_{a,q}$ is an explicit finite computable rational block. For fixed a, the error tends to zero as q grows. In the quantified condition below, however, the depth is prescribed, so increasing q is not a free way to satisfy the separation hypothesis.

Remark 9.20 (Scope of the implication). A cofinal $1/32$-separation of the raw approximants at the prescribed indices $q = q_a$ suffices for #249. Allowing arbitrary depths would be a different hypothesis. The cited short-window theorem gives nonintegrality of Ω_a at $a = 4, 6$ (Proposition 6.94). It neither asserts the $1/32$ margin at those exponents nor says that these are the only successful exponents. Indeed, the diagonal table also gives short-window certificates $(a, L) = (2, 7)$ and $(3, 15)$, since $H_4 = 12$, $H_8 = 840$, and $7 < 8$, $15 < 16$. All these are finite nonintegrality examples, not the cofinal approximation-separation condition.

Definition 9.21 (The additional hypothesis)**.** The *cofinal approximation-separation condition*, which is unproved, is

$$\forall a_0\ \exists a \geq \max(2, a_0)\ \exists q, \quad q = q_a \wedge$$
$$\forall z \in \mathbb{Z}, \quad \tfrac{1}{32} + \varepsilon_{a,q} \leq |\Omega_a - z|.$$

This hypothesis is about the real quantity Ω_a, not just a finite rational calculation. It implies $|\rho_{a,q_a} - z| > 1/32$ for every integer z, by the displayed error bound and the triangle inequality. A cofinal $1/32$-separation condition on the finite approximants is another sufficient input, through the residue-band implication. These sufficient conditions are not asserted to be equivalent.

9.3.5 The short-window examples through exponent 6

Theorem 9.22.

$$\forall a_0 \leq 6,\ \exists a, L,\ a_0 \leq a \wedge L < 2 \cdot 2^a \wedge \mathbb{C}(H_{2^a}, H_{2^a}, L)$$

The single witness $(a, L) = (6, 93)$ works for every threshold $a_0 \leq 6$. Its certificate is verified by exact integer arithmetic on the finite totient windows. This proves the displayed bounded statement. It does not construct witnesses for unbounded thresholds; further finite checks would extend the verified range, not establish the universally quantified condition below.

Definition 9.23 (The additional hypothesis)**.** The *cofinal short-window condition*, which is unproved, is

$$\forall a_0\ \exists a, L,\ a_0 \leq a \wedge L < 2 \cdot 2^a \wedge \mathbb{C}(H_{2^a}, H_{2^a}, L).$$

The depth restriction gives a useful description of the offsets: if $1 \leq j < 2 \cdot 2^a$ and $j \nmid H_{2^a}$, then j is a prime power greater than 2^a. Indeed, some prime-power divisor p^r

of j exceeds 2^a; if $j/p^r \geq 2$, then $j > 2 \cdot 2^a$, a contradiction. This describes the possible offsets, not the size of their weighted sum.

The condition restricts the depth in the ordinary diagonal certificate criterion. It is not established by pointwise completeness, which permits arbitrarily large depths. Nor does a successful short-window certificate by itself yield the fixed $1/32$ separation required by the preceding approximation condition. These are distinct sufficient hypotheses; no implication from this condition to that fixed-margin condition is proved here.

9.3.6 The diagonal certificate table

Theorem 9.24.

$$\forall t \in \{1,2,3,4,5,7,8,9,11,13,16,17\},\ \exists L,\ C\ (H\ t)\ (H\ t)\ L$$

at depths $\{6,5,7,7,9,14,15,14,21,22,23,26\}$ *respectively, extended by separate computations to 28 historical values through* $t = 64$. *The later aggregate theorem closes every scale* $t \leq 82$ *with no holes. Each witness is checked by exact integer arithmetic on two finite totient windows. For example,* $H_{17} = 12252240$ *and* $L = 26$ *use* $\varphi(H_{17} + i)$ *and* $\varphi(2H_{17} + i)$ *for* $1 \leq i \leq 26$, *not every totient between the endpoints* 12252241 *and* 24504506. *These computations verify the listed cases; they do not give a formula producing a certificate for every* t.

Remark 9.25 (The observed depths are finite data). The necessary depth inequality of Lemma 6.17 forces $2(2H_t+L+2) < 2^L$, that is, $L \gtrsim \log_2(4H_t)$ at any certified depth. Comparing that bound with the recorded depths shows that every recorded certificate depth lies within a small additive constant of that minimum, at every tested scale. This is an observation from finite computation, and no general theorem asserts it.

Proposition 9.26 (A sufficient extension)**.**

$$\exists C\ \forall t_0\ \exists t \geq t_0\ \exists L \leq \log_2(4 \cdot H\ t) + C, \quad C\ (H\ t)\ (H\ t)\ L.$$

This asks for a certificate within a fixed additive constant of the necessary logarithmic depth bound. The uniform constant is an additional requirement, not a consequence of the finite data.

9.3.7 One common certificate for sixteen shifts

Theorem 9.27.

$$C(h, 14, 9) \qquad (h \in \{1, \ldots, 16\}).$$

The basepoint 14 *and depth* 9 *are common to all sixteen shifts. Consequently,* $S \neq r$ *whenever* $r \in \mathbb{Q}$ *and* $\operatorname{den}(r) \mid 2^{14}(2^h - 1)$ *for some integer* h *with* $1 \leq h \leq 16$. *The exact computation uses two nine-term windows for each shift and reduces their discrepancy modulo* 2^9. *The earlier eight-shift computation uses basepoint* 12 *and depth* 16.

Remark 9.28 (Finite simultaneity and the unbounded condition). The displayed computation concerns only $N = 14$, $L = 9$ and $1 \le h \le 16$. It does not provide certificates beyond every basepoint threshold. Simultaneous certification of finitely many shifts at a sufficiently large, unrestricted depth is not an additional obstacle: under irrationality it follows from the same truncation estimate, as shown next. The argument below does not prescribe the common depth in terms of the basepoint and number of shifts.

Proposition 9.29 (Simultaneous certificates with unrestricted depth)**.** *The existence of a function $f : \mathbb{N} \to \mathbb{N}$ with $f(N) \to \infty$ such that*

$$\forall N_0\ \exists N \ge N_0\ \exists L\ \forall h \in \{1, \dots, f(N)\}, \quad C(h, N, L)$$

is equivalent to irrationality of S. Under irrationality, one can take $f(N) = N + 1$, and every N has a suitable depth.

Proof. The displayed condition supplies Proposition 6.71's quantified certificate condition: for fixed $h \ge 1$, choose the basepoint threshold so large that $f(N) \ge h$ beyond it. That proposition gives irrationality.

Conversely, suppose S is irrational. For every $h \ge 1$ and $N \ge 0$,

$$R_{N+h} - R_N = 2^N(2^h - 1)S - (\Phi_{N+h} - \Phi_N)$$

is nonintegral. Fix N and a positive integer M. Choose L so large that

$$\frac{2(N + M + L + 2)}{2^L} < \min_{1 \le h \le M} \operatorname{dist}(R_{N+h} - R_N, \mathbb{Z}).$$

The finite minimum is positive, whereas the left side tends to zero. For each $h \le M$, the truncation error between $D(h, N, L)/2^L$ and $R_{N+h} - R_N$ is at most $(N + h + L + 2)/2^L$. The triangle inequality therefore puts $D(h, N, L)/2^L$ at distance greater than $(N + h + L + 2)/2^L$ from every integer, which is precisely $C(h, N, L)$. Taking $M = N + 1$ proves the converse. □

This proof gives no quantitative lower bound for the finite minimum as the basepoint or number of shifts grows. It selects a common depth after those parameters are fixed; it does not impose depth equal to the period.

9.3.8 The fixed Farey bound

Theorem 9.30. *If S is rational, its reduced denominator exceeds* 7.963 964 664 670 137 532 335 577 487 583 105 3 $\times 10^{34}$. *This is the finite denominator exclusion in the corpus, logically independent of the certificate-supply reduction.*

The proof combines the general Farey mediant lemma (Lemma 5.4) with an exact finite computation at the window $K = 240$ (Proposition 5.5).

Remark 9.31 (What a larger Farey bound requires). Unbounded valid denominator exclusions would imply $S \notin \mathbb{Q}$. For the single window $K = 240$, however, the first-failure computation (Lean source) proves that the residue inequality fails at the mediant denominator $b + d = Q_0 + 1$. This exact first failure prevents a larger cutoff from that same

inequality at that same window. It is not an obstruction to other windows or to other tests using the finite data.

The existing proofs for individual windows use explicit totient residues and rational endpoint calculations. Extending that implementation calls for new finite data; a uniform proof of a growing bound need not use hard-coded convergents. If S had reduced denominator q, every valid exclusion bound would be smaller than q. Thus unbounded exclusions suffice for irrationality; rationality would bound them, but would not force eventual constancy. No converse or comparison with the continued-fraction denominators of S is proved here.

Observation 9.32 (The limited gain from a unit-gap refinement). The finite unit-gap condition of Observation 6.114 (Lean source) and its quantified version (Lean source) are definitions; the latter is not asserted to hold. Two separate results (Lean source, Lean source) show that, for p prime and $e \geq 1$, a window at denominator p^e that *passes the refined test* contains at most one candidate integer. A window with two or more candidates fails the test. This is a condition for successful certification, not a bound on all windows or on the improvement in the denominator cutoff. It does not apply to all odd denominators: Observation 6.114 gives an actual totient window with two admissible nonunit candidates modulo 15.

Proposition 9.33 (A sufficient extension)**.** *Suppose $g(K) \to \infty$ and, for every K, the $(N = 1, K)$ gap check excludes every rational of reduced denominator at most $g(K)$. Then S is irrational: any rational value of S would have a fixed finite denominator, contradicted at a sufficiently large K. Only unbounded exclusion bounds along a sequence of windows are needed for this argument. No equivalence with a bound for the continued-fraction denominators of S is asserted; such a comparison would require its own proof connecting those convergents to the certified intervals.*

9.3.9 A two-adic congruence that does not give a certificate

Theorem 9.34. *The theorem is unconditional and holds for every K, H, B. For every $K \geq 2$, every $H > K$, and every bound B, there are primes $p > B$ with a length-$(K-1)$ zero prefix and a terminal half-modulus, giving*

$$D(H, p-K, K) \equiv 2^{K-1} \pmod{2^K}.$$

Under eventual integrality this transfers to an integer z with $(z : \mathbb{R}) = R(p+H) - R_p$ and $z \equiv 2^{K-1} \pmod{2^K}$.

Remark 9.35 (Why the central residue still fails the test). This places the residue at the centre of the residue interval: 2^{K-1} is as far as possible from 0 modulo 2^K. Taking $N{=}p{-}K$, $h{=}H$, $L{=}K$ gives radius $N{+}h{+}L{+}2 = p{+}H{+}2$, so C requires $2^{K-1} > p{+}H{+}2$. But $z \equiv 2^{K-1} \pmod{2^K}$ forces $|z| \geq 2^{K-1}$, while the directed tail bound forces $|z| < p{+}H{+}2$; and the construction's own congruence $p \equiv 1{+}2^{K-1} \pmod{2^K}$ (equivalently $v_2(p{-}1) = K{-}1$) forces $p \geq 1{+}2^{K-1}$. So $p{+}H{+}2 > 2^{K-1}$ always, and no choice of p rescues it. The quantifier order is inverted: the theorem gives, for fixed K, cofinally many *large* p; the certificate needs a p *small* relative to 2^{K-1}, impossible for a single totient difference since $|\varphi(N{+}H) - \varphi(N)| < N{+}H$. The depth is not the obstacle: the construction

already works uniformly for every K, combining by the Chinese remainder theorem one congruence for $p+H$ and two for each of the $K-1$ preceding positions (Proposition 6.92).

Proposition 9.36 (A sufficient accumulated-residue condition)**.** *Suppose that for every integer $h \geq 1$ and every $N_0 \in \mathbb{N}$ there are $N \geq N_0$ and $L \geq 1$ such that*

$$D(h,N,L) \equiv 2^{L-1} \pmod{2^L}, \qquad N + h + L + 2 < 2^{L-1}.$$

Then $S \notin \mathbb{Q}$: these two inequalities imply $C(h,N,L)$, and the full quantified certificate condition applies. The half-modulus condition here is imposed on the accumulated weighted sum $D(h,N,L)$, not on a single totient difference. An unbounded set of triples with no control of the shift does not supply the displayed quantifiers. The required family remains unproved.

9.3.10 The residue margin at an LCM jump

Theorem 9.37 (Arbitrarily large prime-power LCM jumps)**.** *For every $t_0 \in \mathbb{N}$ there is $t \geq t_0$ with $H(t) < H(t+1)$. One may take $t = p - 1$ for any prime $p > t_0$. The positions $t = 2^a - 1$ are also strict LCM jumps for $a \geq 1$: 2^a is the next required power of 2. The restriction excludes $a = 0$, since $H(0) = H(1) = 1$. The second condition in the next definition asks for the additional residue margin along this explicit sequence. Knowing these jump positions does not establish that margin.*

Definition 9.38 (The remaining residue margin)**.** For $t \geq 3$, put $H = H_t$ and define

$$\begin{aligned} m &= \lfloor \log_2 H \rfloor + 10, \\ d &= (D(H, H+1, m) - D(H, H, m)) \bmod 2^m, \\ \sigma_t &= \min\{d - 2^{m-5},\ 2^m - 2^{m-5} - d\}. \end{aligned}$$

Thus d is the difference of two adjacent window numerators, reduced modulo 2^m, not the residue of $D(H,H,m)$ alone. The integer σ_t is nonnegative exactly when $2^{m-5} \leq d \leq 2^m - 2^{m-5}$. The choice of m makes the error bound $3H + 2m + 5$ smaller than 2^{m-5}; nonnegativity is the additional arithmetic requirement, not a consequence of that choice. Lean source Lean source Lean source

The first unproved condition asks for every t_0 for an index $t \geq \max(3, t_0)$ with $H_t < H_{t+1}$ and $\sigma_{t+1} \geq 0$. The second restricts these endpoints to powers of two: for every a_0, there must be $a \geq \max(2, a_0)$ with $\sigma_{2^a} \geq 0$. They are defined in Lean (Lean source, Lean source). Each condition implies $S \notin \mathbb{Q}$ (Lean source, Lean source). The implications are proved; both conditions are unproved.

Remark 9.39 (Finite numerical evidence). All 40 strict jumps up to endpoint 113 pass, closest margin ≈ 0.000221 of the modulus at $t = 100$; the power-of-two endpoints $4, 8, 16, 32$ pass with slack fractions $0.41, 0.036, 0.40, 0.19$. This is a finite computation; both conditions remain unproved at arbitrarily large endpoints. A strict jump has $t + 1 = p^k$ for a prime p and $k \geq 1$, with $H_{t+1} = pH_t$ (Lean source). Thus σ_{t+1} uses the new height pH_t, not the pre-jump height H_t. Estimating its change would require information about the adjacent-window residue, not just the LCM ratio. Moreover $\sigma_t = \sigma_u$ whenever $H_t = H_u$ (Lean source). Thus the value changes only when the LCM height changes. This reduces the number of distinct values to study; it does not determine their signs.

Proposition 9.40 (A sufficient inequality)**.** *If, for every $a_0 \in \mathbb{N}$, there is a $\geq \max(2, a_0)$ with $\sigma_{2^a} \geq 0$, then $S \notin \mathbb{Q}$. This is the sufficient condition defined above, not a claim that the inequality holds at arbitrarily large indices. Equality at the edge is allowed: the condition is nonnegativity, not strict positivity.*

9.3.11 A certificate for a four-tail combination

Theorem 9.41. *Use $J(H,p)$, $W(H,p,L)$ and $B(H,p,L)$ from Theorem 6.60. For all $H,p,L \in \mathbb{N}$,*

$$B(H,p,L) < W(H,p,L) \bmod 2^L < 2^L - B(H,p,L) \quad \Longrightarrow \quad J(H,p) \notin \mathbb{Z}.$$

The bound $B(H,p,L) = 3pH + (p+1)(L+2)$ follows by grouping the nonnegative remainders as in that theorem. Compared with the $4pH + (p+1)(L+2)$ bound obtained by treating the two diagonal differences separately, it saves pH. This is a strict improvement when $pH > 0$; optimality for totient tails is not asserted.

Theorem 9.42 (One explicit witness)**.** *For $H = H(4) = 12$, $p = 5$ and $L = 15$, direct integer evaluation gives*

$$W(12,5,15) = 149906, \qquad W(12,5,15) \bmod 32768 = 18834, \qquad B(12,5,15) = 282.$$

Thus $282 < 18834 < 32486$, and $J(12,5) \notin \mathbb{Z}$. This is the one instance singled out in Theorem 6.60, not a claim of uniqueness or of the smallest possible height. The implication from a quantified family is Theorem 6.60.

Remark 9.43 (Scope of the witness). The quantified hypothesis asks, for every $t_0 \in \mathbb{N}$, for $t \geq t_0$, $p \geq 1$ and $L \geq 0$ satisfying the displayed residue condition with $H = H(t)$. The multiplier p need not be prime, and need not be tied to the next LCM jump. No comparison of the difficulty of this condition with the prescribed-depth separation condition is proved.

At a strict LCM jump, $H(t+1) = pH(t)$ for a prime p. This gives one natural choice of multiplier, but the prime need not be a new divisor: $H(4) = 2H(3)$ and 2 already divides $H(3) = 6$. The identity of Proposition 6.45 cancels contributions affine in the multiplier. It does not bound the remaining terms of the four-tail combination. The finite example above therefore does not establish the quantified condition.

9.3.12 Denominators of Möbius sums

Theorem 9.44. *For every integer $t \geq 5$, with $\mathcal{P}_t = \{p \text{ prime} : t/2 < p \leq t\}$ as above,*

$$2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{P}_t} (2^p - 1) \leq \operatorname{den}(H_t \beta_{H_t}).$$

Moreover, some $p \in \mathcal{P}_t$ satisfies

$$2^{\lfloor t/2 \rfloor} \leq 2^p - 1 < 2^t, \qquad 2^p - 1 \mid \operatorname{den}(H_t \beta_{H_t}).$$

Indeed, Bertrand's postulate makes $\mathcal{P}_t$ nonempty. Its members satisfy $p - 1 \geq \lfloor t/2 \rfloor$ and $p \leq t$, giving the size bounds. The divisibility follows from Proposition 6.6; divisibility of a positive reduced denominator also gives the product inequality.

Remark 9.45 (Growth of these denominators does not give approximation of S). The number $H_t\beta_{H_t}$ is the finite rational contribution to a totient-tail decomposition. Its complementary term can still cancel its fractional part. The stated denominator bound supplies neither an error estimate for approximation of S nor nonintegrality of $R_{2H_t} - R_{H_t}$.

To apply the diagonal criterion in Theorem 6.49, one must establish this tail nonintegrality at arbitrarily large LCM scales. Denominator growth alone does not supply that condition.

Proposition 9.46 (The additional approximation hypothesis)**.** *A sequence of rationals u_t proves irrationality of S if*

$$u_t \neq S \quad \text{for all sufficiently large } t, \qquad \operatorname{den}(u_t)|S - u_t| \longrightarrow 0.$$

This is the rational-separation criterion of Proposition 6.22: if $S = a/b$ were reduced, every unequal u_t would instead satisfy $\operatorname{den}(u_t)|S - u_t| \geq 1/b$. An upper bound tending to zero for this product is needed. A lower bound for the denominator cannot establish it; improving a lower bound does not change the actual approximation error.

A different argument could deduce a residue certificate from the surviving divisor and further information about the complementary term. The denominator theorem alone contains no such separation statement. Neither sufficient argument is established by its lower bound, and these are not claimed to exhaust possible approaches.

9.3.13 Rationality and carry rank

The totient sections have a dimension lower bound, as do the sections of an integral carry under the rationality hypothesis. An incompatible upper bound would finish a rank argument. The rational comparison sequences show why rationality by itself gives no such bound.

Theorem 9.47. *For every $e \geq 1$, the retained dyadic totient sections form a basis for all sections through level e, of dimension 2^e+1. The proof uses the Chinese remainder theorem and Dirichlet's theorem. The complete level-zero truncation, consisting only of φ, has dimension one.*

Theorem 9.48. *If $S \in \mathbb{Q}$, choose an integer $v > 0$ with $vS \in \mathbb{Z}$ and put $u_N = vR_N$. Then $u_N \in \mathbb{Z}$, $u_{N+1} = 2u_N - v\varphi(N+1)$, and $0 \leq u_N \leq v(N+2)$. For every $e \geq 0$,*

$$\dim_{\mathbb{Q}} \operatorname{span}\{n \mapsto u_{2^j n+r} : 1 \leq j \leq e,\ 0 \leq r < 2^j\} \geq 2^e - 1.$$

The proof is given in Section 10.8.

Remark 9.49 (The generic upper-bound inference is false). The bound $2^e - 1$ holds at every level. An upper bound contradicting it would need additional information about the totient, since the rational $5/4$ comparison has the same lower bound. One proposed system of integer equalities and strict inequalities is inconsistent: there are no integers $Q, v > 0$, $A \neq 0$ and b with $QvA = b$ and $|b| < Qv$, since a nonzero integer multiple of Qv has absolute value at least Qv (Lean source). This excludes those simultaneous requirements, not every argument using a matrix or its adjugate. Separately, the span of all dyadic totient sections is infinite-dimensional (Proposition 1.6).

Proposition 9.50 (An additional hypothesis for a totient-specific rank argument)**.** *A rank bound that contradicts the lower bound* 2^e-1 *would have to use additional arithmetic of the actual totient coefficients. The generic proposal that rationality bounds the rank of every integral carry satisfying* $u(n)/2^n \to 0$ *is disproved by the comparison sequence recorded in the short paper: its carry rank is at least* $2^e - 1$ *at every level. It is therefore not a remaining general lemma from which totient irrationality follows. The conditional Proposition 1.7 records eventual periodicity modulo* v *together with unbounded rational rank. Periodicity of the residue sequences concerns values in a finite quotient; it does not bound the rational span of the integer-valued sections. The* $5/4$ *comparison shows that this distinction persists for a rational coefficient series.*

9.4 Relations between the sufficient conditions

A fixed saving for the real part or norm of the block exponential sum would give a certificate. The endpoint conditions seek a residue or size bound along an LCM sequence instead. Proposition 9.14 relates the sufficient formulations; the arithmetic estimates required by them remain open.

The generic bounded-rank argument fails for a separate reason: the rational comparison sequence has unbounded section rank. A totient-specific upper bound would require hypotheses not shared by that sequence.

These are statements about the displayed conditions and comparisons, not an exhaustive classification of possible proofs. Some conditions are equivalent to irrationality; others are only sufficient. No relative difficulty, truth probability, or independent formal verification follows from their position in the table.

10 Counterexamples to proposed deductions

The next constructions produce rational series from coefficients satisfying selected totient data. Others test a proposed estimate without changing the coefficients. We specify the retained assumptions in each case, so that the failed deduction can be distinguished from an argument using additional properties of φ. None decides the irrationality of S.

10.1 Changing the coefficients after a finite prefix

Proposed argument. The quantified residue condition requires certificates beyond every starting index. A finite computation can check many instances, but one fixed inspection bound cannot establish these quantifiers. The following comparison explains this limitation without making any claim about all possible irrationality arguments. LEAN SOURCE

Counterexample. Theorem 2.4 gives an explicit splice construction; its series value, exact reduced denominator and failure of the separation condition are checked in Lean. Choose an integer $P > B$, set $\gamma(n) = \varphi(n)$ for $n \le B$, and $\gamma(n) = n - \mathbf{1}_{P\mid n}$ for $n > B$. Then $0 \le \gamma(n) \le n$, and the binary series $\sum \gamma(n)/2^n$ is *rational*: it is a finite rational prefix, plus the tail of $\sum_n n2^{-n}$, minus a geometric sum over the multiples of P beyond B. Because γ agrees with φ on every index $\le B$, γ passes every finite certificate that only inspects indices $\le B$; in particular it passes or fails every instance of $\mathcal{C}(h,N,L)$ with $N+h+L \le B$ exactly as φ does. Yet $\sum \gamma(n)/2^n \in \mathbb{Q}$ by construction.

Scope. For each fixed bound B, a different rational comparison sequence can agree with φ on all inspected coefficients. This rules out a deduction from that finite prefix alone. It does not rule out a proof that establishes the required instances for arbitrarily large B with the correct quantifiers. Such a proof would no longer be a single finite inspection. The comparison also does not preserve multiplicativity or every other arithmetic property of φ.

Consequence. The finite-prefix example and the parity example below are distinct constructions. The former preserves a prescribed finite prefix; the latter preserves parity and has separated nonperiodic blocks. Each shows why its own list of assumptions is insufficient. The recorded finite diagonal examples remain useful individual certificates, but their number alone gives no assertion at arbitrarily large scales. LEAN SOURCE

10.2 A rational series with matching totient parity

Matching totient parity and the bound $c(n) \le n$ does not determine the rationality of a binary sum. The example below is bounded and aperiodic, yet its sum is rational. Its paired changes can be placed beyond any threshold with any prescribed minimum separation.

Construction. Start with the sequence $0, 1, 1, 2, 4, 4, 4, \ldots$. At each position $m = 2^{k+3}$, $k \ge 0$, replace the pair $(4, 4)$ at $m, m+1$ by $(6, 0)$. Equivalently,

$$c(n) = \begin{cases} 0, & n = 0, \\ 1, & n = 1, 2, \\ 2, & n = 3, \\ 6, & n = 2^{k+3} \text{ for some } k \ge 0, \\ 0, & n = 2^{k+3} + 1 \text{ for some } k \ge 0, \\ 4, & \text{otherwise.} \end{cases}$$

Each replacement leaves the binary sum unchanged because $2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0$. The increasingly large gaps between replacements prevent eventual periodicity. The strongest recorded result also quantifies their separation:

Proposition 10.1 (A rational series preserving totient parity and the stated separation properties)**.** *There exists $c : \mathbb{N} \to \mathbb{N}$ such that: $c(n) \le 6$ for all n; $c(n) \le n$; $c(n) \equiv \varphi(n) \pmod 2$ for every n; for every N, G, K there are K pairs of coefficients $(6, 0)$ starting beyond N, with starting positions separated by more than G; and $\sum_n c(n)/2^n = 3/2$.*

Scope. The displayed bounds, parity agreement and aperiodicity, including the stated number and separation of the edits, are jointly compatible with a rational binary sum. They therefore do not imply irrationality. The sequence c differs from φ: the example preserves its parity, not all its arithmetic properties. It excludes an argument using only the listed assumptions, not every strengthening of an aperiodicity condition or every use of totient coefficients.

Consequence. The paired edits have zero total contribution, since $2 \cdot 2^{-m} - 4 \cdot 2^{-(m+1)} = 0$. Applied at the specified sparse positions of this particular base sequence,

they preserve the rational sum and the required bounds while preventing eventual periodicity. This does not construct a rational series for every prescribed parity pattern, and it is not the finite-prefix construction of Theorem 2.4. A proposed criterion using only the assumptions of this example must fail; a criterion using further arithmetic information is not addressed.

10.3 Square-CRT factorisations

For a prime p and $m \geq 1$, the identity

$$\varphi(pm) = (p-1)\varphi(m) + \mathbf{1}_{p|m}\varphi(m)$$

separates the coprime and divisible cases. Prescribing a cofactor modulo p by a congruence modulo p^2 can remove the second term over a chosen finite range. It simplifies the product formula, but does not show that differences between the resulting values are nonzero.

The examples in Observation 6.117 show both possibilities. At $n = 52$, the offset pairs $(13, 13)$ and $(26, 18)$ give respectively $\varphi(65) - \varphi(65) = 0$ and $\varphi(78) - \varphi(70) = 0$ (Lean source). All four prime–cofactor products are coprime (Lean source). At $n = 27$, the offset pairs $(3, 3)$ and $(6, 8)$ instead give $\varphi(30) - \varphi(30) = 0$ and $\varphi(33) - \varphi(35) = -4$ (Lean source), again with coprime factors (Lean source). These finite equalities are checked by exact arithmetic. No minimality claim about either witness is needed. The general existence theorem Lean source supplies a common base for the stated finite congruence requirements, not a nonvanishing assertion.

The vanishing example disproves the implication from correction-free factorisation to nonvanishing; the other example shows that this factorisation does not force vanishing either. This does not establish formal independence of two propositions, or exclude an argument using additional hypotheses. The prime-dilation identity Lean source and the CRT constructions Lean source, Lean source remain available with their own prime and residue hypotheses.

A separate elementary identity in the same file concerns alternating sums. Let I be finite, let $i \notin I$, and suppose an integer-valued function f on finite subsets satisfies $f(T \cup \{i\}) = f(T)$ for every $T \subseteq I$. Pairing T with $T \cup \{i\}$ gives

$$\sum_{T \subseteq I \cup \{i\}} (-1)^{|T|} f(T) = 0.$$

This identity is proved in Lean (Lean source). It has no totient hypothesis, but an application must still verify invariance under the indicated insertion. Neither it nor the congruence construction supplies the missing nonzero totient difference at a prescribed LCM height.

10.4 The cost of clearing a rational denominator

Clearing a denominator changes the multiplier as well as the fraction. The example $4(5/12) = 5/3$ illustrates the divisibility relation below for arbitrary reduced fractions; no totient identity is needed.

Lemma 10.2 (Complement divisibility after multiplication)**.** *Write $x = a/b$ in lowest terms, where $a \in \mathbb{Z}$ and $b \geq 1$. Let $c \in \mathbb{Z}$ and let H be a positive divisor of b. If the reduced denominator of cx divides H, then*

$$b/H \mid c.$$

Indeed, the hypothesis makes $Hcx = Hca/b$ integral. Since $\gcd(a, b) = 1$, it follows that $b \mid Hc$, hence $b/H \mid c$. Writing $c = (b/H)t$ also gives $Hcx = ta$; this equality is also proved in Lean (Lean source). If $c \neq 0$, the divisibility implies $|c| \geq b/H$. The nonzero hypothesis is essential for this size bound: $c = 0$ has reduced denominator one and satisfies the divisibility, but has no positive size cost. The lemma is about multiplication of one rational value, not arbitrary cancellations among several values.

A related elementary inequality compares a positive numerator with its denominator. If $a > 0$, $r \geq 0$, $n \geq 1$, $2^r \mid a$, and $a/b < 2/(2^n - 1)$, then

$$2^r(2^n - 1) \leq a(2^n - 1) < 2b.$$

Lean proves this Mersenne instance (Lean source) and the version with a general positive denominator parameter (Lean source). Positivity of a is what turns numerator divisibility into a lower bound; it cannot be omitted.

The accompanying linear-algebra statement has a one-dimensional quotient. Let V, W be vector spaces over $\mathbb{Q}$, let $\mathrm{ev} : V \to \mathbb{Q}$ and $\Lambda : V \to W$ be linear, and suppose $\ker \mathrm{ev} \subseteq \ker \Lambda$. Choose $e \in V$ with $\mathrm{ev}(e) = 1$. Then, for every $v \in V$,

$$v - \mathrm{ev}(v)e \in \ker \mathrm{ev}, \qquad \Lambda(v) = \mathrm{ev}(v)\Lambda(e).$$

This is factorisation through the scalar quotient $V/\ker \mathrm{ev}$ (Lean source). A surjection onto a higher-dimensional space would imply only a linear factorisation through that space, not multiplication by one fixed vector.

The denominator calculations use two lemmas proved in Lean (Lean source, Lean source). For integers a and $D \geq 1$, the reduced denominator of a/D is $D/\gcd(|a|, D)$. Hence a divisor of D coprime to a divides the reduced denominator. Coprimality is sufficient, not necessary: $2/8 = 1/4$ retains the divisor 2. More generally, if C is a positive divisor of D, $\gcd(C, a) = 1$, and $h \geq 0$ is an integer, then $C/\gcd(C, h)$ divides the reduced denominator of ha/D. At $h = 0$ this assertion retains only the divisor one.

For a nonzero multiplier, a lower bound for b/H therefore gives a lower bound for its absolute value. For an integer $t \geq 5$, the product estimate $2^{\lfloor t/2 \rfloor} \leq \prod_{p \in \mathcal{P}_t}(2^p - 1)$ (Lean source) can be used when that product is shown to divide b/H. Neither the product estimate nor the scalar lemma proves a quadratic growth law for arbitrary approximation schemes.

The finite Farey exclusion for S is a separate use of reduced denominators. Its certified interval contains no rational whose reduced denominator is at most

$$79{,}639{,}646{,}646{,}701{,}375{,}323{,}355{,}774{,}875{,}831{,}053$$

(Lean source). That numerical exclusion requires the interval and Farey calculation; it is not a consequence of the scalar lemma alone.

10.5 Signed moments and parity

A signed determinant can serve in rational approximation only with both a nonzero numerator and a small error. Here parity proves the first requirement for the particular finite signed sum.

Lemma 10.3 (Nonvanishing from a unique largest denominator exponent)**.** *Let I be a finite set, let $u_i \in \mathbb{Z}$ and $e_i \in \mathbb{N}$ for $i \in I$, and suppose that $m \in I$ satisfies $e_i < e_m$ for every $i \neq m$. If u_m is odd, then*

$$2^{e_m} \sum_{i \in I} \frac{u_i}{2^{e_i}} = \sum_{i \in I} u_i 2^{e_m - e_i} \equiv 1 \pmod 2.$$

In particular, the sum is nonzero.

Every term except u_m is even, which proves the assertion. For example, $1/2 - 1/8 = 3/8$ has odd cleared numerator. Uniqueness is necessary for this argument: two odd contributions with the same largest exponent can cancel.

Lean also proves a separate determinant expansion from the Leibniz formula (Lean source) and defines the truncated signed moments $\sum_{N=1}^{Y} a_N 2^{-N(k+1)}$ and their Hankel determinants (Lean source, Lean source). The parity lemma is not derived from that determinant identity. A determinant application must first identify a sum to which the parity hypotheses apply.

For the totient series, one would still need suitable configurations at arbitrarily large scales and a denominator-weighted error tending to zero. Nonvanishing alone provides neither. No such family is supplied by these two algebraic statements; compare the separate rank discussion in §10.8. In another base, the analogous argument would require divisibility by a prime factor of that base rather than an unchanged appeal to parity.

10.6 Column rescaling and cancellation

A nonzero determinant need not prevent one row from being constant. For example,

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

The first and last matrices have determinants -2 and 2, respectively; the second row of the resulting matrix is constant. This simple column rescaling matters when a determinant is proposed as evidence for cancellation in the first-harmonic sums.

Proposition 10.4 (A nonzero minor survives inverse-phase column weights)**.** *Let $d \geq 1$ be an integer, let $e_0, \dots, e_{d-1}$ be nonnegative integers, and let $z_0, \dots, z_{d-1}$ be nonzero complex numbers. Write $P_{ij} = z_j^{e_i}$, and suppose $e_{i_0} = 1$ for some i_0. Multiplying column j by $W_j = z_j^{-1}$ gives*

$$(P \operatorname{diag}(W))_{i_0 j} = 1, \qquad \det(P \operatorname{diag}(W)) = \det(P) \prod_{j=0}^{d-1} z_j^{-1}.$$

Thus a nonzero determinant remains nonzero while row i_0 becomes constant. If all $|z_j| = 1$, its absolute value is also unchanged.

The entry formula and the determinant of a product prove the statement. If row-dependent weights are allowed, the choice $W_{ij} = z_j^i / z_j^{e_i}$ gives $W_{ij}P_{ij} = z_j^i$ and makes row zero constant. It produces the ordinary Vandermonde matrix, which is nonsingular when the z_j are pairwise distinct (Lean source, Lean source).

These are statements about complex matrices, not arithmetic properties of the totient sequence. In particular, determinant nonvanishing, even with unit-modulus weights, does not by itself exclude complete alignment in the indicated row. An application of the first-harmonic criterion (Lean source) needs additional information that excludes such weights or controls the resulting weighted sum. The example does not require that information to take the form of an identity coupling different rows, and it does not rule out determinant arguments that use appropriate additional hypotheses.

10.7 Tail integrality on an LCM grid

The comparison. The diagonal condition (Lean source) uses the pair $(H_t, 2H_t)$. Under rationality, the same fractional part occurs at every positive multiple of H_t once t is sufficiently large. A finite set of these multiples therefore gives another possible place to test for a nonintegral difference. The criterion below does not assert a smaller successful depth than the diagonal test.

Eventual equality of fractional parts.

Proposition 10.5 (Tail integrality on an LCM grid)**.** $S \in \mathbb{Q} \implies \exists t_1,\ \forall t \geq t_1,\ \forall q, m : \mathbb{N},\ 0 < q \implies R(q \cdot H(t) + m \cdot H(t)) - R(q \cdot H(t)) \in \mathbb{Z}$.

Rationality makes the fractional parts of R_{kH_t} equal for all $k \geq 1$ at each scale $t \geq t_1$. The corresponding implication, Theorem 6.53, shows that a certificate $C(mH_t, qH_t, L)$ with $q \geq 1$ at arbitrarily large t implies $S \notin \mathbb{Q}$.

(b′) A finite test on the grid.

Proposition 10.6 (A finite-grid certificate gives a nonintegral pair)**.** *Let $Q \subseteq \mathbb{N}_{>0}$ be finite and nonempty. For each $q \in Q$, put $A_q = \sum_{j=1}^{L} \varphi(qH + j)2^{L-j}$ and $B_q = qH + L + 2$. Suppose $B_q < 2^L$ for every $q \in Q$ and*

$$\forall q_i \in Q\ \exists q_j \in Q, \qquad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L.$$

Then $R_{q_jH} - R_{q_iH} \notin \mathbb{Z}$ for some $q_i, q_j \in Q$. The proof and a four-point example are given in Theorem 6.56.

Scope. Eventual integrality on the LCM grid is a necessary consequence of rationality, not a claim about the actual series made without that hypothesis. The finite-grid test uses the individual one-sided bounds $B_q < 2^L$, rather than charging the larger radius to both ends of a pairwise interval. This compares modulus bounds, not binary depths. A finite nonempty set and every stated size inequality are needed; three or more points and pairwise compatibility are not.

Neither statement supplies the unbounded family required to decide #249. Exhibiting a nonintegral tail difference on the LCM grid at arbitrarily large t would refute rationality. The finite-grid test is a way to certify such a difference, not a proof that these tests succeed at arbitrarily large scales.

The implication from rationality to grid flatness (Lean source) should not be confused with an assertion of flatness for the actual series. The exact diagonal integrality reformulation and its quantified converse (Lean source, Lean source) give other equivalent forms of the problem. They are not the only such forms; Proposition 1.8 lists further equivalent certificate conditions.

Consequence. Choosing the least deep tail is the step that turns a common-fractional-part assumption into an intersection of arcs. A related finite comparison concerns first and second differences: at $(h, N) = (1, 8)$ the first-difference test holds at depth 8, while no second-difference certificate exists at depth at most 8 (Lean source). This disproves a universal claim of smaller depth for second differences; it does not show that they can never improve a certificate.

10.8 Exact dyadic rank and the limits of a rank argument

At each level the scalar totient identities supply spanning, while affine independence excludes any further relations. This computes the rank, rather than merely proving its unboundedness. The latter already follows from Coons's nonregularity theorem in every base $k \geq 2$ [2, Theorem 3.2]. Martin's stronger theorem supplies the affine independence[4, Theorem 1]; the proportional arguments n and $2n$ require a separate two-evaluation argument. The all-base proof and integral relation calculation appear in the short paper. We give the dyadic form needed for carry rank here.

Construction.

Theorem 10.7 (Exact dyadic rank and infinite-dimensionality)**.** *For every $e \geq 1$, the family*

$$\{n \mapsto \varphi(n),\ n \mapsto \varphi(2n)\} \cup \{n \mapsto \varphi(2^j n + r) : 1 \leq j \leq e,\ 0 < r < 2^j,\ r \text{ odd}\}$$

is linearly independent over $\mathbb{Q}$. It contains $2^e + 1$ sequences and spans the sections through level e by the zero- and even-residue reductions.

The two zero-residue sections need separate treatment because their arguments n and $2n$ are proportional. Restrict a proposed relation to $n = 2m + 1$, where $\varphi(2n) = \varphi(n)$. The odd-residue sections, together with $\varphi(n)$, now have pairwise nonproportional affine arguments. The evaluation-matrix construction chooses one input per row. At that input a selected primitive form is prime, while each competing form has a prime divisor whose predecessor is divisible by a fixed auxiliary prime. The matrix is diagonal and nonzero on the diagonal modulo that auxiliary prime. The check that the selected prime progression is reduced is part of the construction.

Independence removes the positive-residue coefficients and leaves $A + B = 0$. The same remaining relation at $n = 2$ gives $A + 2B = 0$, so both coefficients vanish. This is the dyadic specialisation of the short paper's proof. The Lean proof is unconditional; it uses the Chinese remainder theorem and Dirichlet's theorem on primes in arithmetic progressions from Mathlib. Consequently the full infinite family spans an infinite-dimensional $\mathbb{Q}$-space (Lean source). A separate elementary observation concerns a proposed integer identity. There are no $Q, v \in \mathbb{N}_{>0}$ and $A, b \in \mathbb{Z}$ with $QvA = b$, $A \neq 0$, and $|b| < Qv$: the identity forces $|b| = Qv|A| \geq Qv$ (Lean source).

The Lean independence theorem is also stated for $e = 0$, but its auxiliary canonical index then still contains the two zero-residue sections $\varphi(n)$ and $\varphi(2n)$. It is therefore not the actual truncation through level zero, which consists only of $\varphi(n)$ and has dimension one.

Integral relations. Rational independence becomes an integral normal form because every omitted section is an *integer* multiple of a retained one. Keep the formal coefficient vectors separate from the sequences: a relation is nonzero in the former group and evaluates to zero in the latter. Fix $e \geq 1$, write $F_{j,r}(n) = \varphi(2^j n + r)$, and let $I_e = \{(j,r) : 0 \leq j \leq e,\ 0 \leq r < 2^j\}$. Retain $F_{0,0}$, $F_{1,0}$, and the odd-residue sections at levels $1, \dots, e$; call this retained index set J_e. The totient identities reduce every omitted section to $F_i = a_i F_{j(i)}$ for a retained section $j(i)$ and an integer $a_i \geq 0$. Thus the retained family generates the lattice $L_e = \sum_{i \in I_e} \mathbb{Z} F_i$. Its rational independence gives integer independence as well: every element of L_e has unique integer coordinates in these $2^e + 1$ retained sections.

To describe *all* integral relations, let E_i denote the formal coordinate vector in $\mathbb{Z}^{I_e}$ and use the evaluation map

$$\mathrm{ev}_e : \mathbb{Z}^{I_e} \longrightarrow L_e, \qquad (c_i) \longmapsto \sum_{i \in I_e} c_i F_i.$$

For each omitted index put $R_i = E_i - a_i E_{j(i)}$. Then

$$\ker(\mathrm{ev}_e) = \bigoplus_{i \in I_e \setminus J_e} \mathbb{Z} R_i, \qquad \operatorname{rank}_{\mathbb{Z}} \ker(\mathrm{ev}_e) = 2^e - 2.$$

For a vector c, subtract $\sum_{i \notin J_e} c_i R_i$. All omitted coordinates disappear, and the retained coordinate at j is $c_j + \sum_{i \notin J_e,\, j(i) = j} a_i c_i$. The vector is a relation exactly when every retained coordinate is zero. In that case its relation coefficients are already known: they are the original omitted coordinates c_i. Uniqueness follows because R_i has coefficient 1 at its own omitted index and 0 at every other one. These unit coefficients explain why no division, and no passage from $\mathbb{Q}$ back to $\mathbb{Z}$, is required. The depth-two base-six example in the short paper illustrates these two-term rows. The distinction between the retained basis and the redundant spanning family of all sections already matters for k-regular sequences that are not eventually zero: their growth exponent is the base-k logarithm of the joint spectral radius of their section matrices in a basis, whereas a redundant spanning family gives only an upper bound (Coons [3, Theorem 1, Proposition 4 and Corollary 7]). The totient is not regular, so that theorem does not apply; the short paper records how the section maps act on the retained bases from one level to the next.

The integral normal form and the full dyadic rational basis are formally checked in Lean. The formal statement gives the same construction for every integer base $k \geq 2$, with relation rank $\sum_{j=1}^{e-1} k^j$. The full dyadic assembly separately gives a basis for the infinite rational span and for its finitely supported rational relations. These finite and infinite statements should not be confused with a claim about infinite sums of relations.

Relation to the series value. Infinite section rank does not imply irrationality of S. The result quantifies the nonregularity already cited above: it gives the rank, the retained sections and their integral relations. Martin's simultaneous inequalities supply

another proof for pairwise nonproportional affine forms with positive slopes [4, Theorem 1]. His theorem does not separate n from $2n$; the two evaluations above do that. The linked Lean proof is independent of Martin's theorem.

The same independence argument also bounds the rank of an integral carry. If $S \in \mathbb{Q}$, choose an integer $v > 0$ with $vS \in \mathbb{Z}$ and put $u_N = vR_N$, where $R_N = \sum_{j\ge1} \varphi(N + j)2^{-j}$. Then $u_N \in \mathbb{Z}$,

$$u_{N+1} = 2u_N - v\varphi(N+1), \qquad 0 \le u_N \le v(N+2), \qquad u_N/2^N \longrightarrow 0.$$

For every $e \ge 0$,

$$\dim_{\mathbb{Q}} \operatorname{span}\{n \mapsto u_{2^j n+r} : 1 \le j \le e,\ 0 \le r < 2^j\} \ \ge\ 2^e - 1.$$

This is the finite family in the linked statement (Lean source); it contains all carry residues at levels $1, \dots, e$, not just odd residues or an unspecified retained subfamily. Indeed, for each odd r with $0 < r < 2^j$, the recurrence gives

$$v\varphi(2^j n + r) = 2u_{2^j n+r-1} - u_{2^j n+r}.$$

These $2^e - 1$ totient sections are independent and belong to the span of the displayed carry sections after multiplication by $v \neq 0$. The rank bound follows. The boundary condition $u_N/2^N \to 0$ identifies the canonical carry, but is not used in this rank-transfer argument.

A contradiction would require an upper bound on the same carry rank. Rationality and the recurrence supply no such bound: Section 12.6 constructs a rational 5/4 series whose unchanged odd totients force exactly this rank transfer. The changed even coefficients break other totient identities. Hence the comparison rules out a rank argument based only on rationality, linear growth and the recurrence, while leaving arguments using those additional identities untouched.

A separate finite-dimensional calculation concerns the matrix $U_N(i,j) = \mu((i+1)/(j+1))$ when $(j+1) \mid (i+1)$ and zero otherwise, for $0 \le i, j < N$. It is lower triangular with diagonal entries one. Hence $\det U_N = 1$ and the map $x \mapsto U_N x$ is injective (Lean source). This proves injectivity of that particular map, not the impossibility of every finite-dimensional argument for irrationality.

The evaluation-matrix argument. For functions $f_1, \dots, f_s$ with values in a field, points $n_1, \dots, n_s$ for which $\det(f_j(n_i)) \neq 0$ prove linear independence: evaluating a relation gives an invertible linear system for its coefficients. This elementary criterion is Proposition 6.24. It does not, by itself, construct the points. Here the totient formula, CRT and Dirichlet's theorem provide them for nonproportional affine forms, followed by the separate zero-residue argument. Another arithmetic function would require its own argument for nonvanishing of the evaluation determinant.

10.9 Further counterexamples and sharp bounds

The table records further counterexamples and sharp bounds. For each proposed deduction it lists the retained hypotheses and the conclusion that fails, together with the source for that argument.

Approach	Argument and source	Scope and remaining question
Complete terminal divisibility pattern	Under $a \geq 8$, $m \leq K$, $J + K + a + 6 < 2 \cdot 2^a$ and $2H + J + K + 2 < 2^m$, the full set of divisibilities in Observation 6.107 is impossible: the last difference would be a positive multiple of 2^m smaller than 2^m. Lean source.	The obstruction is specific to that pattern and those bounds. If the last divisibility is omitted, the additional terminal bound of Proposition 6.86 forces the penultimate difference to be 2^{m-1}. It does not prove that such partial patterns exist. Lean source.
Finite linear combinations of shifted carry recurrences	The explicit sequence in Observation 6.115 has $a_{(q-1)H-1} = \varphi(H)$ for $2 \leq q < t$ and a bounded state satisfying $a_i = 2c_i - c_{i+1}$. Every finite integer combination of its shifts retains a recurrence of this form, including products of differences along multiples of H. Its bounds acquire the factor $\sum_j \lvert\lambda_j\rvert$; they are not uniform in the coefficients of the linear combination. The explicit theorem and ordinary proof are given in Observation 6.115.	The constructed coefficients are not asserted to be actual totient differences. Preserving the displayed recurrence under shifts therefore does not distinguish them from the coefficients required in the proposed argument. The full totient divisor identity Lean source retains information absent from the recurrence alone. The counterexample does not require every future proof to use that particular identity.
Adding three multiplicative scales to an integrality test	For $H \geq 0$ and positive integers p, q, requiring $R_{2kH} - R_{kH} \in \mathbb{Z}$ for every $k \in \{1, p, q, pq\}$ is equivalent to requiring only $R_{2H} - R_H \in \mathbb{Z}$. Neither multiplier need be prime. The affine transport formula along $H \mapsto kH$ carries integrality at the first scale to each of the other three. Lean source.	Adding these three integrality assertions does not strengthen the first one. A different test involving part of the tail would need an estimate for the part it discards; this equivalence provides no such estimate. The result concerns the displayed four conditions, not every test using several scales.

Approach	Argument and source	Scope and remaining question
Prescribing valuations and finite unit residues	For the valuations ν_j and odd residues a_j of Proposition 2.11(iii), the free integers z_j can always be chosen so that $e_{j+1} = 2e_j + 2^{\nu_j}(a_j + 2^u z_j)$ lies in the stated centred interval at every step. LEAN SOURCE.	The specified residues do not exclude all finite carry sequences while the higher quotients z_j remain unrestricted. An argument would need further constraints on those quotients or on the actual coefficients. The lemma does not assert that its freely chosen coefficients are totient differences.
The information retained by an endpoint residue	For an integer sequence a and integer initial values u_0, v_0, let $u_{n+1} = 2u_n - a(n+1)$ and $v_{n+1} = 2v_n - a(n+1)$. Induction gives $u_L - v_L = 2^L(u_0 - v_0)$ for every $L \geq 0$. LEAN SOURCE.	The endpoints agree modulo 2^L, but they are distinct integers whenever $u_0 \neq v_0$. Indeed, if the forcing terms are known, then $$u_0 = \frac{u_L + \sum_{j=1}^{L} 2^{L-j} a(j)}{2^L}.$$ Only reduction modulo 2^L discards the initial value. The identity does not exclude an argument that retains the full endpoint or other information about the sequence.
Bounded finite-state / autonomous successor decoder for the binary carry	For the family in Proposition 2.11(i), with $m \geq 2$, all histories before m agree but the next tail takes $\lfloor (m+1)/2 \rfloor + 1$ distinct values. A common state cannot determine that tail; a set of labels permitting exact recovery must have at least that many elements. LEAN SOURCE.	This counterexample concerns states determined only by the common history of the displayed family. It does not exclude a state using additional information specific to the totient sequence. The bound counts labels, not bits; it is not a complexity theorem about the binary digits of S.

Approach	Argument and source	Scope and remaining question
Zeros on a progression versus nonzero coefficients arbitrarily far out	Let $\chi_4(n)$ be 0 for even n, 1 for $n \equiv 1 \pmod 4$, and -1 for $n \equiv 3 \pmod 4$, and put $c(n) = \sum_{d\mid n} \chi_4(d)$ for $n \geq 1$. Divisor pairing gives $c(n) = 0$ when $n \equiv 3 \pmod 4$. Lean source. Nevertheless $c(2^k) = 1$ for every $k \geq 0$, since 1 is the only odd divisor of 2^k.	This rules out nonvanishing on an arbitrarily prescribed progression, not nonzero coefficients arbitrarily far out. In fact c is multiplicative. Its prime-power values are 1 at powers of 2, $a+1$ at p^a with $p \equiv 1 \pmod 4$, and 1 or 0 at p^a with $p \equiv 3 \pmod 4$ according as a is even or odd. Thus $c(n) \geq 0$ for every $n \geq 1$. These elementary calculations verify both coefficient hypotheses of Lean source: $\sum_{a\geq 1} \chi_4(a)/(b^a - 1)$ is irrational for every integer $b \geq 2$. This application concerns the displayed weighted Lambert series, not the totient series S.
Using a finite exclusion test to prove membership	finite Mersenne-weight exclusion$(x, \text{level}, \text{lookahead})$ is a decidable, finite-depth certificate proving non-membership only (Lean source); a finite computation excludes 3/4 at level 1 (Lean source).	The cited test has a sound exclusion implication. The analogous soundness statement for the residue test of this record, Proposition 6.18, also does not make a finite unsuccessful search a proof of integrality. This is a limit of what follows from those tests, not an impossibility theorem for rationality or membership proofs using additional identities.
Reconstructing one totient value from a two-tail absolute-value linear combination	For any finite $w : \iota \to \mathbb{Q}$, $x : \iota \to \mathbb{N}$ with $\sum w_i \varphi(x_i) = 1$ (exact isolation), the crude two-tail cost $\sum \lvert w_i \rvert (2(x_i + 1) + (x_i + 2)) \geq 3$, using only $\varphi(x) \leq x$. Lean source (see also Lean source).	The displayed termwise absolute-value estimate cannot be less than 1. The same inequality holds for any coefficients $0 \leq c(n) \leq n$ with $\sum_i w_i c(x_i) = 1$. It does not rule out a smaller error bound that uses cancellation between the tails, or a different approximation identity.

Approach	Argument and source	Scope and remaining question
Homogeneous Mersenne-primitive prime factor alone forces a contradiction	For the completely multiplicative sequence $c(n) = n$, if $0 < K < q$ and $q \mid 2^K - 1$, then the scaled-tail difference is the integer K at every starting index, but $q \nmid K$. Lean source.	A factor of $2^K - 1$ does not, from the recurrence alone, have to divide the inhomogeneous tail difference. The example $K = 3$, $q = 7$ already has a primitive prime factor. An application to another series would need a correspondence between its coefficients and this recurrence; Mersenne denominators alone do not provide one.
The unit-gap test at a prime-power denominator	For p prime and $e \geq 1$, if every candidate integer in the interval at denominator p^e is a nonunit modulo p^e, then there is at most one candidate. Equivalently, the notation of Observation 6.114 satisfies $L = 0$, or $L = 1$ with $p \mid J + 1$. Lean source (for the equivalent characterisation, Lean source).	This candidate-count bound does not apply to every odd denominator; the example modulo 15 in Observation 6.114 has two candidates. It also does not bound the denominator cutoff obtainable by changing the window or approximation. It supplies no proof that the fixed exclusion in Section 10.4 extends to arbitrarily large denominators.
Extra divisibility of a three-term totient difference	At $H = 2^{n+2}$, $j = 3 \cdot 2^{n+1}$, $\varphi(3H + j) - 2\varphi(2H + j) + \varphi(H + j) = -2^{n+1}$ for every $n \geq 0$. Lean source.	The exact valuation concerns this scaled family, not the LCM family with $j^2 \leq 2^a$; see Observation 6.108. The independent coefficient calculation says that integer solutions of $c_1 + c_2 + c_3 = c_1 + 2c_2 + 3c_3 = 0$ are exactly the multiples of $(1, -2, 1)$ (Lean source). It classifies these three coefficients; it does not determine the valuation of their totient combination on every restricted family.

10.10 Signed interpolation with entire Dirichlet defects

The small block

$$z^2 - 2z^4 + z^6 = z^2(1 - z^2)^2$$

has coefficient sum $1 - 2 + 1 = 0$ and first moment $2 - 8 + 6 = 0$, but its value at $z = \beta^{-1}$ is positive for $\beta \geq 2$. It can therefore alter an interior value while preserving these two moments. This is a model for one block, not a construction satisfying all the conditions below; translation, spacing and increasing the difference order supply the additional requirements.

This construction assigns different jobs to the shape and amplitude of a correction. High-order finite differences annihilate low-order moments; their integer amplitudes adjust the series value. As the orders increase, later completed blocks contribute zero to each fixed moment, so its defect stabilises between blocks. The prescribed initial collection of moment defects is zero, not merely constant. Spacing the blocks controls their density and analytic convergence. All corrections occur at even indices, leaving every odd totient fixed. The three numbered remarks below have ordinary proofs; none has yet been formalised in Lean.

For a sequence a, write $F_a(z) = \sum_{n\geq 1} a(n)z^n$. For integers $q \geq 1$, $0 \leq c < q$ and $d \geq 0$, put

$$M_{q,c,d}(a;N) = \sum_{\substack{1\leq n<N \\ n\equiv c \pmod q}} n^d a(n).$$

Let $\mu_{a,X,m}$ be the empirical distribution of the length-m blocks starting at the integers $X \leq N < 2X$, with mass $1/X$ per start. Upper Banach density means the limiting upper density over all intervals of a given length, with their starting points unrestricted.

Remark 10.8 (Sparse interpolation around the totient). Fix integers $\beta \geq 2$, $K \geq 1$, $P \geq 2$, and any function $f : \mathbb{N}_{>0} \to (0, \infty)$ with $f(n) \to \infty$. There exist $T > 0$ and target-independent sets $S \subseteq H \subseteq \{n > K\}$, with every element of S even, such that every

$$r \in [F_\varphi(\beta^{-1}) - T, F_\varphi(\beta^{-1}) + T]$$

equals $F_b(\beta^{-1})$ for an integer sequence $b = \varphi + e$ satisfying:

(i) $0 \leq b(n) \leq n$, $|e(n)| \leq f(n)$, and $\operatorname{supp}(e) \subseteq S$. In particular the prefix through K and every odd coefficient are unchanged.

(ii) For every fixed $q \geq 1$, both $e(n)$ and $\sum_{i<n} e(i)$ are divisible by q eventually, with a cutoff independent of r.

(iii) For every fixed (q, c, d), the defect $M_{q,c,d}(b;N) - M_{q,c,d}(\varphi;N)$ equals an integer $C_{q,c,d,r}$ for all sufficiently large $N \notin H$. The cutoff is independent of r. If $q \leq P$ and $d < P$, this defect is zero at every $N \notin H$.

(iv) For each root of unity ζ, the radial function $x \mapsto F_b(x\zeta) - F_\varphi(x\zeta)$ extends smoothly to $x = 1$. Its Taylor coefficients there lie in $\mathbb{Z}[\zeta]$. The coefficients of orders $u < P$ are zero if the order of ζ is at most P.

(v) For every fixed (q, c) the absolutely convergent series

$$D_{q,c,r}(s) = \sum_{\substack{n\geq 1\\ n\equiv c \pmod q}} e(n)n^{-s}, \qquad \operatorname{Re} s > 3/2,$$

extends to an entire function, and $D_{q,c,r}(-d) = C_{q,c,d,r}$.

The set S has upper Banach density zero. The set H has ordinary density zero and upper Banach density one. Define

$$g(n) = \min\left(\inf_{k\geq n} f(k), (n/2)^{1/4}\right), \qquad \ell(X) = 2\left\lfloor \frac{\log_\beta g(\lfloor X/4\rfloor)}{8}\right\rfloor.$$

For large X, $\ell(X) > 0$, and one constant C, independent of r, gives

$$d_{\mathrm{TV}}(\mu_{\varphi,X,m}, \mu_{b,X,m}) \leq \min\left(1, C\left[\frac{m}{\ell(X)} + \ell(X)^{-1/2}\right]\right), \qquad 1 \leq m \leq X.$$

Thus every block length $m = o(\ell(X))$ is preserved asymptotically.

Ordinary proof; not formalised in Lean.

The allowance can grow as slowly as desired and need not be monotone. For $f(n) = (\log(n + 3))^\epsilon$, $\epsilon > 0$, the allowed block scale is $o(\log\log X)$; for $f(n) = (n + 3)^\alpha$, $\alpha > 0$, it is $o(\log X)$. The exact Taylor series, growing moduli, zeros and regular parts of Dirichlet functions are not preserved. The correction has both signs, so the earlier one-sided perturbation constraint is not retained. The construction makes no rationality assertion about $F_\varphi(1/2)$. The two exceptional sets also have different roles. The support S contains the individual altered indices. The larger set H contains the intervals spanned by correction blocks, where a prefix may have included only part of a cancelling block. Its upper Banach density is one because it contains arbitrarily long intervals, even though its ordinary density is zero. Thus the moment conclusion outside H does not assert exact cancellation at every sufficiently large cutoff.

Proof. We construct the blocks first, independently of the target value. Their increasing order enforces the moment and boundary conditions; only their amplitudes will be chosen afterwards to obtain that value. All blocks lie beyond the prescribed prefix.

The tail infimum in g is positive, nondecreasing and tends to infinity. The cap gives $g(n) \leq (n/2)^{1/4}$, while $g(n) \leq f(k)$ for every $k \geq n$. Choose a large even $n_{-1} > K$ so that $g \geq 1$ thereafter and the first spacing below is at least $\max(16, (2PP!)^2)$. Define

$$\begin{aligned} s_j &= 2\left\lfloor \tfrac{1}{8}\log_\beta g(n_{j-1})\right\rfloor, & n_j &= n_{j-1} + s_j,\\ k_j &= \max\{k \geq 1 : (2kk!)^2 \leq s_j\}, & M_j &= k_j!, \quad t_j = 2k_j!, \quad W_j = k_j t_j,\\ A_j &= \beta^{s_j+2}. \end{aligned}$$

These parameters serve two different constraints. The inequality $W_j \leq \sqrt{s_j} < s_{j+1}$ keeps one block clear of the next. The larger digit range $|d_j| \leq A_j$ will make the possible values overlap, but its coefficient cost must still fit within g. The spacings and orders are

nondecreasing and tend to infinity, with $s_j = O(\log n_{j-1})$. For digits $d_j \in \{-A_j, \dots, A_j\}$, form

$$P_j(z) = d_j M_j z^{n_j}(1 - z^{t_j})^{k_j}, \qquad E(z) = \sum_{j \geq 0} P_j(z).$$

The factor $(1 - z^{t_j})^{k_j}$ supplies cancellation of moments; M_j supplies divisibility; and the even n_j, t_j leave every odd coefficient unchanged. The digit d_j remains free to adjust the value at $z = 1/\beta$. Let S consist of all $n_j + lt_j$, $0 \leq l \leq k_j$, and let H be the union of their full integer hulls $[n_j, n_j + W_j]$. These hulls are disjoint and all points of S are even.

Bounds for the coefficients. If $s \geq 16$ and $2kk! \leq \sqrt{s}$, then $4k!2^k \leq 2\sqrt{s}\,2^{\sqrt{s}/2} \leq 2^s$. It follows for $\beta \geq 2$ that $\beta^2 k! 2^k \leq \beta^s$. Therefore

$$\|P_j\|_1 \leq M_j \beta^{s_j+2} 2^{k_j} \leq \beta^{2s_j} \leq \sqrt{g(n_{j-1})}. \tag{1}$$

At every modified index n this is at most $f(n)$ and at most $(n_{j-1}/2)^{1/8} \leq \sqrt{n/2}$. Both signs of the correction are permitted at even indices, because the product formula gives

$$\frac{\varphi(n)^2}{n} = \prod_{p^a \| n} p^{a-2}(p-1)^2 \geq \frac{1}{2},$$

because all odd-prime factors are at least one and the smallest possible factor at two is $1/2$. Also $\varphi(n) \leq n/2$ for even n. Thus $b = \varphi + e$, where e is the coefficient sequence of E, stays between zero and n. The coefficient bound ensures absolute convergence of E inside the unit disk.

An interval of values. At $z = 1/\beta$, the contribution of block j is $d_j w_j$, where $a_j = (1 - \beta^{-t_j})^{k_j}$ and $w_j = M_j \beta^{-n_j} a_j$. We must check that the later blocks can fill the gaps between successive choices of d_j. Bernoulli's inequality gives $3/4 \leq a_j \leq 1$. Since $M_{j+1} \geq M_j$,

$$A_{j+1} w_{j+1} = \beta^2 M_{j+1} \beta^{-n_j} a_{j+1} \geq 3 M_j \beta^{-n_j} \geq 3 w_j.$$

The sum $T = \sum_j A_j w_j$ is positive and finite, by (1). After choosing d_j, the remaining digits have total allowance $T_j = \sum_{i>j} A_i w_i$. Consecutive choices of d_j change the current contribution by w_j, while $T_j \geq A_{j+1} w_{j+1} \geq 3w_j$. Hence the intervals $[dw_j - T_j, dw_j + T_j]$, $-A_j \leq d \leq A_j$, overlap and cover $[-A_j w_j - T_j, A_j w_j + T_j]$. Successively choose a digit whose interval contains the remaining target. The residual tends to zero with T_j, so the digits represent every value in $[-T, T]$. This is the finite-choice interval principle of Crmarić and Kovač [36, Lemma 4(a), Remark 5], here verified for signed finite differences. It also shows T can be made arbitrarily small, while positive, by increasing the initial cutoff.

Congruences and moments. For every polynomial p of degree less than k,

$$\sum_{l=0}^{k} (-1)^l \binom{k}{l} p(n + lt) = 0. \tag{2}$$

This follows by iterating the forward-difference operator, which lowers polynomial degree. Eventually $q \mid M_j$, so every later coefficient and every partial block sum vanish

modulo q. Complete blocks have mass zero, giving the cumulative assertion. Eventually also $q \mid t_j$ and $k_j > d$. Each such block lies in one residue class modulo q and its moment of degree d is zero by (2). The restriction on the cutoff is needed here: outside H it cannot split a block, so the moment sum contains only complete blocks. All sufficiently late complete blocks cancel, leaving just the integer contribution of the finitely many early blocks. If $q \le P$ and $d < P$, there are no exceptional early blocks. These cutoffs depend on the block parameters, not on the later choice of digits.

Sparsity and blocks. Within the jth block the gaps in S are $t_j \to \infty$; between blocks they are at least $s_{j+1} - \sqrt{s_{j+1}} \to \infty$. Uniform spacing after a finite prefix implies upper Banach density zero. To estimate the hulls without assuming regular variation of f, take any hull meeting $[X, 2X + m)$, $m \le X$. Its center lies in $[X/2, 3X]$ eventually and its preceding center is at least $X/4$, since its width and spacing are $O(\log X)$. Thus $s_j \ge \ell(X)$ for every such block. There are $O(X/\ell(X) + 1)$ of them, and

$$W_j + 1 \le 2\sqrt{s_j} \le \frac{2s_{j+1}}{\sqrt{\ell(X)}}.$$

The following gaps telescope to $O(X)$ over these consecutive centers. Consequently the total hull lengths are $O(X/\sqrt{\ell(X)})$. A hull can meet length-m blocks at at most $m + W_j + 1$ starts. Coupling the two empirical laws at the same start proves the stated total-variation bound; the extra m/X is absorbed by $m/\ell(X)$. The hull estimate at $m = 1$, followed by dyadic summation, gives ordinary density zero for H. But $W_j \to \infty$, so H contains arbitrarily long solid intervals and its upper Banach density is one.

Radial smoothness. The number of differences increases with the block index, so any fixed derivative order is eventually controlled. Fix a root ζ of order q, and integers $u, A \ge 0$. Choose $B \ge u + A + 2$. For all sufficiently late blocks, $q \mid t_j$ and $k_j \ge B$. Factor

$$P_j(x\zeta) = (1 - x)^B R_j(x), \qquad R_j(x) = d_j M_j \zeta^{n_j} x^{n_j} (1 + x + \cdots + x^{t_j - 1})^B (1 - x^{t_j})^{k_j - B}.$$

Its coefficient norm is at most $\sqrt{g(n_{j-1})}\, t_j^B = O_B(n_j^{1/2})$, because $g(n) \le (n/2)^{1/4}$ and $t_j = O(\sqrt{\log n_j})$. All exponents lie between n_j and $2n_j$ eventually. For $x \in [1/2, 1)$,

$$\sum_j |R_j^{(v)}(x)| \le C_{B,v} \sum_{n \ge 1} n^{v+1/2} x^n \le C'_{B,v} (1 - x)^{-v-3/2}.$$

The series and derivatives converge normally on compact subintervals. After differentiating $(1 - x)^B \sum_j R_j(x)$ u times, every term is $O((1 - x)^{B-u-3/2}) = O((1 - x)^{A+1/2})$. Thus, after subtracting a finite initial polynomial, all prescribed derivative orders vanish to any prescribed order at the boundary. Integration of the continuous derivative limits gives a single smooth extension. For an integer polynomial $\sum c_n z^n$, its radial Taylor coefficient of order u at ζ is $\sum c_n \binom{n}{u} \zeta^n \in \mathbb{Z}[\zeta]$. Every such coefficient stabilizes after finitely many blocks. The first P vanish at roots of order at most P.

Entire Dirichlet continuation. Here we apply the same finite differences to y^{-s} rather than to a polynomial. Fix (q, c) and a compact set of complex s, with $\operatorname{Re} s \ge -R$. Choose an integer $B > R + 3/2$. All sufficiently late blocks align modulo q and have $k_j \ge B$. The derivative of y^{-s} of order B is $(-1)^B (s)(s + 1) \cdots (s + B - 1) y^{-s-B}$. Repeated integration

expresses its Bth finite difference as an integral over $[0,t_j]^B$. The remaining differences contribute at most 2^{k_j-B}, so the absolute Dirichlet contribution of one polynomial is at most

$$C|d_j|M_j 2^{k_j-B} t_j^B n_j^{R-B} \le C' n_j^{R-B+1/2}.$$

This is summable over the distinct positive centers. The Dirichlet polynomials of the blocks therefore converge normally on every compact set and define an entire function. In $\operatorname{Re} s > 3/2$ they agree with the absolutely convergent ungrouped series. At $s = -d$, all late blocks vanish by their moment identity, leaving exactly $C_{q,c,d,r}$. These are analytic-continuation values; the original ungrouped series need not converge at negative integers. □

Remark 10.9 (Observations that do not determine irrationality). The following data, taken together, do not determine whether $F_b(\beta^{-1})$ is rational: the fixed prefix and odd coefficients; every fixed eventual coefficient and cumulative congruence; the block laws on scales $o(\ell(X))$; fixed progression moments modulo eventual integer constants outside H; progression Dirichlet functions modulo entire functions; and radial germs at all roots of unity modulo smooth functions with Taylor coefficients in $\mathbb{Z}[\zeta]$. This remains true with exact equality of the first P moment degrees and first P radial Taylor coefficients at moduli and root orders at most P.

Ordinary proof; not formalised in Lean.

Proof. Every constructed sequence has the specified data of φ, including the common exceptional sets, while the nondegenerate interval of attained values contains both rational and irrational numbers. □

Taking finite linear combinations of the Dirichlet series on arithmetic progressions shows that every fixed periodic twist changes by an entire function. An entire summand has no negative Laurent coefficients: it therefore preserves every pole and its principal part wherever the original series has meromorphic continuation. The regular part and zeros need not be preserved. In the untwisted case the classical identity gives

$$\sum_{n\ge 1} b(n) n^{-s} = \frac{\zeta(s-1)}{\zeta(s)} + D_{1,0,r}(s), \qquad \operatorname{Re} s > 2.$$

The version used by Coons [37, Theorem 3.3] supplies this function identity, not an irrationality theorem for one value of F_φ. Likewise, Habiro's injectivity of the exact Taylor map for a cyclotomic completion [38, Theorem 5.2] concerns a different, stronger datum. The full sequence of Taylor coefficients is not fixed here: the second Taylor coefficient of $Mz^n(1-z^t)^2$ at one is Mt^2.

The change of sign is indispensable for this construction. An infinitely supported nonnegative integer correction has unbounded Abel limit at one. The Dirichlet series of the correction also cannot extend to an entire function if its defining series converges in a right half-plane. Indeed, choose a positive real σ inside that half-plane. Termwise differentiation gives $(-1)^k D^{(k)}(\sigma) = \sum_n e(n) n^{-\sigma} (\log n)^k \ge 0$. The entire Taylor series at zero and Tonelli's theorem then give

$$D(0) = \sum_{k\ge 0} \frac{\sigma^k}{k!} \sum_n e(n) n^{-\sigma} (\log n)^k = \sum_n e(n) < \infty.$$

Integer nonnegativity forces finite support, and a finite correction at an integer reciprocal base cannot change rationality. Signed differences are what allow the stronger observation class above.

Remark 10.10 (The first-harmonic test on the rational members). Let $b(n)$ be integers with $0 \le b(n) \le n$ and $F_b(1/2) = u/(2^c v)$, where $c \ge 0$ and v is odd and positive. Choose $h \ge 1$ with $v \mid 2^h - 1$. If $X \ge \max(c, 1)$ and $16(2X + h + L + 2) \le 2^L$, then for every integer $N \in [X, 2X)$,

$$\cos\left(2\pi\, 2^{-L} \sum_{j=0}^{L-1} [b(N + h + 1 + j) - b(N + 1 + j)] 2^{L-1-j} \right) \ge \cos(\pi/8) > \frac{9}{10}.$$

Ordinary proof; not formalised in Lean.

Proof. The actual tail $R_N = \sum_{j\ge 1} b(N + j)2^{-j}$ lies in $[0, N + 2]$ and satisfies $R_N \equiv 2^N F_b(1/2) \pmod{\mathbb{Z}}$. Thus $R_{N+h} - R_N$ is an integer for $N \ge c$. If D denotes the integer window sum in the statement, direct truncation gives $2^L(R_{N+h} - R_N) - D = R_{N+h+L} - R_{N+L}$. The right side has absolute value at most $N + h + L + 2$, so $D/2^L$ is within $1/16$ of an integer. The cosine bound follows; its strict comparison with $9/10$ follows from $\cos^2(\pi/8) = (2 + \sqrt{2})/4 > 17/20 > 81/100$. □

This applies to every nonempty subset of the block, including the *actual totient-selected* set $\mathcal{G}$ of Theorem 3.5(b), at $s = 26$, $\eta = 1/1000$ and its minimal admissible depth $L(X)$. This set is held fixed rather than recomputed from b. Wherever it has more than $67X/100$ elements, the real first-harmonic sum exceeds $603X/1000$. The interpolation interval has an explicit dyadic rational member: for sufficiently large J, the finite prefix $\sum_{n\le J} \varphi(n)2^{-n}$ lies inside it, since the omitted tail is at most $(J + 2)2^{-J}$. For this member one may take $h = 1$. Consequently the preserved observations cannot imply the good-base saving for all admissible sequences. No counterexample to the saving for the actual totient is asserted; additional structure, such as its exact multiplicative relations at the relevant growing scales, is still required.

11 Hypotheses of the conditional implications

Uniformity in a parameter does not remove a hypothesis. An implication can hold at every scale although the required arithmetic input is known at only finitely many scales, or at none. We examine that distinction for the displayed implications, without drawing conclusions about all arguments that use the same objects.

Implications concerning the totient series

Cancellation versus a certificate. The first-harmonic implication converts the stated cosine bound on one block into a finite residue certificate (Lean source). To prove irrationality, the bound must hold with the quantifiers in Section 3: for every shift and

beyond every block threshold. A finite identity for the sum, or a computation at one block, does not establish this assertion.

Sign versus distance from an integer. The positivity theorem for the LCM tail difference (Lean source) identifies the relevant endpoint under the stated size and modulus bounds. Nonintegrality follows only after excluding the corresponding residue interval (Lean source). Likewise, the terminal-dominance condition is a sufficient hypothesis, not a construction of the required parameters (Lean source). Under integrality its inequality contradicts positivity of the terminal carry; the contradiction establishes sufficiency, not the inequality itself.

An error bound versus a separated approximation. The finite-tail approximation has a uniform error radius (Lean source). The cited exact check supplies nonintegrality at the two exponents $a = 4, 6$ (Lean source); it is not an assertion that no other exponent works. To obtain irrationality by this criterion, separation must occur at arbitrarily large LCM scales. In the nondivisor approximation of Section 6.6.8, the error bound is also proved when $D \geq 2H$. Its remaining requirement is finite rational separation, not an additional convergence hypothesis.

Denominator growth versus approximation quality. The lower bound for the product of the upper-half Mersenne factors (Lean source) concerns a denominator expression. An irrationality argument needs a comparison with the error of the corresponding approximation. Growth of that denominator expression alone does not give either a successful residue test or a nonintegral totient tail.

A rank lower bound versus a contradiction. The coefficient independence and hypothetical-carry rank bounds hold uniformly in the level (Lean source, Lean source). A contradiction would require an upper bound for the actual carry that is strictly smaller than the lower bound at some level. The rational 5/4 comparison disproves the proposed generic upper bound. The separate integer boundary conditions in the cited adjugate calculation are inconsistent (Lean source), but that calculation does not exclude matrix arguments with different boundary conditions.

Related source comparisons for Problem #257

The following comparisons concern Problem #257 and are included with the hypotheses of their cited sources. No statement in this subsection is an input to a proof about S.

The comparison of the middle-carry estimate (Lean source) with the remainder estimate (Lean source) distinguishes the direction of an inequality from control of the required deviation. No unconditional proof of the associated sign claim is supplied, and a sign alone would not give a lower bound for the magnitude.

The row-capacity and later-skip implications retain their respective local deficit and skip hypotheses (Lean source, Lean source). The finite-shadow comparison distinguished an existential horizon, obtained from convergence, from the particular horizon $J = k - 2$ needed in that application (Lean source). The terminal-strip formulation also imposes its own support and carry restrictions (Lean source); a bound on carry size does not automatically replace a restriction on the positions of the nonzero coefficients.

For finite sums of $1/(2^n-1)$, the common denominator is odd. Their reduced denominators are therefore odd, so they cannot equal a rational number with even reduced denominator. This is the elementary reason behind the finite-support exclusion and its denominator source (Lean source, Lean source). The cited infinite-skipping theorem is stated for the target $1/2$ (Lean source). A proposed extension to other targets requires checking that theorem's hypotheses and proof; the finite-denominator observation alone does not prove such an extension or the needed infinite-orbit estimate.

Finally, the support-fraction statement and the more general collision bound have distinct hypotheses (Lean source, Lean source). A broader use would need an additional mean-residue estimate, which is not supplied, and the endpoint proofs of the dichotomy in the linked Lean source have not been checked for a changed target (Lean source). No such extension is used here.

Quantifiers must be checked independently. For example, a statement of the form $\exists N\, \exists L\, \forall h$ does not provide the required $\forall h\, \forall N_0\, \exists N \geq N_0\, \exists L$. The first assertion controls many shifts at one location; the second requires arbitrarily late locations for each fixed shift. Relabelling a source statement does not bridge this difference.

12 Additional mathematics required by the sufficient conditions

We turn from the proved reductions to the estimates needed to apply them. The doubling identity relates window phases to exact tails; the subsequent decompositions separate the weighted sums for which cancellation is required. The suggested analytic applications remain proposals where those estimates are missing. Failure of one application does not exclude the analytic method.

The section contains a doubling-orbit formulation with explicit thresholds, a lacunary comparison sequence for the strength of a block estimate, and sufficient estimates for the prime-factor decomposition. The discussion of a rank upper bound must also be read with the rational $5/4$ comparison sequence: rationality alone does not give bounded rank. These are the scope distinctions needed to interpret the individual statements, not priority claims.

The proved reductions identify explicit sufficient hypotheses. Those hypotheses remain to be established; the reductions do not prove them.

12.1 The doubling identity

Write $S = \sum_{n\geq 1} \varphi(n)/2^n$, $R_N = \sum_{m\geq 1} \varphi(N+m)/2^m$ for the local tail (Lean source), and $\Phi_N = \sum_{n\leq N} \varphi(n)2^{N-n} \in \mathbb{Z}$ for the integer prefix, so that $2^N S = \Phi_N + R_N$. For $h \geq 1$ set

$$\alpha_h \;:=\; (2^h - 1)\, S.$$

Lemma 12.1 (The doubling identity)**.** *For all $N \geq 0$ and $h \geq 1$,*

$$R_{N+1} = 2R_N - \varphi(N+1), \qquad R_{N+h} - R_N \;=\; 2^N \alpha_h - (\Phi_{N+h} - \Phi_N),$$

so $R_{N+h} - R_N \equiv 2^N \alpha_h \pmod 1$, *and consequently the exact first character of the tail difference is the* $\times 2$ *orbit of one real number:*

$$e(R_{N+h} - R_N) = e(2^N \alpha_h), \qquad e(x) := \exp(2\pi i x).$$

Hence, for fixed h, *the phases* $\{ R_{N+h} - R_N \bmod 1 \}_{N \geq 0}$ *are the forward orbit of* $\alpha_h \bmod 1$ *under* $x \mapsto 2x$.

The recurrence $R_{N+1} = 2R_N - \varphi(N+1)$ is one line from the definition and is used below only for intuition.

These identities express the phases as a lacunary exponential sum in one real variable. The next proposition transfers an estimate for that orbit to the truncated phases, accounting for the truncation error. The identities themselves are already in the cited sources.

Proposition 12.2 (Transfer to the doubling orbit)**.** *Suppose that*

$$\forall h \geq 1 \ \forall X_0 \ \exists X \geq \max(X_0, 1) : \quad \sum_{N=X}^{2X-1} \cos(2\pi \, 2^N \alpha_h) \leq \tfrac{89}{100} X.$$

This hypothesis implies that S *is irrational.*

Proof. Fix $h \geq 1$ and X_0, and take X as supplied. Choose $L \geq h$ with $2^L \geq 1024\,(2X + h + L + 2)$; such L exists because $2^L/L \to \infty$, and it satisfies the size inequality $16(2X + h + L + 2) \leq 2^L$ a fortiori. For every $N < 2X$ the proved truncation bound (Lean source) gives $\| E(h, N, L) - e(2^N \alpha_h) \| < 2\pi(N + L + h + 2)/2^L \leq 2\pi/1024 < 1/100$, so, by the corresponding cosine comparison (Lean source), $\operatorname{Re} E(h, N, L) \leq \cos(2\pi 2^N \alpha_h) + 1/100$. Summing over the X values $N \in [X, 2X)$ gives $\sum_N \operatorname{Re} E(h, N, L) \leq \frac{89}{100} X + \frac{1}{100} X = \frac{9}{10} X$, so Theorem 9.4 produces $N \in [X, 2X)$ with $C(h, N, L)$, and $N \geq X \geq X_0$. This is exactly the quantified certificate condition of Theorem 6.47. □

The proof above makes the truncation margin explicit. The full implication is also proved in Lean (Lean source). It uses a real-part estimate for the exact phases and spends $1/100$ on truncation. Compare the sufficient norm condition of Theorem 6.59. This is a separately sufficient condition. It is not asserted to be equivalent to the $21/25$ norm bound or to the four separate estimates stated above. The constants account for different quantities and different error margins.

Corollary 12.3 (A digit version of the analytic condition)**.** *Let* $\rho_h(X)$ *denote the proportion of* $N \in [X, 2X)$ *with* $\| 2^N \alpha_h \|_{\mathbb{R}/\mathbb{Z}} \geq 1/4$. *If for every* $h \geq 1$ *there are cofinally many* X *with* $\rho_h(X) \geq 11/100$, *then* S *is irrational. For nondyadic* α_h, *the condition counted by* $\rho_h(X)$ *is exactly a change between binary digits* $N + 1$ *and* $N + 2$. *Thus the same sufficient hypothesis asks for at least* $11X/100$ *such changes, counted with* $X \leq N < 2X$, *on arbitrarily large blocks for every* h.

Proof. The inequality $\|x\|_{\mathbb{R}/\mathbb{Z}} \geq 1/4$ gives $\cos(2\pi x) \leq 0$. Every remaining term is at most 1, so

$$\sum_{N=X}^{2X-1} \cos(2\pi 2^N \alpha_h) \le (1-\rho_h(X))X \le \frac{89}{100}X.$$

Proposition 12.2 now applies. For the digit reading, for a nondyadic α_h the fractional part of $2^N\alpha_h$ is in $[1/4, 3/4]$ exactly when those two binary digits differ. The nondyadic qualification excludes the endpoint ambiguity at $3/4$. □

The exact hypothesis is the digit-change inequality in the corollary. It is not a claim about an average run length: the two boundary runs of a finite block need not be complete, and their lengths are not specified by the number of interior digit changes. The finite observations below satisfy the displayed bound at their sampled parameters. They do not establish it on arbitrarily large blocks.

h	X	$\rho_h(X)$	$X^{-1}\sum_{N\in[X,2X)}\cos(2\pi 2^N\alpha_h)$	required
1	256	0.4766	0.0037	≤ 0.89
1	2048	0.5073	−0.0043	≤ 0.89
3	256	0.4648	0.0296	≤ 0.89
3	2048	0.5020	0.0027	≤ 0.89
8	2048	0.5112	−0.0088	≤ 0.89
13	2048	0.5049	−0.0195	≤ 0.89

Table 6: Finite block statistics for $\alpha_h = (2^h-1)S$. The exact digit-change counts in the six rows are 122, 1039, 119, 1028, 1047 and 1034; the proportions and cosine averages are rounded to four decimal places. The interval calculation described below certifies these counts and displayed roundings.

For the numerical enclosure, compute $P = \sum_{n=1}^{6080} \varphi(n) 2^{6080-n}$. The elementary tail bound gives

$$P \le 2^{6080} S \le P + 6082.$$

The endpoints have the same quotient on division by 2^{80}, which certifies $\lfloor 2^{6000} S \rfloor$; an error bound smaller than one would not alone certify a floor near an integer boundary. Multiplying this interval by $(2^h-1)2^{N-6080}$ encloses each phase. None of the resulting intervals crosses a digit-test boundary, and interval evaluation of the cosine certifies the rounded averages. Among the first 5800 fractional digits, the numbers of runs for $h = 1, 3, 8$ are 2864, 2964, 2927, and the longest runs have lengths 13, 12, 14. These count the two boundary runs as observed, whether or not they continue outside the window. No randomness assumption or asymptotic conclusion is part of these finite calculations.

Observation 12.4. Lemma 12.1 expresses each tail phase in terms of the doubling orbit of $\alpha_h = (2^h-1)S$. The block conditions average those phases, the prime condition samples prescribed prime-indexed positions, and the full-depth condition requires a finite residue certificate for each (d, N) at some multiple of d. Corollary 12.3 gives one sufficient digit-change test for the real-part estimate; it is not an equivalence with the norm bound or with the four separate budgets. Likewise, a bound on a binary run length is not the exact prime-gap condition. Section 12.7 compares the original inequalities.

12.2 The full-block exponential-sum estimate

The required inequality.

$$\forall h \geq 1\ \forall X_0\ \exists X, L: \quad \max(X_0, 1) \leq X, \quad 16(2X + h + L + 2) \leq 2^L, \quad \Big\| \sum_{N=X}^{2X-1} e(D(h,N,L)/2^L) \Big\| \leq \tfrac{21}{25} X,$$

where $D(h,N,L) = \sum_{j<L} (\varphi(N+h+1+j) - \varphi(N+1+j)) 2^{L-1-j}$. This inequality is unproved; it implies irrationality (Lean source, Lean source). By Proposition 12.2 the real-part bound with constant 89/100 for the *exact doubling phases* suffices; its transfer to the truncated phases uses the separate error estimate in that proposition.

The discrepancy in terms of the doubling map. For $L \geq h$, dividing by 2^L and reindexing the two ranges gives

$$\frac{D(h,N,L)}{2^L} = \underbrace{-\sum_{t=1}^{h} \frac{\varphi(N+t)}{2^t}}_{\text{head}} + \underbrace{(2^h - 1) \sum_{t=h+1}^{L} \frac{\varphi(N+t)}{2^t}}_{\text{body}} + \underbrace{\sum_{t=L+1}^{L+h} \frac{\varphi(N+t)}{2^{t-h}}}_{\text{tail}}. \tag{3}$$

The phase is a geometrically weighted sum of totients at $L+h$ consecutive arguments. In the first two ranges, $1 \leq t \leq L$, the coefficient has denominator 2^t and odd numerator. Its contribution is therefore integral exactly when $v_2(\varphi(N+t)) \geq t$. In the last range, $L < t \leq L+h$, the denominator is instead 2^{t-h}, so the corresponding threshold is $t-h$. Here $v_2(a)$ denotes the exponent of 2 in the positive integer a. The selected offset in the four-term decomposition lies in the middle range, where the odd multiplier is $2^h - 1$.

Relevant comparison estimates. The following comparisons identify the estimates still needed; they do not rule out these methods.

Weyl differencing and van der Corput. The van der Corput inequality applies to arbitrary bounded complex sequences [23, Theorem 2.1], but its use here requires estimates for shifted correlations. For the exact phases $x_N = 2^N \alpha_h$ of Lemma 12.1,

$$x_{N+q} - x_N = (2^q - 1) x_N.$$

Thus differencing gives another doubling orbit, not a polynomial phase of lower degree. The finite phase $D(h,N,L)/2^L$ approximates this orbit modulo 1; it is not the exact orbit itself. No bound for the resulting correlations is established here.

The large sieve. A proposed application would need a suitable family of separated frequencies and a way to recover a bound for each fixed h. Averaging only over h would not suffice: the condition asks for a block estimate for every h, not for most of them.

Bilinear sums over prime factors. Factoring $N+t = mp$ and using $\varphi(mp) = \varphi(m)(p-1)$ makes the selected totient contribution linear in the prime p. Section 12.3 gives this decomposition and identifies the remaining weight. A bound for that weight is still needed before a Vaughan- or Vinogradov-type argument can be applied.

Erdős's 1948 digit method. The comparison series is $E = \sum_{n\geq 1} 1/(2^n - 1) = \sum_{N\geq 1} d(N)/2^N$, whose irrationality Erdős proved [11]. For the totient series, the corresponding Möbius expansion is exact: $\varphi = \mu * \mathrm{id}$ gives

$$S = \sum_{d \geq 1} \mu(d) \frac{2^d}{(2^d - 1)^2} \tag{4}$$

The identity follows by Möbius inversion; a numerical computation gives $S = 1.3676308019850223\ldots$ to sixteen digits.

Erdős combines congruences forcing increasing powers of 2 to divide a block of divisor counts with an averaged tail bound, producing long zero blocks in the binary expansion [11, pp. 63–66]. To adapt that argument here, one would need congruences for the actual totient coefficients together with a compatible tail estimate. The identities alone provide neither combination. In (4), the coefficients $\mu(d)$ can have either sign, so positivity of the original totient series does not give positivity term by term in this expansion. This is a limitation of the proposed direct transfer, not a proof that a digit argument or a different use of the Möbius expansion cannot work.

The unresolved estimate. For general coefficients $0 \le c(n) \le n$, irrationality alone does not imply the block norm condition. The following example has very sparse nonzero coefficients. It is irrational, but almost all phases in each sufficiently large block lie close to 1.

Theorem 12.5 (The block norm condition is stronger in the class $0 \le c(n) \le n$)**.** *Let $c(n) = 1$ if $n = k!$ for some $k \ge 1$ and $c(n) = 0$ otherwise, so $0 \le c(n) \le n$ for all $n \ge 1$, and let $\beta = \sum_{n\ge1} c(n)/2^n = \sum_{k\ge1} 2^{-k!}$. Then β is irrational, and for every $h \ge 1$ and every $X \ge 81(h+5)$,*

$$\sum_{N=X}^{2X-1} \cos(2\pi\, 2^N (2^h - 1)\beta) \;>\; \tfrac{9}{10}X.$$

Consequently the block norm condition fails for β at every sufficiently large scale, for every admissible truncation depth, even though β is irrational. Thus irrationality, nonnegative integer coefficients and the bound $c(n) \le n$ do not imply the block gap. A proof for S needs an additional property not shared by this example; the comparison does not specify which additional property will suffice.

Proof. The binary expansion of β has infinitely many ones separated by unbounded strings of zeros. It is not eventually periodic, so β is irrational. Write $\gamma = (2^h - 1)\beta$, and let $k!$ be the least factorial greater than N. The earlier terms become integers after multiplication by 2^N, so

$$2^N\gamma \equiv (2^h - 1)\sum_{j\ge k} 2^{N-j!} \pmod{\mathbb{Z}}.$$

If $N \le k! - h - 5$, the expression on the right lies between 0 and $2 \cdot 2^{N+h-k!} \le 1/16$. Thus $\|2^N\gamma\|_{\mathbb{R}/\mathbb{Z}} \le 1/16$ and $\cos(2\pi 2^N\gamma) \ge \cos(\pi/8) > 9238/10000$. The remaining N lie in $(k! - h - 5,\ k!]$, at most $h + 5$ values per factorial. The relevant factorials have ratio at least 3, and $h + 5 < X$, so $[X, 2X)$ meets at most one such interval. Hence the sum is greater than

$$\frac{9238}{10000}(X - h - 5) - (h + 5) \ge \frac{9}{10}X + \frac{h+5}{250}$$

when $X \ge 81(h + 5)$.

To pass from these exact phases to the finite residue test, note that $0 \le c(n) \le 1$ puts each scaled tail in $[0,1]$. The error in the truncated tail difference is therefore at most 2^{-L}, so each complex phase changes by at most $2\pi 2^{-L}$. At an admissible depth, $2^L \ge 16(2X+h+L+2) > 32X$, and the total change in the block sum is less than $\pi/16$. Its real part is consequently greater than $9X/10 - \pi/16 > 21X/25$ for $X \ge 81(h+5)$. Its norm cannot satisfy the $21X/25$ bound either. □

This example rules out the implication from irrationality and the stated coefficient bound to the block norm condition. It does not say that the totient sum is Liouville-like, or that every argument based on its digits fails. A proof of the block estimate must use information that distinguishes the actual totient phases from this comparison sequence.

The size of the required saving. Under Lebesgue measure on the phase circle, the binary digit maps are independent fair bits. The digit-change proportion therefore tends to $1/2$ for almost every phase. For almost every phase, the real-part block sum is also $O(\sqrt{X \log\log X})$, by the law of the iterated logarithm of Erdős and Gál for lacunary series [29, p. 65]. The block bound follows by subtracting two initial partial sums. An almost-everywhere theorem supplies no conclusion at the particular phase $\alpha_h = (2^h - 1)S$. The requirement is $\rho_h \ge 0.11$ and a saving of a constant factor. The finite values in Table 6 have ρ_h near 0.50 and small block averages. They are compatible with square-root cancellation, but do not establish its asymptotic order. The sufficient condition asks only for the stated fixed saving on arbitrarily large blocks. No estimate in this record supplies it for the totient phases.

12.3 The four-term decomposition

The simultaneous inequalities. Use the four finite sums defined in Section 3.2; in particular, each group mean averages the selected prime phase, not the residual weight. For every $h \ge 1$ there must exist $s \ge 1$ and $\eta \in (0,1)$ such that for every X_0 there are $X \ge \max(X_0, 1)$ and L with $h \le L - s$, $16(2X+h+L+2) \le 2^L$, and all four of

$$\operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) \le \tfrac{14}{25} X,$$

$$\left\| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right\| \le \tfrac{1}{100} X,$$

$$\left\| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E(h,N,L) \right\| \le \tfrac{1}{100} X,$$

$$\left\| \sum_{\substack{X \le N < 2X \\ N \notin \mathcal{A}}} E(h,N,L) \right\| \le \tfrac{8}{25} X.$$

Their simultaneous validity remains unproved; together they imply irrationality (Lean source, Lean source). The decomposition is an exact identity (Lean source). The last two sums are bounded by counting indices, since every phase has modulus

one. The mean contribution is bounded by the triangle inequality once a uniform group-mean estimate is available. The remaining sum retains a varying weight after centring, and requires a separate real-part estimate. The parameters s, η must be fixed before X_0, with the estimates valid for the same X, L; separate successful blocks do not suffice. For $X \geq 4$ and $s = L - h$, the single cofactor group $m_N = 1$, not the full assigned set, consists of the indices for which $N + h + 1$ is prime (Lean source).

An upper bound for the unassigned terms. Only an upper bound is needed here. We enclose the unassigned arguments in a set of smooth numbers and count that larger set. There is no claim that every smooth argument is unassigned.

Proposition 12.6 (A one-sided bound for the unassigned terms)**.** *Fix h, s and choose the admissible depth L minimally for each large X. Put $t = L - s + 1 = O_{h,s}(\log X)$ and $y_X = 4\sqrt{X} + 2t/\sqrt{X}$. If $n = N + t$ is unassigned, then its largest prime factor satisfies $P(n) \leq y_X$. Consequently*

$$\begin{aligned}\#\{N \in [X, 2X) : N \notin \mathcal{A}\} &\leq \Psi(2X + t - 1, y_X) - \Psi(X + t - 1, y_X) \\ &= (1 - \log 2 + o(1))X < \tfrac{8}{25}X\end{aligned}$$

for all sufficiently large X. Here $\Psi(x, y)$ counts the positive integers at most x whose prime factors are all at most y. This is an upper bound for the unassigned count, not an asymptotic equality for that count.

Proof. Let $p = P(n)$ and $m = n/p$. Since $n > 1$, failure of the assignment condition means either $p \leq 2\lfloor\sqrt{X}\rfloor$ or $m > \lfloor\sqrt{X}/2\rfloor$. In the second case $m > \sqrt{X}/2$, so

$$p = \frac{n}{m} < \frac{2(2X + t)}{\sqrt{X}} = y_X.$$

The first case also gives $p \leq y_X$. Counting the resulting smooth integers in $[X + t, 2X + t)$ proves the first inequality.

For completeness, the classical fixed-parameter asymptotic $\Psi(x, x^{1/u}) \sim x\rho(u)$ suffices; no additional short-interval estimate is needed. Indeed, for either $x = X + t - 1$ or $x = 2X + t - 1$, $\log y_X / \log x \to 1/2$. For every fixed $0 < \epsilon < 1$, eventually

$$x^{1/(2+\epsilon)} \leq y_X \leq x^{1/(2-\epsilon)}.$$

Monotonicity in y bounds $\Psi(x, y_X)/x$ between quantities tending to $\rho(2 + \epsilon)$ and $\rho(2 - \epsilon)$. Letting $\epsilon \to 0$ and using continuity gives the limit $\rho(2)$. Subtracting the two endpoint asymptotics yields $(\rho(2) + o(1))X$. Since $\rho(2) = 1 - \log 2 < 8/25$, the claimed bound follows. □

The Lean proof uses no smooth-number asymptotic. For large X, $y_X^2 > 2X + t - 1$, so an integer in $(X + t - 1, 2X + t - 1]$ that is not y_X-smooth has exactly one prime factor above y_X, and

$$\Psi(2X + t - 1, y_X) - \Psi(X + t - 1, y_X) = X - \sum_{y_X < p \leq 2X + t - 1} \left(\left\lfloor \frac{2X + t - 1}{p} \right\rfloor - \left\lfloor \frac{X + t - 1}{p} \right\rfloor \right).$$

Mertens' theorems, proved in Lean with explicit constants, then give $(1-\log 2)X + O(X/\log X)$. The written proof above cites instead the fixed-u asymptotic [20, (1.1), p. 268] and the value $\rho(u) = 1 - \log u$ on $[1,2]$ [20, (1.3), p. 268]. No effective crossover scale or exact density for the unassigned indices is asserted. The estimate uses the minimal admissible depth; it does not supply either weighted phase bound at that depth.

Proposition 12.7 (The excluded-cofactor estimate)**.** *Fix h, s and use the minimal admissible depth L, as in Proposition 12.6; thus $t = L - s + 1 = O_{h,s}(\log X)$. For $\eta \in (0,1)$ let $B(\eta) = \{m \geq 1 : \varphi(m) < \eta m\}$, with natural density $D(\eta)$; by Schoenberg's theorem D exists, is continuous, and $D(0+) = 0$ [21, §17, p. 193], in the framework of [22, Theorem 1, pp. 318–319, and §8, p. 323]. Then*

$$\#\{N \in \mathcal{A} : m_N \in B(\eta)\} \;\leq\; (D(\eta) + o(1))X,$$

so a single choice of η with $D(\eta) < 1/200$ meets the $\frac{1}{100}X$ bound for all large X. This choice fixes η before X_0, as required. It supplies only the excluded-cofactor bound: the mean and mean-subtracted estimates must still hold for this same η, and do not follow from making η smaller.

The Lean proof assumes the prime number theorem. It also takes the existence of the natural density $D(\eta)$ as a hypothesis, which the statement supplies from Schoenberg's theorem.

Proof. For a fixed cofactor $m \leq \sqrt{X}/2$, each assigned index gives a prime p with $X + t \leq mp < 2X + t$. Dropping the other assignment restrictions bounds the number of indices by the number of primes in this interval. The prime number theorem, uniformly for $X/m \geq 2\sqrt{X}$, and $t = O_{h,s}(\log X)$ give the upper bound

$$(2 + o(1))\frac{X}{m \log X}.$$

The possible endpoint contribution is $O(1)$ for each m, hence $o(X)$ after summing over $m \leq \sqrt{X}/2$. Natural density and partial summation give

$$\sum_{\substack{m \leq \sqrt{X}/2 \\ m \in B(\eta)}} \frac{1}{m} = \left(\frac{D(\eta)}{2} + o(1)\right)\log X.$$

Multiplying the two estimates proves the stated bound. If $D(\eta) < 1/200$, its right side is less than $X/100$ for all sufficiently large X. □

The existence of a usable cutoff can also be made explicit by an elementary estimate. For $x \geq 1$,

$$\begin{aligned}\frac{1}{x}\sum_{n \leq x}\frac{n}{\varphi(n)} &\leq \sum_{d \geq 1}\frac{\mu(d)^2}{d\varphi(d)} = \prod_p\left(1 + \frac{1}{p(p-1)}\right) \\ &\leq \exp\left(\sum_{n \geq 2}\frac{1}{n(n-1)}\right) = \exp(1) < 3.\end{aligned}$$

The first step uses $n/\varphi(n) = \sum_{d \mid n}\mu(d)^2/\varphi(d)$ and $\lfloor x/d \rfloor \leq x/d$; the Euler product has nonnegative terms and converges by the displayed majorant. Markov's inequality

gives $\#(B(\eta) \cap [1,x]) < 3\eta x$, hence $D(\eta) \le 3\eta$. In particular, $\eta = 1/1000$ gives $D(\eta) \le 3/1000 < 1/200$, an explicit sufficient choice for the excluded-cofactor bound. This does not establish either weighted phase estimate for that choice, and it is not claimed to be an optimal cutoff. Lean checks the estimate $\#(B(\eta) \cap [1,x]) < 3\eta x$ with no further input (Lean proof) and the excluded-cofactor bound at $\eta = 1/1000$ with no input at all (Lean proof). For that bound Chebyshev's bound replaces the prime number theorem in the proof above: the primes p with $a+1 < p \le 2a+1$ divide $\binom{2a+1}{a} \le 4^a$, so $\#\{p \text{ prime} : a < p \le 2a+1\} \log a \le a \log 4 + \log a$, the fibre bound becomes $(2\log 4 + o(1))X/(m \log X)$, and the count is at most $(3\log 4/1000 + o(1))X$ with $3\log 4/1000 < 0.0042$. The proof uses $\eta = 1/1000$ only through $3\eta \log 4 < 1/100$, so Lean checks the bound with no input for every η with $0 < \eta < 1/(300 \log 4)$ (Lean proof).

The first element of $B(1/5)$ is $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, and $\varphi(30030)/30030 = 5760/30030 \approx 0.191808$. Enumeration through $3 \cdot 10^5$ gives 38 elements of $B(1/5)$ and

$$\frac{1}{\log(3 \cdot 10^5)} \sum_{\substack{m \le 3 \cdot 10^5 \\ m \in B(1/5)}} \frac{1}{m} \approx 2.82 \cdot 10^{-5}.$$

This finite weighted proportion is not the limiting density $D(1/5)$. Using $\eta = 1/5$ in the asymptotic bound would require a rigorous bound for $D(1/5)$, including the integers beyond the enumeration; the finite count does not supply one.

The growing dyadic modulus. The selected offset is $t = L - s + 1$ in (3). The four-term condition fixes s after h but before the threshold X_0, whereas

$$2^L \ge 16(2X + h + L + 2) > 32X, \qquad 2^t > 2^{6-s}X.$$

On a group with cofactor m, the coefficient of $p-1$ in the phase has reduced denominator $2^{\max(t - v_2(\varphi(m)),0)}$, because $2^h - 1$ is odd. Assignment gives $\varphi(m) \le m \le \sqrt{X}/2$. Thus, for fixed s and all sufficiently large X, the reduced denominator is greater than $2^{7-s}\sqrt{X}$. In particular, the assignment conditions do not reduce these moduli to a fixed power of $\log X$. This is the modulus for variation in p, not a lower bound for the denominator of the phase value at each individual prime.

Choosing $t = O(\log \log X)$ would require s to grow with X, contrary to the stated quantifiers. Likewise, the formal observation that the phase is 1 when $v_2(\varphi(m)) \ge t$ cannot provide such groups at large X under these size bounds. The proposed small-modulus prime-distribution argument therefore does not establish the mean estimate. The prime-number-theorem argument in Section 3.3 instead uses prime counts in proportional intervals, not equidistribution in these residue classes. Failure of the small-modulus argument does not disprove the four-term condition.

The remaining prime-weighted correlation. Propositions 12.6 and 12.7 give the unassigned and excluded-cofactor bounds at minimal admissible depth. They must be combined with mean and correlation estimates at the same X, L, s, η; bounds obtained at unrelated depths cannot be substituted.

With the notation of Section 3.2, the correlation estimate still required is

$$\operatorname{Re}\sum_{N\in\mathcal{G}} w_N\left(e\left(\frac{(2^h-1)\varphi(m_N)(p_N-1)}{2^t}\right)-\bar{z}_{m_N}\right)\le\frac{14}{25}X.$$

The sum is over the assigned indices with good cofactors, not all primes and cofactors in the interval. By (3), w_N is the phase with the selected offset t removed.

Regrouping by $m_N = m$ gives weighted sums over the primes $p_N = p$. A bilinear argument would need to separate the m- and p-dependence or estimate the resulting two-variable weight directly. For $N = mp - t$, that weight depends on $\varphi(mp + j - t)$ at the other offsets $1 \le j \le L + h$, $j \ne t$. The identity $\varphi(mp) = \varphi(m)(p - 1)$ controls the selected value, not these neighbouring totients. No bound for the resulting correlation is established here; this is the missing step, not a limitation of every bilinear method.

A fixed-offset example illustrates the correlation at issue. At $h = 1$ and offset $t = 2$, a prime $p \ge 5$ satisfies $\varphi(p - 1) \equiv 0 \pmod 2$ and $\varphi(p) = p - 1$. The phase of R_{p-2} is therefore $\chi_{-4}(p)$ times the phase of the following weighted totients, since $e((p-1)/4) = i^{p-1} = \chi_{-4}(p)$. One possible fixed-saving estimate, with a fixed $\delta > 0$, would be

$$\operatorname{Re}\sum_{X<p\le 2X}\chi_{-4}(p)\,e\Big(\frac{\varphi(p+1)}{8}+\frac{\varphi(p+2)}{16}+\cdots\Big)\ \le\ \Big(\frac{9}{10}-\delta\Big)\,(\pi(2X)-\pi(X)).$$

This fixed-offset example is not Definition 3.4: there s is fixed while $t = L - s + 1$ grows. The example isolates a correlation between a quadratic character at p and the binary phases of totients at $p + 1, p + 2, \ldots$. It is analogous in this respect to shifted correlations studied in the Chowla and Elliott problems, but the analogy does not supply an applicable theorem. The cited theorems have different hypotheses and types of average: Matomäki and Radziwiłł compare short and long averages of real multiplicative functions bounded by 1, outside an exceptional set of short intervals [24, Theorem 1]; Tao proves logarithmically averaged two-point Chowla and a corrected Elliott statement, the latter under an explicit non-pretentiousness hypothesis on one of the bounded multiplicative functions [25, Theorems 1.2–1.3]; and Tao and Teräväinen prove the odd-order logarithmically averaged Chowla conjecture [26, Theorem 1.1]. The phase here depends on φ at a growing window of shifts, so none of these applies directly.

Averaging over the block parameter. For each fixed h, an average of the nonnegative, normalised block norms can establish the full-block condition. It is enough to prove

$$\sum_{k=1}^{K} 2^{-k}\left|\sum_{N=2^k}^{2^{k+1}-1} e(D(h,N,L_k)/2^{L_k})\right| = o(K),$$

where every L_k satisfies the size inequality for $X = 2^k$. Indeed, the number of $k \le K$ for which the normalised norm exceeds $21/25$ is at most $25/21$ times the displayed sum, hence is $o(K)$. There are therefore arbitrarily large admissible blocks satisfying Definition 3.1. This argument does not establish the four-term condition: that condition requires all four separate budgets with common X, L, s, η.

The displayed average is a Cesàro average of norms on dyadic blocks. It is not the signed logarithmically weighted correlation appearing in the cited Chowla–Elliott theorems. The position of the absolute value matters: cancellation between different blocks

would not bound this sum of norms. Invoking entropy decrement would therefore require an additional argument yielding this particular estimate.

The finite phase depends on the $L + h$ totient values at offsets $1, \dots, L + h$, with the coefficients in (3). Although its absolute value is 1, it is not a multiplicative function of N. Both the number of shifts and their dyadic denominators grow with L. The cited fixed-shift correlation theorems do not supply the uniform bounds for this growing family. No such extension is proved here.

12.4 Separation along prime positions

The quantified prime condition.

$$\forall h \geq 1\ \forall N_0\ \exists p \text{ prime}: \quad \max(N_0 + h + 1, h + 5) \leq p\ \wedge\ \mathrm{Re}(e(R_{p-1} - R_{p-h-1})) < \tfrac{9}{10}.$$

This condition is unproved; it implies irrationality (Lean source, Lean source).

Its meaning for the doubling map. $\mathrm{Re}\, e(2^{p-h-1}\alpha_h) < 9/10$ says $\|2^{p-h-1}\alpha_h\|_{\mathbb{R}/\mathbb{Z}} > \arccos(9/10)/2\pi = 0.0717831\ldots$, so the requirement is exactly: *for each h, the indicated prime-indexed $\times 2$ orbit point stays more than $0.0717831\ldots$ from the nearest integer, for cofinally many primes p.* A short-run condition on the binary digits can be a sufficient proxy for this circle-distance inequality, but it is not equivalent to it; a run-length description alone does not determine the residual position inside the dyadic cylinder.

What the prime condition adds. The preceding condition asks for a separated point at a prime-indexed position. The full-block average does not by itself locate such a point on a prescribed sparse set. Conversely, the prime condition does not assert cancellation over the whole interval. They are different sufficient hypotheses; no general comparison of their difficulty is proved here.

Finite comparison. Among the 501 primes in $[1000, 5000)$, the strict inequality holds at 432, 426 and 430 primes for $h = 1, 2, 5$, respectively. The corresponding proportions are 0.8623, 0.8503 and 0.8583 to four decimal places, and each sample has a cosine value below -0.9999. These counts were recomputed using the same interval for S as in Table 6, with no interval meeting the threshold $9/10$. They do not establish separation at arbitrarily large primes.

An average over a prescribed prime sample. The subset implication, Theorem 9.5, allows a nonempty finite set T. If T is chosen after a separated point is known, the subset formulation may add no arithmetic content: the equivalence proved using selected pairs, Proposition 2.7(b), illustrates this issue. To avoid that particular choice, fix the sample in advance by

$$T_{h,X} = \{N \in [X, 2X) : N + h + 1 \text{ is prime}\}.$$

For $X \geq 4$ and $L \geq h$, this is precisely the assigned group with cofactor $m_N = 1$ of §3.2, taken with $s = L - h$ (Lean source).

A sufficient, unproved condition is

$$\begin{aligned}&\forall h \ge 1\ \forall X_0\ \exists X \ge \max(X_0, 4)\ \exists L \ge h :\\ &\qquad T_{h,X} \ne \emptyset, \qquad 16(2X + h + L + 2) \le 2^L,\\ &\qquad \sum_{N \in T_{h,X}} \operatorname{Re} E(h, N, L) \le \frac{9}{10}|T_{h,X}|.\end{aligned} \tag{5}$$

The subset theorem then gives a certificate at some $N \ge X \ge X_0$. Nonemptiness matters: for the empty set the displayed sum inequality would hold automatically and would produce no certificate. The bounds $X \ge 4$ and $L \ge h$ are also the hypotheses of the identification with that cofactor group.

On this sample the selected cofactor is 1, so its coefficient is the odd integer $2^h - 1$ and its modulus is 2^{h+1}, fixed once h is fixed. The unweighted prime average has zero Ramanujan main term, since $c_{2^{h+1}}(2^h - 1) = \mu(2^{h+1}) = 0$. That observation does not establish (5): the prime phase is multiplied by a residual phase built from the other totient values. Their correlation is the missing estimate. Restricting the sample removes the complement and unwanted cofactor terms, but not this residual correlation.

Remark 12.8. Fixing the sample rules out the particular argument that selects a pair only after locating separation. It does not prove that (5) is inequivalent to irrationality, or that it is a strictly stronger condition. No such comparison is established here.

12.5 Depth equal to a period multiple

The quantified residue condition.

$$\text{full-depth condition} :\equiv \forall d \ge 1\ \forall N\ \exists t \ge 1 :\ C(td, N, td),$$

unpacked, $(N + 2td + 2 : \mathbb{Z}) < D(td, N, td) \bmod 2^{td} < 2^{td} - (N + 2td + 2)$. This condition is unproved. It is equivalent to irrationality (Lean source, Lean source); both directions are proved, and the remaining gap is a nonintegral tail difference $R_{N+d} - R_N$ for every $d \ge 1$ and basepoint N.

The condition in terms of binary digits. Writing $h = td$ and using $D(h, N, L)/2^L = (R_{N+h} - R_N) - (R_{N+L+h} - R_{N+L})/2^L$ at $L = h$, together with the proved strip $|R_{M+h} - R_M| < M + h + 2$ (Lean source), applied at $M = N + h$, one gets:

Proposition 12.9. *$C(h, N, h)$ holds whenever $\|2^{N+h}S - 2^N S\|_{\mathbb{R}/\mathbb{Z}} > 2(N + 2h + 2)/2^h$.*

Proof. $\|D(h, N, h)/2^h\| \ge \|R_{N+h} - R_N\| - |R_{N+2h} - R_{N+h}|/2^h > 2(N + 2h + 2)/2^h - (N + 2h + 2)/2^h = (N + 2h + 2)/2^h$, and $R_{N+h} - R_N \equiv 2^N(2^h - 1)S = 2^{N+h}S - 2^N S \pmod 1$ by Lemma 12.1. A residue at distance more than $(N + 2h + 2)/2^h$ from $\mathbb{Z}$ is exactly a residue certificate at depth h. □

Proposition 12.9 gives a sufficient analytic criterion for the condition with depth equal to the shift: for each $d \ge 1$ and $N \ge 0$, it is enough to find $h = td$, $t \ge 1$, for which the displayed distance exceeds the stated threshold. The criterion is not asserted to be equivalent to each individual finite certificate. It involves the phase of S itself, and

the threshold tends to zero exponentially as t grows. The propagation theorem in Appendix A.1 explains why one nonintegral tail difference produces late certificates. It does not supply such tail differences for all d, N.

Comparison with digit complexity. The phase inequality above is a quantitative statement at specified positions and period multiples. A general block-complexity assertion is a different quantified statement, of the kind treated by the Adamczewski–Bugeaud method for algebraic numbers [27, Theorem 1, p. 549]. No implication from such an assertion to the required inequality is established here. In particular, this record does not justify describing one existing method as the only possible route, or algebraicity as the sole missing input. The arithmetic nonintegrality obligation must be stated and proved in its own right.

Finite examples. At $N = 300$, the least t with $C(td, 300, td)$ exists and is small for every $d \le 24$: $t = 10$ for $d = 1$, $t = 5$ for $d = 2$, $t = 4$ for $d = 3$, $t = 3$ for $d = 4$, $t = 2$ for $5 \le d \le 9$, and $t = 1$ for $10 \le d \le 24$; in every case at or near the first depth admitted by the lower bound on the depth. The residues are not marginal: $r/2^h = 0.3594$ at $d = 1$, 0.5977 at $d = 3$, 0.6853 at $d = 7$, 0.2620 at $d = 17$, 0.5887 at $d = 23$, against edge radii $3.1 \cdot 10^{-1}$, $8.0 \cdot 10^{-2}$, $2.0 \cdot 10^{-2}$, $2.6 \cdot 10^{-3}$, $4.1 \cdot 10^{-5}$. This is a finite verification at the displayed basepoint and shifts. It neither bounds the required multiplier for arbitrary d, N nor establishes the universally quantified condition.

A possible heuristic, not an estimate. If a phase U were uniform on $\mathbb{R}/\mathbb{Z}$, then $\Pr(\|U\| \le \epsilon) = \min(1, 2\epsilon)$ for $\epsilon \ge 0$. At $\epsilon = 2(N + 2h + 2)2^{-h}$ this gives the marginal model $\min(1, 4(N + 2h + 2)2^{-h})$. The phases $2^N(2^{td} - 1)S$ for fixed d are arithmetically dependent; no independence model or uniform distribution theorem for this orbit is proved here. Consequently the marginal estimate cannot be multiplied across depths, and no summability conclusion over all basepoints follows from it. The finite experiments are finite evidence; the condition for every d and N remains unproved.

12.6 A rational series with unbounded carry rank

For an integer sequence u, write $\mathrm{rk}_e(u)$ for the rational dimension of the span of all sequences $n \mapsto u_{2^j n + r}$ with $0 \le j \le e$ and $0 \le r < 2^j$. Rationality does not force these ranks to be bounded, even for one fixed coefficient sequence. More precisely, the assumptions

$$0 \le c(n) \le n, \qquad \sum_{n \ge 1} c(n) 2^{-n} \in \mathbb{Q}, \qquad v \in \mathbb{N}_{>0}, \qquad u_n \in \mathbb{Z},$$

$$u_{n+1} = 2u_n - vc(n+1), \qquad u_n/2^n \longrightarrow 0$$

do not imply $\sup_e \mathrm{rk}_e(u) < \infty$. Nor do they imply an upper bound that is $o(2^e)$. Such a bound for the actual totient carry would contradict the lower bound of Proposition 1.6, but the following rational comparison already has that lower bound.

The limit condition fixes the value of the series. Dividing the recurrence by 2^{n+1} and summing gives

$$v\sum_{n=1}^{N} c(n)2^{-n} = u_0 - 2^{-N}u_N.$$

Thus $u_N/2^N \to 0$ identifies the sum as u_0/v; it imposes no finite-dimensional description of the carry sections. In the construction below $u_0 = 5$ and $v = 4$. For fixed c and v this boundary condition also makes the carry unique: two solutions of the recurrence differ by $C2^N$, whose quotient by 2^N tends to C. Uniqueness therefore leaves no freedom to choose a simpler carry. The comparison shows that this unique carry can nevertheless have unbounded section rank.

The comparison with value $5/4$**.** We alter only even coefficients, whose weights are $2^{-2m} = 4^{-m}$. Thus we need a base-four expansion of $5/4 - S$, with signed digits small enough to keep the corrected coefficients between zero and their indices. The digits $-2, -1, 0, 1$ will do this; the first even coefficient will require its own check. The elementary bounds $9/8 \le S \le 3/2$ follow respectively from the first four terms and from $\varphi(n) \le n - 1$ for $n \ge 2$. Put $x_0 = 5/4 - S$ and, for $m \ge 1$, define

$$d_m = \left\lfloor 4x_{m-1} + \frac{2}{3} \right\rfloor, \qquad x_m = 4x_{m-1} - d_m.$$

The interval $[-2/3, 1/3)$ contains x_0 and is preserved by this recurrence, so $d_m \in \{-2, -1, 0, 1\}$. Iteration gives

$$x_0 = \sum_{m=1}^{M} d_m 4^{-m} + 4^{-M}x_M, \qquad |x_M| \le \frac{2}{3}.$$

Consequently $x_0 = \sum_{m\ge1} d_m 4^{-m}$. Define

$$c(0) = 0, \qquad c(2m-1) = \varphi(2m-1), \qquad c(2m) = \varphi(2m) + d_m \quad (m \ge 1).$$

These are nonnegative integers: for $m \ge 2$, $\varphi(2m) \ge 2$; for $m = 1$, the sharper bound $x_0 \ge -1/4$ gives $d_1 \ge -1$. The inequalities $d_m \le 1$ and $\varphi(2m) \le 2m-1$ also give $c(n) \le n$. Thus $|c(n) - \varphi(n)| \le 2$ and

$$\sum_{n\ge1} c(n)2^{-n} = S + \sum_{m\ge1} d_m 4^{-m} = \frac{5}{4}.$$

This is the centred base-four construction in the Lean comparison-series theorem.

The construction has separated value from rank: the even coefficients set the value, while the unchanged odd coefficients force large rank. For the latter, take $v = 4$ and $u_N = 4\sum_{j\ge1} c(N + j)2^{-j}$. The rational sum makes u_N integral; the coefficient bound gives $0 \le u_N \le 4(N + 2)$. For odd r, the carry recurrence and $c(2^j n + r) = \varphi(2^j n + r)$ give

$$4\varphi(2^j n + r) = 2u_{2^j n+r-1} - u_{2^j n+r}.$$

The same $2^e - 1$ independent totient sections therefore lie in the span of the carry sections at levels $1, \dots, e$, proving $\mathrm{rk}_e(u) \ge 2^e - 1$ for every $e \ge 0$. The corresponding formal statement is the unbounded-rank theorem for this comparison. This example is distinct from the bounded $3/2$ parity comparison in Proposition 1.9. It disproves an inference from the listed assumptions alone, not one that also uses identities of the actual totient coefficients.

The rank quantities. Finite-dimensional span of all dyadic sections is the definition of a 2-regular sequence in the sense of Allouche and Shallit [1, Definition 2.1]. For integer-valued sequences this is equivalent to finite generation of the $\mathbb{Z}$-module generated by the sections: if their span has dimension d, some d evaluations embed it in $\mathbb{Q}^d$ and that module in $\mathbb{Z}^d$. Coons [2, Theorem 3.2] proved that φ is not k-regular for any $k \geq 2$. The unconditional Lean theorem (Lean source) proves the case $k = 2$ independently. For integer coefficient and carry sequences, the recurrence transfers non-2-regularity of the coefficients to non-2-regularity of the carry. A proof must include the zero-residue sections, not just the positive residues. The following argument gives an explicit dimension bound.

For a rational-valued sequence f, let $V_e(f)$ be the span of all its sections through levels $0, \dots, e$. Define the backward shift by $(Bf)(0) = 0$ and $(Bf)(n) = f(n-1)$ for $n \geq 1$, and let δ_0 be 1 at 0 and 0 elsewhere. A level-j section of Bf is a section of f if its residue is positive. At residue zero it is the backward shift of $n \mapsto f(2^j n + 2^j - 1)$. Hence

$$V_e(Bf) \subseteq V_e(f) + B(V_e(f)).$$

For integer sequences c, u and a positive integer v, the recurrence $vc(n+1) = 2u(n) - u(n+1)$ gives

$$c = \frac{2Bu - u}{v} + \left(c(0) + \frac{u(0)}{v} \right) \delta_0.$$

Since every section of δ_0 is 0 or δ_0, it follows that

$$V_e(c) \subseteq V_e(u) + B(V_e(u)) + \mathbb{Q}\delta_0, \qquad \dim_{\mathbb{Q}} V_e(c) \leq 2 \dim_{\mathbb{Q}} V_e(u) + 1.$$

In particular, bounded carry ranks would force bounded coefficient ranks. Non-2-regularity of c therefore forces unbounded carry ranks. This is an ordinary proof, not a claim of a new Lean declaration. It also gives the generic bound $\dim_{\mathbb{Q}} V_e(u) \geq 2^{e-1}$ for $c = \varphi$ and $e \geq 1$, using the coefficient rank $2^e + 1$. The factor 2 is not an additive $O(1)$ loss, and this bound concerns all sections, including level zero.

The theorem for φ in Proposition 1.6 is sharper: the carry sections at all residues of levels $1, \dots, e$ have rank at least $2^e - 1$. It uses the independence of the odd-residue totient sections. Neither lower bound is an irrationality proof, since rational coefficient series can have nonregular carries. The separate 5/4 comparison verifies this directly. Any proposed rank upper bound must use additional totient-specific hypotheses not satisfied by the 5/4 comparison sequence.

12.7 Finite observations and quantified conditions

The finite observations can be compared with the quantified conditions, but they do not establish an asymptotic digit distribution. They also supply no basis for ranking possible proof methods.

Comparison of conditions. The following table records the conditions themselves, rather than sufficient digit proxies. Lemma 12.1 explains how their phases are related.

Condition	Required assertion	Indices and quantifiers
Block norm	Norm at most $21X/25$ for the truncated phase sum	All $N \in [X, 2X)$ at one depth; arbitrarily large X for each h
Four terms	The four separate budgets of Definition 3.4	One common X, L, s, η; s, η fixed before X_0
Prime gap	Distance from $\mathbb{Z}$ greater than $\arccos(9/10)/(2\pi)$	The phase $2^{p-h-1}\alpha_h$ at arbitrarily large primes for each h
Full depth	$C(td, N, td)$ for some $t \geq 1$	Every $d \geq 1$ and every $N \geq 0$
Generic rank bound (false)	A uniform or subexponential upper bound for carries of rational coefficient series	All finite dyadic levels; refuted by the $5/4$ comparison

The first four conditions concern the same constant but impose different averaging, sampling and separation requirements. The $11/100$ digit-change test in Corollary 12.3 is sufficient for a real-part bound; it is not a replacement for the norm or four-term rows. Only the full-depth row is proved here to be equivalent to irrationality. The last row records a false generic upper bound, not an additional open hypothesis.

Quantitative estimates. The counterexamples identify missing assumptions in the specific arguments tested here. They do not imply that the binary digits of S are random, normal, or unusually structured. A constructed counterexample can disprove an implication without resembling S in other respects. Likewise, failure of one congruence construction does not exclude every use of the Chinese remainder theorem.

Additional arithmetic information. Table 6 records finite observations: the sampled digit-change proportions are near 0.50, above the sufficient threshold 0.11, and the displayed normalized cosine sums are below 0.89. These computations are compatible with the proposed conditions. They do not establish those conditions for every shift and arbitrarily large blocks, or prove a fair-coin model for the digits. The relevant quantified estimates remain unproved here.

The recorded small margins near a forbidden endpoint also require careful interpretation. One small margin disproves a proposed lower bound larger than that margin at that index. It does not rule out every positive margin on a suitable subsequence, or every estimate with scale-dependent bounds. No numerical census in this record determines which of these possibilities holds asymptotically.

Scope of the comparison. Corollary 12.3 specifies one sufficient digit statistic: for every fixed h, the digit-change proportion must reach at least $11/100$ on arbitrarily large blocks. A statement for one shift, a smaller positive proportion, or finitely many blocks is not this hypothesis. An upper bound $o(N)$ on the longest run alone would not imply irrationality; a periodic alternating expansion has bounded runs. The prime-index con-

dition in (5) is another proposed sufficient estimate. Its exact algebraic decomposition does not establish the required correlation bound.

13 Remaining arithmetic questions

The remaining questions ask for witnesses with unbounded indices. We collect them in residue and exponential-sum form, retaining the quantifiers needed to imply irrationality of $S = \sum_{n\geq 0} \varphi(n)2^{-n}$. Erdős Problem #249 remains open.

This section collects quantified conditions on the scaled tail $R_N = \sum_{j\geq 0} \varphi(N+1+j)2^{-(j+1)}$ that suffice for irrationality. The implication theorems and the arithmetic assumptions have different status: the former are proved in the indicated sources, while the latter are not supplied for the actual totient sequence. Lean source

Each condition is stated with its quantifiers and linked to its formal source. The comparison distinguishes proved equivalences from one-way sufficient conditions. It is not an exhaustive list of possible methods. A finite list of certificates does not stand in for a condition beyond every threshold, and an implication does not establish its antecedent.

Recall that $H_t = H(t) = \operatorname{lcm}(1,\dots,t)$ (Lean source). For nonnegative integers h, N, L, put

$$D(h,N,L) = \sum_{j=0}^{L-1} (\varphi(N+h+1+j) - \varphi(N+1+j))2^{L-1-j}.$$

Lean source The certificate $C(h,N,L)$ is the pair of strict inequalities

$$N+h+L+2 < D(h,N,L) \bmod 2^L < 2^L - (N+h+L+2).$$

Lean source This condition is decidable. Thus the least nonnegative residue must lie more than $N+h+L+2$ from either endpoint. For fixed h, N, this excluded radius grows linearly with L, while its proportion of the modulus tends to zero.

13.1 Certificate completeness

For each fixed $h, N \geq 0$, a certificate at some *unrestricted* depth exists exactly when the tail difference is nonintegral:

$$(\exists L \geq 0,\ C(h,N,L)) \quad \Longleftrightarrow \quad R_{N+h} - R_N \notin \mathbb{Z}.$$

Lean source (The same equivalence is also stated at Lean source.)

This equivalence removes the existential depth from the unrestricted certificate conditions below, including the diagonal and LCM-grid forms. It does not by itself supply a depth bound, a prescribed separation margin, a common depth for a family, or an exponential-sum estimate. Such requirements must be retained unless a separate argument removes them. In particular, completeness does not establish any of the quantified nonintegrality assertions needed for irrationality.

13.2 The original quantified residue condition

The threshold quantifier excludes every possible starting point for a rational tail period.

The quantified residue condition.

$$S \notin \mathbb{Q} \quad \Longleftrightarrow \quad \forall h \geq 1\ \forall N_0\ \exists N \geq N_0\ \exists L, \quad C(h, N, L).$$

Here $C(h, N, L)$ is the finite residue test defined earlier; the universal quantifiers over h and N_0 belong to the displayed assertion, not to that finite test. LEAN SOURCE The equivalence is proved; the quantified existence assertion is not proved.

The reverse direction is by contradiction against the tail-period law: if S were rational, the eventual-period theorem (LEAN SOURCE, unconditional) supplies a period $h > 0$ and pre-period N_0 with $R_{N+h} - R_N \in \mathbb{Z}$ for all $N \geq N_0$; the certificate hypothesis instantiated at that (h, N_0) produces an $N \geq N_0$ where $R_{N+h} - R_N \notin \mathbb{Z}$, a contradiction via nonintegrality from a certificate (LEAN SOURCE).

For the forward direction, irrationality gives pointwise non-integrality of every positive tail shift and certificate completeness supplies a witness already at $N = N_0$.

For fixed h, the required certificate may choose both $N \geq N_0$ and L. It is not required at every N or every depth. Its error radius is $N + h + L + 2$, so for fixed h, N it grows only linearly in L, whereas the modulus is 2^L. The missing assertion is that an admissible residue lies outside both endpoint intervals beyond every threshold, for every positive h. The reformulations below specify other ways to choose such witnesses; none establishes this assertion for the actual totient sequence.

13.3 Multiples, diagonals, and LCM grids

Completeness allows the irrationality question to be expressed using multiples of a period, diagonal tails or an LCM grid. The implications below prove these equivalences. Each formulation permits a different estimate to be attempted, but none supplies the witnesses it requires.

Multiples of a period.

$$\forall h_0 > 0,\ \forall N_0,\ \exists m > 0,\ \exists N \geq N_0,\ \exists L,\ C(m \cdot h_0, N, L) \implies S \notin \mathbb{Q}.$$

LEAN SOURCE

The preceding residue condition implies this one by taking $m = 1$. Conversely, a hypothetical period h_0 also makes every positive multiple mh_0 a period of the tail fractional parts. A certificate for one such multiple contradicts that consequence. The freedom to choose m changes the allowed witnesses, while the quantified assertion remains equivalent to irrationality.

The diagonal.

$$S \notin \mathbb{Q} \iff \forall t_0,\ \exists t \geq t_0,\ \exists L,\ C(H_t, H_t, L).$$

LEAN SOURCE

The two parameters in a hypothetical rational tail period are absorbed by one LCM index. For $t \geq \max(h_0, N_0)$, both $h_0 \mid H_t$ and $H_t \geq t \geq N_0$ hold. A diagonal certificate then contradicts periodicity through the multiple-period argument; the precise intermediate implication is proved in Lean (Lean source). Conversely, irrationality and pointwise completeness give a diagonal certificate already at $t = t_0$.

The finite assertion $P(t) :\equiv \exists L\, C(H_t, H_t, L)$ is verified for every $t \leq 82$ (Lean source). The historical list contained 28 explicit values through $t = 64$, with depths $[6, 5, 7, 7, 9, 14, 15, 14, 21, 22, 23, 26, \dots]$ for $t \in \{1, 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, \dots\}$ (Lean source; endpoint Lean source). These are finite computations. The current contiguous range is still finite and does not establish witnesses at arbitrarily large indices; no certificate at $t = 83$ is claimed here.

The LCM grid.

$$\forall t_0,\ \exists t \geq t_0,\ \exists q, m, L,\ 0 < q \ \wedge\ C(m \cdot H_t,\ q \cdot H_t,\ L) \implies S \notin \mathbb{Q}.$$

Lean source The diagonal choice is $q = m = 1$. Allowing other positive grid points therefore enlarges the choice of finite witnesses. To prove sufficiency, suppose rationality supplies a period h and pre-period N_0. For $t \geq \max(h, N_0)$, all positive multiples of H_t are past the pre-period and their differences are multiples of h. Consequently their tails have the same fractional part: $R_{qH_t + mH_t} - R_{qH_t} \in \mathbb{Z}$ for every $q > 0$ and $m \geq 0$ (Lean source). A certificate for any such difference is a contradiction.

The finite-grid condition of Theorem 6.56 offers a second test. At $H = H_t$, choose a finite nonempty $Q \subseteq \mathbb{N}_{>0}$ with $B_q = qH_t + L + 2 < 2^L$ for every q. It requires

$$\forall q_i \in Q\ \exists q_j \in Q, \qquad B_{q_j} < (A_{q_i} - A_{q_j}) \bmod 2^L, \quad A_q = \sum_{j=1}^{L} \varphi(qH_t + j) 2^{L-j}.$$

For every threshold t_0, such data at some $t \geq t_0$ would imply irrationality. This quantified condition, including the nonemptiness and size assumptions, remains unproved (Lean source). The modulus bound is one-sided at each point; it does not imply a factor-of-two reduction in the number of binary digits.

Equivalent nonintegrality statements Via the completeness iff above, the diagonal and cone forms restate, *exactly*, with no reference to C at all:

$$\forall t_0,\ \exists t \geq t_0,\ R_{2H_t} - R_{H_t} \notin \mathbb{Z} \iff \text{diagonal certificate condition} \implies S \notin \mathbb{Q},$$

Lean source (cone analogue at Lean source)

Thus one can ask directly whether $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$ for arbitrarily large t. Finite instances are known in this record; the universally quantified assertion is not proved here.

13.4 The block exponential-sum condition

A bound for an exponential sum is sufficient for the quantified residue condition. With $e(x) = \exp(2\pi i x)$, the assertion is

$$\text{block norm condition} :\equiv \forall h > 0,\ \forall X_0,\ \exists X, L,\quad \max(X_0, 1) \le X \ \wedge\ 16\,(2X + h + L + 2) \le 2^L \ \wedge\ \Big\| \sum_{N \in [X, 2X)} e(D(h, N, L)/2^L) \Big\| \le \tfrac{21}{25} X.$$

LEAN SOURCE The universally quantified assertion is not proved.

$$\text{block norm condition} \implies S \notin \mathbb{Q}.$$

LEAN SOURCE The threshold is X_0, while X and L are chosen witnesses and may depend on both h and X_0. Taking $X_0 = N_0$ gives a certificate at some $N \in [X, 2X)$, hence at $N \ge N_0$. The elementary reason is that, if every residue failed the test, each would be within $(2X + h + L + 2)/2^L \le 1/16$ of an integer. Every summand would then have real part at least $\cos(\pi/8) > 21/25$, contradicting the norm bound. The implication is proved; the bound on suitable blocks beyond every threshold remains to be established.

The underlying finite lemma is more general: it applies to any nonempty finite $T \subseteq [0, 2X)$ with the stated size bound and $\| \sum_{N \in T} e(D(h, N, L)/2^L) \| \le (21/25)|T|$. Taking $T = [X, 2X)$ gives the full-block implication. Allowing a chosen subset weakens the hypothesis; it does not strengthen it. The Lean proof (LEAN SOURCE) uses the elementary block lemma of Theorem 9.4, which is unconditional as an implication. The real-part version uses $\mathrm{Re} \sum_{N \in T} e(D(h, N, L)/2^L) \le (9/10)|T|$. Its proof uses $\cos(\pi/8) > 9/10$ and averaging. An unspecified saving below $|T|$ is not the stated hypothesis.

What supplying this form would take: genuine cancellation for the additive character $e(D(h, N, L)/2^L)$ at the growing dyadic moduli permitted by the size bound. The quantifiers require suitable blocks beyond every threshold for each fixed $h > 0$; they do not require one threshold that works simultaneously for all h. A determinant identity (LEAN SOURCE) gives a comparison in which every weighted phase equals 1 while the specified determinant remains nonzero. Thus nonvanishing of that determinant alone cannot prove cancellation. A successful use of a determinant would need further information connecting its entries to the phase sum; the comparison does not exclude every such argument.

13.5 The actual tail difference on the LCM diagonal

For $H = H_{2^a}$, the actual difference $\Omega_a = R_{2H} - R_H$ removes the auxiliary rational approximation from the LCM criterion. The equivalence below restricts attention to this sparse sequence of tails. Its required nonintegrality is still an arithmetic assertion to prove.

The exact equivalence.

$$S \notin \mathbb{Q} \iff \forall a_0,\ \exists a \ge a_0,\ \Omega_a \notin \mathbb{Z}.$$

LEAN SOURCE This is an unconditional equivalence, where $\Omega_a := 2^H(2^H - 1)S - (\Phi_{2H} - \Phi_H)$, $H = H_{2^a}$ (LEAN SOURCE). The right-hand side is an equivalent reformulation of irrationality, not a weaker hypothesis already established by the finite examples. The following sufficient conditions are attempts to prove it.

An impossible complete divisibility pattern Let $a \geq 8$, $H = H_{2^a}$ and $J, K, m \in \mathbb{N}$, with

$$m \leq K, \qquad J + K + a + 6 < 2 \cdot 2^a, \qquad B := 2H + J + K + 2 < 2^m.$$

It is impossible that

$$2^{r+1} \mid \delta_{2^a}(J + K - m + r + 1) \quad \text{for every } 0 \leq r < m.$$

Lean source Indeed, the last requirement would make $\delta_{2^a}(J + K)$ a multiple of 2^m, whereas the short-window bounds give $0 < \delta_{2^a}(J + K) < B < 2^m$. The positivity input is

$$\delta_{2^a}(j) > 0 \qquad (a \geq 8,\ 0 < j < 2 \cdot 2^a),$$

Lean source, without a rationality hypothesis.

Omitting only the last divisibility requirement leaves the partial pattern in Proposition 6.86. It additionally requires $\delta_{2^a}(J + K) \leq 2^m - B$. The resulting necessary equality is *penultimate*:

$$\delta_{2^a}(J + K - 1) = 2^{m-1}, \qquad B < 2^m < 2B.$$

Lean source The terminal difference is not pinned to 2^{m-1}. Neither this necessary restriction nor the full-pattern impossibility supplies an unbounded family of successful partial patterns.

The remaining upper-endpoint residue

$$\mathcal{G}_a(J, K, m) :\equiv m \leq K \ \wedge\ 2H + J + K + 2 < 2^m \ \wedge\ D(H, H + J, K) \bmod 2^m \leq 2^m - (2H + J + K + 2).$$

Lean source No unbounded family satisfying this condition is known.

The positivity theorem gives

$$R_{2H+J} - R_{H+J} > 0 \quad \text{if } a \geq 8 \text{ and } J + a + 6 < 2 \cdot 2^a.$$

Lean source For $a \geq 8$, the residue test at depth K uses the stronger bound $J + K + a + 6 < 2 \cdot 2^a$. Under these hypotheses, an integral initial tail difference would give an integer terminal carry $e = R_{2H+J+K} - R_{H+J+K}$ with $0 < e < 2H + J + K + 2$. The tail identity is

$$D(H, H + J, K) = 2^K (R_{2H+J} - R_{H+J}) - e.$$

Hence, when $2H + J + K + 2 < 2^K$, it is the residue of $D(H, H + J, K)$, not of e, that equals $2^K - e$ (Lean source). More generally, the bounds $m \leq K$ and $2H + J + K + 2 < 2^m$ give

$$D(H, H + J, K) \bmod 2^m = 2^m - e > 2^m - (2H + J + K + 2),$$

because 2^m divides 2^K and $0 < e < 2^m$. This contradicts $\mathcal{G}_a(J, K, m)$. Consequently, for $a \geq 8$ and $J + K + a + 6 < 2 \cdot 2^a$,

$$\mathcal{G}_a(J,K,m) \quad \Longrightarrow \quad R_{2H+J} - R_{H+J} \notin \mathbb{Z}.$$

Lean source The one-sided inequality excludes the upper-endpoint interval forced by integrality. Positivity alone does not establish that inequality.

The cofinal target built from it is

$$\text{cofinal upper-endpoint condition} :\equiv \ \forall a_0,\ \exists a,K,m,\quad a_0 \le a \ \wedge\ 8 \le a \ \wedge\ K + (a{+}6) < 2\cdot 2^a \ \wedge\ \mathcal{G}_a(0,K,m).$$

Lean source This condition is unproved; its implication is proved:

$$\text{cofinal upper-endpoint condition} \Longrightarrow S \notin \mathbb{Q}.$$

Lean source

Five sufficient conditions for that residue inequality The next five quantified conditions each imply irrationality. Conditions (i)–(iv) reach the upper-endpoint test; (v) gives nonintegrality directly. Their proven relations are summarised in Proposition 6.65.

(i) For every a_0, there are $a \ge \max(a_0, 8)$ and $m \ge 0$ with $H = H_{2^a}$ such that

$$m + 1 + a + 6 < 2\cdot 2^a, \qquad 2H + m + 3 < 2^m,$$
$$2H + m + 2 \le (D(H,H,m) + \delta_{2^a}(m+1)) \bmod 2^m \le 2^m - (2H + m + 3),$$

where $\delta_{2^a}(j) = \varphi(2H + j) - \varphi(H + j)$. The size condition includes the larger depth $m + 1$, so the conclusion remains valid whichever of the two adjacent depths gives the certificate. This condition is unproved (Lean source); its sufficiency is proved (Lean source, Lean source).

(ii) For every a_0, there is $a \ge \max(a_0, 14)$ with the following property. Put $H = H_{2^a}$ and choose q so that $2q + 1$ is the least odd integer not smaller than $\lfloor \log_2 H \rfloor + 10$. Define integers $W_{a,j}$ recursively by

$$W_{a,0} = 0,$$
$$W_{a,j+1} = 4W_{a,j} + \frac{\delta_{2^a}(2j+3) - 2\delta_{2^a}(2j+2) + \delta_{2^a}(2j+4)}{2}.$$

The increments $\delta_{2^a}(s)$ are even, so the recurrence is integral. The required inequality is

$$H + q + 2 \le W_{a,q} \bmod 4^q \le 4^q - (H + q + 2).$$

At depth $m = 2q + 1$, the residue in (i) equals $2(W_{a,q} \bmod 4^q)$. Dividing its two endpoint bounds by 2 and using integrality gives exactly the threshold $H + q + 2$. This condition is unproved (Lean source). The width is $H + q + 2$ and the depth is the prescribed depth; both must be retained when comparing this with a fixed-fraction band. Its sufficiency is proved (Lean source).

(iii) Use exactly the quantifiers and prescribed depth of (ii). Let $r = W_{a,q} \bmod 4^q$ and define its centred representative by

$$u_{a,q} = \begin{cases} r, & 2r \le 4^q, \\ r - 4^q, & 2r > 4^q. \end{cases}$$

Thus $-4^q/2 < u_{a,q} \le 4^q/2$; when a midpoint occurs, the positive representative is used. The equivalent inequality is

$$H + q + 2 \le |u_{a,q}|.$$

This condition is unproved (Lean source). At the prescribed depth the size bound $2(H+q+2) \le 4^q$ holds. For a residue in $[0, 4^q)$, the band in (ii) is therefore exactly this absolute-value condition. The equivalence, including its quantifiers, is proved (Lean source).

(iv) Allow the odd depth to vary, but retain both size bounds. For every a_0, require $a \ge \max(a_0, 8)$ and $q \in \mathbb{N}$ with

$$2q + 2 + a + 6 < 2 \cdot 2^a, \qquad 2(H + q + 2) \le 4^q,$$
$$H + q + 2 \le |u_{a,q}|, \qquad H = H_{2^a}.$$

This condition is unproved (Lean source). The first bound keeps the adjacent depth inside the short window; the second makes the required residue band nonempty. One implication passes through (i), and another gives the two-sided carry exclusion below (Lean source, Lean source).

(v) Retain the full quantifiers and the two size bounds in (iv), but replace its absolute-value inequality by

$$\delta_{2^a}(2q + 2) \le 2u_{a,q}.$$

This compares the signed centred residue with the terminal totient difference. It is not an unrestricted assertion at an arbitrary q. This condition is unproved (Lean source); its sufficiency is proved (Lean source).

A further sufficient terminal-carry condition With $H = H_{2^a}$, the unproved quantified condition is

$$\begin{aligned} &\forall a_0\ \exists a \ge \max(a_0, 8)\ \exists q : \\ &\quad 2q + 2 + (a + 6) < 2 \cdot 2^a, \qquad 2(H + q + 2) \le 4^q, \\ &\quad 2u_{a,q} \le \delta_{2^a}(2q + 2) - (2H + 2q + 3) \quad \text{or} \quad \delta_{2^a}(2q + 2) \le 2u_{a,q}. \end{aligned}$$

Lean source Either alternative gives nonintegrality at that scale, so the quantified condition implies irrationality (Lean source).

The proof assumes an integer $z = \Omega_a$ and uses

$$2u_{a,q} = \delta_{2^a}(2q+2) - c_{H,H,z}(2q+1).$$

Lean source The carry is then a positive tail difference, strictly smaller than $2H + 2q + 3$. The negative endpoint representative is *minus* this carry (Lean source). The first alternative contradicts the upper bound; the second contradicts positivity, which is the terminal-dominance argument (Lean source).

Both arguments are conditional contradictions, not evidence against either unproved estimate. No converse from nonintegrality to either finite inequality is asserted. In particular, the exact identity must not be described as making this sufficient condition necessary.

Certificates in short windows For every $t, j \in \mathbb{N}$, the arithmetic difference on the LCM progression is exactly

$$\delta_t(j) = \varphi(2H_t + j) - \varphi(H_t + j).$$

No coprimality restriction on j is needed (Lean source). Consequently the corresponding window numerator is

$$D(H_t, H_t, L) = \sum_{j=1}^{L} \delta_t(j) 2^{L-j},$$

and applying the same residue and error bounds gives precisely the test $C(H_t, H_t, L)$ (Lean source). Thus the quantified assertion

$$\forall a_0\ \exists a \geq a_0\ \exists L < 2 \cdot 2^a, \qquad C(H_{2^a}, H_{2^a}, L)$$

implies irrationality (Lean source, Lean source); the assertion itself is unproved. It restricts the diagonal indices to powers of two and imposes the additional bound $L < 2 \cdot 2^a$. Pointwise completeness allows an unspecified finite depth; it does not imply a witness within this short window.

The displayed certified examples have $a = 4, L = 23$ and $a = 6, L = 93$ (Lean source); these are finite computations. The bounded statement of Theorem 9.22 uses the single witness $(a, L) = (6, 93)$ to cover all thresholds $a_0 \leq 6$. It makes no assertion for larger thresholds, or that these are the only successful exponents.

A sufficient approximation bound Put $H = H_{2^a}$ and define the finite rational and its error bound by

$$\rho_{a,q} = \frac{D(H, H, 2q+1) + \delta_{2^a}(2q+2)}{2^{2q+1}}, \qquad \varepsilon_{a,q} = \frac{2H + 2q + 3}{2^{2q+1}}.$$

For every $a, q \in \mathbb{N}$,

$$|\Omega_a - \rho_{a,q}| < \varepsilon_{a,q}.$$

Lean source Let $2q_a + 1$ be the least odd integer not smaller than $\lfloor \log_2 H \rfloor + 10$, as in condition (ii). The sufficient quantified separation condition is

$$\forall a_0\ \exists a \geq \max(2, a_0)\ \forall z \in \mathbb{Z}, \qquad \tfrac{1}{32} + \varepsilon_{a,q_a} \leq |\Omega_a - z|.$$

This condition is unproved; its sufficiency is proved (Lean source). The triangle inequality then gives $|\rho_{a,q_a} - z| > 1/32$ for every integer z. The hypothesis concerns the real tail Ω_a, not merely its computable approximant. The truncation estimate transfers the stated separation, but does not itself establish it. Also, the depth is prescribed by a; it is not a free parameter in this hypothesis.

13.6 An equivalent test using two leading residue bits

Proposition 6.87 gives an equivalent certificate condition: there are $s, b \in \mathbb{N}$ such that either $C(h, N+s, b+1)$ holds, or both

$$N + s + h + b + 4 < 2^b, \qquad 2^b \leq D(h, N+s, b+2) \bmod 2^{b+2} < 3 \cdot 2^b.$$

Lean source The forward proof selects b using the depth L of the first certificate, so this is an encoding of a witness rather than a bound on how far one must search. Existence for the actual totient sequence remains the missing assertion. A use for another coefficient sequence would require its own recurrence and tail bounds.

13.7 Farey growth and the failed generic rank argument

The Farey growth law is an open sufficient hypothesis. The generic rank bound listed beside it is false for rational comparison sequences, and is retained to explain that failed input rather than as another open problem.

The hypothetical rank bound. Unconditionally, for every level $e \geq 1$, the dyadic totient basis of 2^e+1 sequences is linearly independent over $\mathbb{Q}$ (via CRT and Dirichlet's theorem on primes in arithmetic progression) (Lean source). Consequently, if $S \in \mathbb{Q}$, there is an integer $v > 0$ with $vS \in \mathbb{Z}$. Its integral carry $u_N = vR_N$ satisfies, for every $e \geq 0$,

$$\dim_{\mathbb{Q}} \operatorname{span}\{n \mapsto u_{2^j n + r} : 1 \leq j \leq e,\ 0 \leq r < 2^j\} \geq 2^e - 1.$$

It obeys $u_{N+1} = 2u_N - v\varphi(N+1)$ and $0 \leq u_N \leq v(N+2)$. Lean source

A bound independent of e for these same carry-section ranks would contradict the displayed lower bound; so would an upper bound that is $o(2^e)$. Rationality and linear coefficient growth alone do not give either conclusion: the fixed sequence constructed in Section 12.6 has sum $5/4$ and the same rank lower bound. An incompatible upper bound for the actual totient carry would have to use information not shared by that comparison sequence.

The most direct generic construction is also impossible: no integer certificate satisfying the following conditions ($Q \cdot v \cdot A =$ boundary with $|\text{boundary}| < Q \cdot v$ and $A \neq 0$) can exist (Lean source); and the full family is proved infinite-dimensional in span, the case $k = 2$ of Coons's non-regularity theorem (§10.8; Lean source).

Proposition 1.7 gives a common eventual period modulo v for the sections of this same u, together with their unbounded rational ranks. These conclusions concern different properties of the same carry; they supply no rank upper bound.

Growth of the Farey denominator bound. The finite denominator exclusion for S comes from a Farey-gap denominator exclusion at a single fixed window $K = 240$:

$$S \neq a/b \quad \text{for all } a \in \mathbb{Z}, \quad 1 \leq b \leq 796396466467013753233557748758310 53, \quad \gcd(a,b) = 1.$$

Lean source This bound is *sharp* at $K = 240$; the mediant denominator $q = 796396466467013753233557748758310 54$ is the exact first failing denominator (Lean source); so the same interval cannot exclude every rational at the next denominator. A different window requires a new certificate or a uniform argument; the fixed-window calculation supplies neither.

A sufficient extension is a function $g(K) \to \infty$ for which every $(N = 1, K)$ gap check excludes all rationals of reduced denominator at most $g(K)$. More generally, unbounded valid exclusion bounds along a sequence of windows suffice. This is a statement about the certified intervals. The record does not prove it equivalent to a lower bound on the continued-fraction denominators of S. This extension is unproved.

For p prime and $e \geq 1$, a successful unit-gap certificate at denominator p^e has zero candidates or a single candidate divisible by p (Lean source). The success hypothesis is essential: the bound is not a statement about every window at that denominator. Nor does it bound the eventual denominator cutoff. The prime-power restriction cannot be omitted, as the modulo-15 example in Observation 6.114 shows. No obstruction to a uniform estimate as K varies is proved here.

13.8 Summary of implications

For each condition, the second column states its relation to the quantified residue test or to the named intermediate assertion.

Reading the table. "Sufficient" records only the indicated implication. It does not assert a converse or a strict comparison of strength. "Equivalent" records two proved directions, either directly or through the base equivalence with $S \notin \mathbb{Q}$. Logical equivalence does not by itself provide a direct construction from the finite witnesses in one formulation to those in another.

13.9 Remaining estimates

Two explicit estimates for φ remain: weighted cancellation in the exponential sums and a residue inequality involving the final totient difference. Neither is derived here from the cited methods. Other approaches to the irrationality question are not excluded.

Exponential sums. For each fixed $h > 0$ and every threshold X_0, the block norm condition asks for some $X \geq \max(X_0, 1)$ and L satisfying both the modulus bound and

$$\left| \sum_{N \in [X, 2X)} e(D(h, N, L)/2^L) \right| \leq \frac{21}{25} X.$$

The saving $21/25$ is fixed, but the chosen block and depth may depend on h and X_0; uniformity of these choices in h is not required. The preceding four-term decomposition and the proved counting bounds identify contributions to this sum. By Theorem 3.5(b) the two remaining weighted estimates may be replaced by the single bound

$\operatorname{Re}\sum_{N\in\mathcal{G}} E(h,N,L(X)) \le 603X/1000$ at $s = 26$, $\eta = 1/1000$ and minimal depth, for arbitrarily large X; that bound is open. The record therefore gives an exact reduction and some of its estimates; the displayed cancellation bound remains unproved.

The last totient difference. The endpoint criteria compare

$$\delta_{2^a}(2q+2) = \varphi(2H+2q+2) - \varphi(H+2q+2), \qquad H = H_{2^a},$$

with twice the centred residue $u_{a,q}$. An appropriate inequality, proved for arbitrarily large a with the stated bounds on q, would contradict the carry interval required by integrality. The exact identity explains why such an inequality would suffice; it does not supply the inequality. The finite examples through $t = 82$, and the short-window examples at $a = 4, 6$, do not establish an asymptotic pattern.

A Complementary proofs for the short paper

The basis theorem, the integral relation theorem and the bounded residue theorem have complete proofs in the short paper. This appendix supplies the longer arguments behind its discussion of S: propagation from a nonintegral tail, Mersenne residues, rational comparisons and the sharp rank-one bound. Throughout, $K(h,N,L)$ is the residue condition $C(h,N,L)$ defined in the main text.

A.1 Propagation of nonintegral tail differences

Write

$$S=\sum_{n\geq 1}\varphi(n)2^{-n},\qquad R_N=\sum_{j\geq 1}\varphi(N+j)2^{-j},\qquad \Delta_h(N)=R_{N+h}-R_N.$$

The identity

$$2^N S=\sum_{n=1}^{N}\varphi(n)2^{N-n}+R_N$$

has an integer finite sum. Subtracting the identities at $N+h$ and N gives

$$\Delta_h(N)\equiv 2^N(2^h-1)S \pmod{\mathbb{Z}}. \tag{6}$$

In particular, a nonintegral tail difference excludes every rational value whose denominator divides $2^N(2^h-1)$.

The finite test. For $h,N,L\in\mathbb{N}$, let

$$D_{h,N,L}=\sum_{j=0}^{L-1}(\varphi(N+h+1+j)-\varphi(N+1+j))2^{L-1-j},\qquad B=N+h+L+2.$$

All residues below are least nonnegative residues. Write $K(h,N,L)$ for

$$B<D_{h,N,L} \bmod 2^L<2^L-B.$$

The omitted terms form another difference of scaled tails:

$$2^L\Delta_h(N)-D_{h,N,L}=R_{N+h+L}-R_{N+L}.$$

Since $0\leq R_M\leq\sum_{j\geq 1}(M+j)2^{-j}=M+2$, their absolute difference is bounded by $N+h+L+2$. Thus

$$|2^L\Delta_h(N)-D_{h,N,L}|\leq B. \tag{7}$$

If $K(h,N,L)$ holds, the interval $[D_{h,N,L}-B,D_{h,N,L}+B]$ lies strictly between two consecutive multiples of 2^L. It contains $2^L\Delta_h(N)$, so $\Delta_h(N)\notin\mathbb{Z}$. Conversely, write $\|x\|_{\mathbb{R}/\mathbb{Z}}=\min_{z\in\mathbb{Z}}|x-z|$ for the distance to the nearest integer. If $\Delta_h(N)\notin\mathbb{Z}$, choose L so that $2^L\|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}}>2B$. Such a choice exists: h,N are fixed, the distance is positive, and $B=N+h+L+2$ grows only linearly in L. The error estimate then puts the residue of $D_{h,N,L}$ more than B from either endpoint. Thus nonintegrality is equivalent to a certificate at some unrestricted depth. This is a one-sided finite test: success proves nonintegrality, but failure at a particular depth does not prove integrality. The converse guarantees eventual success only after a fixed nonintegral difference has been supplied.

Theorem A.1 (Propagation of one nonintegral tail difference)**.** *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

$$\exists t \geq 1 : \; K(td, N, td) \quad \Longleftrightarrow \quad \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

Unrestricted completeness cannot simply set L equal to the shift: it supplies a depth only after the shift is fixed. The theorem couples them by varying both along multiples of d. The recurrence below prevents two adjacent failures. Its starting hypothesis is still just one nonintegral difference; only the requirement for *every* d, N is equivalent to irrationality.

Proof. Set $F_t = \Delta_{td}(N)$. Equation (6) yields $F_{t+1} \equiv 2^d F_t + F_1 \pmod{\mathbb{Z}}$. This recurrence expresses the fixed F_1 in terms of two adjacent differences. A failed certificate makes one of those differences close to an integer: if $K(td, N, td)$ fails, (7) gives

$$\|F_t\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon_t, \qquad \epsilon_t = 2(N + 2td + 2)2^{-td}.$$

If the test also fails at $t+1$, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ and the recurrence give

$$\|F_1\|_{\mathbb{R}/\mathbb{Z}} \leq \|F_{t+1}\|_{\mathbb{R}/\mathbb{Z}} + 2^d \|F_t\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon_{t+1} + 2^d \epsilon_t.$$

The recurrence has transferred two shrinking errors to the same fixed number F_1. Its distance from $\mathbb{Z}$ is positive, whereas $\epsilon_{t+1} + 2^d \epsilon_t \to 0$; hence adjacent failures are impossible for large t. Conversely, integral F_1 makes every F_t integral by the recurrence, so soundness excludes every certificate.

Finally, (6) is nonintegral for every d, N when S is irrational. If $S = a/(2^c v)$ with v odd, take $N = c$ and $d = \varphi(v)$ to make it integral. □

For instance $D_{1,12,16} = -143140$, with residue 53468 modulo 2^{16} and $B = 31$, verifies the hypothesis at $d = 1, N = 12$. More explicitly,

$$\Delta_1(12) \in \left[\frac{-143171}{65536}, \frac{-143109}{65536} \right] \subset (-3, -2).$$

This enclosure proves nonintegrality without evaluating the infinite tails. The theorem gives infinitely many certificates with depth equal to the shift for this pair, not for every d, N.

A.2 Residues modulo divisors of Mersenne numbers

Fix a hypothetical denominator $2^c v$, with $c \geq 0$ and odd $v \geq 1$. For a positive multiple H of $\varphi(v)$, Euler's theorem makes

$$M = \frac{2^H - 1}{v}, \qquad B_{H,c} = \sum_{j=0}^{H-1} \varphi(c + 1 + j) 2^{H-1-j}$$

an integer modulus and an integer prefix, respectively. The test below asks whether the residue of $-B_{H,c}$ stays away from the endpoints by more than the tail error.

Equivalent formulation A.2 (A residue test modulo a divisor of $2^H - 1$)**.** *The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies*

$$c + H + 1 < (-B_{H,c}) \bmod M < M - (c + H + 1). \tag{8}$$

Proof. The identity

$$B_{H,c} = 2^H R_c - R_{c+H} = M(vR_c) - \Delta_H(c)$$

is exact. For $H \geq 1$, $|\Delta_H(c)| < c + H + 1$. Indeed $R_N \leq N + 1$ for $N \geq 1$ and $0 < R_0 \leq 3/2$; the strict inequalities follow by subtracting a positive tail. If $S = a/(2^c v)$, then $vR_c \in \mathbb{Z}$ and $\Delta_H(c) \in \mathbb{Z}$. The residue of $-B_{H,c}$ is therefore within distance less than $c + H + 1$ of $0 \pmod M$, so (8) fails.

If S is irrational, fix c, v and put $\delta = \operatorname{dist}(vR_c, \mathbb{Z}) > 0$. As H grows through positive multiples of $\varphi(v)$, choose it large enough that $|\Delta_H(c)|/M < \delta/3$ and $(c+H+1)/M < \delta/3$. Since $-B_{H,c}/M = -vR_c + \Delta_H(c)/M$, its distance from $\mathbb{Z}$ is then greater than $2\delta/3$, while the forbidden endpoint width is below $\delta/3$. This proves (8) for all sufficiently large allowed H. □

For fixed c, v, one successful test excludes every value $S = a/(2^c v)$, regardless of the numerator a. Conversely, irrationality gives success for every sufficiently large allowed H for each fixed c, v. This converse does not supply successful tests for every c, v without the irrationality assumption.

A formal declaration for the equivalence is available here, with the modulus written as $vM = 2^H - 1$. The finite residues also satisfy

$$\rho_{H,N+1,M} = (2\rho_{H,N,M} - \varphi(N + H + 1) + \varphi(N + 1)) \bmod M, \quad \rho_{H,N,M} = (-B_{H,N}) \bmod M,$$

which is an exact recurrence for searches directed at (8).

A.3 LCM tails and rational comparisons

We first specialise the tail congruence to LCM shifts, then compare it with rational series preserving selected coefficient data. The source declarations accompany the formulas; references point to the complete constructions and their hypotheses in the preceding sections.

1. **Tail differences at least common multiples.** Let $H_a = \operatorname{lcm}(1, \ldots, 2^a)$. Then S is irrational if and only if, beyond every threshold a_0, some $a \geq a_0$ satisfies $\Delta_{H_a}(H_a) \notin \mathbb{Z}$. This is the exact cofinal nonintegrality criterion for the actual LCM diagonal, not a quantitative separation estimate and not a proof that such indices occur; the forward implication is recorded separately.

2. **Positive tail differences need not be nonintegral.** For $a \geq 8$ and $J + (a+6) < 2 \cdot 2^a$, one has $\Delta_{H_a}(H_a + J) > 0$. Suppose this difference is an integer. If a truncation depth K satisfies

$$J + K + (a + 6) < 2 \cdot 2^a, \qquad 2H_a + J + K + 2 < 2^K,$$

then

$$D_{H_a,H_a+J,K} \bmod 2^K = 2^K - \Delta_{H_a}(H_a + J + K)$$

lies strictly between $2^K - (2H_a + J + K + 2)$ and 2^K. The later tail difference is a positive integer; its *negative* is the representative near zero. Thus positivity determines which endpoint contains the residue, but does not exclude an integral tail difference. Both displayed bounds on K are required for this conclusion. *Source declarations:* positive tail differences, negative endpoint representative, residue near the upper endpoint.

3. **Divisibility and finite linear combinations do not suffice.** A dyadic coboundary is a sequence of the form $2u_j - u_{j+1}$; its weighted sum telescopes. For every $t \geq 3$, with $H = \operatorname{lcm}(1, \ldots, t)$, the linked source constructs a nonzero comparison sequence of this form. Its coefficients reproduce the actual totient differences at $t - 2$ prescribed indices and are divisible by $\varphi(H)$. Every finite integer linear combination of forward shifts preserves the telescoping identity and divisibility by $\varphi(H)$ and the corresponding lower-level factors. The bounds remain uniform in the index, but are multiplied by the sum of the absolute values of the combination's coefficients. A variant preserves sparse prescribed values along entire arithmetic progressions. These properties alone do not force a contradiction: the comparison need not consist of actual totient differences. The theorem concerns linear combinations of shifts, not nonlinear operations, and does not supply certificates at arbitrarily large scales. *Source declarations:* obstruction for finite integer combinations of shifts.

If $S = a/v$ with integers a and $v \geq 1$, the scaled tails $u_N = vR_N$ are integers and satisfy

$$u_{N+1} = 2u_N - v\varphi(N+1), \qquad 0 \leq u_N \leq v(N+2).$$

In particular $u_N/2^N \to 0$. Its dyadic sections $n \mapsto u_{2^j n+r}$ have rational rank at least $2^e - 1$ through level e and a common eventual period modulo v. Both properties are the corresponding theorem for the same carry; they are compatible, and neither supplies a contradiction.

Section 12.6 constructs a rational comparison sequence with $c(n) = \varphi(n)$ for odd n, $|c(n) - \varphi(n)| \leq 2$ for even n, $0 \leq c(n) \leq n$, and

$$\sum_{n \geq 1} c(n)2^{-n} = \frac{5}{4}.$$

This comparison sequence has an integral carry satisfying $u_N/2^N \to 0$, with dyadic rank at least $2^e - 1$ at every level. Thus the same lower bound for a hypothetical totient carry is compatible with rationality. For one fixed prime p and $g : \mathbb{N} \to \mathbb{Z}$, the exact laws $g(pn) = pg(n)$ when $p \mid n$ and $g(pn) = (p-1)g(n)$ otherwise, together with $g - \varphi = o(n)$, force $g = \varphi$: a nonzero defect keeps a nonzero ratio to the argument along $p^j n$. The comparison sequence fails these dilation laws. The unchanged odd-residue sections also give coefficient rank at least $2^e - 1$ at every level. This is a lower bound, not an exact formula for the full coefficient rank.

A.4 The sharp rank-one quotient bound

The quotients below are a particular proposed family of rational approximants to a Möbius-weighted value. We determine their minimum and show that they remain a fixed distance above that value. Put

$$\Theta_2 = \sum_{d\ge1} \frac{\mu(d)}{(2^d-1)^2} = S - \frac{1}{2}, \qquad Q(e,Y) = \frac{\left(\sum_{d=1}^{Y} \mu(d)/(2^d-1)^{e+2}\right)^2}{\sum_{d=1}^{Y} \mu(d)/(2^d-1)^{2e+2}}.$$

The divisor-convolution identity $\varphi = \mu * \mathrm{Id}$ gives $S = \sum_{d\ge1} \mu(d)2^d/(2^d-1)^2$. Now $2^d = (2^d-1)+1$ and $\sum_{d\ge1} \mu(d)/(2^d-1) = 1/2$, the latter following from $\sum_{d\mid n} \mu(d) = 0$ for $n > 1$ and 1 for $n = 1$. These identities give the displayed value of Θ_2; all the sums are absolutely convergent. Amiram Eldar posted the corresponding formula $S = \frac{1}{2} + \sum_{d\ge1} \mu(d)/(2^d-1)^2$, together with its coprimality interpretation, to OEIS A256936 on 15 March 2026 [8], and Steve Fan posted the formula on the problem's forum thread on 16 May 2026 [9]. The result below gives the exact extremum for this family, refining the earlier lower bound $1/480$ in Section 6.6.13.

Theorem A.3 (A lower bound for the rank-one quotients)**.** *For $e \ge 1$ and $Y \ge 4$, the denominator of $Q(e,Y)$ is positive, and the unique minimiser is $(e,Y) = (1,5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1,5)$.

Proof. We first bound every denominator away from zero. We then compare $Q(1,5)$ with the whole range $e \ge 2$ using one uniform estimate, and with $e = 1$ using a seven-term prefix. Put

$$P_Y(r) = \sum_{d=1}^{Y} \frac{\mu(d)}{(2^d-1)^r}, \quad \Theta_r = \sum_{d\ge1} \frac{\mu(d)}{(2^d-1)^r}, \quad E_r(Y) = \frac{2^{-rY}}{1-2^{-r}}.$$

Since $|\mu(d)| \le 1$ and $2^d - 1 \ge 2^{d-1}$,

$$|\Theta_r - P_Y(r)| \le E_r(Y), \qquad |P_Y(r) - P_Z(r)| \le E_r(Z) \quad (Y \ge Z).$$

For $r \ge 4$, separating the terms $d = 1,2$ gives $\Theta_r = 1 - 3^{-r} + T_r$, where $|T_r| \le [2^r(2^r-1)]^{-1}$. Thus

$$\alpha := \frac{6373}{6480} = 1 - \frac{1}{81} - \frac{1}{240} \le \Theta_r < 1, \qquad E_r(Y) \le \varepsilon := \frac{1}{61440} \quad (Y \ge 4).$$

For the strict upper bound, the tail bound is smaller than 3^{-r}: $2^r(2^r-1) > 3^r$ for $r \ge 4$. In particular $P_Y(2e+2) \ge \alpha - \varepsilon > 0$ for every admissible e, Y, proving denominator positivity.

Write $q_* = Q(1,5)$. Using $(\mu(1), \dots, \mu(7)) = (1,-1,-1,0,-1,1,-1)$ gives

$$q_* = \frac{35076077250375200}{37573118933633199}.$$

For $e \geq 2$, both exponents in $Q(e, Y)$ are at least four, so the uniform bounds just proved apply to the numerator and denominator. For $e = 1$, the numerator instead contains $P_Y(3)$; we use the seven-term prefix to bound it. This leaves the two cases $Y = 4, 6$ to compare with $Y = 5$. The four exact rational comparisons are

$$\begin{aligned}
\frac{(\alpha-\varepsilon)^2}{1+\varepsilon} &> q_* && (e \geq 2),\\
\frac{79049881}{84671559} = Q(1,4) &> q_* && (e = 1,\ Y = 4),\\
\frac{1043700953468949889}{1117992059749781919} = Q(1,6) &> q_* && (e = 1,\ Y = 6),\\
\frac{(P_7(3) - 1/1835008)^2}{P_7(4) + 1/251658240} &> q_* && (e = 1,\ Y \geq 7).
\end{aligned}$$

Each comparison follows by multiplication by positive denominators. The first left-hand side bounds every quotient in its range from below. So does the last, by the tail estimates with $Z = 7$; the numerator lower bound is positive before it is squared. The middle two rows give the quotients themselves. These ranges cover every admissible pair except $(1, 5)$, proving its unique minimality.

It remains to compare the minimum with Θ_2. An upper bound for Θ_2 proves the stated gap; a lower bound disproves the stronger proposed gap. The same tail estimate at $r = 2$ gives

$$P_6(2) - \frac{1}{3072} \leq \Theta_2 \leq P_7(2) + \frac{1}{12288}.$$

The exact prefixes are

$$P_6(2) = \frac{3309559}{3814209}, \qquad P_7(2) = \frac{53376062902}{61519376961}.$$

Comparison of fractions gives

$$q_* - \left(P_7(2) + \frac{1}{12288}\right) > \frac{21}{320}, \qquad q_* - \left(P_6(2) - \frac{1}{3072}\right) < \frac{1}{15}.$$

The minimum proves the lower gap for every quotient; the second inequality proves failure of the proposed $1/15$ bound at $(1, 5)$. Multiplying the strict lower gap by positive weights, summing over a nonempty finite set and dividing by the positive total weight proves the assertion for weighted averages. □

This theorem treats averages after the quotients are formed. Signed coefficients, coupling before the quotient, and higher-rank kernels require separate arguments. For rational approximants p_j/q_j in lowest terms, with $q_j > 0$, a sufficient arithmetic target is

$$0 < q_j \left|\Theta_2 - p_j/q_j\right| \longrightarrow 0.$$

The denominator sets the scale of the required error. Convergence alone does not control their product. For the elementary dyadic prefixes, the available bound is only $q_j|S - p_j/q_j| \le N + 2$ at depth N, not a quantity tending to zero. A prescribed approximation family therefore needs a sharper error or reduced-denominator bound. When approximants may be chosen freely, continued-fraction convergents show that the criterion is equivalent to irrationality, just as (8) is. The theorem above excludes the displayed rank-one quotients and their positive averages, not other approximation families.

B Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The links below identify formal declarations at the supplied source revisions. A checked conditional implication retains its hypothesis; the linked source alone does not establish that input for φ. The record gives the available Lean and Comparator status and does not report a new replay or independent review of the printed proofs.

Bases and relations.

reduction of sections with residue zero; reduction of even residues; independence; span equality; basis; independence; dimension; the exact finite-truncation rank; infinite-dimensionality directly; coordinate specification; injective residues; unique representation; level count; section-index count; index as two plus the odd-residue count; full index count; a nonzero evaluation determinant; an invertible minor for affine totient sections; power-multiplication identity; affine cross-gcd identity; nonzero reduction scalar; all-base residue step; canonical spanning; span equality; rank in every base from independence; all-base truncation rank from independence; independence in every integer base; the rank of the retained all-base family; the rank of all sections through a finite level.

Tail identities, residue tests and carries.

the finite-prefix and tail identity; definition; certificates for shifts 1 through 8; nonintegral tail differences at the checked shifts; the integer carry recurrence and its growth bound; residue recurrence; comparison of the two residue conditions; exact equivalence; a certificate in each sufficiently late pair of multipliers; pointwise equivalence; global equivalence; equivalence after setting depth equal to the shift; irrationality and arbitrarily large certificates; the carry-rank lower bound under rationality; carry rank at least $2^e - 1$ from independent totient sections; modular periodicity and the rank lower bound for the same carry; forcing vanishing; geometric reduction; identity; sign; primes; prime powers; unboundedness; exact centre decomposition; explicit error bound; separation criterion; target equivalence; endpoint nonnegativity; foreign-channel identity; five-point loss bound; simultaneous finite family; finite horizon; vanishing block; nonzero block; centrality equivalence; equivalence of the two conditions; conditional

irrationality theorem; finite complement bound; exact channel split; conditional target-miss theorem; the periodic theorem; $\sum \varphi(n)/(2^n-1) = 2$; rationality and integer carries with controlled growth; diagonal split; definition; definition of the scaled tail; the proposed cyclotomic residue condition; the checked equivalence; checked conditional support/period counterexample; the checked binary-layer theorem; certificates with depth equal to the period multiple.

Finite denominator exclusions.

exclusion; reduced-denominator form; first failure; the conditional implication; farey gap; the denominator bound at window 240; denominator exclusion from a finite interval; the earlier denominator bound at window 120; denominator exclusion for the coprimality probability; list; verification; certificates at every scale through 82.

Least common multiples and limits of the tested methods.

certificate completeness; tail-difference integrality; irrationality iff every tail difference is nonintegral; exact frontier; forward implication; positive tail differences; negative endpoint representative; residue near the upper endpoint; divisibility preserved by finite combinations of shifts; the counterexample with sparse prescribed values; pointwise; cofinal at each shift; lcm diagonal; the certificate at $t = 2^4$ and depth 23; the certificate at $t = 2^6$ and depth 93; actual-orbit non-integrality at both heights; certificates beyond every threshold $a_0 \le 6$; an explicit approximation error; the approximation as an integer over 4^q; distance at least $1/32$ from every integer; the cofinal separation hypothesis; the conditional endpoint theorem; the staircase is impossible under the stated room hypothesis; terminal-remainder identity; the limitation of fixed-precision valuation data; rationality from an integral tail difference.

Möbius identities and rational approximation.

unique minimizer; uniform lower bound $21/320$; lower bound $1/16$; $1/15$ failure; positive weighted averages of the quotients; rational linear-form obstruction; divisor-count layer; first gcd moment; no fixed index; the primorial divisibility; irrationality of the Erdős–Borwein series; $\sum \mu(n)/(2^n-1) = 1/2$; the squared-Mersenne expression for the totient series; the coprimality-probability expression; the Lambert-weighted sum over coprime positive pairs; the Stern–Brocot recursion with stopping; $s = 1$; $s = 2$; first difference; second difference.

Proposed exponential-sum and prime-factor estimates.

window-separated pairs; the unproved quantitative separation condition at primes; a positive adaptive truncation budget; S is irrational; density hypothesis; block-gap hypothesis; density-to-gap implication; exact squaring recurrence; iterate formula; initial-phase normal form; initial-phase equivalence; dyadic-root obstruction; prime-alignment implication; mixed difference; separable case; fixed-stencil uniqueness; a four-point divisor-layer identity; a centred representative for a residue class modulo an even modulus; eventual nontriviality and coprimality of the layers; order divisibility with bounded extension degree; exclusion of any fixed finite prime support; unbounded prime divisors; membership; injectivity; image equality; checked counterexample; the exact decomposition into four prime-factor contributions; the conditional irrationality

theorem; the four inequalities on a common block; the first-harmonic bound from those four inequalities.

Declarations

Further background. Erick Wong posted a proof that $\sum_{n\geq 1}(\varphi(n) \bmod k)k^{-n}$ is irrational for every integer $k > 2$ on Mathematics Stack Exchange on 29 March 2015 [5]; there the base equals the modulus. Erdős's 1957 criterion already gives every base (Remark 1.13). The Möbius-square form of S and its coprimality interpretation were posted publicly before this record, as credited in §5.4. Kovač and Tao show that suitable subsums of Lambert series in distinct integer bases can be rational [18, Theorem 2.3]; the freedom to choose those subsums is absent for S. The Lambert-series factorisations of Merca [17, Theorem 1.2] and of Merca and Schmidt [16] are transform background for the Möbius–Mersenne coordinate. Balasubramanian, Giri and Srivastav prove asymptotics for finite shifted convolutions of divisor-convolution functions, including $\varphi(n)/n$, uniformly for $|h| \leq x/2$ at cutoff x [19, Theorem 2.2]. Their partial-summation extension uses differentiable weights with bounded derivative on the summation interval [19, Rem. 2.9]. Applying that observation here would require control of the weighted error for the present phases; no such estimate is established in this record. In the totient specialisation the divisor-convolution kernel is $\mu(d)/d$ and both decay exponents in their theorem are 1, giving an $O(\log^2 x)$ error for the two-point product. The present quantity is instead an exponential of a window of overlapping totient values, with a denominator growing with L. Partial summation would carry the variation of its weight into the error term; it does not remove the need to bound that variation and the resulting weighted error. The cited results are not themselves the dyadic residue separation required by Sep. The remaining sources are cited where their hypotheses are used.

At the generating-function level, the published height criterion of Adamczewski, Bell and Smertnig gives k-regularity for a k-Mahler series with integer coefficients of polynomial growth [35, Thm. 1.2(b), p. 2528]. For the totient coefficients the logarithmic heights are $\log\max(1, \varphi(n)) \leq \log n$ for $n \geq 1$, so Coons's nonregularity already rules out k-Mahler equations in every base. The later multiplicative classification of Bell and Smertnig provides another route [32, Thm. 1.3 and p. 3]. The algebraic and D-finite classifications of Bell, Bruin and Coons belong to the same function-level context [33, Thms. 1.5–1.6]. They do not by themselves decide the value at $z = 1/2$.

Automatic prime-sampling theorems address a different issue. Adamczewski, Drmota and Müllner identify a computable local coprimality condition governing the relevant logarithmic densities, with a conditional ordinary-density transfer [34, Thm. 1.4 and Rem. 1.5]. Their theorem does not assert that every output state has positive density on a prescribed prime progression, nor does it prove independence with automatic coefficient weights. Any use for that purpose needs its own compatibility argument.

Sources and data. The evidence record lists every Lean declaration, version and check behind the margin marks.

Verification scope. Kernel checking validates a formal proof of its stated proposition relative to the declared axioms. It does not check that every prose summary has the same hypotheses or verify the literature comparisons. In particular, a checked implication with an unproved hypothesis is not a proof that Problem 249 is solved.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.3	Lean · Comparator	Proposition 1.4	Lean · Comparator
Proposition 1.5	Lean · Comparator	Proposition 1.6	Lean · Comparator
Proposition 1.7	Lean · Comparator	Proposition 1.8	Lean · Comparator
Proposition 1.9	Lean · Comparator	Proposition 1.10	Lean · Comparator pending
Proposition 1.11	Lean · Comparator pending	Remark 12, p. 8	Verification record
Lemma 2.1	Lean · Comparator	Lemma 2.2	Lean · Comparator
Theorem 2.4	Lean · Comparator	Corollary 2.5	Lean · Comparator
Proposition 2.7	Lean · Comparator	Proposition 2.9	Lean · Comparator
Proposition 2.11	Lean · Comparator	Proposition 2.12	Lean · Comparator
Proposition 2.13	Lean · Comparator	Theorem 3.5	Lean · Comparator pending
Proposition 5.3	Lean · Comparator	Lemma 5.4	Lean · Comparator
Proposition 5.5	Lean · Comparator	Theorem 5.6	Lean · Comparator
Proposition 5.9	Lean · Comparator	Proposition 5.10	Lean · Comparator
Theorem 5.11	Lean · Comparator	Theorem 5.12	Lean · Comparator
Proposition 5.14	Lean · Comparator	Proposition 5.16	Lean · Comparator
Proposition 5.17	Lean†	Proposition 5.18	Lean · Comparator
Theorem 6.1	Lean · Comparator	Theorem 6.2	Lean · Comparator
Theorem 6.3	Lean · Comparator	Proposition 6.4	Lean · Comparator
Proposition 6.5	Lean · Comparator	Proposition 6.6	Lean · Comparator
Proposition 6.7	Lean · Comparator	Proposition 6.8	Lean · Comparator
Proposition 6.9	Lean · Comparator	Proposition 6.10	Lean · Comparator
Proposition 6.11	Lean · Comparator	Proposition 6.12	Lean · Comparator
Proposition 6.16	Lean · Comparator	Lemma 6.17	Lean · Comparator

Statement	Formal proof	Statement	Formal proof
Proposition 6.18	Lean · Comparator	Theorem 6.19	Lean · Comparator
Proposition 6.21	Lean · Comparator	Proposition 6.22	Lean · Comparator
Proposition 6.23	Lean · Comparator	Proposition 6.24	Lean · Comparator
Proposition 6.25	Lean · Comparator	Proposition 6.26	Lean · Comparator
Corollary 6.27	Lean · Comparator	Proposition 6.28	Lean · Comparator
Proposition 6.30	Lean[†]	Proposition 6.31	Lean · Comparator
Proposition 6.33	Lean · Comparator	Proposition 6.34	Lean · Comparator
Proposition 6.35	Lean · Comparator	Proposition 6.36	Lean · Comparator
Proposition 6.37	Lean · Comparator	Proposition 6.38	Lean · Comparator
Proposition 6.39	Lean · Comparator	Proposition 6.40	Lean · Comparator
Proposition 6.41	Lean · Comparator	Proposition 6.42	Lean · Comparator
Proposition 6.43	Lean · Comparator	Proposition 6.45	Lean · Comparator
Proposition 6.46	Lean · Comparator	Theorem 6.47	Lean · Comparator
Theorem 6.48	Lean · Comparator	Theorem 6.49	Lean · Comparator
Lemma 6.50	Lean · Comparator	Proposition 6.51	Lean · Comparator
Theorem 6.52	Lean · Comparator	Theorem 6.53	Lean · Comparator
Corollary 6.54	Lean · Comparator	Proposition 6.55	Lean · Comparator
Theorem 6.56	Lean · Comparator	Theorem 6.57	Lean · Comparator
Proposition 6.58	Lean · Comparator	Theorem 6.59	Lean · Comparator
Theorem 6.60	Lean · Comparator	Proposition 6.61	Lean · Comparator
Proposition 6.62	Lean · Comparator	Proposition 6.63	Lean · Comparator
Proposition 6.64	Lean · Comparator	Proposition 6.65	Lean · Comparator
Proposition 6.66	Lean · Comparator	Proposition 6.67	Lean · Comparator
Proposition 6.68	Lean · Comparator	Proposition 6.69	Lean · Comparator
Proposition 6.70	Lean · Comparator	Proposition 6.71	Lean · Comparator
Proposition 6.72	Lean · Comparator	Proposition 6.73	Lean · Comparator
Proposition 6.74	Lean · Comparator	Proposition 6.75	Lean · Comparator
Proposition 6.76	Lean · Comparator	Proposition 6.77	Lean · Comparator
Proposition 6.78	Lean · Comparator	Proposition 6.79	Lean · Comparator
Proposition 6.80	Lean · Comparator	Proposition 6.81	Lean · Comparator
Proposition 6.82	Lean · Comparator	Proposition 6.83	Lean · Comparator
Proposition 6.84	Lean · Comparator	Proposition 6.85	Lean · Comparator
Proposition 6.86	Lean · Comparator	Proposition 6.87	Lean · Comparator
Proposition 6.88	Lean · Comparator	Proposition 6.89	Lean · Comparator
Proposition 6.90	Lean · Comparator	Proposition 6.91	Lean · Comparator
Proposition 6.92	Lean · Comparator	Proposition 6.93	Lean · Comparator
Proposition 6.94	Lean · Comparator	Proposition 6.95	Lean · Comparator

Statement	Formal proof	Statement	Formal proof
Proposition 6.96	Lean · Comparator	Proposition 6.97	Lean · Comparator
Proposition 6.98	Lean · Comparator	Proposition 6.99	Lean · Comparator
Proposition 6.100	Lean · Comparator	Proposition 6.101	Lean · Comparator
Proposition 6.102	Lean[†]	Proposition 6.103	Lean · Comparator
Proposition 6.104	Lean · Comparator	Proposition 6.105	Lean · Comparator
Theorem 6.121	Lean · Comparator	Theorem 6.122	Lean · Comparator
Theorem 6.123	Lean · Comparator	Theorem 6.125	Lean · Comparator
Proposition 6.126	Lean · Comparator	Proposition 6.128	Lean · Comparator
Proposition 6.129	Lean · Comparator	Proposition 6.131	Lean · Comparator
Proposition 6.132	Lean · Comparator	Theorem 6.133	Lean · Comparator
Proposition 6.135	Lean · Comparator	Theorem 6.136	Lean · Comparator
Corollary 6.137	Lean · Comparator	Proposition 6.139	Lean · Comparator
Proposition 6.140	Lean · Comparator	Lemma 6.141	Lean · Comparator
Proposition 6.142	Lean · Comparator	Proposition 6.143	Lean · Comparator
Theorem 6.144	Lean · Comparator	Proposition 6.145	Lean · Comparator
Theorem 6.146	Lean · Comparator	Theorem 6.149	Lean · Comparator
Theorem 6.151	Lean · Comparator	Theorem 6.152	Lean · Comparator
Theorem 6.155	Lean · Comparator	Theorem 6.156	Lean · Comparator
Proposition 6.157	Lean · Comparator	Theorem 6.159	Lean · Comparator
Theorem 9.4	Lean · Comparator	Theorem 9.5	Lean · Comparator
Theorem 9.6	Lean · Comparator	Theorem 9.10	Lean · Comparator
Theorem 9.12	Lean · Comparator	Proposition 9.14	Lean · Comparator
Theorem 9.16	Lean · Comparator	Proposition 9.18	Lean · Comparator
Theorem 9.19	Lean · Comparator	Theorem 9.22	Lean · Comparator
Theorem 9.24	Lean · Comparator	Proposition 9.26	Lean · Comparator
Theorem 9.27	Lean · Comparator	Proposition 9.29	Lean · Comparator
Theorem 9.30	Lean · Comparator	Proposition 9.33	Lean · Comparator
Theorem 9.34	Lean · Comparator	Proposition 9.36	Lean · Comparator
Theorem 9.37	Lean · Comparator	Proposition 9.40	Lean · Comparator
Theorem 9.41	Lean · Comparator	Theorem 9.42	Lean · Comparator
Theorem 9.44	Lean · Comparator	Proposition 9.46	Lean · Comparator
Theorem 9.47	Lean · Comparator	Theorem 9.48	Lean · Comparator
Proposition 9.50	Lean · Comparator	Proposition 10.1	Lean · Comparator
Lemma 10.2	Lean · Comparator	Lemma 10.3	Lean · Comparator
Proposition 10.4	Lean · Comparator	Proposition 10.5	Lean · Comparator
Proposition 10.6	Lean · Comparator	Theorem 10.7	Lean · Comparator
Remark 273, p. 112	Verification record	Remark 274, p. 116	Verification record

Statement	Formal proof	Statement	Formal proof
Remark 275, p. 117	Verification record	Lemma 12.1	Lean · Comparator
Proposition 12.2	Lean · Comparator	Corollary 12.3	Lean · Comparator
Theorem 12.5	Lean · Comparator	Proposition 12.6	Lean · Comparator
Proposition 12.7	Lean[†]	Proposition 12.9	Lean · Comparator
Theorem A.1	Lean · Comparator	Theorem A.3	Lean · Comparator